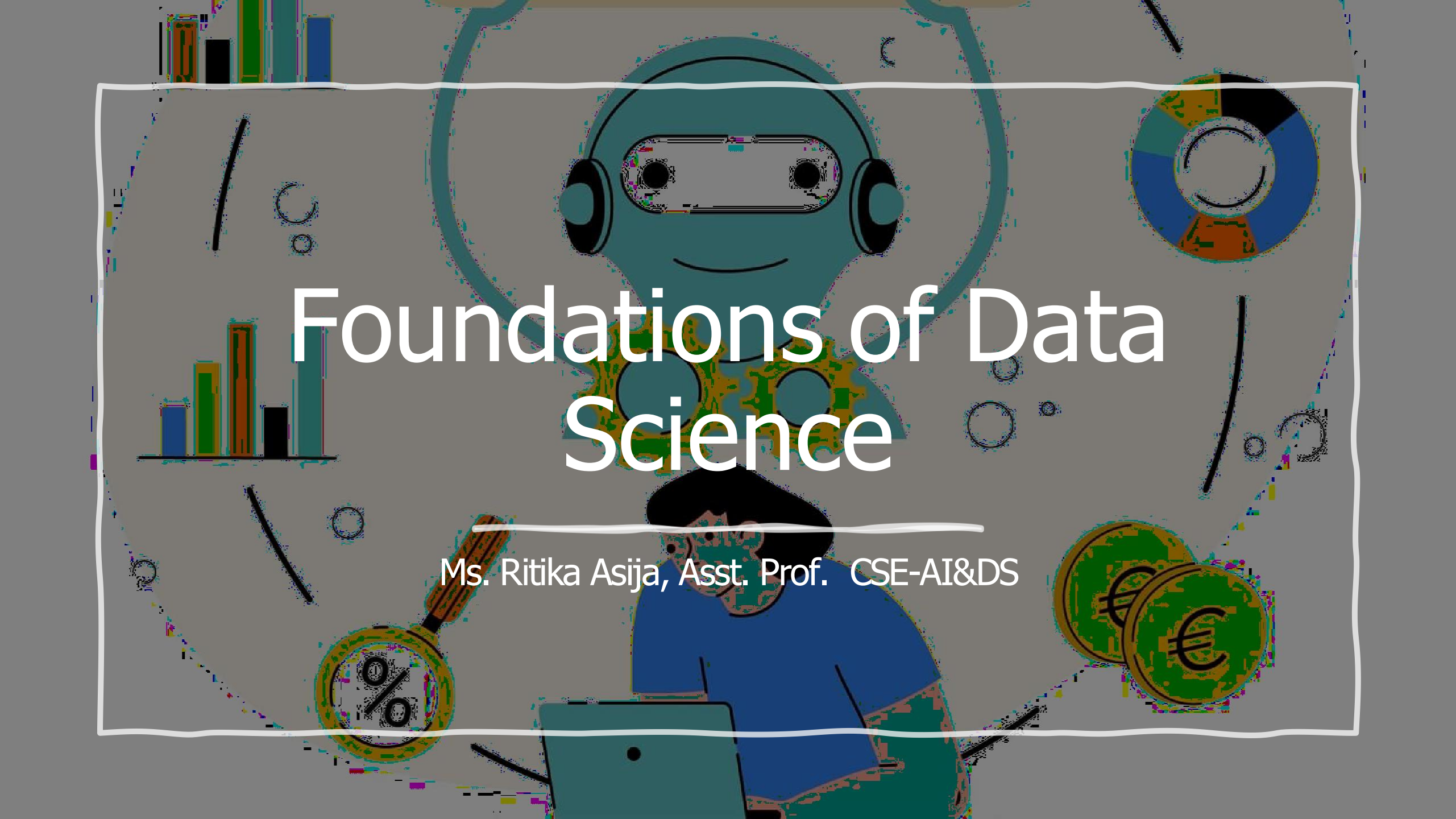


Foundations of Data Science

Ms. Ritika Asija, Asst. Prof. CSE-AI&DS

SYLLABUS

Course Code	Course Title	L	T	P	TW/SL	Credits
BT-AIDS-205A	Foundations of Data Science	45	-	-	45	3
		CIE	SEE			Total
		40	60			100

Course Outcomes: After undergoing the course, Students will be able to	
CO1	Understand the fundamentals and processes of data science.
CO2	Learn to describe data effectively within the data science workflow.
CO3	Explore relationships between different data variables.
CO4	Utilize Python libraries for data wrangling tasks.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	-	-	-	-	-	-	-	1	3	2
CO2	3	3	3	2	-	-	-	-	-	-	-	1	3	2
CO3	2	1	2	1	-	-	-	-	-	-	-	1	3	2
CO4	3	1	3	1	-	-	-	-	-	-	-	2	2	3

COURSE CURRICULUM

Course Outline

Unit: I	Introduction To Data Science	Contact Hours:12

Benefits and uses of Data Science, Facets of data, Overview of the Data Science Process, Defining research goals, Data retrieval and preparation, Exploratory Data Analysis (EDA), Model building and application development Data Mining and Data Warehousing.

Unit: II	Describing Data	Contact Hours:10

Types of data and variables, Descriptive statistics: tables, graphs, averages, Measures of variability, Normal distributions and z-scores, measure of dispersion, Data distribution and shape, Frequency Distributions and Tabulation.

Unit: III	Relationships Descriptions	Contact Hours:12

Correlation and scatter plots, Computational formula for correlation coefficient, Regression analysis: regression lines, least squares method, Standard error of estimate, Interpretation of R^2 (coefficient of determination), Multiple regression equations, Regression towards the mean.

Unit: IV	Python Libraries for Data Wrangling	Contact Hours:11

Basics of NumPy arrays, Aggregations and computations on arrays, Comparisons, masks, and Boolean logic, Fancy indexing and structured arrays, Data manipulation with Pandas: indexing, selection, operations, handling missing data Hierarchical indexing, combining datasets, Aggregation, grouping, and pivot tables.

Term Work and Self-Learning (TW/SL):
Total TW/SL Hours = 45 Hours
1.Assignments / Coding-Based Assignments (Total Hours: 20 Hours)
Students must complete four (04) Assignments/Coding-Based Assignments, each of four (05) hours.
2. Optional Activity (Any One) (Total Hours: 10 Hours)
a) Technical Video-Based Learning

B.Tech. CSE - AI & DS (Hons.): Approved in 2nd meeting of Academic Council held on 28.06.2025

Dr. B.K. VERMA
HOD, CSE-AIDS
PIET, SAMALKHA

[Video Learning: 5 Hours + Report Writing: 5 Hours]
Students shall watch technical videos related to the course from platforms such as **upGrad, Coursera, edX, MTutor**, or equivalent platforms and submit a learning report.
OR
b) Industry/Research Laboratory Visit
[Visit: 5 Hours + Report Preparation: 5 Hours]
Students shall visit an industry, startup, incubation center, or research laboratory and submit a report on observations and learning outcomes.
OR
c) Case Study-Based Learning
[Analysis: 5 Hours + Report Preparation: 5 Hours]
Students shall analyze a real-world case study **or** develop a mini-application, simulation, dashboard, or software prototype based on the course concepts and submit a report.
3. Skill Enhancement Activity (Total Hours: 7 Hours)
a) Poster / PowerPoint Preparation
[Content Preparation: 5 Hours, Presentation/Demonstration: 2 Hours]
Students shall prepare a Poster or PowerPoint Presentation on an emerging topic related to the course.
OR
b) Expert Lecture/Webinar Learning:
[Attendance: 2 Hours, Report Preparation: 5 Hours]

Students shall attend a technical expert lecture or webinar related to the course and submit a detailed report highlighting the key concepts, learning outcomes, and practical applications.
. Assessment Activity (Total Hours: 8 Hours)
Unit-wise Quizzes / Complex Problem Solving
Students shall complete **two quizzes from each unit of 1 hour each** to evaluate conceptual understanding.
OR
Students shall identify and solve **one complex real-world problem** related to the course and submit a detailed solution report.

Textbook/s:

1. Foundations of Data Science — Foundations of Data Science, Year: 2020, ISBN: 978-1108485067.
2. Foundations of Data Science with Python — Foundations of Data Science with Python, Year: 2024, ISBN: 978-1003838548.

Reference Books:

1. Foundations of Statistics for Data Scientists (With R and Python) — Foundations of Statistics for Data Scientists, Year: 2024, ISBN: 978-0367748456.
2. Statistical Foundations of Data Science — Statistical Foundations of Data Science, Year: 2021, ISBN: 978-1466510845.

Online Learning Resources/URLs:

1. https://www.coursera.org/browse/data-science
2. nptel.ac.in/courses/111106525

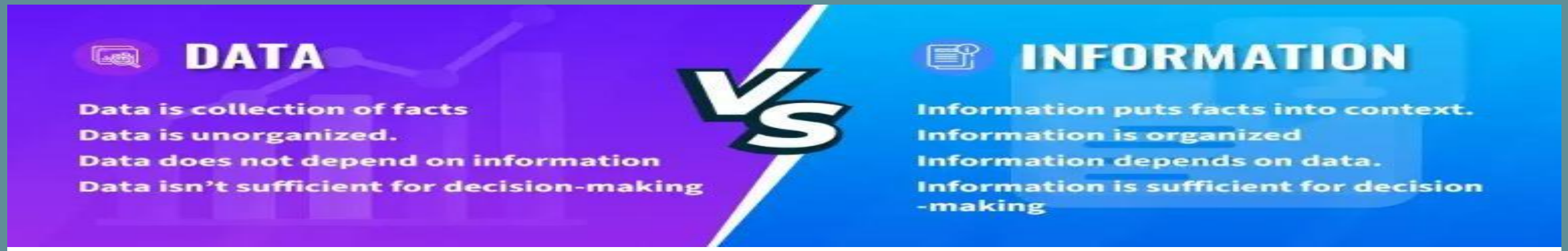
Data

- **Data is a collection** of facts, numbers, words, observations or other useful information. Through data processing and data analysis, organizations transform raw data points into valuable insights that improve decision-making and drive better business outcomes.

Data

There are two types of Data:

- **Quantitative:** Quantitative data refers to numerical information like weight, height, etc.
- **Qualitative:** Qualitative data refers to non-numeric information like opinions, perceptions, etc.
- **Information:** Information is data that has been processed, organized, or structured in a way that makes it meaningful, valuable, and useful.
- It is data that has been given context, relevance, and purpose

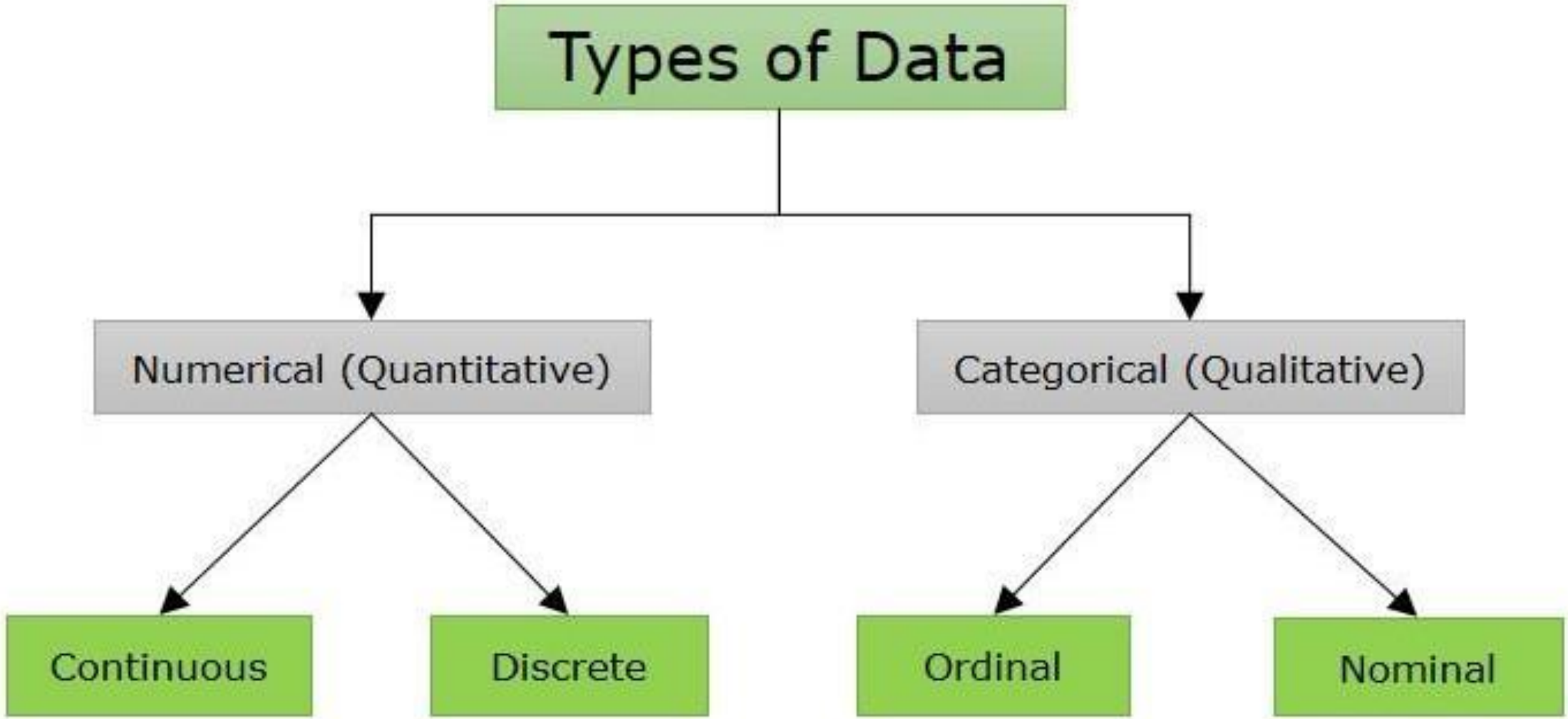


Data Example

- **Temperature Readings: Numbers representing temperature throughout the day, such as** "72°F", "68°F", "75°F"**.**
- **Student Grades**: A list of numerical scores obtained by students on a test, like "85", "92", "78".
- **Stock Prices**: Daily closing prices of a company's stock, such as "$50.25", "$48.90", "$52.10".

Information Example

- **Weather Report:** Based on the temperature reading a weather report can be generated.
- **Grade Average:** Based on the student grades, the average grade of class can be derived.
- **Market Analysis:** The stock market showed a slight increase today derived from stock prices.


Types of Data
Numerical (Quantitative)
Categorical (Qualitative)
Continuous
Discrete
Ordinal
Nominal

Types of Data

- ***Qualitative (*Categorial Data):** In categorical data, the data has a defined category.

For example:
  - Marital Status
  - Political Party
  - Eye colour

- **Nominal Scale:** A nominal scale classifies data into several distinct categories in which no ranking criteria are implied.

For example: Gender, Marital Status.

- ***Quantitative(*Numerical Data) *:***

- **Numerical Data:** Numerical data can further be classified into two categories:
  - **Discrete Data:** Discrete data contains the data which have discrete numerical values

    For example: Number of Children, Defects per Hour, etc.
  - **Continuous Data:** Continuous data contains the data that have continuous numerical values.

    For example: Weight, Voltage etc.

Some of the most common types of data include:

- Quantitative data/ **Numerical Data**
- Qualitative data/ **Categorial Data**
- Structured data
- Unstructured data
- Semi-structured data
- Metadata
- Big data

Quantitative Data

- Quantitative data consists of values **that can be measured numerically**. Examples of quantitative data include discrete data points (such as the number of products sold) or continuous data points (such as temperature or revenue figures).
- **Quantitative data is often structured**, making it easy to analyze using mathematical tools and algorithms.
- Common use cases of **quantitative data include trend forecasting, statistical analysis, budgeting, pattern identification and performance measurement**.

Qualitative data

- **Qualitative data is descriptive and non-numerical,** capturing characteristics, concepts or experiences that numbers cannot measure. Examples include customer feedback, product reviews and social media comments.
- Qualitative data can be **structured** (such as coded survey responses) or **unstructured** (such as free-text responses or interview transcripts).
- Common use cases for qualitative data include **understanding customer behavior, market trends and user experiences.**

Data Never Sleeps 9.0

How much data is generated *every minute*?

The 2020 pandemic upended everything, from how we engage with each other to how we engage with brands and the digital world. At the same time, it transformed how we eat, how we work and how we entertain ourselves. Data never sleeps and it shows no signs of slowing down. In our 9th edition of the "Data Never Sleeps" infographic, we bring you a glimpse of how much data is created every digital minute in our increasingly data-driven world.



As of July 2021, the internet reaches 65% of the world's population and now represents 5.17 billion people—a 10% increase from January 2021. Of this total, 92.6 percent accessed the internet via mobile devices. According to Statista, the total amount of data consumed globally in 2021 was 79 zettabytes, an annual number projected to grow to over 180 zettabytes by 2025.

Global Internet Population Growth

(IN BILLIONS)



As the world changes, businesses need to change too—and that requires data. Domo gives you the power to make data-driven decisions at any moment, on any device, so that you can make smart choices in a rapidly changing world. Every click, swipe, share, or like tells you something about your customers and what they want, and Domo is here to help you and your business make sense of all of it.

Learn more at domo.com

SOURCES: LOCAL IQ, BUSINESS OF APPS, DUSTIN STOUT, HOOTSUITE, EXPANDED RAMBLINGS, INTERNET WORLD STATS, STATISTA, CNBC, BRANDWATCH, KILL THE CABLE BILL, YOUTUBE, KINSTA, THE VERGE, MANAGEMENT COMMUNICATIONS: A CASE ANALYSIS APPROACH, INTERNET LIVE STATS, SODA, STATISTA

Big Data

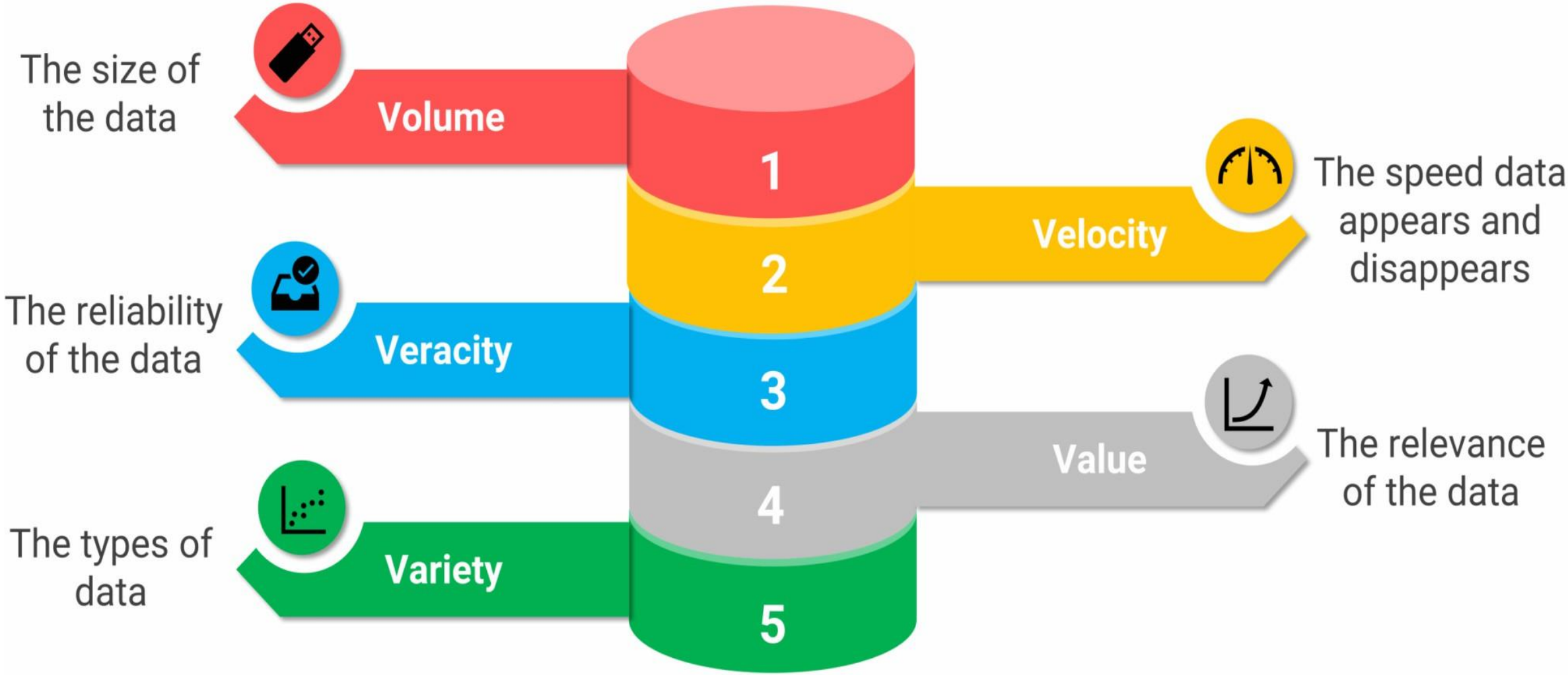
- Big Data refers to vast and rapidly growing volumes of data that are too large and complex for traditional data processing tools to manage.
- This data comes in many forms structured (e.g., tables), semi-structured (e.g., JSON, XML), and unstructured (e.g., text, images, video).



Big Data

- **Big Data** refers to massive, complex data sets that traditional systems can't handle. It includes both structured and unstructured data from sources such as sensors, social media and transactions etc.
- **Big Data** analytics helps organizations process and analyze these large data sets to **systematically extract valuable insights.** It often requires advanced tools.
- Common use cases for big data include customer behavior analysis, fraud detection and predictive maintenance.

The 5 Vs of Big Data



Big Data Technologies













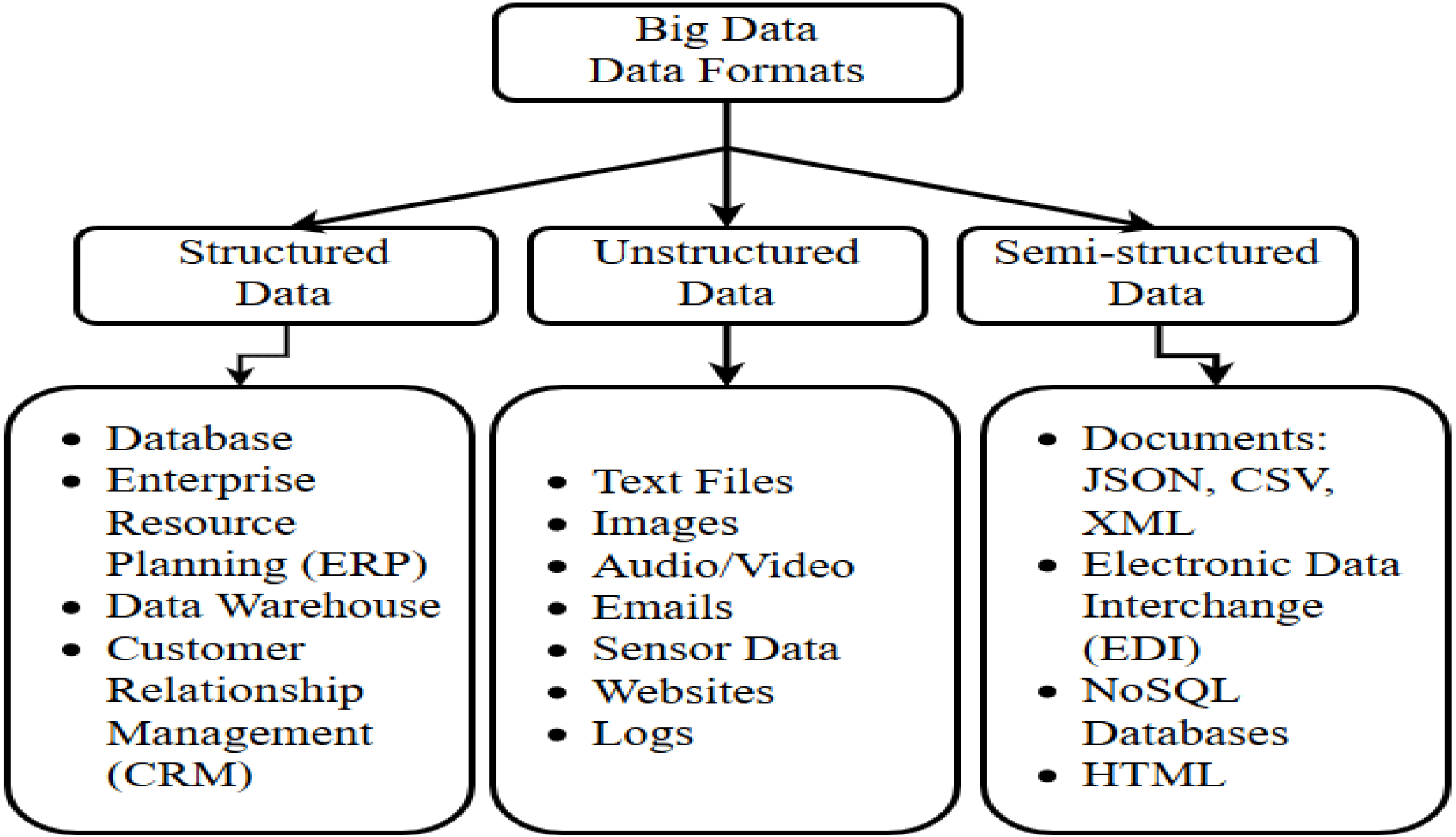


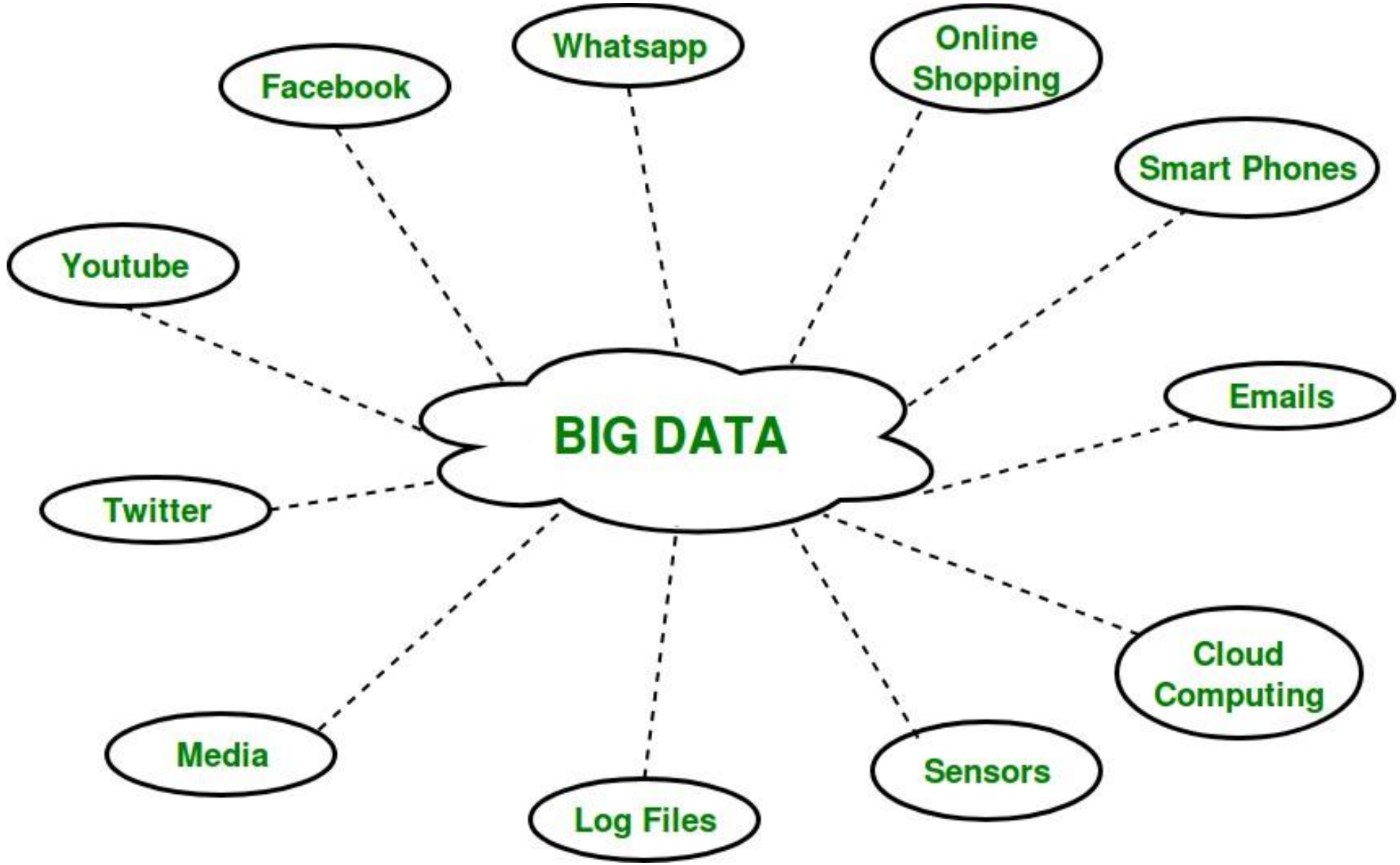









Big Data
Data Formats
Structured Data
Unstructured Data
Semi-structured Data
Database
Enterprise Resource Planning (ERP)
Data Warehouse
Customer Relationship Management (CRM)
Text Files
Images
Audio/Video
Emails
Sensor Data
Websites
Logs
Documents: JSON, CSV, XML
Electronic Data Interchange (EDI)
NoSQL Databases
HTML


BIG DATA
Facebook
Whatsapp
Online Shopping
Smart Phones
Youtube
Emails
Twitter
Cloud Computing
Media
Sensors
Log Files

Structured data

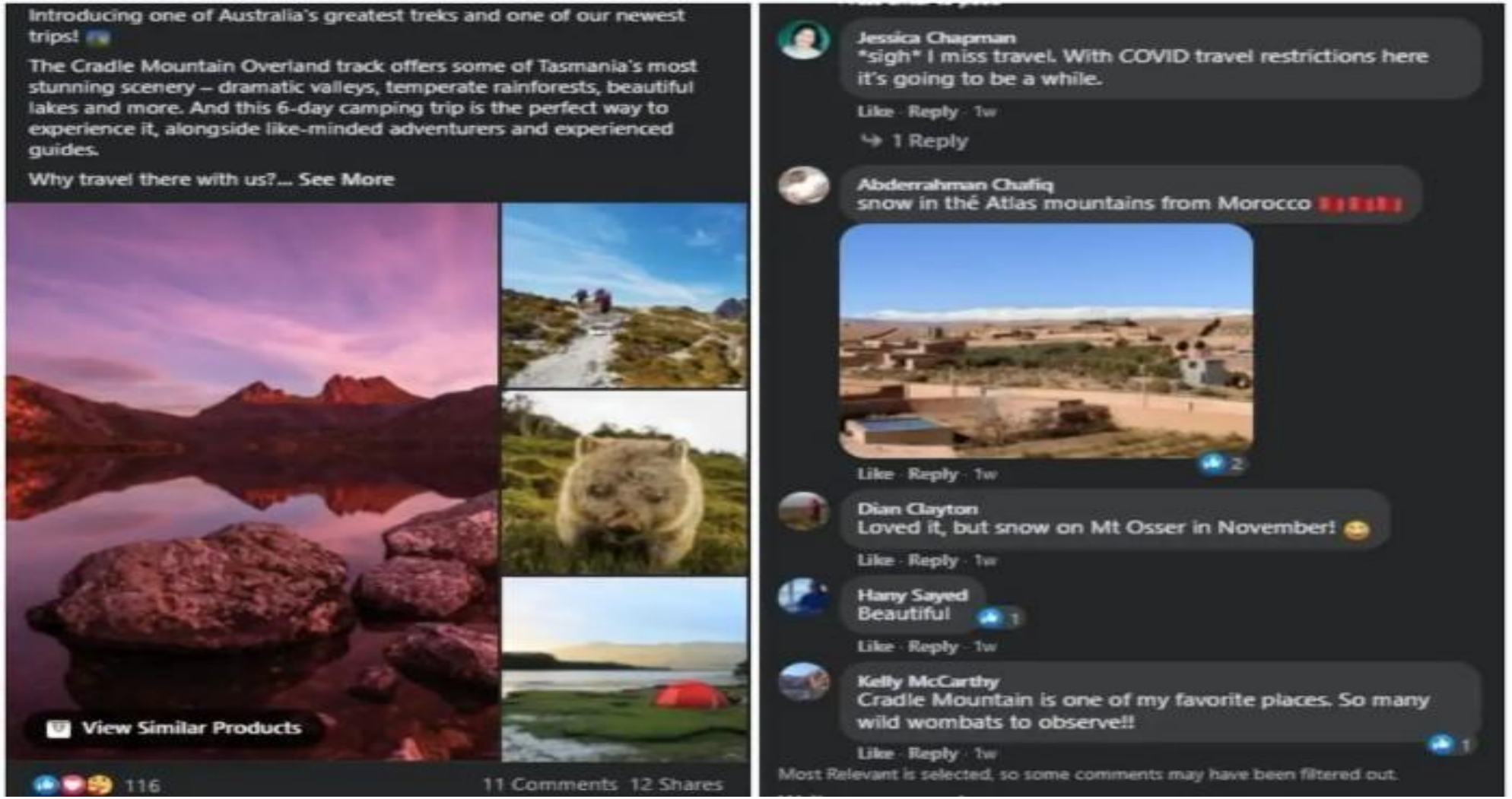
- **Structured data** is organized in a clear, defined format, often stored in **relational databases or spreadsheets.**
- It can consist of both quantitative (such as sales figures) and qualitative data (such as categorical labels like “yes or no”).
- **Examples of structured data** include customer records and financial reports, where **data fits neatly into rows and columns with predefined fields**.
- The highly organized nature of structured data allows for **quick querying and data analysis, making it useful for business intelligence systems and reporting processes**.

	A	B	C	D	E	F	G
1	Purchase ID	Last name	First name	Birthday	Country	Date of purchase	Amount of purchase
2	1	Davidson	Michael	04/03/1986	United States	10/12/2016	37
3	2	Vito	Jim	09/01/1994	United Kingdom	02/02/2016	85
4	3	Johnson	Tom	23/08/1972	France	02/11/2016	83
5	4	Lewis	Peter	18/10/1979	Germany	22/11/2016	27
6	5	Koenig	Edward	13/05/1983	Argentina	26/03/2015	43
7	6	Preston	Jack	16/06/1991	United States	06/11/2016	77
8	7	Smith	David	11/03/1965	Canada	15/11/2016	23
9	8	Brown	Luis	03/09/1997	Australia	03/07/2015	74
10	9	Miller	Thomas	07/01/1980	Germany	07/11/2016	13
11	10	Williams	Bill	26/07/1960	United States	20/11/2015	80
12	11	Gemini	Alexia	12/09/1995	Canada	11/03/2017	35
13	12	Bond	James	25/02/1975	United Kingdom	12/08/2017	40
14	13	Burgle	Patricia	01/12/1990	United States	18/01/2015	55
15	14	Reding	Michelle	07/04/1985	Canada	23/02/2017	28
16	15	Harvey	Billy	14/07/1971	United Kingdom	12/01/2016	41
17							

Unstructured data

- Unstructured data lacks a strictly defined format. **It often comes in complex forms such as text documents, images and videos. Unstructured data can include both qualitative information** (such as customer comments) and quantitative elements (such as numerical values embedded in text).
- Examples of unstructured data **include emails, social media content and multimedia files**.
- Unstructured data **doesn't easily fit into traditional relational databases**, and organizations often use techniques such **as NoSQL, Hadoop, Natural language processing (NLP) and machine learning to streamline analysis of unstructured data**.
- Unstructured data often plays a **key role in sentiment analysis, complex pattern recognition** and other advanced analytics projects.

Example:


Introducing one of Australia's greatest treks and one of our newest trips!
The Cradle Mountain Overland track offers some of Tasmania's most stunning scenery – dramatic valleys, temperate rainforests, beautiful lakes and more. And this 6-day camping trip is the perfect way to experience it, alongside like-minded adventurers and experienced guides.
Why travel there with us?... See More
View Similar Products
116
11 Comments 12 Shares
Jessica Chapman
sigh I miss travel. With COVID travel restrictions here it's going to be a while.
Like · Reply · 1w
1 Reply
Abderrahman Chafiq
snow in thé Atlas mountains from Morocco
Like · Reply · 1w
Dian Clayton
Loved it, but snow on Mt Osser in November!
Like · Reply · 1w
Hany Sayed
Beautiful
Like · Reply · 1w
Kelly McCarthy
Cradle Mountain is one of my favorite places. So many wild wombats to observe!!
Like · Reply · 1w
Most Relevant is selected, so some comments may have been filtered out.

Semi-structured data

- Semi-structured data blends elements of **structured and unstructured data**. It **doesn't follow a rigid format** but can include tags or markers that make it easier to organize and analyze. Examples of semi-structured data include **XML files and JSON objects.**
- Semi-structured data is widely used in scenarios **such as web scraping and data integration projects** because it offers flexibility while retaining some structure for search and analysis.

Example:

Unstructured data

The university has 5600 students.
John's ID is number 1, he is 18 years old and already holds a B.Sc. degree.
David's ID is number 2, he is 31 years old and holds a Ph.D. degree. Robert's ID is number 3, he is 51 years old and also holds the same degree as David, a Ph.D. degree.

Semi-structured data

```
<University>
  <Student ID="1">
    <Name>John</Name>
    <Age>18</Age>
    <Degree>B.Sc.</Degree>
  </Student>
  <Student ID="2">
    <Name>David</Name>
    <Age>31</Age>
    <Degree>Ph.D. </Degree>
 </Student>
....
</University>
```

Structured data

ID	Name	Age	Degree
1	John	18	B.Sc.
2	David	31	Ph.D.
3	Robert	51	Ph.D.
4	Rick	26	M.Sc.
5	Michael	19	B.Sc.

Metadata

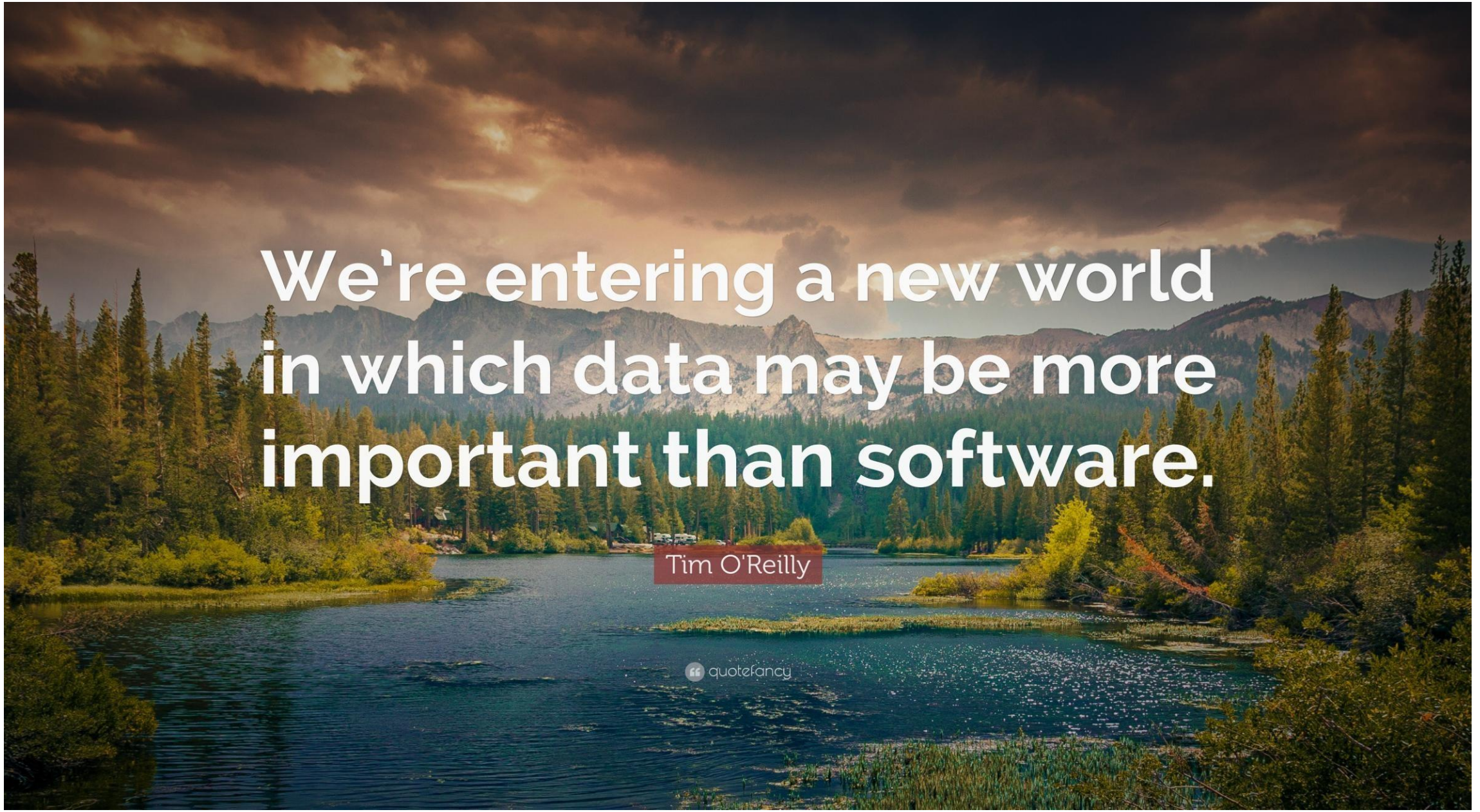
- **Metadata** is data about data. In other words, **it is information about the attributes of a data point or data set, such as file names, authors, creation dates or data types.**
- **Metadata** enhances data organization, searchability and management.
- It is critical to systems such as databases, digital libraries and content management platforms **because it helps users more easily sort and find the data they need.**

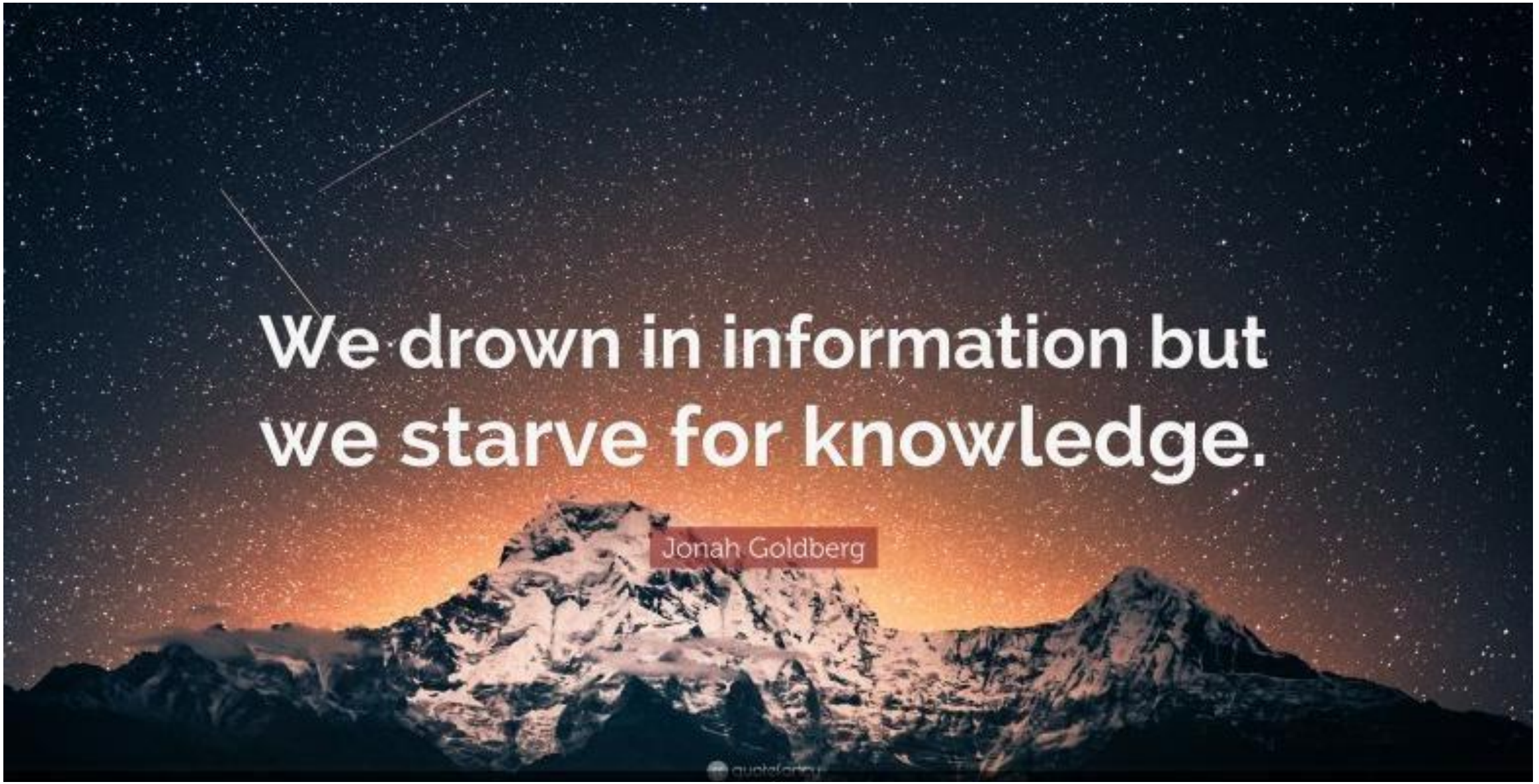
Example:



Filename:	Tadzik.jpg
Author:	Piotr Kononow
Date:	August 15, 2016 6:40:10PM
File:	5,312 × 2,988 JPEG 15.9 megapixels 3,393,448 bytes (3.2 megabytes)
Camera:	Samsung SM-G920F
Lens:	4.3 mm Max aperture f/1.9 (shot wide open) Auto exposure Program AE
Exposure:	1/402 sec f/1.9 ISO 40
Flash:	none

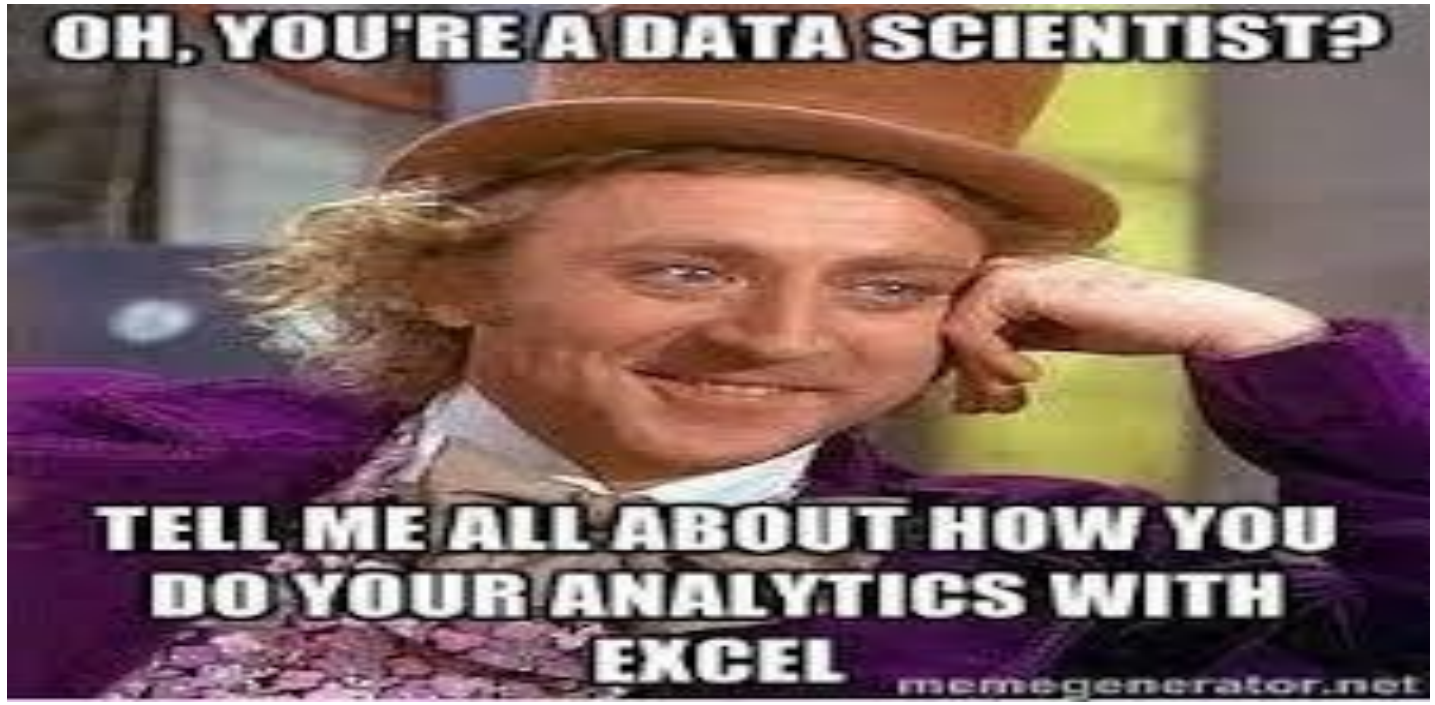



We're entering a new world
in which data may be more
important than software.
Tim O'Reilly
quotefancy

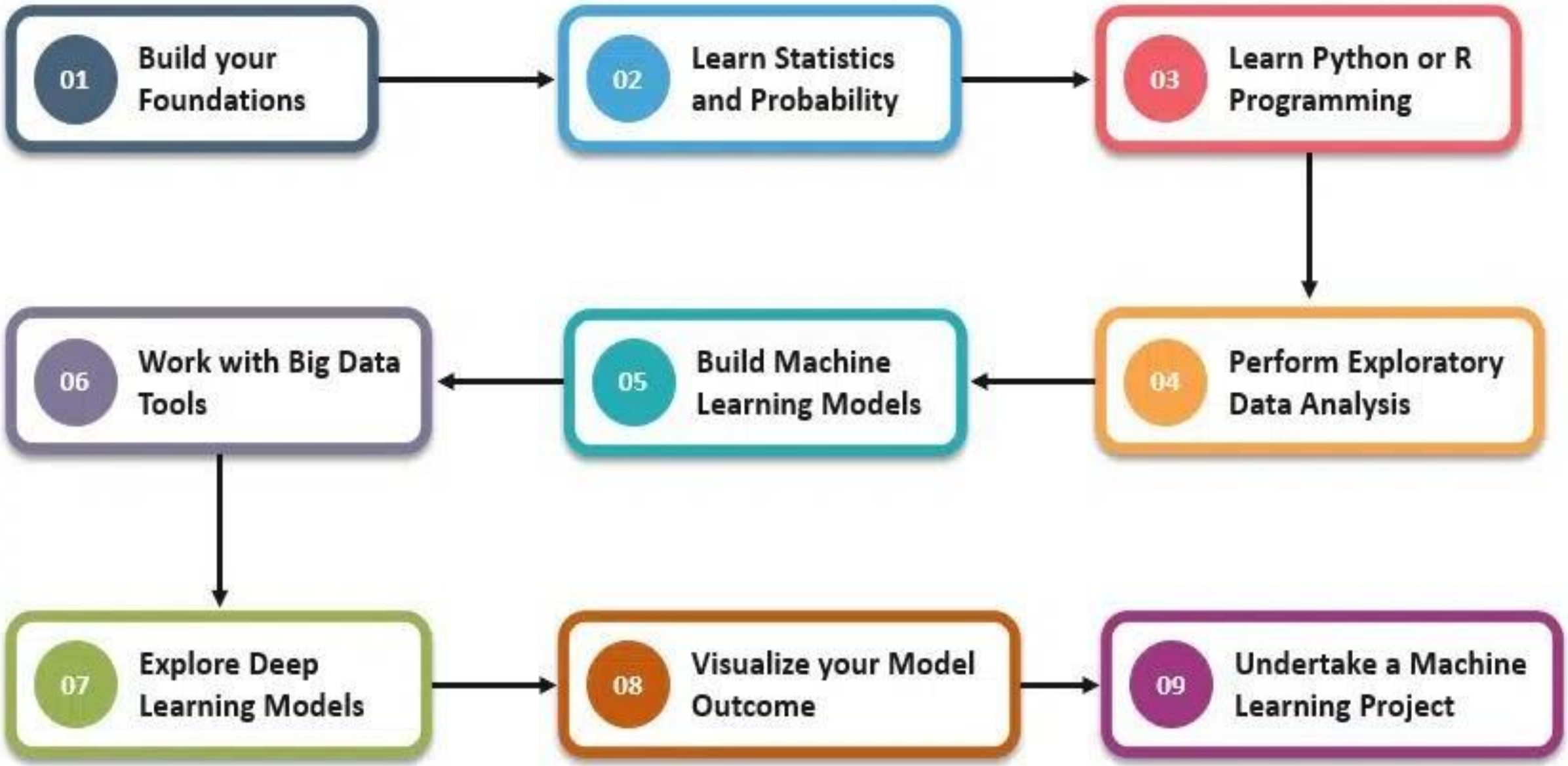

We drown in information but
we starve for knowledge.
Jonah Goldberg

Is Data important?

- **Data** enables organizations to **transform raw information into actionable insights to predict customer behavior**, optimize supply chains and fuel innovation.
- The term "**data" comes from the plural of "datum**", a Latin word meaning "**something given**": a definition that remains just as relevant today.
- Every day, millions of people **provide data to businesses through interactions** such as impressions, **clicks**, **transactions**, **sensor readings** or even just **browsing online**.


OH, YOU'RE A DATA SCIENTIST?
TELL ME ALL ABOUT HOW YOU DO YOUR ANALYTICS WITH EXCEL
memegenerator.net

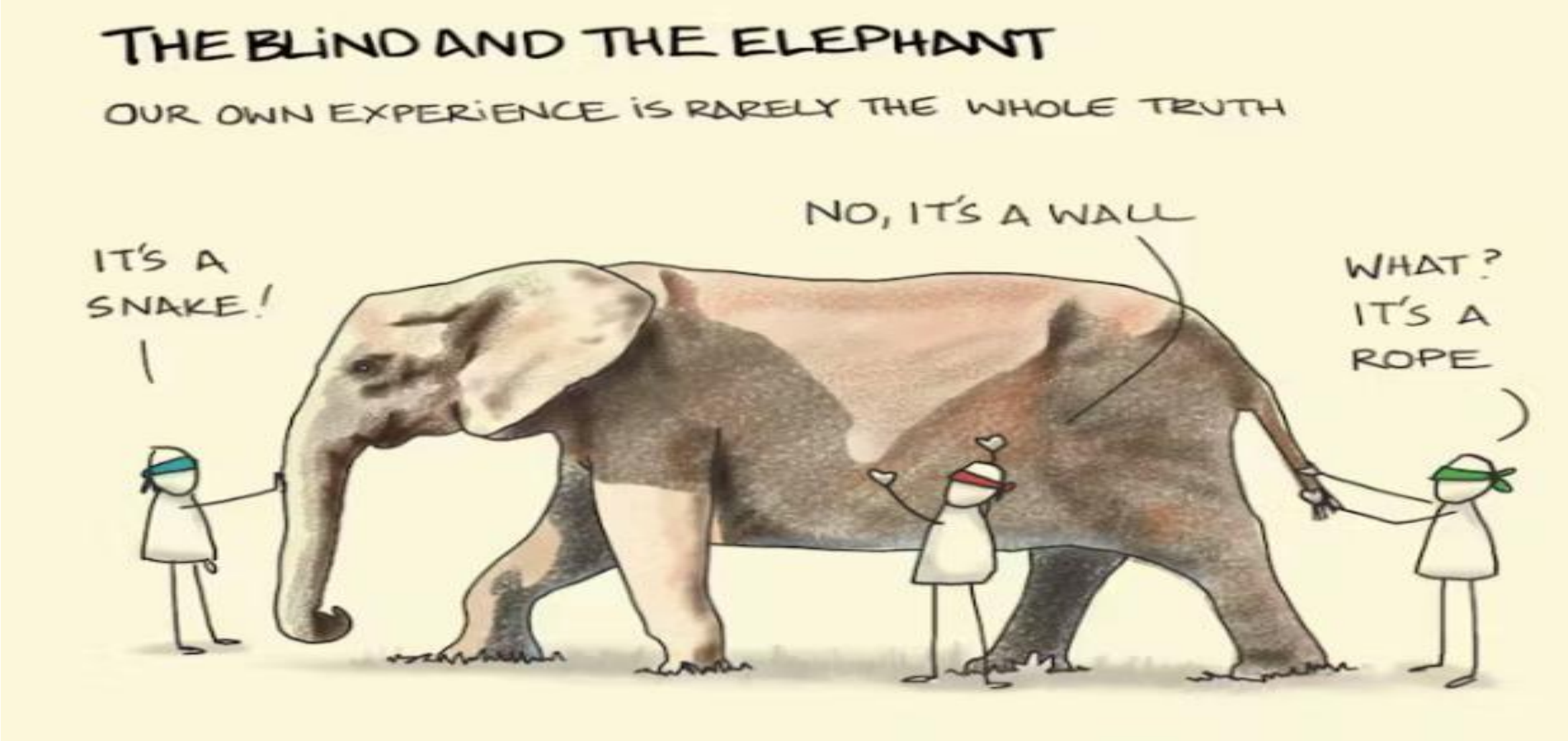
Steps to become an Expert in Data Science



VIDEO

- Link:

What is Data Science?



What is Data Science?

- **Data Science** is about data gathering, analysis and decision-making.
- **Data Science** is about finding patterns in data, through analysis, and make future predictions.

By using Data Science, companies are able to make:

- Better decisions (should we choose A or B)
- Predictive analysis (what will happen next?)
- Pattern discoveries (find a pattern, or maybe hidden information in the data)

Def. of Data Science

- **Data science** is a ***"multidisciplinary field that uses scientific methods, algorithms, and Storage systems to extract knowledge and insights from data".***
- It involves **collecting**, **cleaning**, **analyzing**, and **interpreting data** to identify **patterns**, **trends**, and **actionable information**.

Cont..

- **Data Science** is a "***combination of multiple disciplines that uses statistics, data analysis, and machine learning to analyze data and to extract knowledge and insights from it.***"
- **Data science** is the ***"study of data that helps us derive useful insight for business decision making. "***
- **Data Science** is all ***about "using Technology, Big Data tools, techniques, and creativity to uncover insights hidden within data/Hidden Pattern."***
- ***It combines math(Linear Algebra, Probability, Statistics, Regression), computer science, and domain expertise*** to tackle real-world challenges in a variety of fields.

Cont..

- ***Data Science processes the raw data and solve business problems and even make prediction about the future trend or requirement.***

Example:

- What do customer want?
- How can we improve our services?
- What will the upcoming trend in sales?
- How much stock they need for upcoming festival ?
- In healthcare, a data scientist can predict patient risks and suggest personalized treatments.



Raw Data

- Sales Data
- Customer Feedback
- Web Logs



Data Science Process



Business Insigts

- Sales Prediction
- Product Optimization
- User Behaviour



Data is the nutrition of artificial intelligence. When an AI eats junk food, it's not going to perform very well.

Matthew Emerick

Data science involves these key steps:

- **Data Collection:** Gathering raw data from various sources, such as **databases, sensors, or user interactions.**
- **Data Cleaning:** Ensuring the data is accurate, complete, and ready for analysis.
- **Data Analysis:** Applying statistical and computational methods **to identify patterns, trends, or relationships.**
- **Data Visualization:** Creating **charts, graphs, and dashboards** to present findings clearly.
- **Decision-Making:** Using insights to inform strategies, **create solutions, or predict outcomes.**

Why Is Data Science Important?

- ***In a world flooded with user-data***, data science is crucial for driving progress and innovation in every industry.

Here are some key reasons why it is so important:

- **Helps Business in Decision-Making:** By analyzing data, businesses can understand trends and make informed choices that reduce risks and maximize profits.
- **Improves Efficiency:** Organizations can use data science to **identify areas where they can save time and resources.**
- **Personalizes Experiences:** Data science helps create **customized recommendations and offers that improve customer satisfaction**.
- **Predicts the Future:** Businesses can use data to forecast trends, demand, and other important factors.
- **Drives Innovation: New ideas and products often come** from insights discovered through data science.
- **Benefits Society:** Data science improves public services **like healthcare, education, and transportation by helping allocate resources more effectively**.

Where is Data Science Needed?

- **Data Science** is used in many industries in the world today, e.g. banking, consultancy, healthcare, and manufacturing.

Examples of where Data Science is needed:

- For **Route planning**: To discover the best routes to ship
- To **foresee delays for flight/ship/train** etc. (through predictive analysis)
- To **create promotional offers**
- To find the **best suited time to deliver goods**
- To forecast the **next years revenue for a company**
- To **analyze health benefit of training**
- To **predict who will win elections(Eg. USA Presidential Election)**

Cont..

- **Data Science can be applied in nearly every aspect of a business or sector where data is available.**

Some prominent examples include:

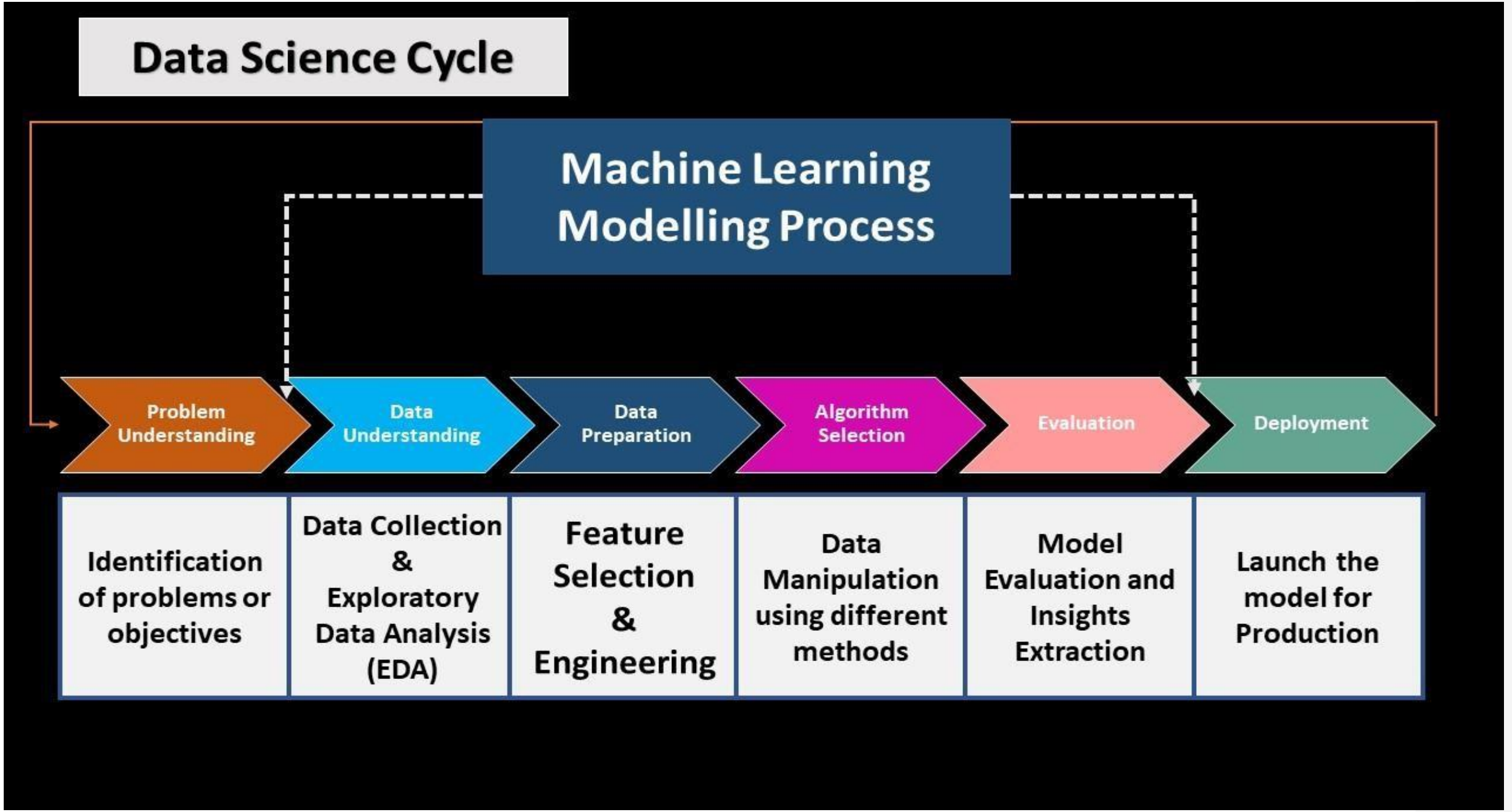
- **Consumer Goods** – understanding customer preferences, demand forecasting, and product optimization
- **Stock Markets** – algorithmic trading, risk prediction, and portfolio optimization
- **Industry/Manufacturing** – predictive maintenance, process optimization, and quality control
- **Politics** – voter behavior analysis, campaign targeting, and sentiment tracking
- **Logistics Companies** – route optimization, supply chain management, and delivery efficiency
- **E-commerce** – personalized recommendations, customer segmentation, and fraud detection
- **Education** – adaptive learning, student performance prediction, and curriculum optimization
- **Healthcare** – disease prediction, drug discovery, and patient care analytics
- **Transportation** – traffic forecasting, autonomous systems, and fleet management

Cont..

- **Predicting customer preferences** for personalized recommendations.
- **Detecting fraud** in financial transactions.
- **Forecasting sales** and market trends.
- Enhancing **healthcare with predictive diagnostics** and personalized treatments.
- **Identifying risks** and opportunities in investments.
- **Optimizing supply chains** and inventory management.

Data Science Life Cycle

- It's passes from many stages and every element is important. One should always follow the proper steps to reach the ladder.
- Every step has its value and it counts in your model.


Data Science Cycle
Machine Learning Modelling Process
Problem Understanding
Data Understanding
Data Preparation
Algorithm Selection
Evaluation
Deployment
Identification of problems or objectives
Data Collection & Exploratory Data Analysis (EDA)
Feature Selection & Engineering
Data Manipulation using different methods
Model Evaluation and Insights Extraction
Launch the model for Production

How Data Science Works



Facets of Data/Various Characteristics

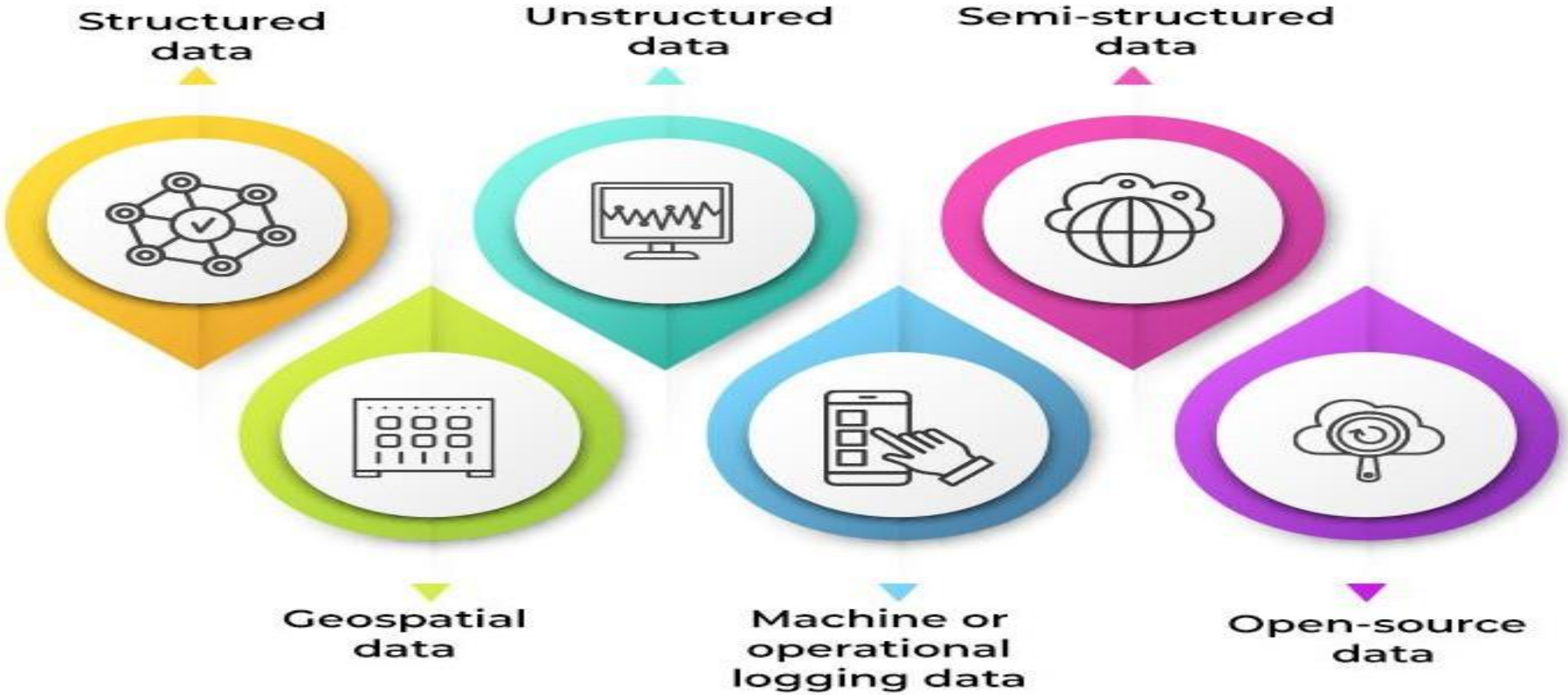
Very large amount of data will generate in big data and data science.

These data is various types and main categories of data are as follows:

a) Structured
b) Natural language
c) Graph-based
d) Streaming
e) Unstructured
f) Machine-generated
g) Audio, video and images


40 ZETTABYTES
of data will be created by 2020, an increase of 300 times from 2005
2.5 QUINTILLION BYTES
of data is created each day
400 MILLION TWEETS
are sent per day by about 200 million monthly active users
As of 2011, the global size of data in healthcare was estimated to be
150 EXABYTES
6 BILLION PEOPLE
have cell phones
WORLD POPULATION: 7 BILLION
30 BILLION PIECES OF CONTENT
are shared on Facebook every month
facebook
Volume
SCALE OF DATA
Variety
DIFFERENT FORMS OF DATA
4 BILLION+ HOURS OF VIDEO
are watched on YouTube each month
Most companies in the U.S. have at least
100 TERABYTES
of data stored
BIG DATA
100 SENSORS
1 IN 3 BUSINESS
don't trust the information they use to make decisions
Velocity
ANALYSIS OF STREAMING DATA
Veracity
UNCERTAINTY OF DATA
The New York Stock Exchange captures
1 TB OF TRADE INFORMATION
27% OF RESPONDENTS
Poor data quality costs the US economy around
$3.1 TRILLION A YEAR
By 2016, it is projected there will be
18.9 BILLION NETWORK CONNECTIONS

TYPES OF BIG DATA



Structured Data

Structured data is arranged in rows and column format. It helps for application to retrieve and process data easily.

Database management system is used for storing structured data.

Structured data is also searchable by data type within content. Structured data is understood by computers and is also efficiently organized for human readers.

- An Excel table is an example of structured data

Unstructured Data

- Unstructured data is data that does not follow a specified format. Row and columns are not used for unstructured data. Therefore it is difficult to retrieve required information.
- Unstructured data has no identifiable structure.
- The unstructured data can be in the form of Text: (Documents, email messages, customer feedbacks), audio, video, images. Email is an example of unstructured data.
- Even today in most of the organizations more than 80 % of the data are in unstructured form.
- This carries lots of information. But extracting information from these various sources is a very big challenge.

Characteristics of Unstructured data:

1. There is no structural restriction or binding for the data.
2. Data can be of any type.
3. Unstructured data does not follow any structural rules.
4. There are no predefined formats, restriction or sequences for unstructured data.
5. Since there is no structural binding for unstructured data, it is unpredictable in nature.

Semi Structured Data

- Semi-structured data is information that doesn’t reside in a relational database but that does have some organizational properties that make it easier to analyze.
- With some process you can store them in relation database (it could be very hard for some kind of semi structured data), but the semi structure exists to ease space, clarity or compute…
- But as Structured data, semi structured data represents a few parts of data (5 to 10%).

- **Examples of semi-structured**: JSON, CSV , XML documents are semi structured documents.

Applications of Data Science

- Data science is used to ***analyze patient data***, predict diseases, develop personalized treatments, and optimize hospital operations.
- It helps detect ***fraudulent transactions***, manage risks, and provide personalized financial advice.
- Businesses use ***data science to understand customer behavior***, recommend products, optimize inventory, and improve supply chains.
- Data science powers innovations like search engines, virtual assistants, and recommendation systems.
- It enables ***route optimization, traffic management***, and predictive maintenance for vehicles.
- Data science helps in designing ***personalized learning experiences***, tracking student performance, and improving administrative efficiency.
- Streaming platforms and content creators use data science ***to recommend shows***, analyze viewer preferences, and optimize content delivery.

Cont..

- **Healthcare:** Data science improves patient outcomes by using predictive analytics to detect diseases early, creating personalized treatment plans and optimizing hospital operations for efficiency.
- **Finance:** Data science helps detect fraudulent activities, assess and manage financial risks, and provide tailored financial solutions to customers.
- **Retail:** Data science enhances customer experiences by delivering targeted marketing campaigns, optimizing inventory management, and forecasting sales trends accurately.
- **Technology:** Data science powers cutting-edge AI applications such as voice assistants, intelligent search engines, and smart home devices.
- **Transportation:** Data science optimizes travel routes, manages vehicle fleets effectively, and enhances traffic management systems for smoother journeys.
- **Manufacturing:** Data science predicts potential equipment failures, streamlines supply chain processes, and improves production efficiency through data-driven decisions.
- **Energy:** Data science forecasts energy demand, optimizes energy consumption, and facilitates the integration of renewable energy resources.
- **Agriculture:** Data science drives precision farming practices by monitoring crop health, managing resources efficiently, and boosting agricultural yields.

Exploratory Data Analysis

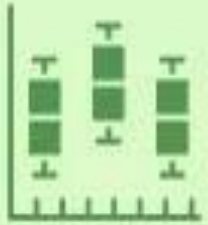
- Exploratory Data Analysis (EDA) is a important step in data science and data analytics as it visualizes data to understand its main features, find patterns and discover how different parts of the data are connected

Exploratory Data Analysis (EDA)



EDA

Explore and summarize data before modeling



Distribution

Use histograms and box plots



Relationships

Use scatter and pair plots



Spot Issues

Find outliers or missing data

Types of EDA



Univariate

Analyzes one variable



Bivariate

Studies relationships between two variables



Multivariate

Looks at interactions between multiple variables

EDA into Three Types:
1. **Univariate Analysis**

- Focuses on studying **one variable to understand its characteristics**.
- It helps to describe data and **find patterns within a single feature.**
- Various common methods like **histograms are used to show data** distribution, **box plots to detect outliers** and understand data spread and **bar charts for categorical data**.
- Summary statistics; like mean, median, mode, variance and standard deviation helps in describing the central tendency and spread of the data.

2. Bivariate Analysis

- Focuses on identifying **relationship between two variables** to find connections, correlations and dependencies.
- It helps to understand how two variables interact with each other.

Some key techniques include:

- **Scatter plots** which visualize the relationship between two continuous variables.
- **Line graphs** are useful for comparing two variables over time in time series data to identify trends or patterns.

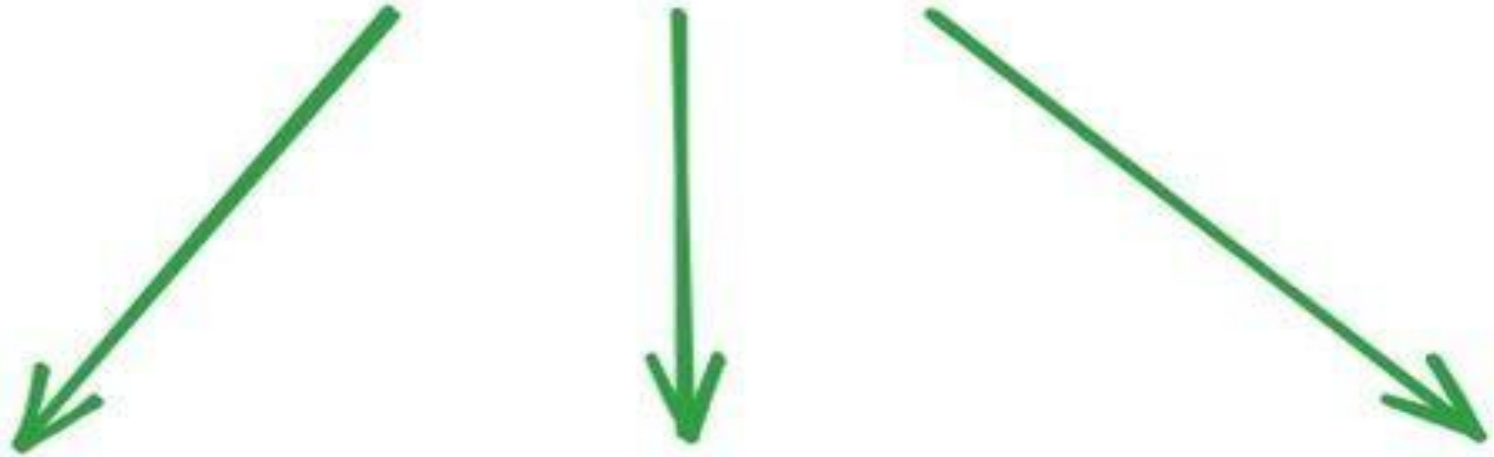
3. Multivariate Analysis

- Multivariate Analysis identify ***relationships between two or more variables in the dataset*** and aims to understand how variables interact with one another which is important for statistical modeling techniques.

It include techniques like:

- **Pair plots** which shows the relationships between multiple variables at once and helps in understanding how they interact.
- Another technique is **Principal Component Analysis (PCA)** which reduces the complexity of large datasets by simplifying them while keeping the most important information.
- **Spatial Analysis** is ***used for geographical data*** by **using maps** and spatial plotting to understand the geographical distribution of variables.
- **Time Series Analysis** is used for datasets that involve **time-based data** and it involves ***understanding and modeling patterns and trends over time***.
- **Common techniques include** line plots, autocorrelation analysis, moving averages and ARIMA models

Data Analysis



Univariate

Univariate statistics summarize only one variable at a time.

It does not deal with causes or relationships and has no dependent variables

Bivariate

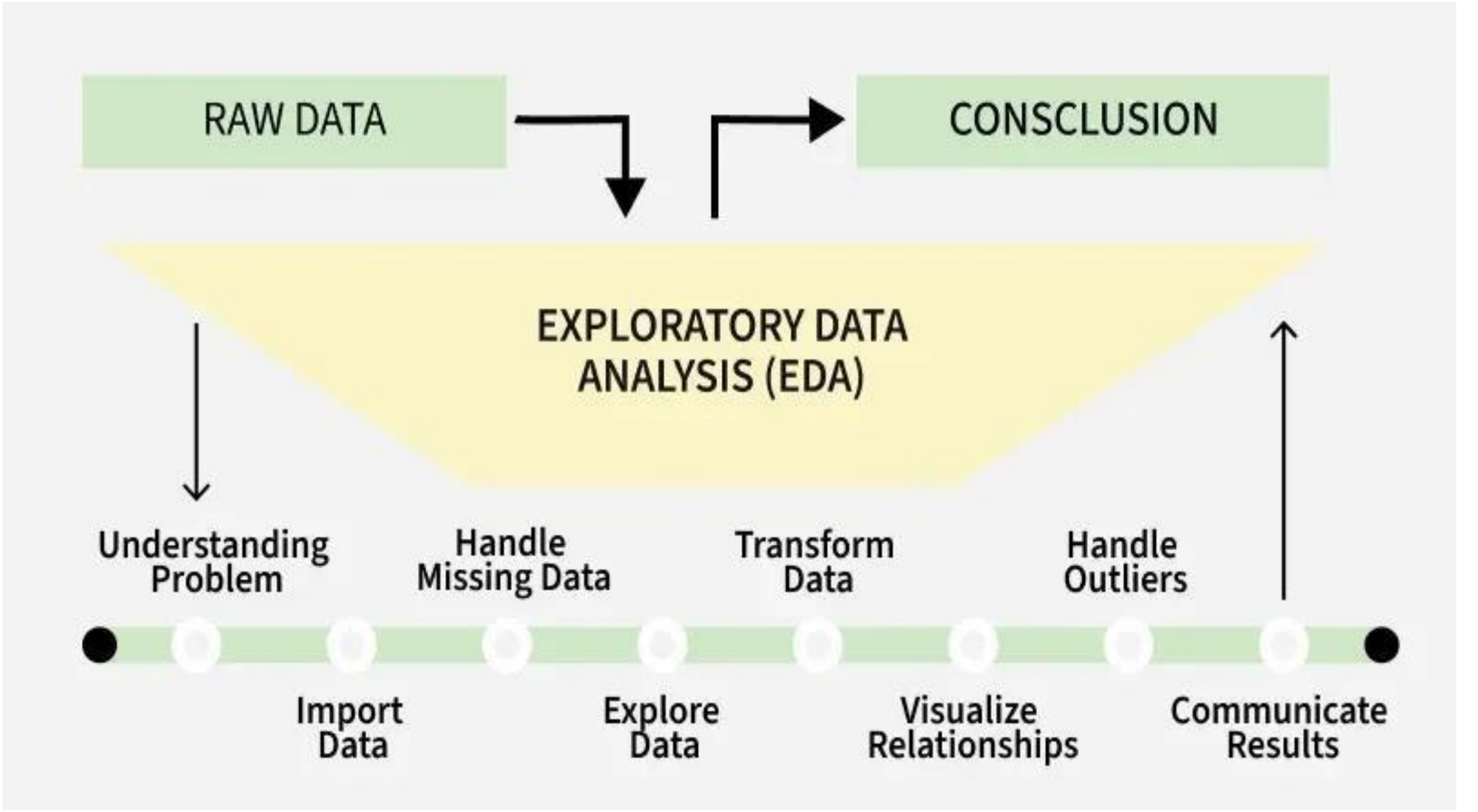
Bivariate statistics compare two variables.

It deals with causes and relationships and the analysis is done and has one dependent variable

Multivariate

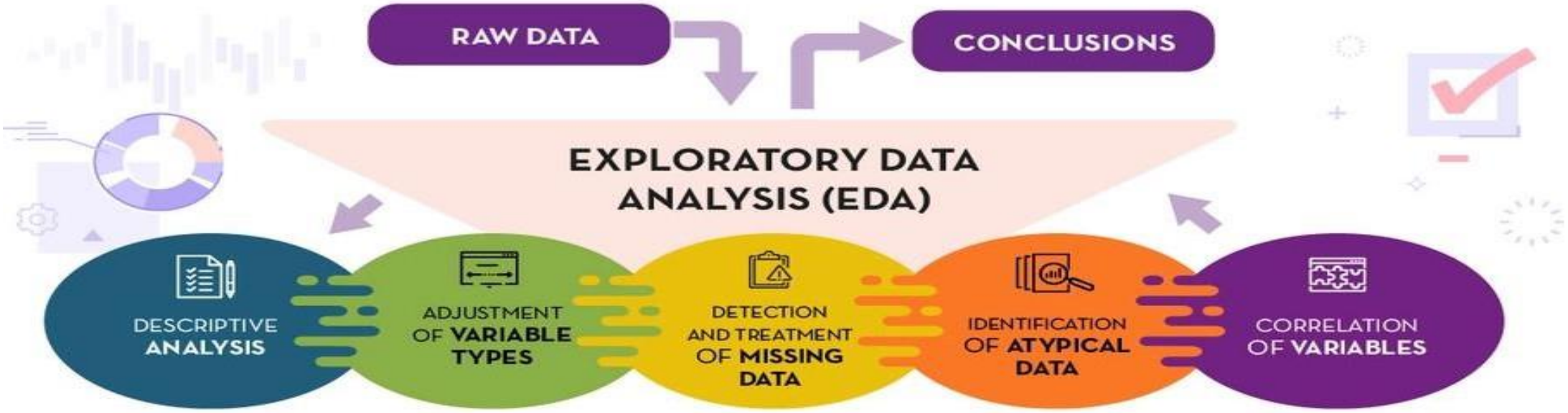
Multivariate statistics compare more than two variables.

It also deals with causes and relationships and the analysis is done but has more than one dependent variables

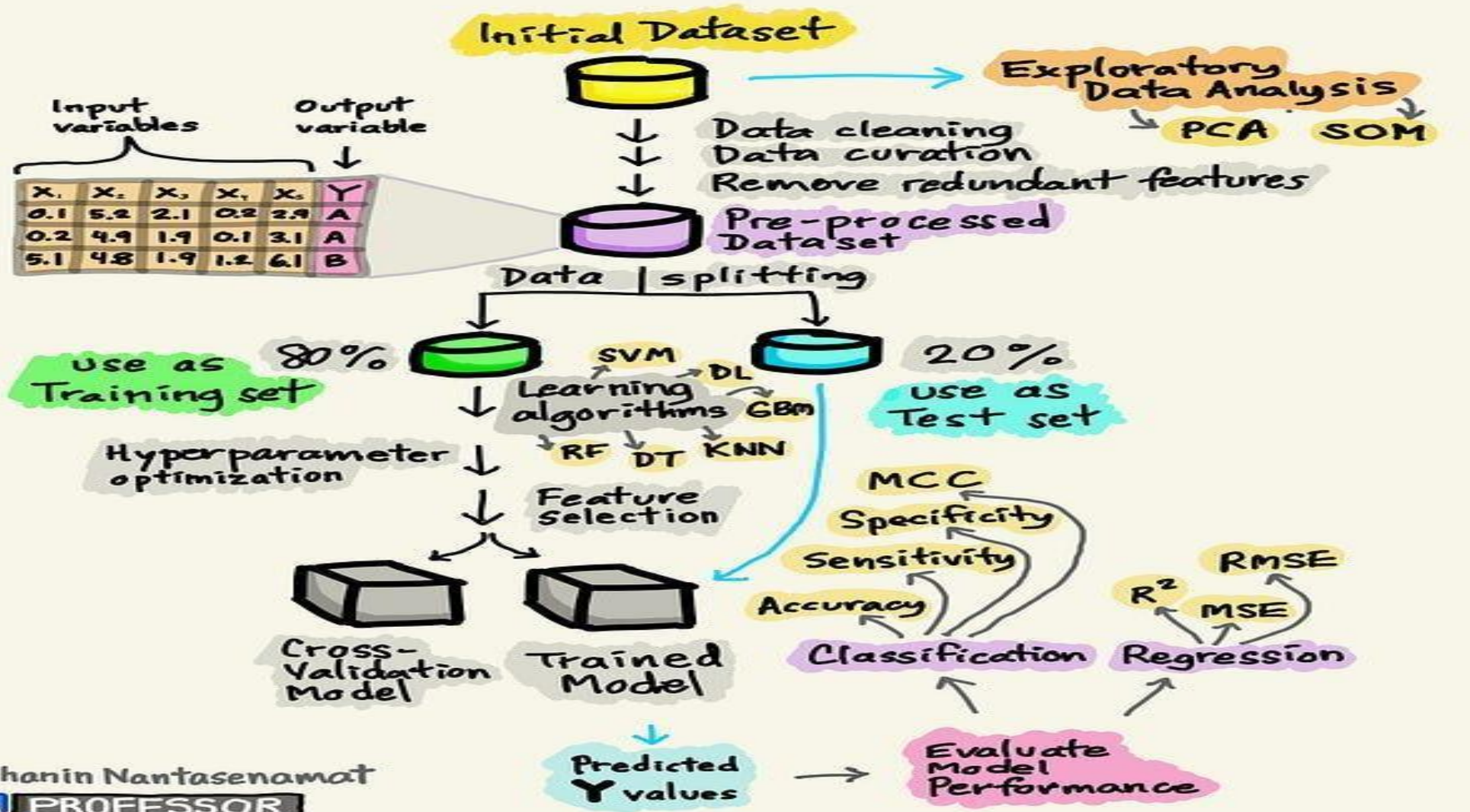

RAW DATA
CONSCLUSION
EXPLORATORY DATA ANALYSIS (EDA)
Understanding Problem
Import Data
Handle Missing Data
Explore Data
Transform Data
Visualize Relationships
Handle Outliers
Communicate Results

EDA(Exploratory Data Analysis)

- Exploratory Data Analysis (EDA) is a crucial initial step in data analysis, where you investigate and summarize the main characteristics of a dataset using statistical and visual methods



BUILDING THE MACHINE LEARNING MODEL



X_1	X_2	X_3	X_4	X_5	Y
0.1	5.2	2.1	0.2	2.9	A
0.2	4.9	1.9	0.1	3.1	A
5.1	4.8	1.9	1.2	6.1	B

By: Chanin Nantasenamat

DATA PROFESSOR

http://youtube.com/dataprofessor

January 1, 2020

Why Exploratory Data Analysis Important?

- Helps to understand the dataset by showing how many features it has, what type of data each feature contains and how the data is distributed.
- Helps to **identify hidden patterns** and relationships between different data points which help us in and model building.
- Allows to **identify errors or unusual data points (outliers)** that could affect our results.
- The insights gained from EDA help us to identify most important features for building models and guide us on how to prepare them for better performance.
- By ***understanding the data, it helps us in choosing best modeling techniques*** and adjusting them for better results.

Model Building and Application

- **Model building** is an essential part of data analytics and is used to extract insights and knowledge from the data to make business decisions and strategies.

To do this we divide our dataset into two parts

- Training dataset
- Test dataset

Dividing The Dataset For Model Building

- To divide the dataset we will use the Python sklearn library which helps us in dividing the dataset into training and testing datasets.
- Here we will choose the ratio by which we want to divide the dataset by default it 3:1 for training and testing.

Difference between Data Science, AI & ML

Data Science

The process of extracting meaningful & useful insights from huge volumes of raw data to solve real-world problems & develop business solutions.



AI

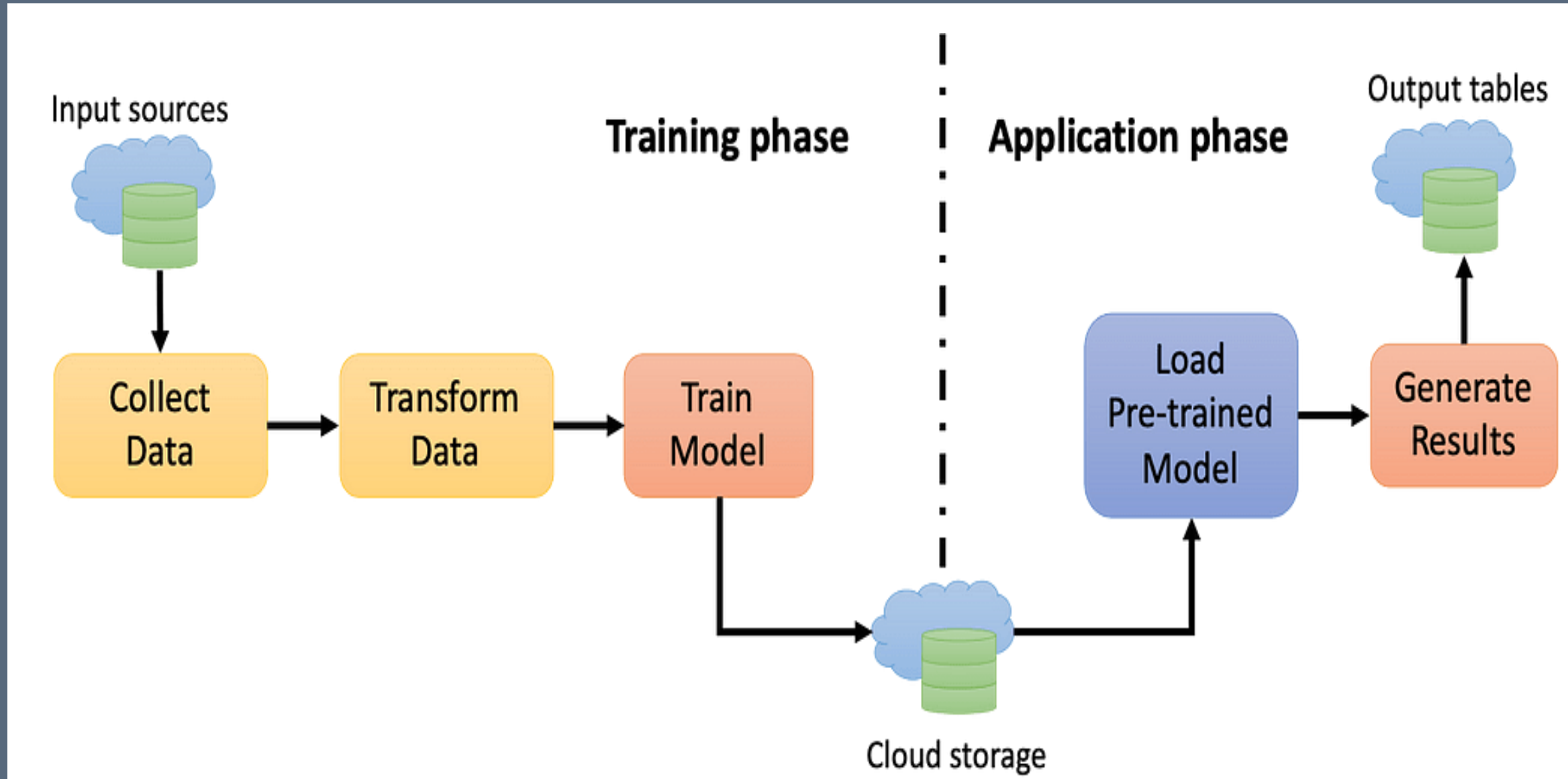
The branch of Computer Science which aims at making machines think and act like humans so that the machine can make decisions on its own.



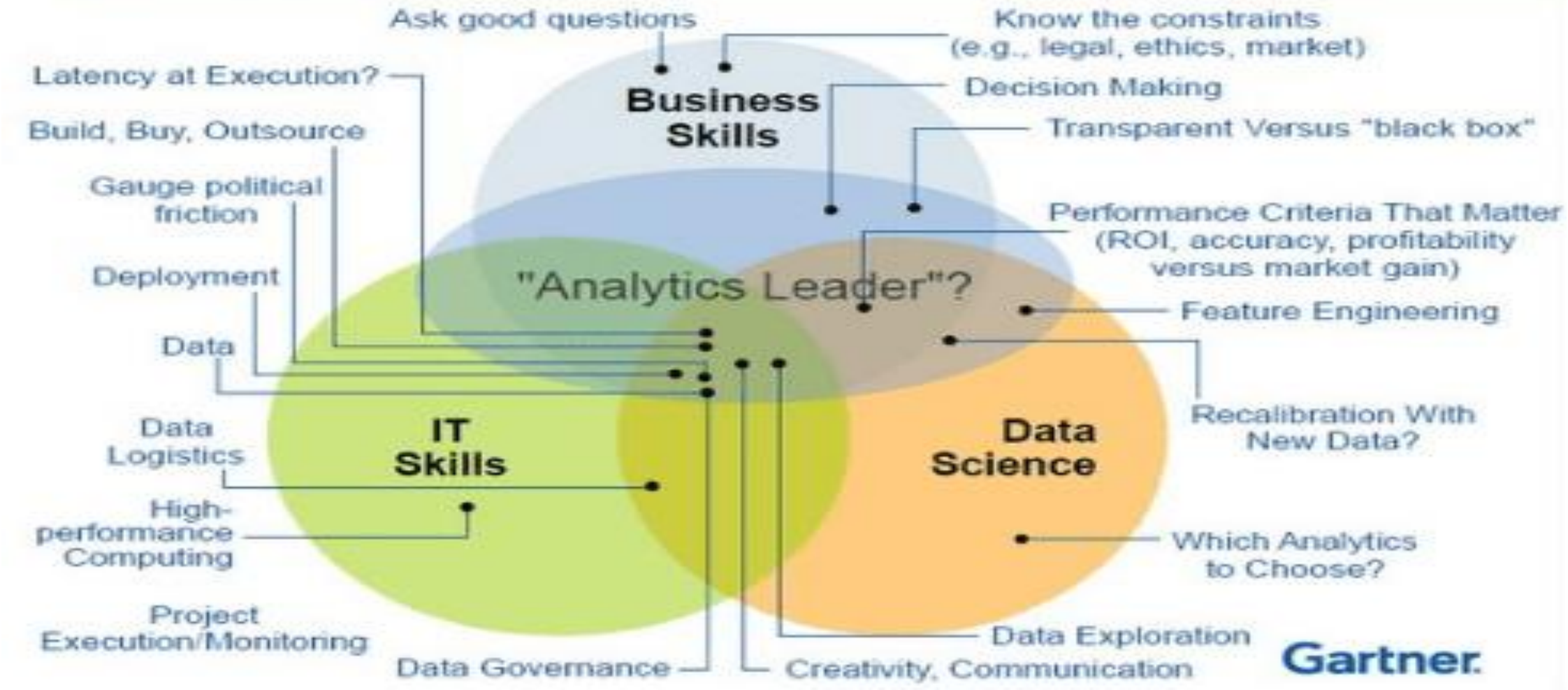
ML

A subset of AI that uses data & algorithms for the machine to autonomously learn and improve with time to help provide the AI make decisions.




Input sources
Training phase
Application phase
Output tables
Collect Data
Transform Data
Train Model
Load Pre-trained Model
Generate Results
Cloud storage

Driving the Success of Data Science Solutions: Skills, Roles and Responsibilities ...



Machine learning

- **Machine learning** is the field of study ***that enables computers to learn from data and make decisions without explicit programming.***
- Machine learning is a subset of artificial intelligence/AI that involves training computer algorithms to automatically learn patterns and relationships in data
- Machine learning models play a pivotal role in tackling real-world problems across various domains by **affecting our approach to tackling problems and decision-making.**
- By using data-driven insights and sophisticated algorithms, machine learning models help us achieve unparalleled accuracy and efficiency in solving real-world problems.

Model

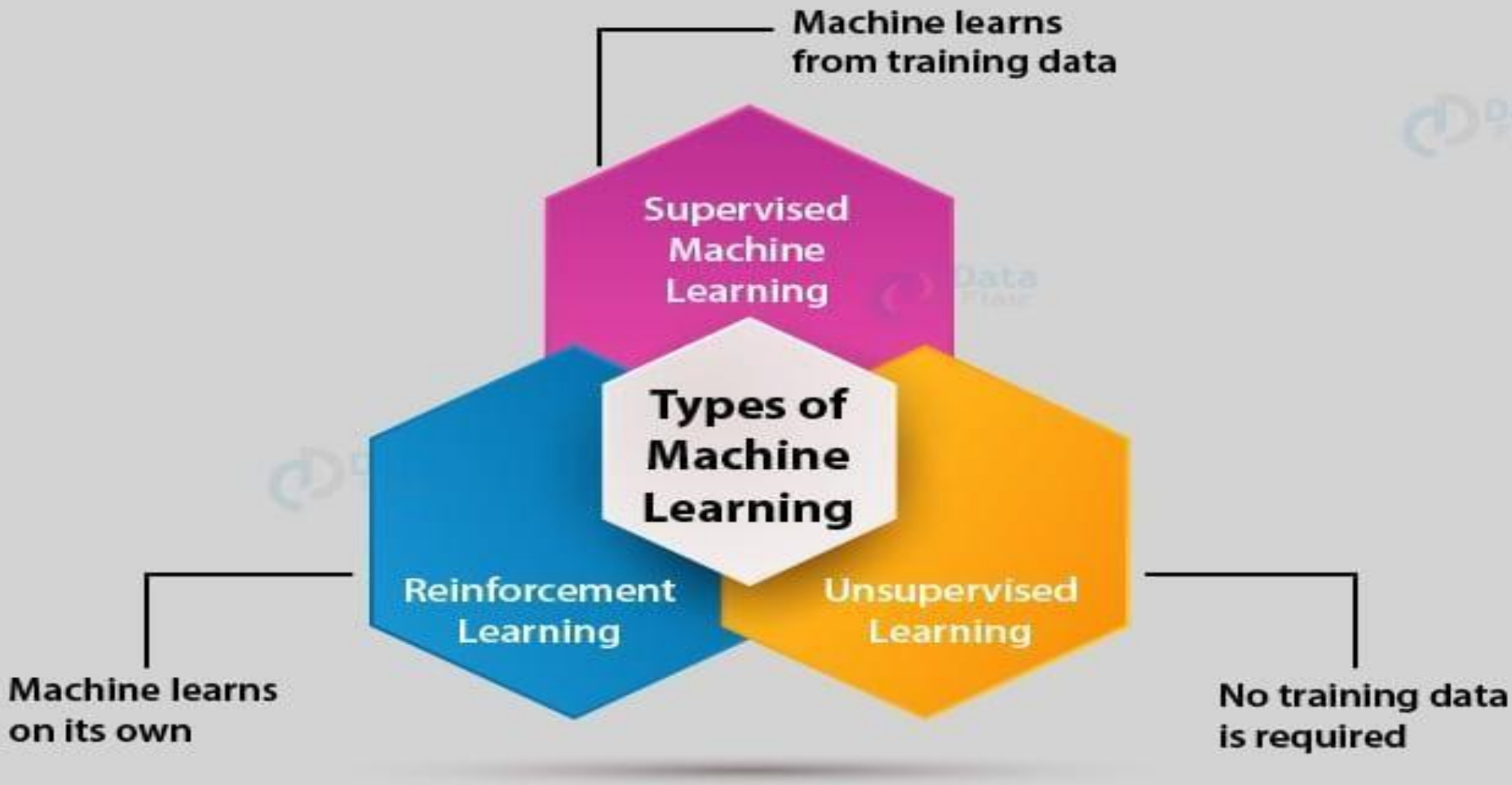
- A machine learning model is a ***mathematical representation of the relationship between the input data (features) and the output (predictions or decisions).*** The model is created using a training dataset and then evaluated using a separate validation dataset. ***The goal is to create a model that can accurately generalize to new, unseen data.***

Training

- ***Training is the process of teaching the machine learning algorithm to make accurate predictions or decisions***. This is done by providing the algorithm with a large dataset and allowing it to learn from the patterns and relationships in the data. During training, the algorithm adjusts its internal parameters to minimize the difference between its predicted output and the actual output.

Testing

- ***Testing is the process of evaluating the performance of the machine learning algorithm on a separate dataset that it has not seen before***. The goal is to determine how well the algorithm generalizes to new, unseen data. ***If the algorithm performs well on the testing dataset, it is considered to be a successful model.***


Machine learns
from training data
Supervised
Machine
Learning
Types of
Machine
Learning
Reinforcement
Learning
Unsupervised
Learning
Machine learns
on its own
No training data
is required

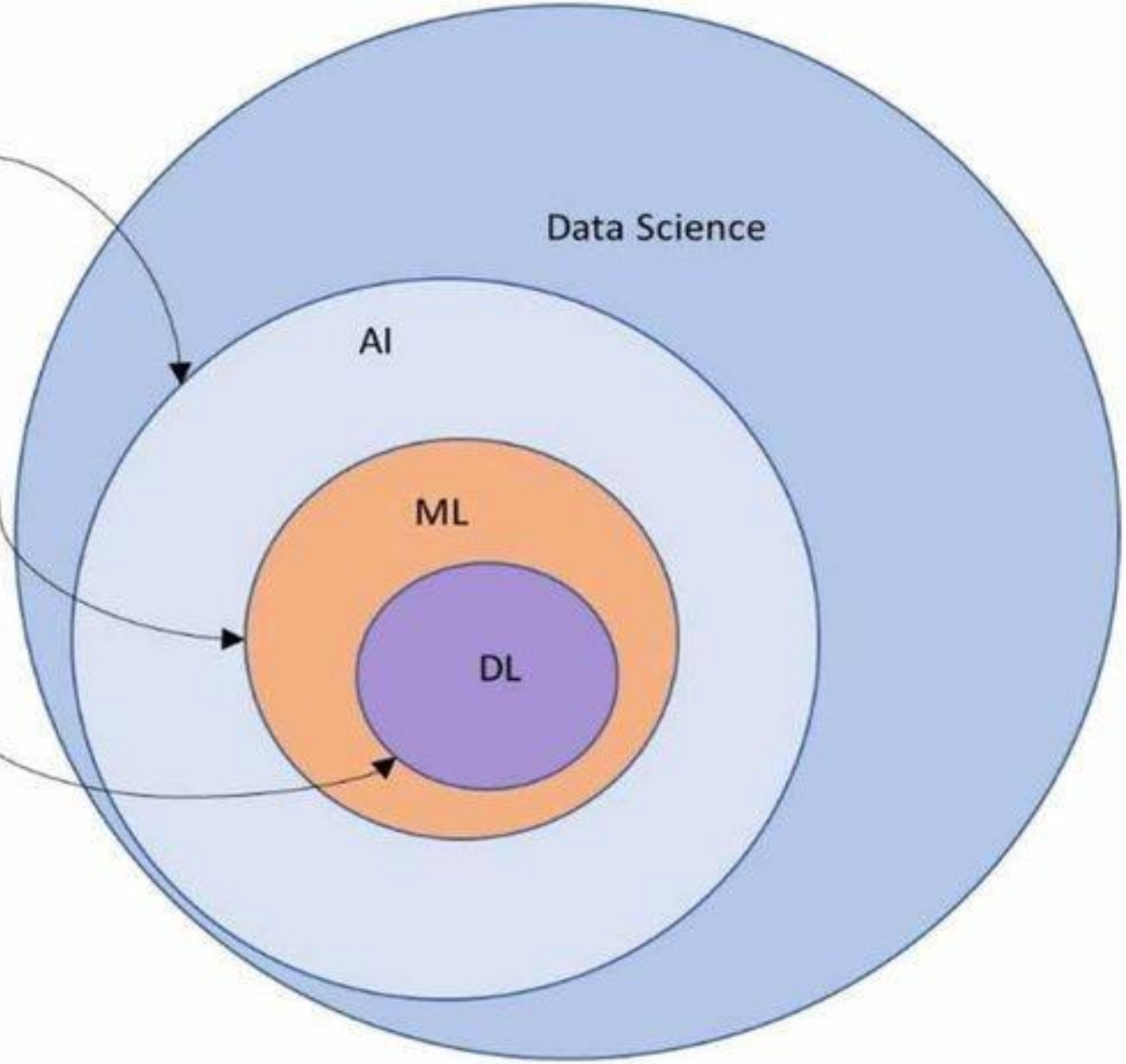
Artificial Intelligence

Machine Learning

- Supervised learning (Labeled historical data)
- Unsupervised learning (unlabeled data)
- Reinforced learning

Deep Learning

- Multilayer Neural Network (Multilayer Perceptron)
- Convolutional NN
- Artificial NN
- Recursive NN
- Recurrent NN
- Shallow NN
- Long Short Term Memories
- Sequence to Sequence Models



Supervised learning

- We have supervised learning when a computer uses given labels as examples to take and sort series of data and thus to predict future events.
- Essentially, ***in supervised learning people teach or train the machine using labeled data***. *Labeled data means it is already tagged with the right answer.*
- The machine algorithm starts from the analysis of a well-known ***training dataset (also called input data)*** and ***then model a function to make predictions about future outcomes.***

Supervised learning

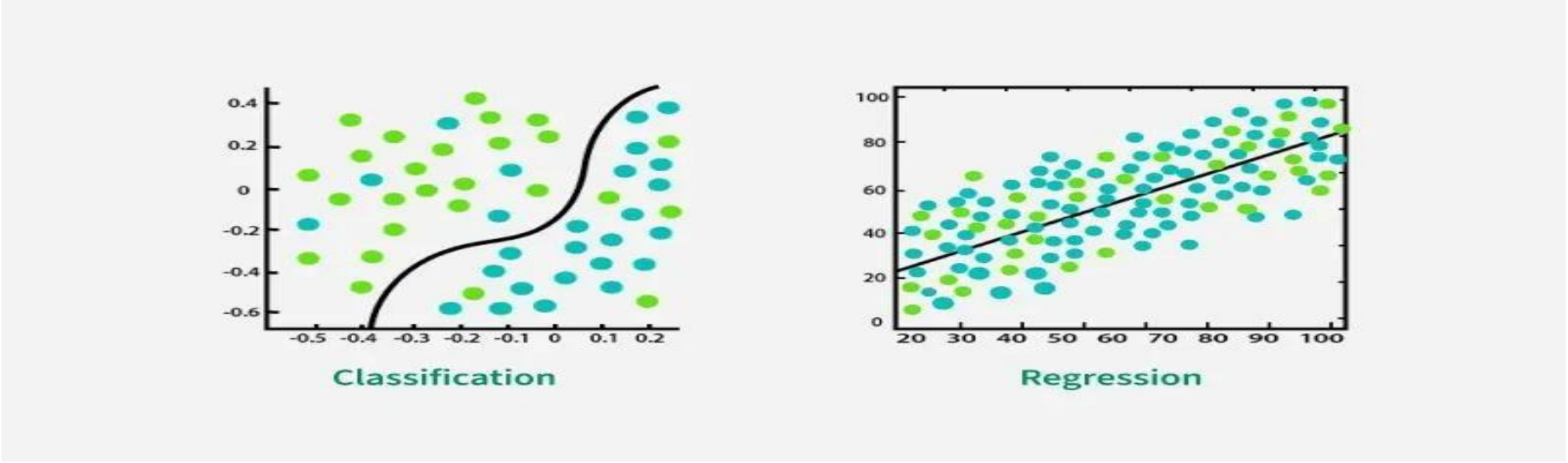
- **Supervised learning** involves ***training a model on labeled data, where the algorithm learns from the input data and its corresponding target (output labels***).

Supervised learning algorithms are generally categorized into **two main types:**

- **Classification** - where the goal is ***to predict discrete labels or categories. Classification deals with discrete outcomes (e.g., yes/no, categories)***
- **Regression** - where the aim is ***to predict continuous numerical values. Regression handles continuous values (e.g., price, temperature).***
- *Both approaches* ***require labeled data for training but differ in their objectives.***
- **Classification aims to find decision boundaries that separate classes, whereas Regression focuses on finding the best-fitting line to predict numerical outcomes.**
- The goal is to map from input to output, allowing the model to learn the relationship and make predictions based on the learnings of new data.
- **Examples:** linear regression, logistic regression, decision trees, and more.

Cont..

- **Classification Example:** it can ***determine whether an email is spam or not***, classify ***images as "cat" or "dog,"*** or ***predict weather conditions like "sunny," "rainy," or "cloudy."*** with decision boundary.
- **Regression models** are used ***to predict house prices based on features like size and location, or forecast stock prices over time with straight fit li***ne.



Regression



What will be the temperature tomorrow?

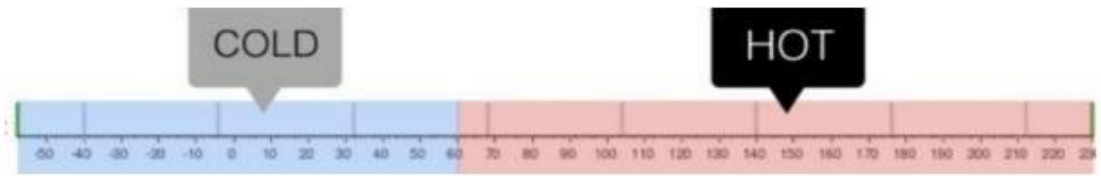


Fahrenheit

Classification



Will it be hot or cold tomorrow?



Fahrenheit

CLASSIFICATION VS REGRESSION



Student Profile

CLASSIFICATION

Predicting Student

Pass Or Fail



Student Profile

REGRESSION

Predicting Student Marks

Percentage

Regression:

Suppose from your past data (**train data**) you come to know that your best friend likes the above movies. ***You also know how many times each particular movie seen by your friend***. Now one new movie **(test data)** released. Now your are going to find how many times this newly released movie will your friend watch. ***It could be 5 times, 6 times,10 times etc…***

If you clearly observe the problem is about finding the **count**, sometimes we can say this as **predicting the value.**

- **If forecasting target class (Classification)**
- **If forecasting a value (Regression)**

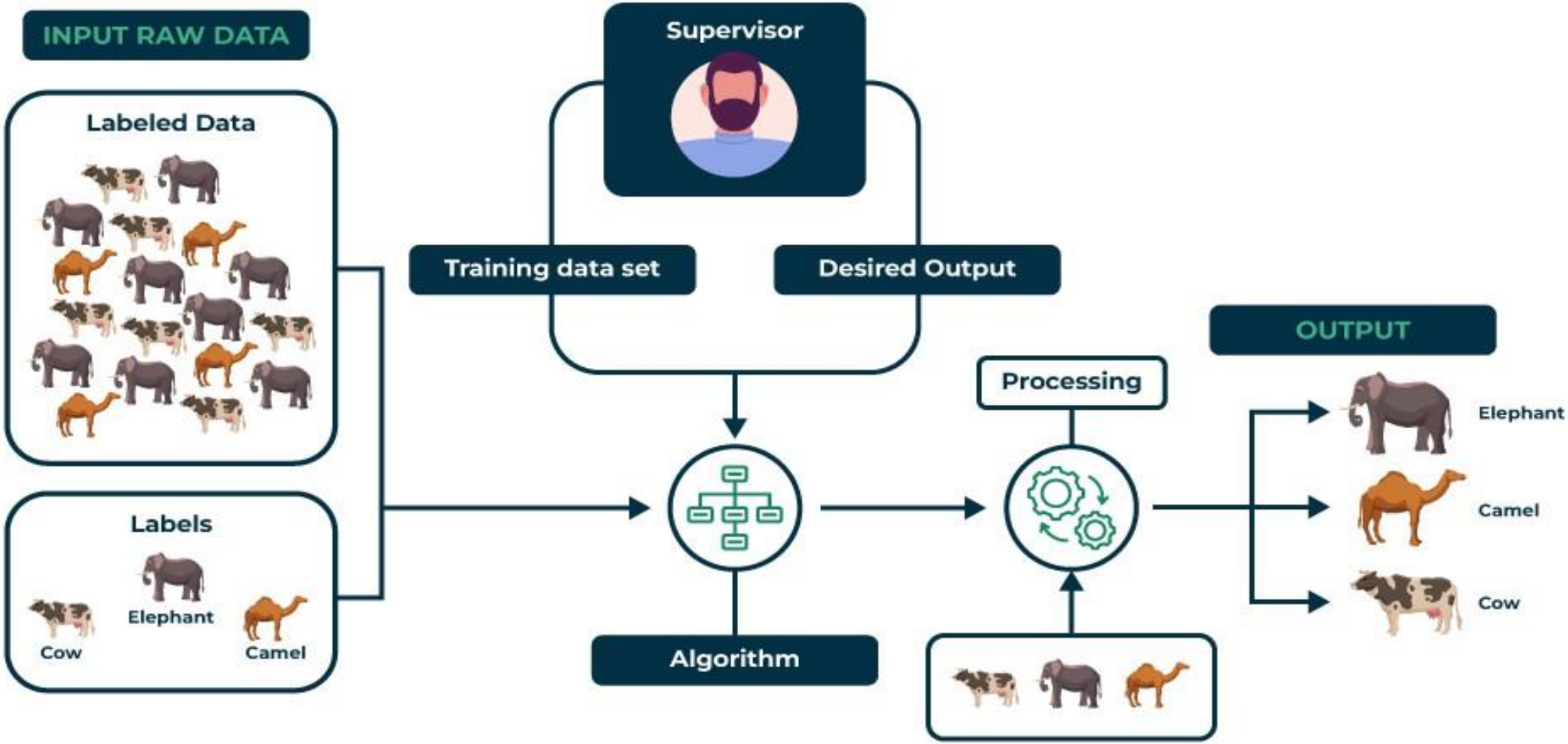


Classification:

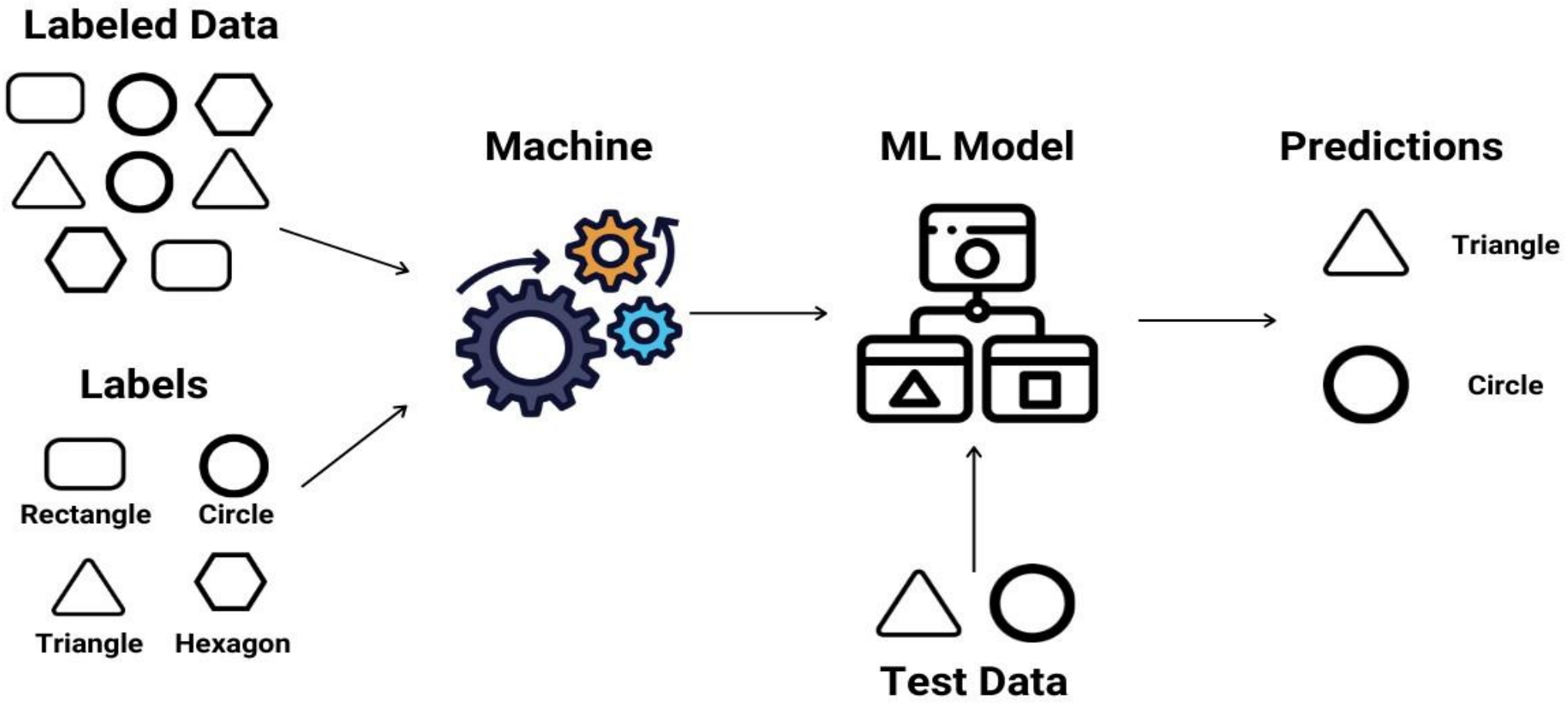
Suppose from your past data (**train data**) you come to know that your best friend likes the above movies. Now one new movie (**test data**) released.

If you clearly observe the problem it is just whether your friend **like or not.** Finding a solution to this type of problem is called as **classification.**

Supervised Learning

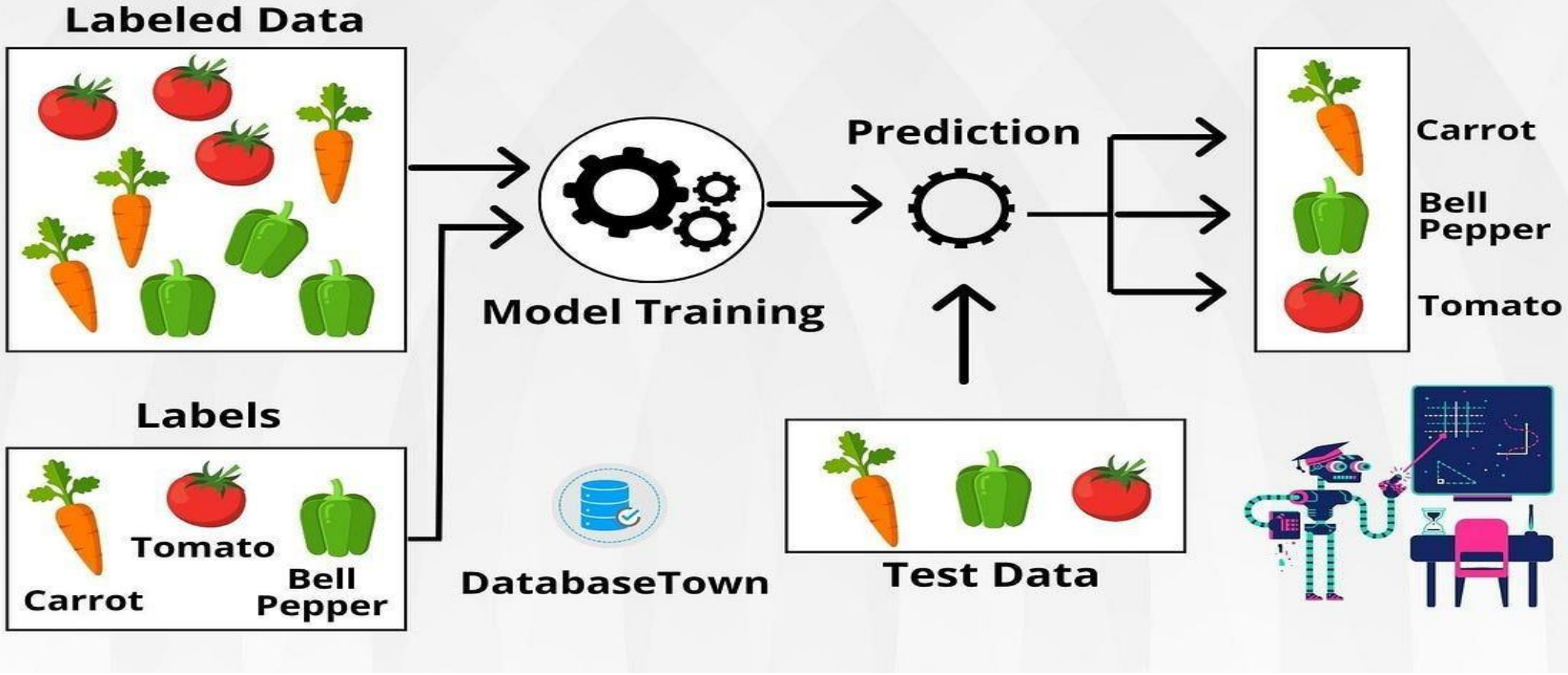


Supervised Learning



SUPERVISED LEARNING

Supervised machine learning is a branch of artificial intelligence that focuses on training models to make predictions or decisions based on labeled training data.





Example:

- ***Suppose a bunch of different kinds of flowers. First, need to train the*** machine on how to classify all different flowers:

You can train it like this:



- **If there are thorns and the head has color Red then it will be labeled as Rose.**
- **If there aren't thorns and the head has color White then it will be labeled as Daisy.**

- After training the data, there is a new separate flower (say Rose/Test Data) from the bunch and you need to ask the machine to identify it.
-

Cont..

- The machine will classify the flower regarding (the presence or absence of thorns) and color and would label the flower name like Rose.

- This is how machines learn from training data (the bunch of flowers in our case) and then use the knowledge to label data.

What is Classification in Machine Learning?

- Classification is used when you want to categorize data into different **classes or groups.**
- **For example,** classifying emails as "spam" or "not spam" or ***predicting whether a patient has a certain disease based on their symptoms.***

Types of classification models:

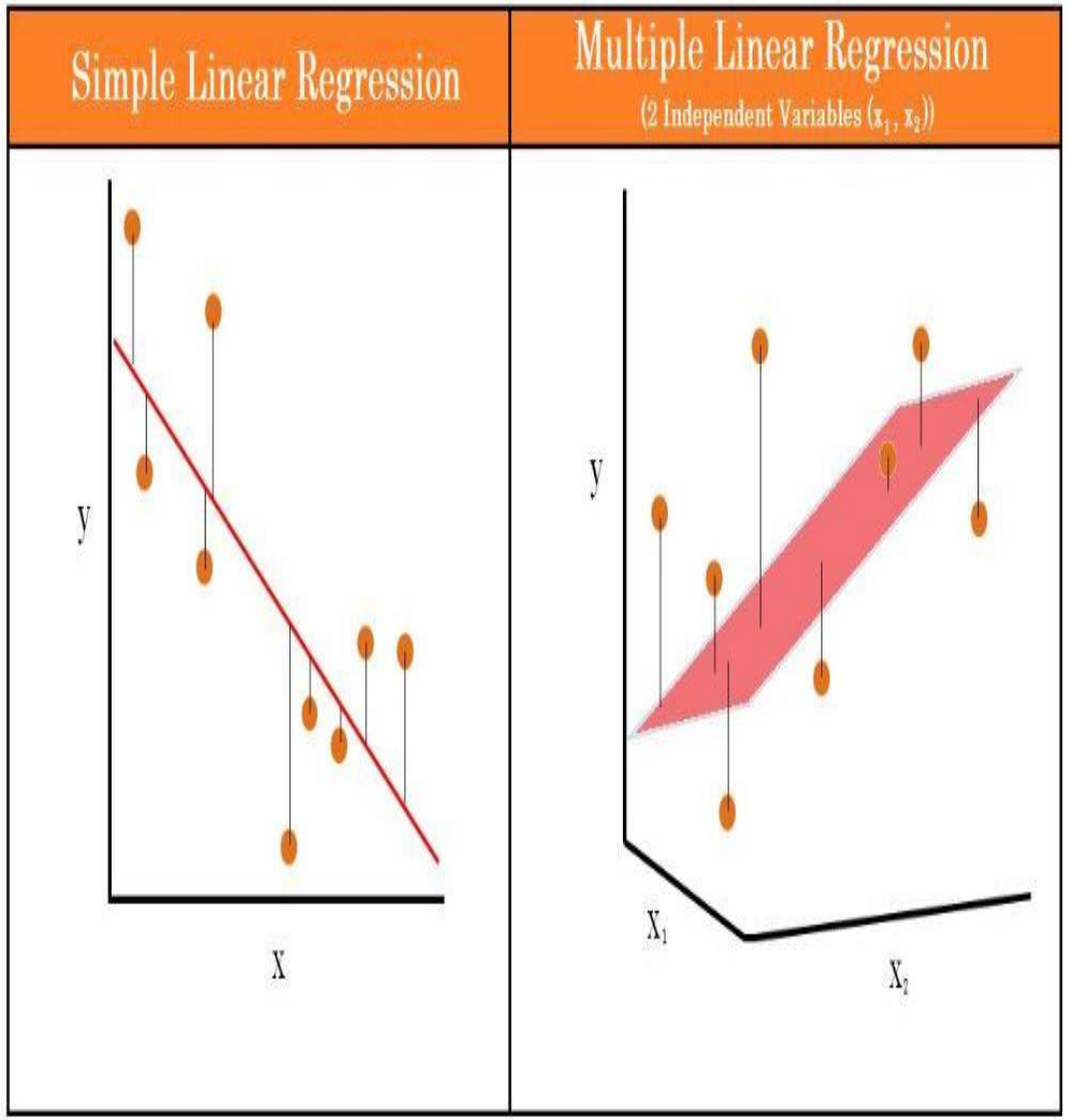
- **Decision Tree Classification**: Builds a tree where each node represents a test case for an attribute, and ***branches represent possible outcomes.***
- **Random Forest Classification**: Uses an ensemble of decision trees to make predictions, improving ***accuracy by averaging the results from multiple decision trees.***
- **K-Nearest Neighbor (KNN)**: ***Classifies data points based on the 'k' nearest neighbors using feature similarity.***

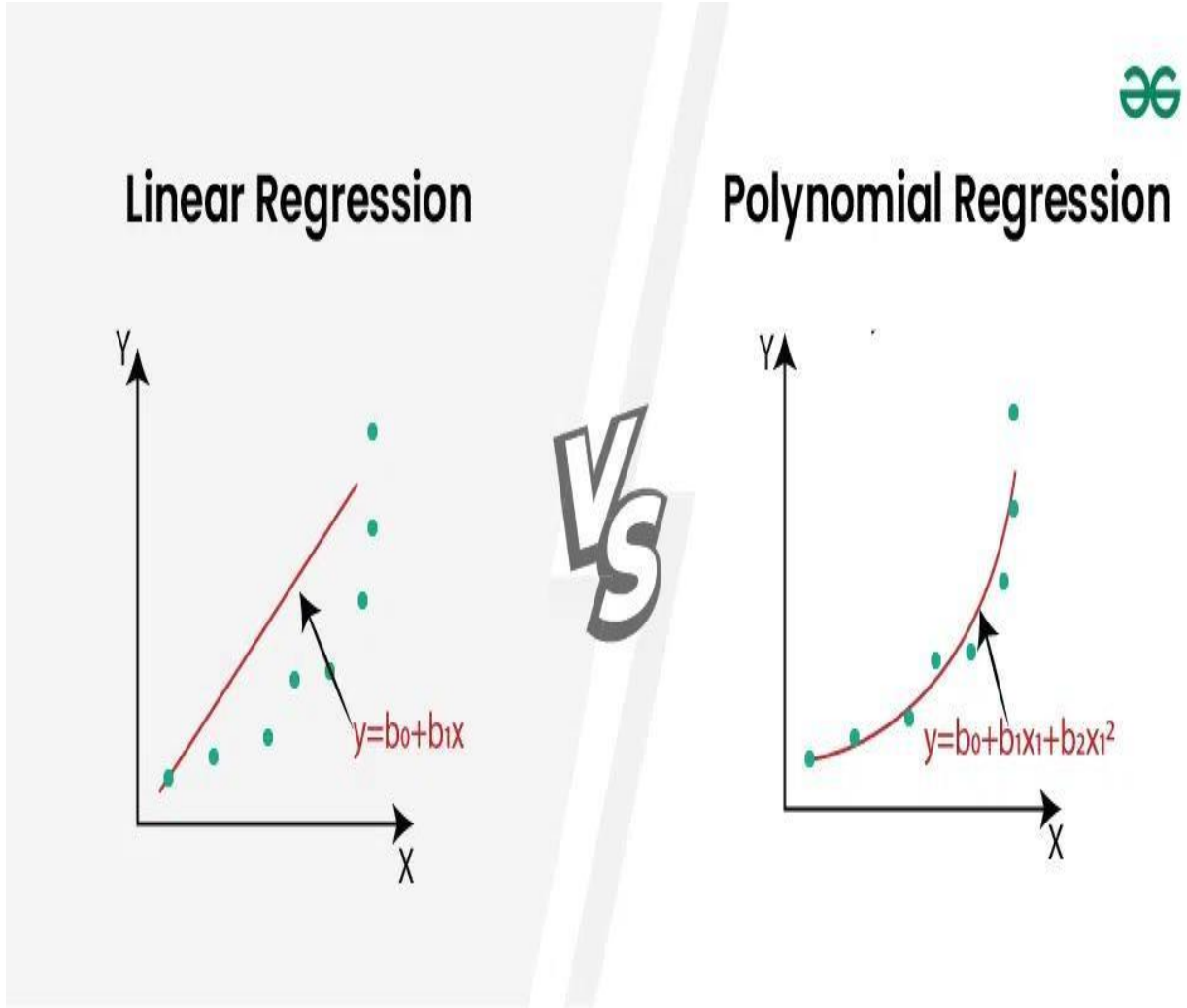
What is Regression in Machine Learning?

- Regression algorithms ***predict a continuous value based on input data.***
- ***When predict numbers such*** as income, height, weight, or even the probability of something happening (like the chance of rain).

Types of Regression are:

- **Simple Linear Regression**: Models the relationship between one independent variable and a dependent variable **using a straight line.**
- **Multiple Linear Regression**: Predicts a dependent variable based on two or more independent variables.
- **Polynomial Regression**: Models nonlinear relationships by fitting a curve to the data.


Simple Linear Regression
Multiple Linear Regression
(2 Independent Variables (x_1, x_2))
y
x
y
X_1
X_2


Linear Regression
Polynomial Regression
Y
X
$y=b_0+b_1x$
VS
Y
X
$y=b_0+b_1x_1+b_2x_1^2$

Feature	Classification	Regression
Output type	In this problem statement, the target variables are discrete. Discrete categories (e.g., "spam" or "not spam")	Continuous numerical value (e.g., price, temperature).
Goal	To predict which category a data point belongs to.	To predict an exact numerical value based on input data.
Example problems	Email spam detection, image recognition, customer sentiment analysis.	House price prediction, stock market forecasting, sales prediction.
Evaluation metrics	Evaluation metrics like Precision, Recall, and F1-Score	Mean Squared Error, R2-Score, , MAPE and RMSE.
Decision boundary	Clearly defined boundaries between different classes.	No distinct boundaries, focuses on finding the best fit line.
Common algorithms	Logistic regression, Decision trees, Support Vector Machines (SVM)	Linear Regression, Polynomial Regression, Decision Trees (with regression objective).

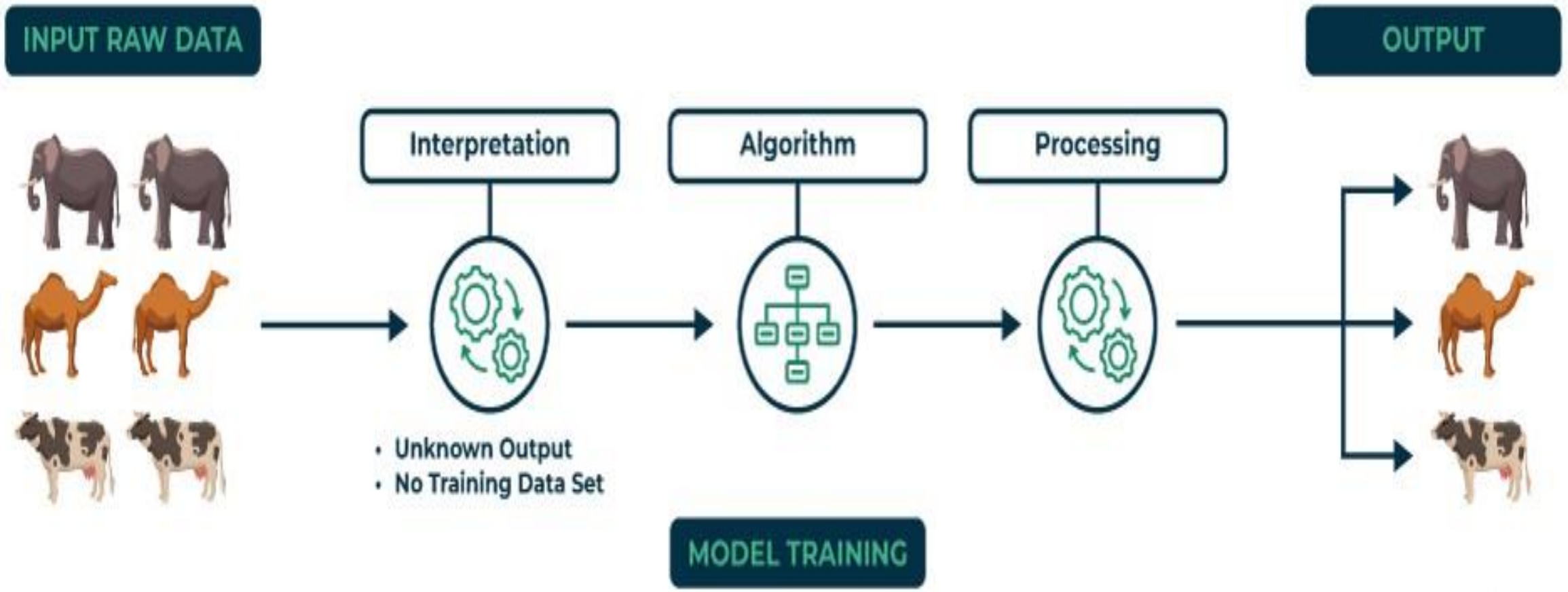
Unsupervised learning

- **Unsupervised learning**, on the other hand, ***deals with the unlabeled dataset where algorithms try to uncover hidden patterns or structures within the data.***
- The model has to discover the hidden patterns or group the data on its own without any human intervention. It is similar to the human brain learning new things.
- **Supervised learning** which ***depends on labeled data*** to create patterns or relationships for further predictions, ***unsupervised learning operates without such guidance.***
- **Examples:** Clustering algorithms like k-means, hierarchical clustering dimensionality reduction algorithms like PCA, and more.

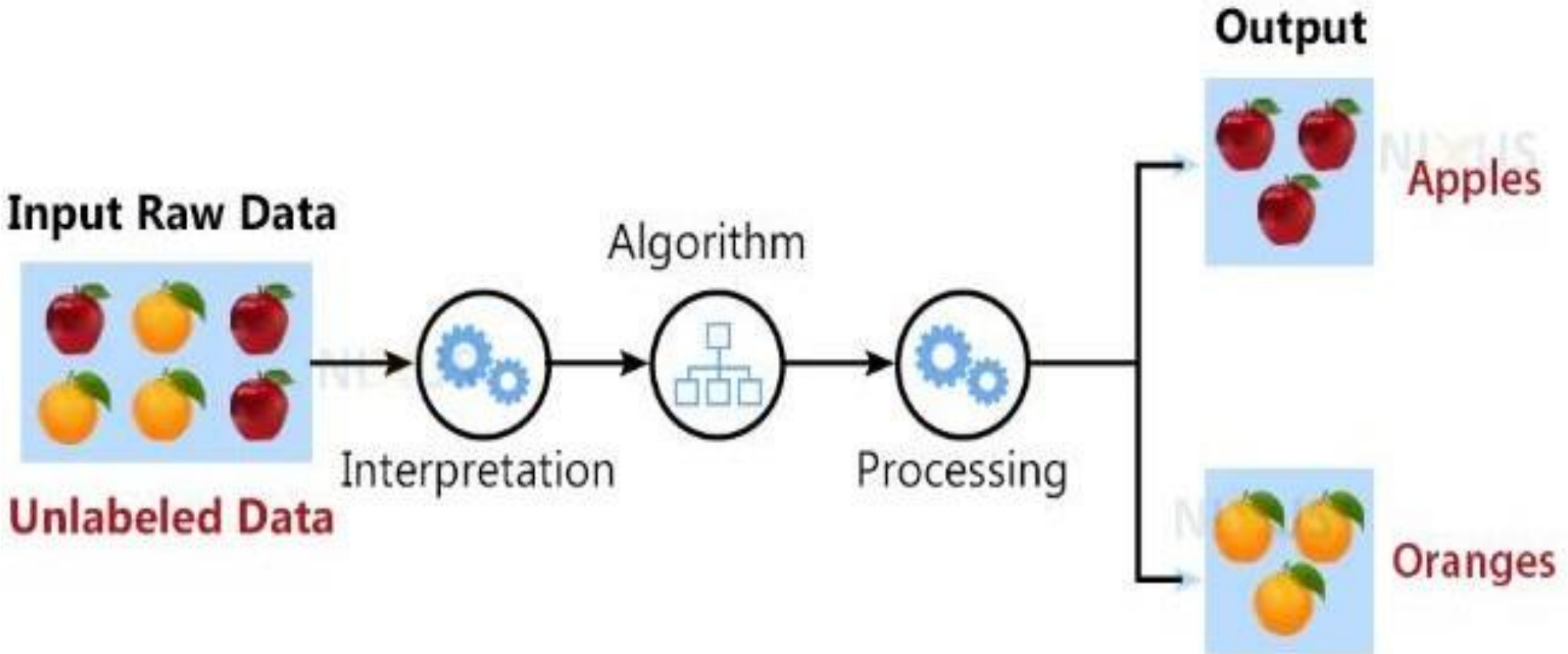
Cont..

- ***The models here do not need labels for their data and sample outputs.*** Instead, they operate independently to identify patterns and trends in the data.
- **Models train on unlabeled data and then operate on it without supervision**
- Unsupervised learning is like ***how a human learns to think via their own experiences, bringing it closer to true AI.***

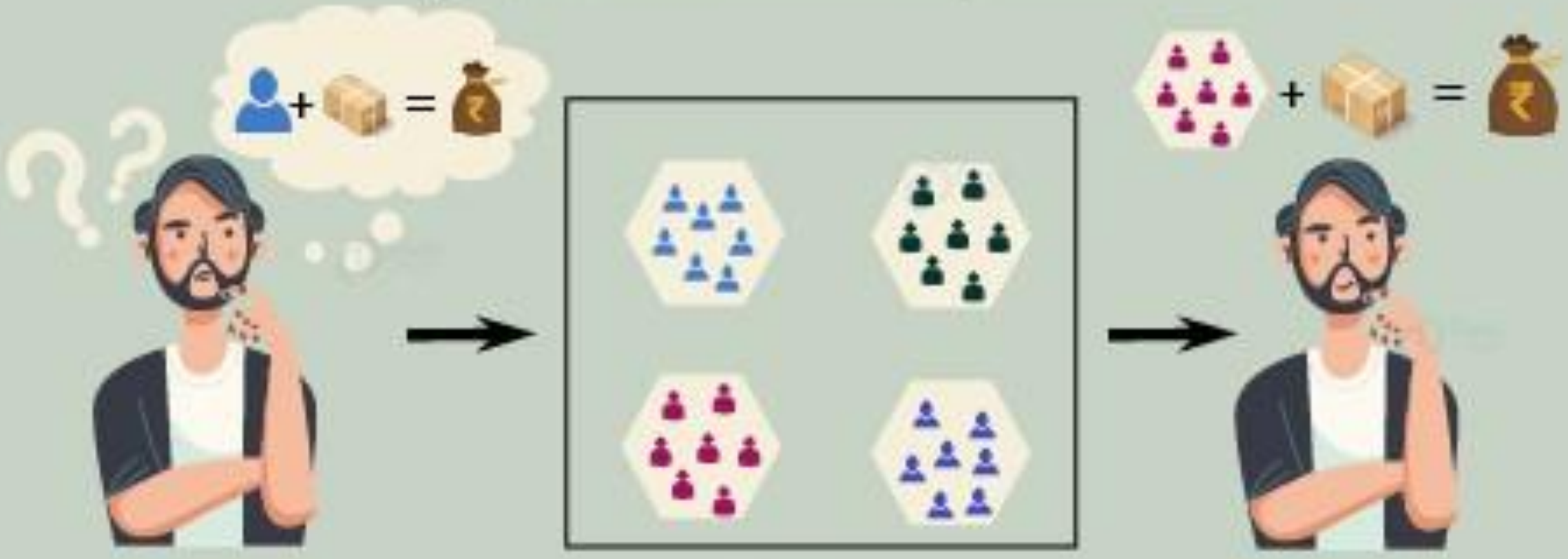
Unsupervised Learning



Unsupervised Machine Learning



Unsupervised Learning in real life

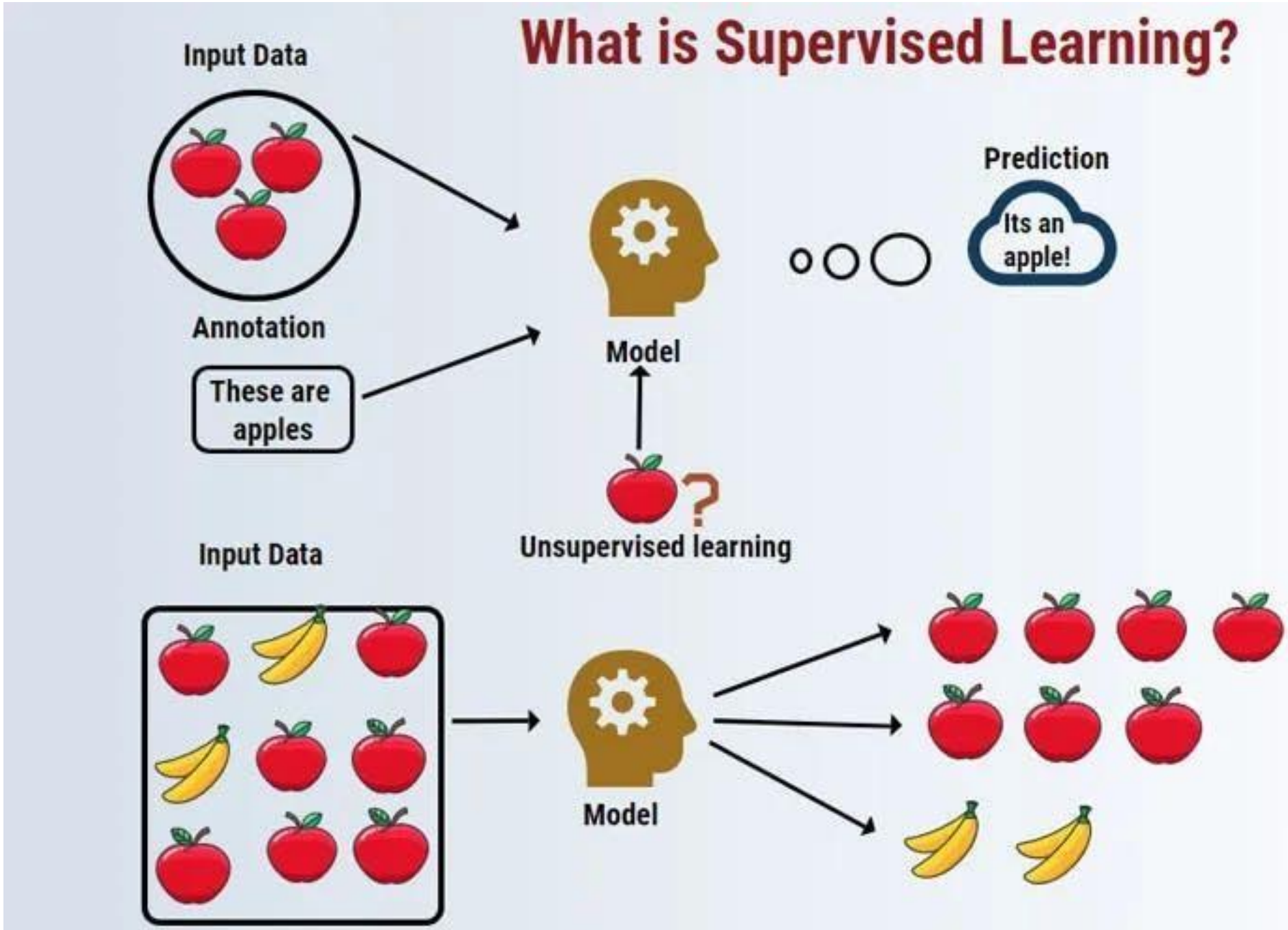


Identifying the potential customer base for selling the product

Implementing Clustering Algorithms to group the customer base

Selling product to the identified customer group

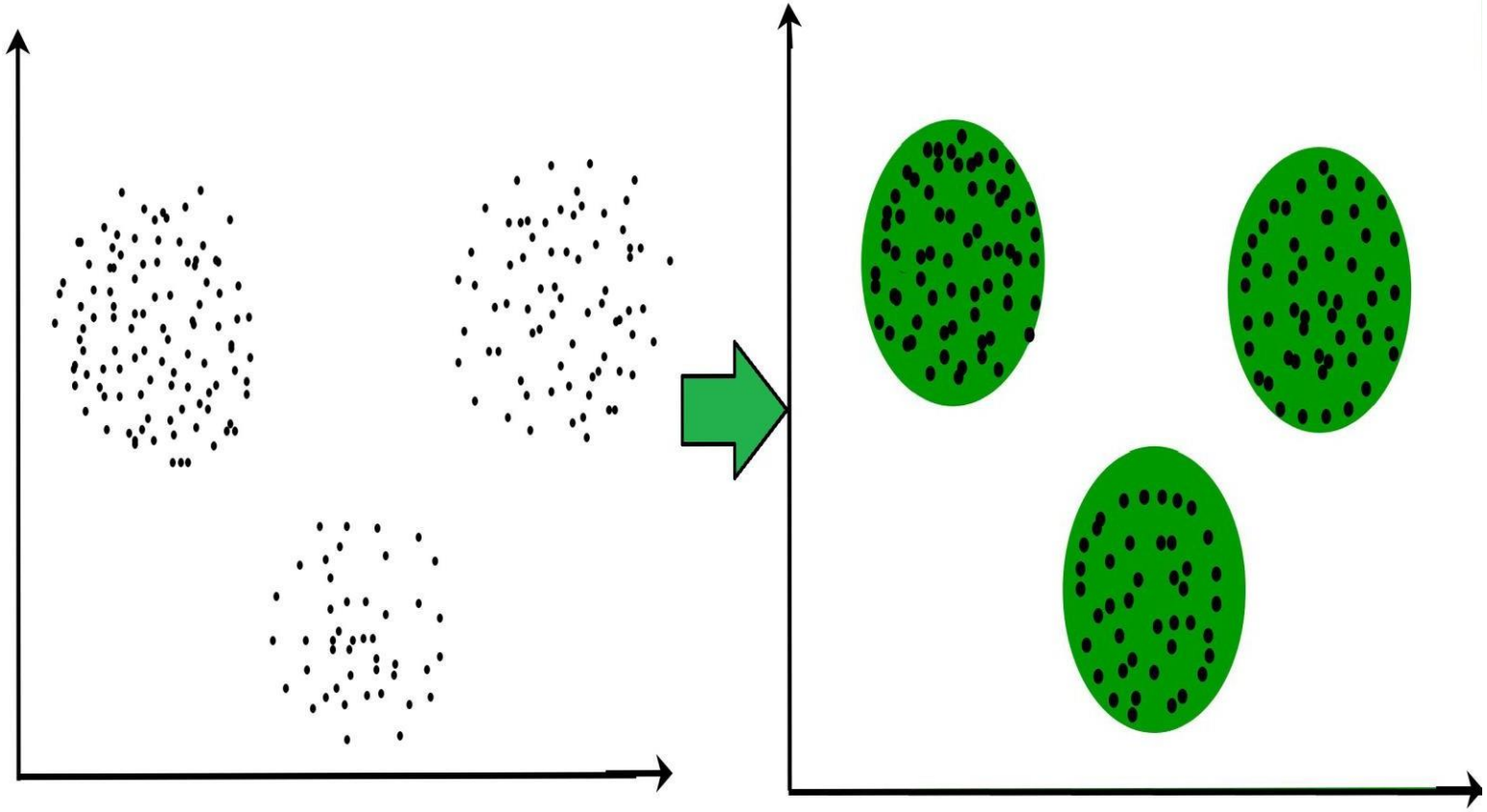
What is Supervised Learning?



Clustering

- The task of **grouping data points based on their similarity with each other is called Clustering or Cluster Analysis**.

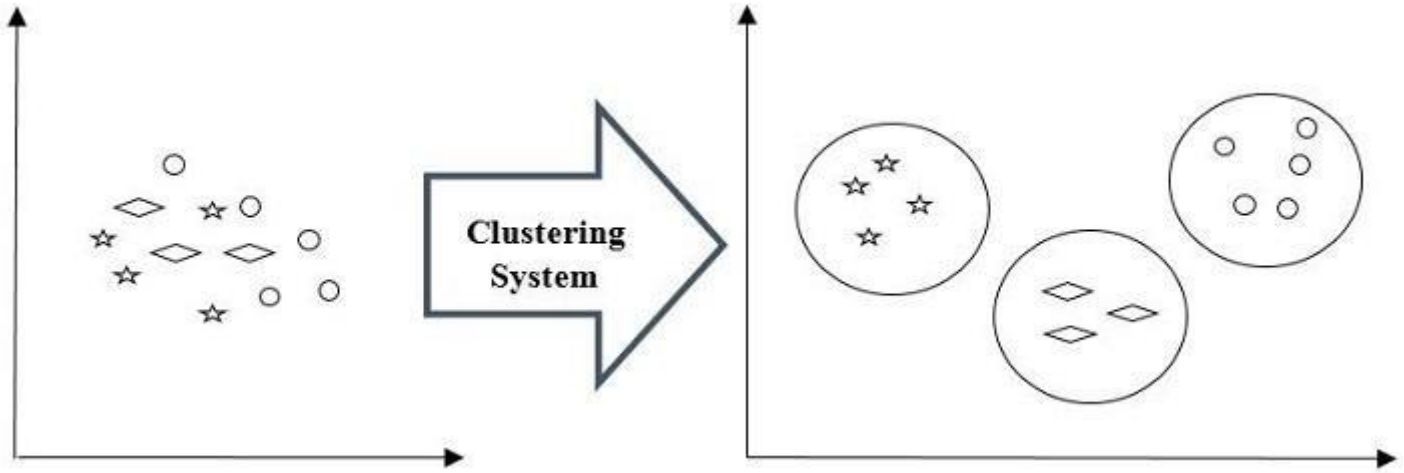
- **Example**: a dataset of customers shopping habits. **Clustering can help you group customers with similar purchasing behaviors, which can then be used for targeted marketing, product recommendations, or customer segmentation**



Clustering Algorithms are one of the most useful unsupervised machine learning methods. **These methods are used to find similarity as well as the relationship patterns among data samples and then cluster those samples into groups having similarity based on features**.

Example: Clustering

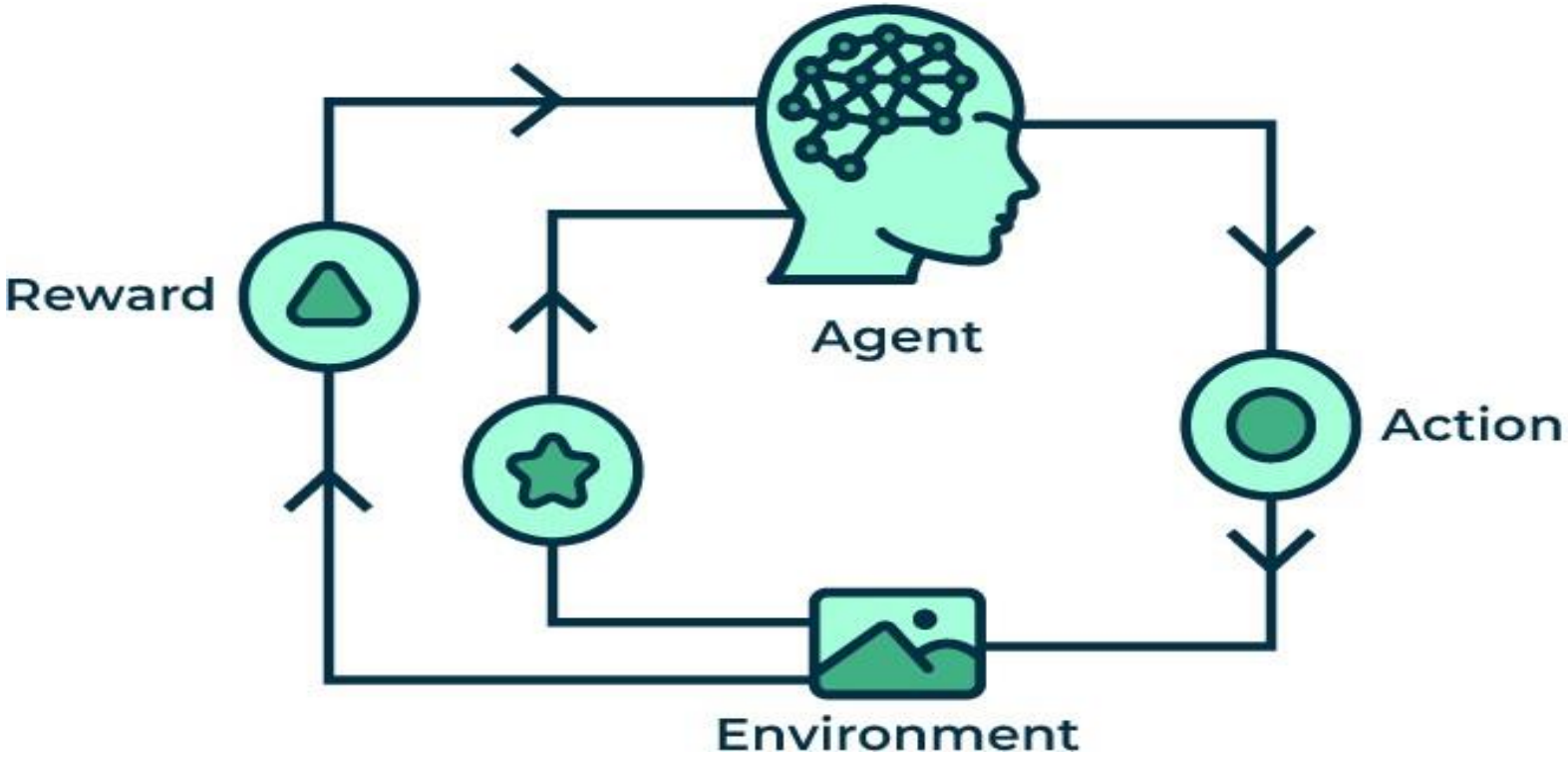
For example, below is the diagram which shows clustering system grouped **together the similar kind of data in different clusters**



Reinforcement learning

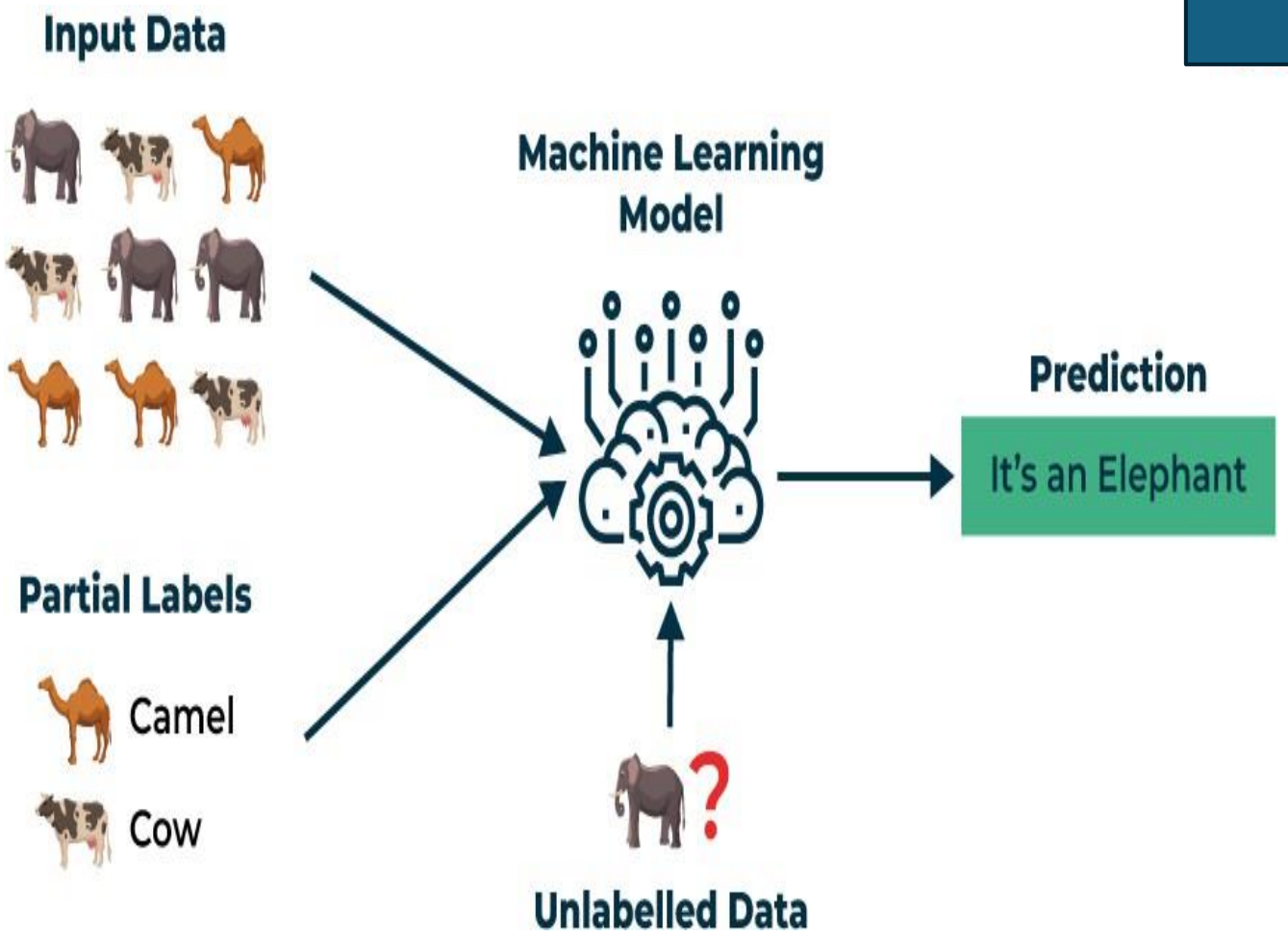
- **Reinforcement learning** is a part of machine learning that ***involves training an agent to interact with an environment and learn optimal actions through trial and error***.
- Reinforcement learning interacts with environment and learn from them based on rewards.
- It employs a reward-penalty strategy, the agent receives feedback in the form of rewards or penalties based on its actions, allowing it to learn from experience and maximize its reward over time.
- **Reinforcement learning** applications in areas such as robotics, games, and more.

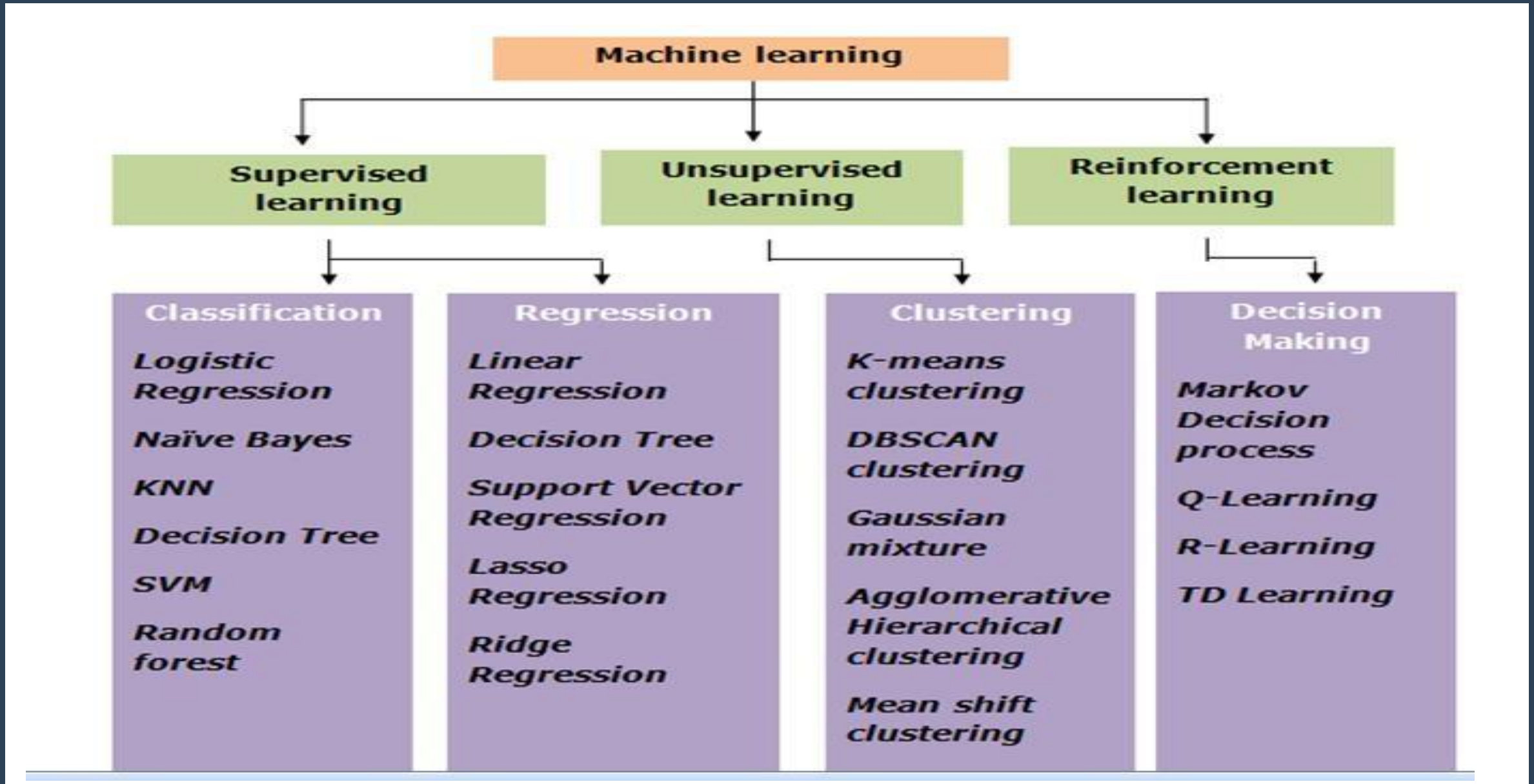
Video link:

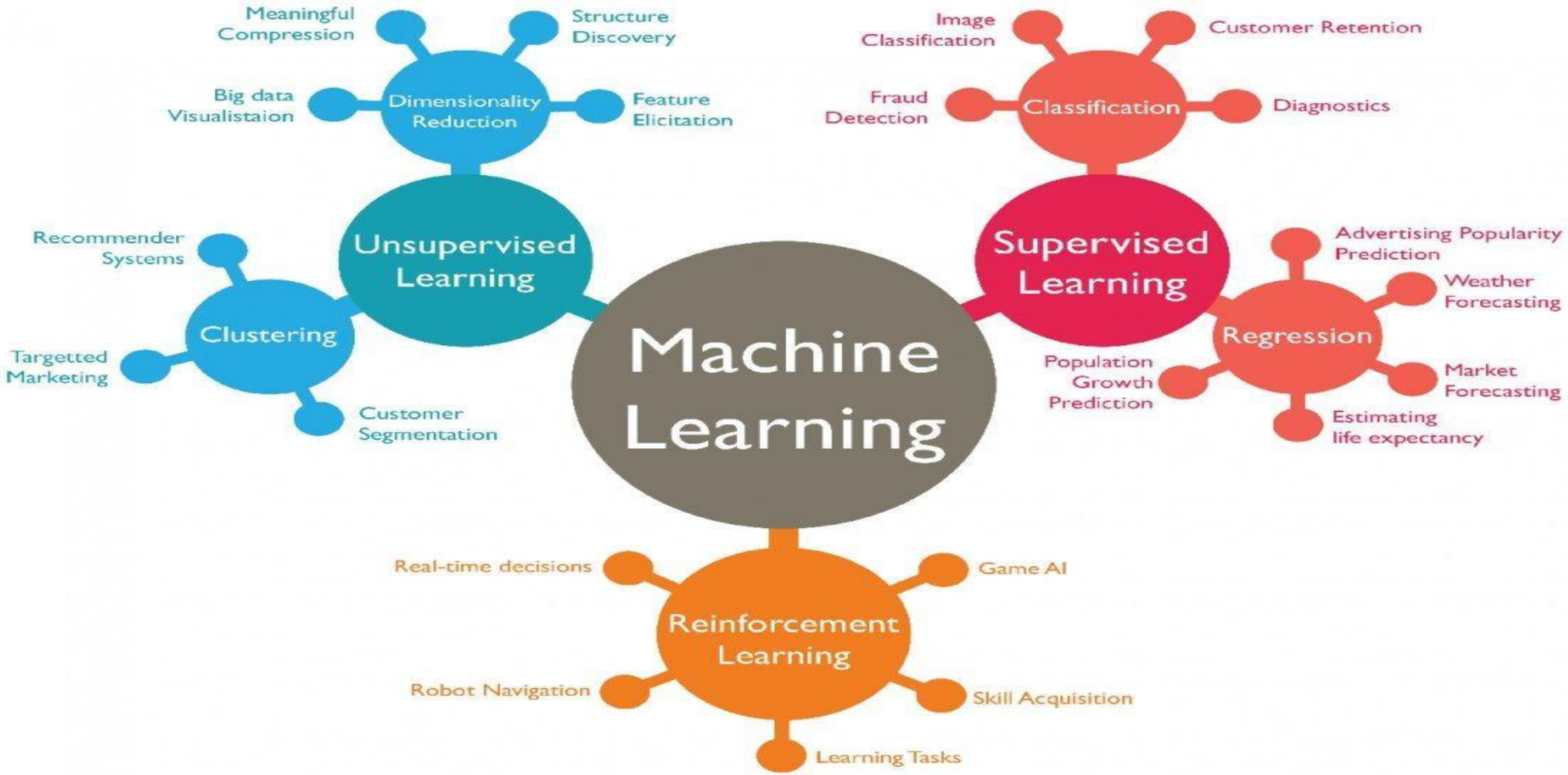


Semi Supervised Learning

- It uses a mix of labeled and unlabeled data making it helpful ***when labeling data is costly or it is very limited.***




Machine learning
Supervised learning
Unsupervised learning
Reinforcement learning
Classification
Logistic Regression
Naïve Bayes
KNN
Decision Tree
SVM
Random forest
Regression
Linear Regression
Decision Tree
Support Vector Regression
Lasso Regression
Ridge Regression
Clustering
K-means clustering
DBSCAN clustering
Gaussian mixture
Agglomerative Hierarchical clustering
Mean shift clustering
Decision Making
Markov Decision process
Q-Learning
R-Learning
TD Learning



Types of Machine Learning

Types of Machine Learning

Supervised Learning

Classification

- Fraud detection
- Email Spam Detection
- Diagnostics
- Image Classification

Regression

- Risk Assessment
- Score Prediction

Unsupervised Learning

Dimensionality Reduction

- Text Mining
- Face Recognition
- Big Data Visualization
- Image Recognition

Clustering

- Biology
- City Planning
- Targetted Marketing

Reinforcement Learning

- Gaming
- Finance Sector
- Manufacturing
- Inventory Management
- Robot Navigation

Key Machine Learning Terminologies:

- **Features:** These are the input variables or attributes used by the model to make predictions.
- **Labels:** The output or target variable that the model predicts in supervised learning.
- **Training Set:** A subset of the data used to train the model by identifying patterns.
- **Validation Set:** Data used to tune the model's hyperparameters and optimize performance.
- **Test Set:** Unseen data used to evaluate the model's final performance.

Most Common Applications of Data Science



Business Analytics

- Customer Analytics
- Market Segmentation
- Churn Prediction

Healthcare

- Disease Prediction
- Drug Discovery
- Healthcare Management

Social Media

- User Behavior Analysis
- Sentiment Analysis
- Content Personalization

Finance

- Risk Assessment
- Algorithmic Trading
- Financial Forecasting

E-commerce

- Recommendation Systems
- Supply Chain Optimization
- Price Optimization

Manufacturing

- Predictive Maintenance
- Quality Control
- Supply Chain Management

Entertainment

- Content Recommendation
- Box Office Prediction
- Gaming

Transportation

- Route Optimization
- Demand Forecasting
- Fleet Management

PHYGITAL INSIGHTS

Few Popular Applications of Artificial Intelligence (AI)

Amazon Delivery Robot



Self-Driving Cars



Robot Helper



Use of Machine Learning in

Recommendation Technology

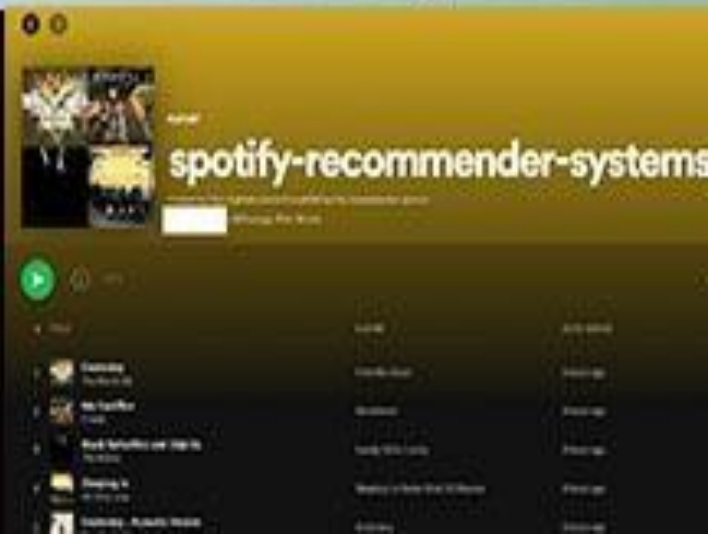
Amazon gives personalised product recommendations to its users using ML



Netflix gives personalised movie recommendations to its users using ML



Spotify gives personalised song recommendations to its users using ML



ROLES IN AI

- Machine Learning Engineer
- Data Scientist
- AI Research Scientists
- Computer Vision Engineer
- NLP Engineer
- DL Engineer
- AI Engineer
- AI Architect
- Robotics Engineer
- AI Product Manager
- AI Ethics Officer
- AI Consultant
- AI Software Developer
- AI Trainer
- AI Analyst



Top 15 Data Science Job Roles

01 Data Scientist

02 Data Analyst

03 Machine Learning Engineer

04 Data Engineer

05 Business Intelligence (BI) Analyst

06 Data Architect

07 Data Scientist Manager/Director

08 Data Science Researcher

09 Data Science Consultant

10 Data Science Educator

11 Data Science Product Manager

12 Data Science Entrepreneur

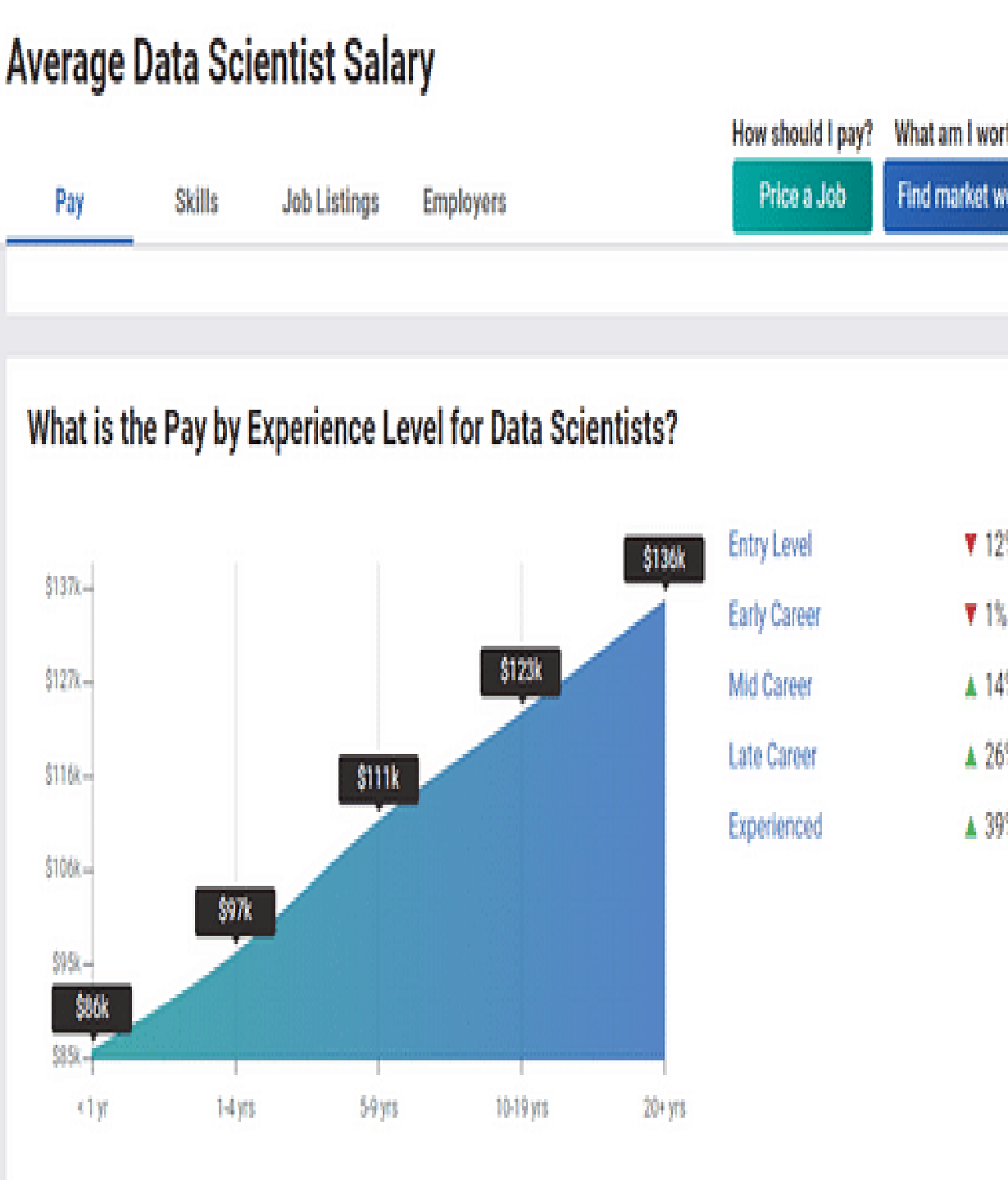
13 Data Science Ethicist

14 Data Science Project Manager

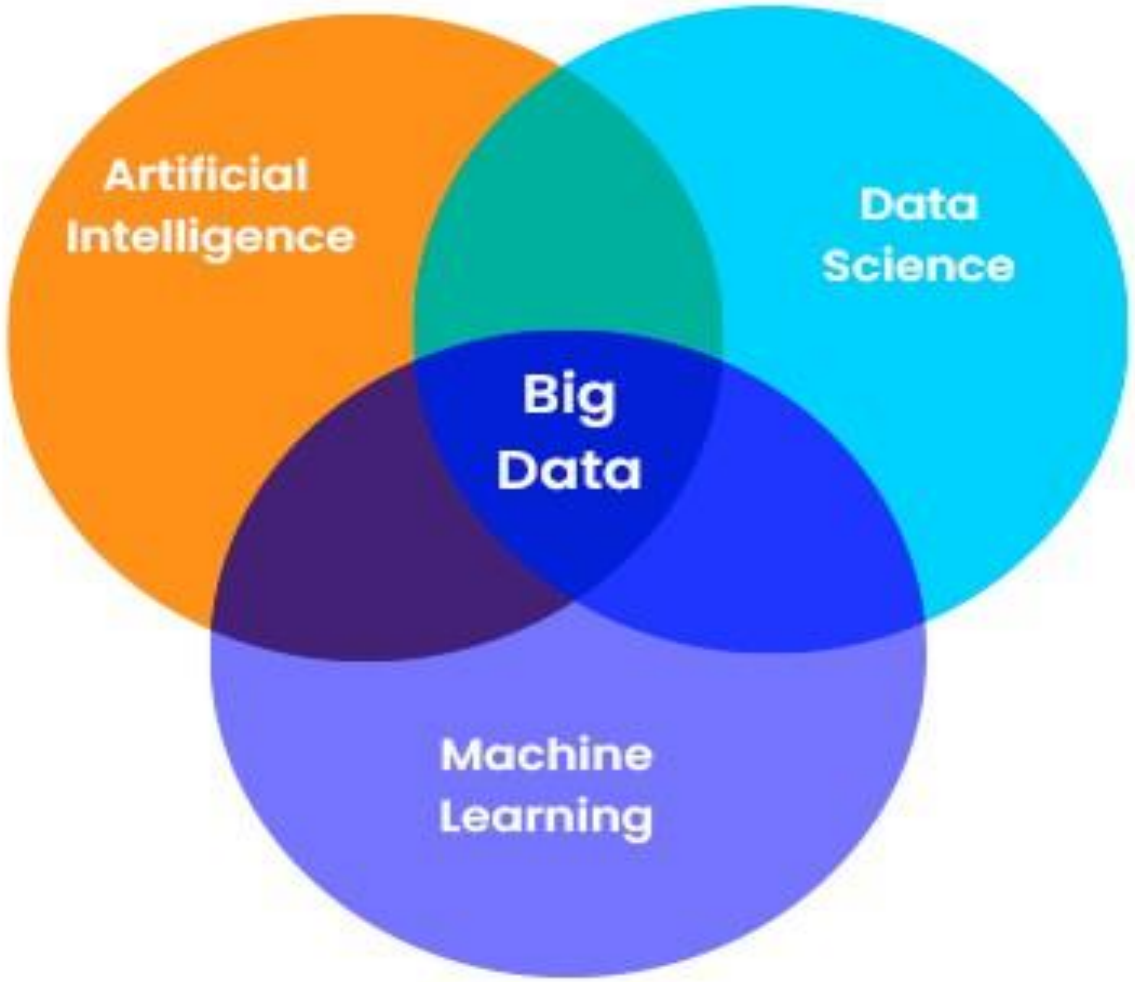
15 Marketing Analyst

•**High pay:** The **average salary** of a data scientist is about $1,00000, according to payscale in USA
•The salary varies as per the experience level, such as:

Entry-level data scientist with 1-4 years of experience earns an average salary of $96,588.



Role	Skills	Salary Range in India (INR)	Salary Range in US (USD)
Data Scientist	Python/R, Statistics, Machine Learning, Data Visualization, Domain Knowledge	₹600,000 - ₹2,500,000 per year	$80,000 - $150,000 per year
Data Analyst	SQL, Data Visualization Tools (Tableau/Power BI), Excel, Basic Statistics	₹400,000 - ₹1,500,000 per year	$50,000 - $100,000 per year
Machine Learning Engineer	Python/Java, TensorFlow/PyTorch, Cloud Platforms (AWS/Azure), Software Engineering	₹800,000 - ₹3,000,000 per year	$90,000 - $180,000 per year
Data Engineer	SQL, Python/Java, Big Data Technologies (Hadoop/Spark), Database Systems	₹600,000 - ₹2,500,000 per year	$80,000 - $150,000 per year
Business Intelligence (BI) Analyst	BI Tools (Tableau/Power BI/Looker), SQL, Data Visualization, Business Acumen	₹500,000 - ₹1,800,000 per year	$60,000 - $120,000 per year
Data Architect	Database Technologies, Data Modeling, Data Integration, Data Security	₹800,000 - ₹3,000,000 per year	$90,000 - $180,000 per year
Data Scientist Manager/Director	Leadership and Management, Technical Expertise in Data Science and Analytics, Communication Skills	₹2,000,000 - ₹5,000,000 per year	$150,000 - $250,000 per year
Data Science Researcher	Advanced Knowledge in Data Science, Research Experience, Publication Record, Analytical Skills	₹800,000 - ₹3,500,000 per year	$100,000 - $200,000 per year
Data Science Consultant	Data Science Methodologies, Business Acumen, Communication Skills, Problem-Solving Abilities	₹800,000 - ₹3,000,000 per year	$90,000 - $180,000 per year
Data Science Educator	Data Science Concepts, Teaching and Instructional Design, Communication Skills, Continuous Learning	₹600,000 - ₹2,000,000 per year	$70,000 - $150,000 per year
Data Science Product Manager	Product Management Methodologies, Data Science Concepts, Project Management Skills, Communication Skills	₹1,200,000 - ₹4,000,000 per year	$120,000 - $200,000 per year
Data Science Entrepreneur	Data Science Expertise, Entrepreneurship Skills, Business Acumen, Leadership Abilities	Varies widely	Varies widely
Data Science Ethicist	Ethical Theories and Principles, Policy Analysis, Communication Skills, Collaboration Abilities	Varies widely	Varies widely
Data Science Project Manager	Project Management Methodologies, Data Science Concepts, Communication Skills, Problem-Solving Abilities	₹800,000 - ₹3,500,000 per year	$90,000 - $180,000 per year
Marketing Analyst	Statistical Analysis, Data Manipulation Tools (Python/R/SQL), Marketing Analytics, Communication Skills	₹600,000 - ₹2,500,000 per year	$70,000 - $140,000 per year


Artificial Intelligence
Data Science
Big Data
Machine Learning

Python code for creating and dividing the dataset

```
from sklearn.model_selection import train_test_split
import numpy as np
import pandas as pd
data = np.random.randint(low=10, high=100,
                         size=2000).reshape(1000, 2)
data = pd.DataFrame(data, columns=('x', 'y'))
X = data[['x', 'y']]
y = np.random.rand(1000)
train_data_x, test_data_x, train_data_y, test_data_y = \
    train_test_split(data, y, test_size=0.25)
```

Scaling Dataset

- ***Scaling the dataset is an important preprocessing step*** **before loading to the Model.**
- Scaling a dataset means adjusting feature values so they are comparable and suitable for machine learning models.

There are several benefits of scaling the data theses are as:

- It prevents features with different scales from dominating the model like **example** suppose column **A** has data ranging from 1 to 1000 and column **B** has data ranging from 0 to 1 in that case, column **A** can influence our model decision even if it is not an important feature. But after scaling all our columns comes in the similar range
- **It speeds up our model convergence.** Many optimization algorithms such as gradient descent are very sensitive to the scale of the data. **By scaling data between 0 to 1 these algorithm converges faster.**

Cont..

- Scaling (or normalization/standardization) is ***the process of adjusting the range or distribution of features in a dataset so that they are comparable.***

- **Different Units Problem**: ***Features may be measured in different units*** (e.g., salary in rupees vs. age in years). Without scaling, large-valued features dominate learning.

- **Distance-based Algorithms**: ***Methods like KNN, K-means, and SVM rely on distance***.

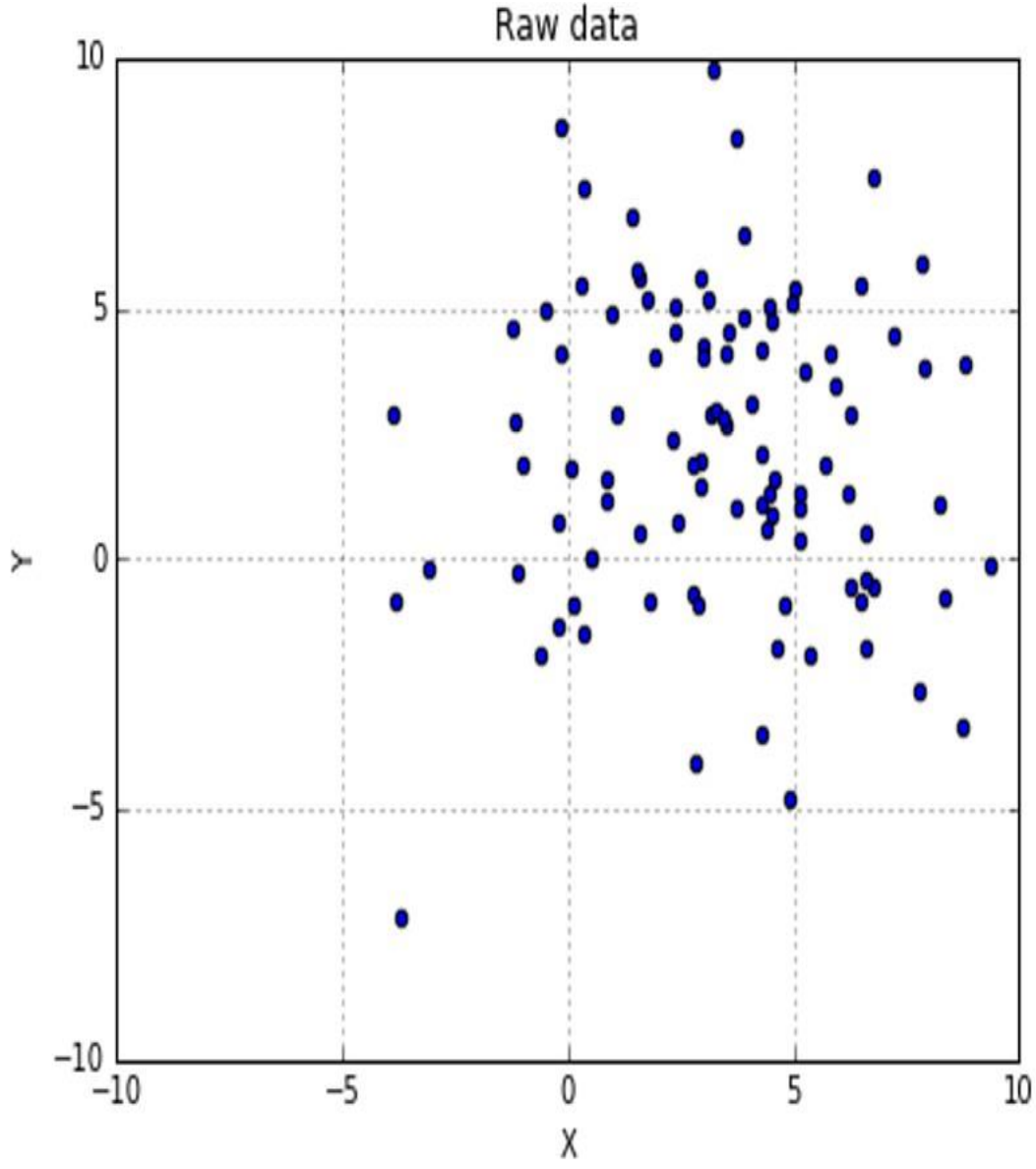
Unscaled data makes results biased toward features with larger ranges.

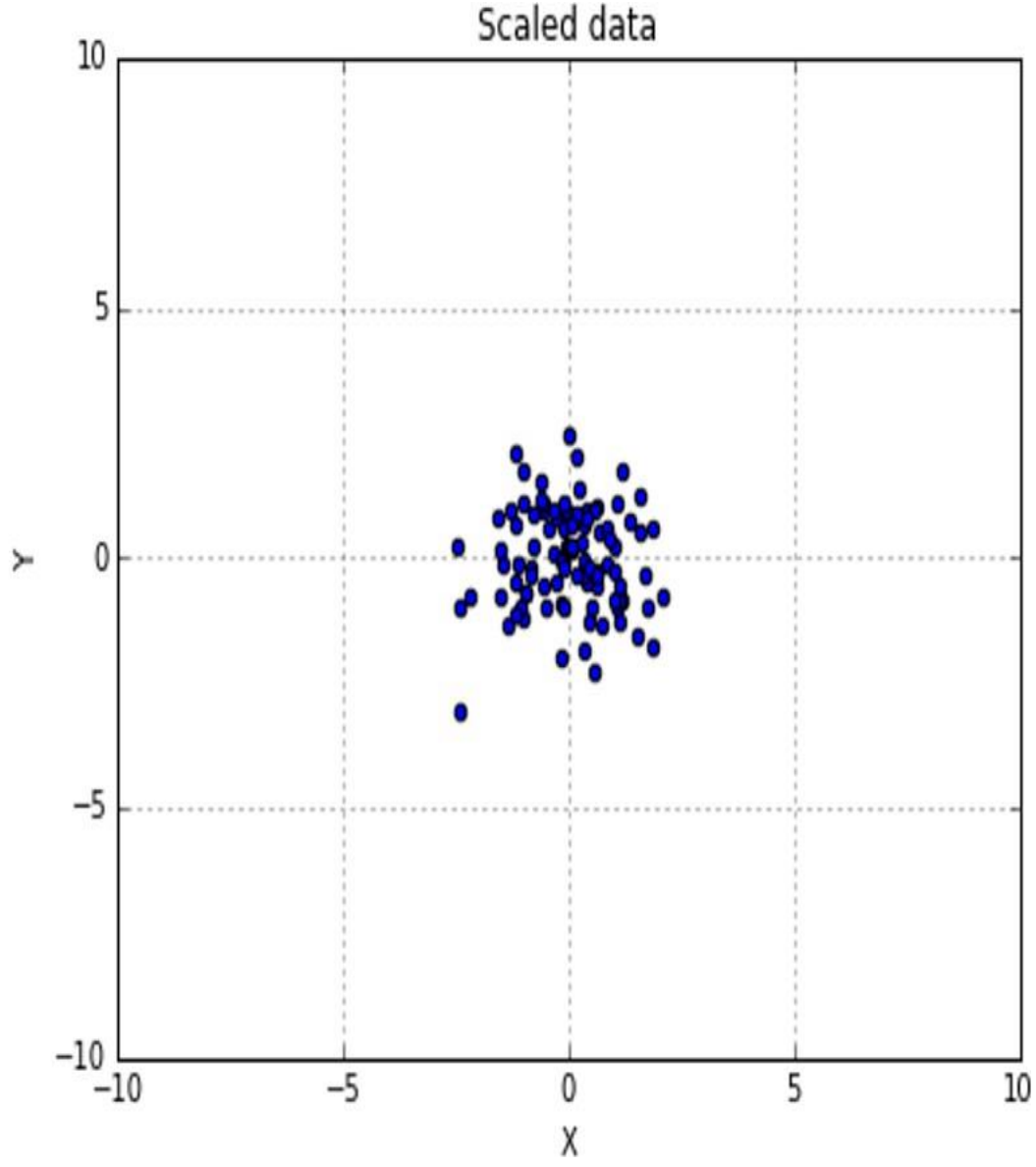
- **Gradient Descent Convergence**: In models like logistic regression, neural networks, etc., scaling helps the optimizer converge faster and avoid oscillations.

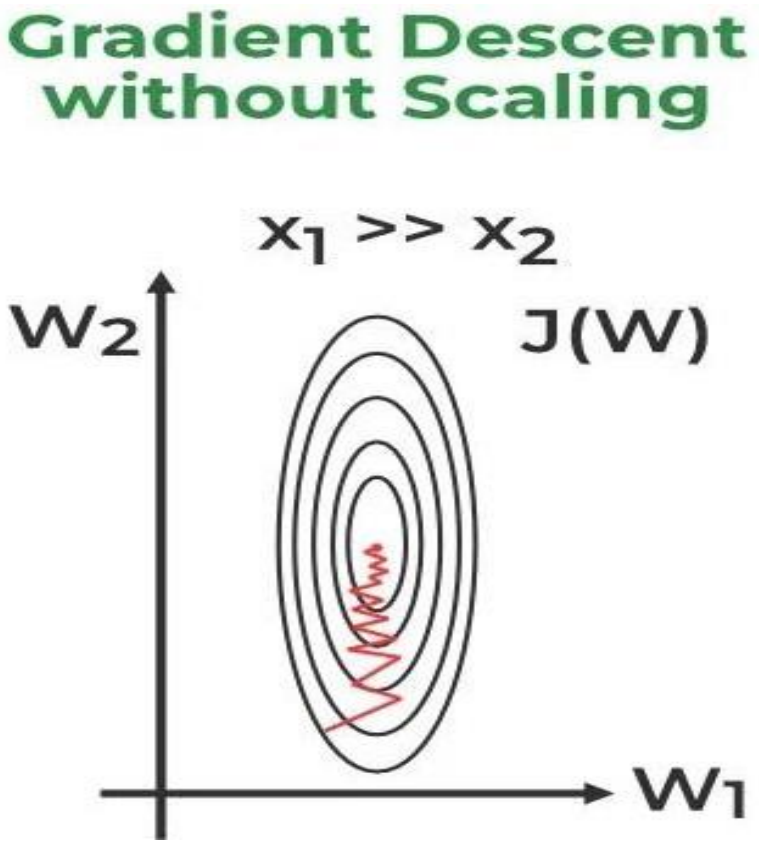
- **Better Interpretability**: ***Scaled features allow easier comparison and more meaningful coefficients.***

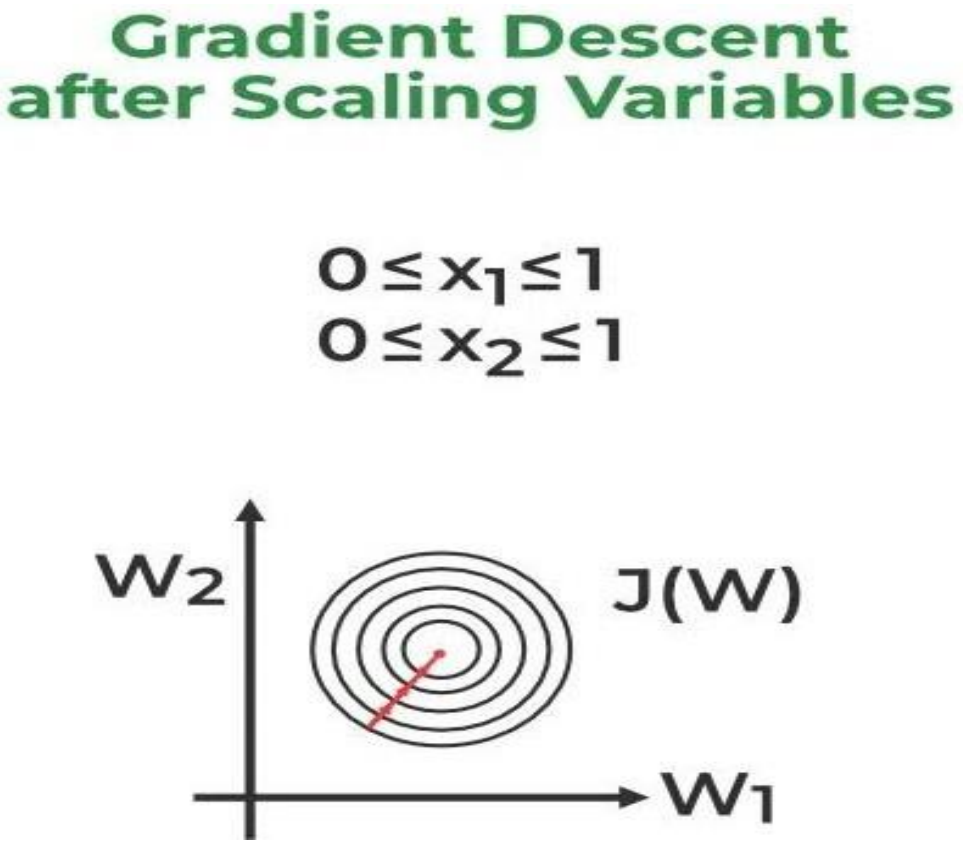
Advantages of Scaling a Dataset

- **Improved Model Accuracy** – especially for algorithms sensitive to feature magnitude.
- **Faster Convergence** – reduces training time in optimization-based models.
- **Equal Weightage to Features** – prevents dominance of high-value attributes.
- **Stability in Numerical Computation** – avoids overflow/underflow in large calculations.
- **Better Visualization** – makes data distributions easier to compare and analyze.


Raw data
10
5
0
−5
−10
Y
−10
−5
0
5
10
X


Scaled data
10
5
0
−5
−10
Y
−10
−5
0
5
10
X





- **Scaling the dataset makes our model more robust to the outliers.**

- **Some algorithms like K-nearest neighbors (KNN), use the distance between data points to make predictions in this case if the columns have different scales then the distance can go higher.**

Example:

- Suppose we have two features of weight and price, as in the below table.
- The "Weight" cannot have a meaningful comparison with the "Price.
- " So the assumption algorithm makes that since "Weight" > "Price,"
- Thus "Weight," is more important than "Price."

Name	Weight	Price
Orange	15	1
Apple	18	3
Banana	12	2
Grape	10	5

Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
df  = pd.DataFrame({'WEIGHT': [15, 18, 12,10],
                    'PRICE': [1,3,2,5]},
    index = ['Orange','Apple','Banana','Grape'])
print(df)
```

Out Put:

	WEIGHT	PRICE
Orange	15	1
Apple	18	3
Banana	12	2
Grape	10	5

Modelling The Data

- ***After scaling and splitting the data it has now become ready for fitting to the model.***
- The ***choice of choosing model totally depends on our problem formulation.***
- ***There are a variety of models present that we can choose from***. However, before choosing the ***model first, we should identify these points in the data***
- ***Whether our problem is a regression problem or a classification problem***
- Whether we want a model which is more explainable or we want a model which has a higher accuracy

Python code for modeling the data

Since our target value is continuous so here we will consider it as a regression problem. For making it simple and explainable we will use the decision tree model.

```
from sklearn.tree import DecisionTreeRegressor,plot_tree
reg = DecisionTreeRegressor(min_samples_split=4,
                            max_leaf_nodes=10)
reg.fit(train_data_x_scaled,train_data_y)
y_pred = reg.predict(test_data_x)
```

After making the model we evaluate the model on the evaluation matrix. In our case, we will mean square error for computing the accuracy of our model.

Python code for evaluation

```
from sklearn.metrics import mean_squared_error
print(mean_squared_error(y_pred,test_data_y))
```

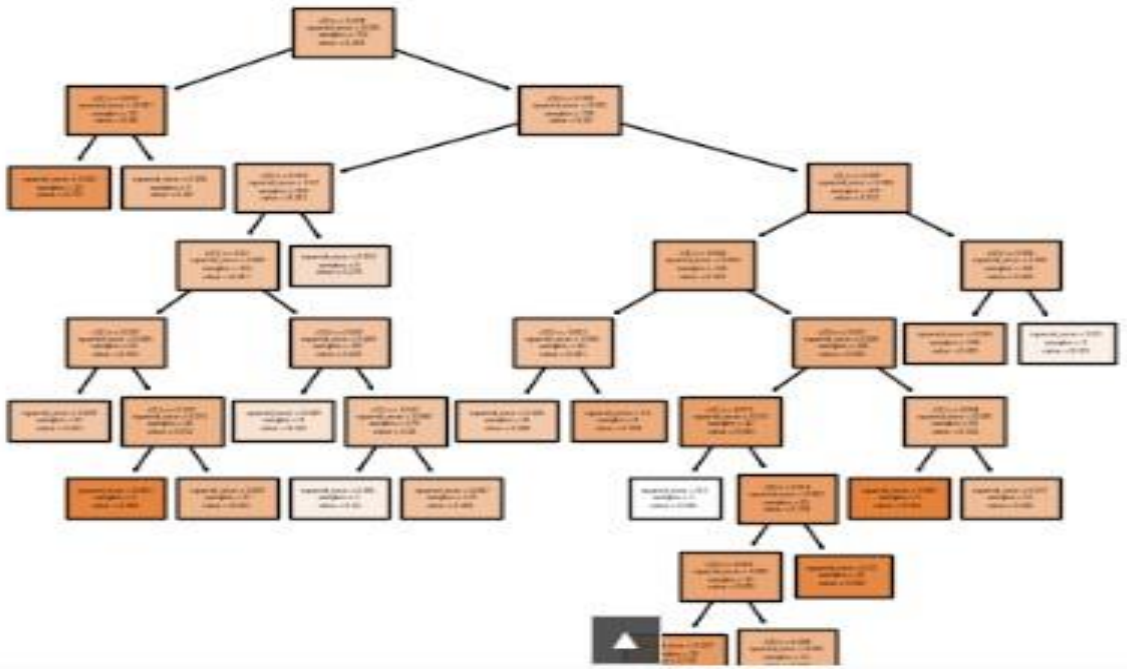
Output

Plotting The Decision Graph

We can use the plot_tree function from the sklearn library to visualize on what basis the decision is made.

```
fig, axes = plt.subplots(nrows=1, ncols=1, figsize=(4,4), dpi=800)
plot_tree(reg, filled=True, ax=axes, fontsize=2)
plt.show()
```

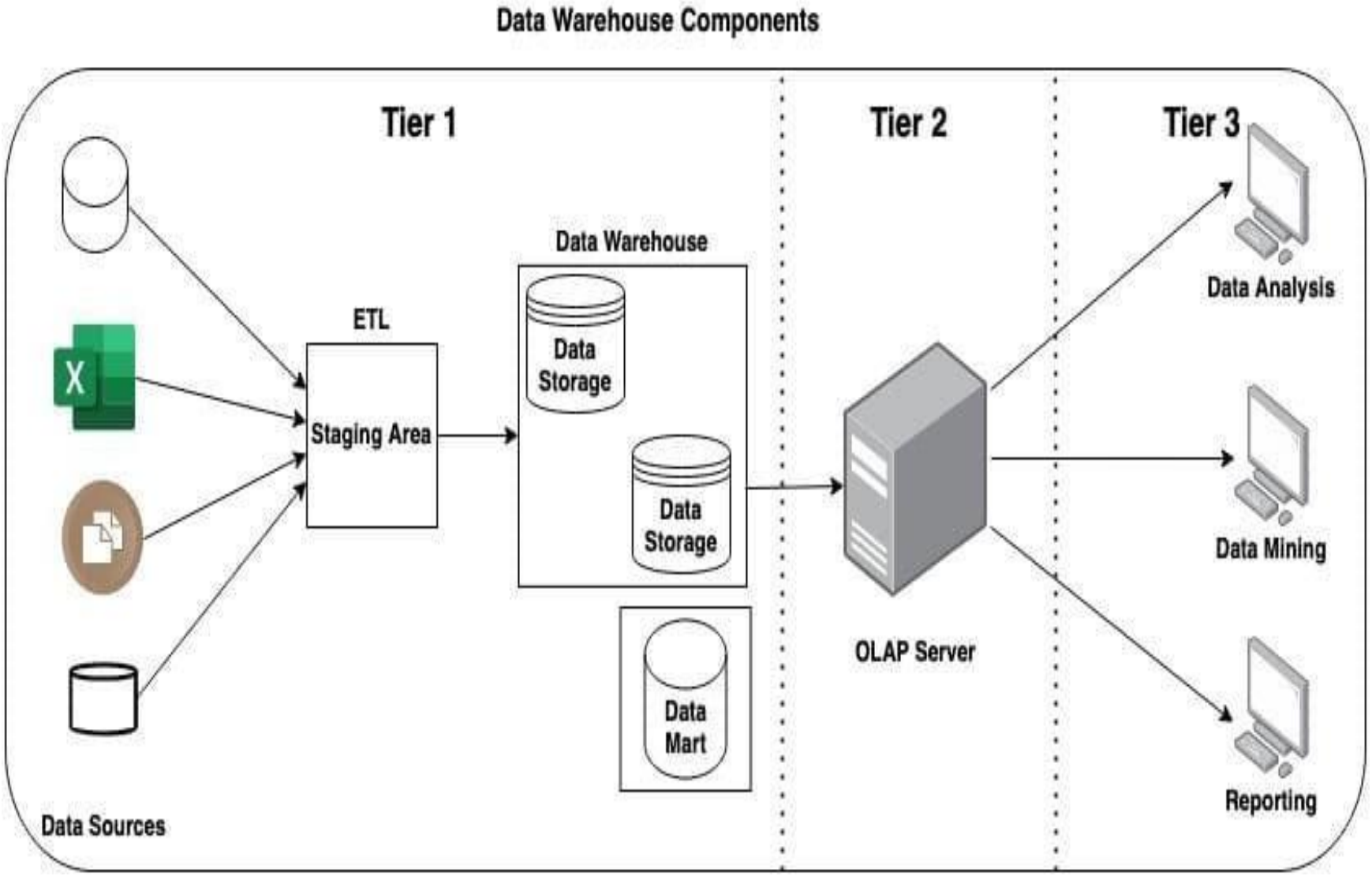
Output:

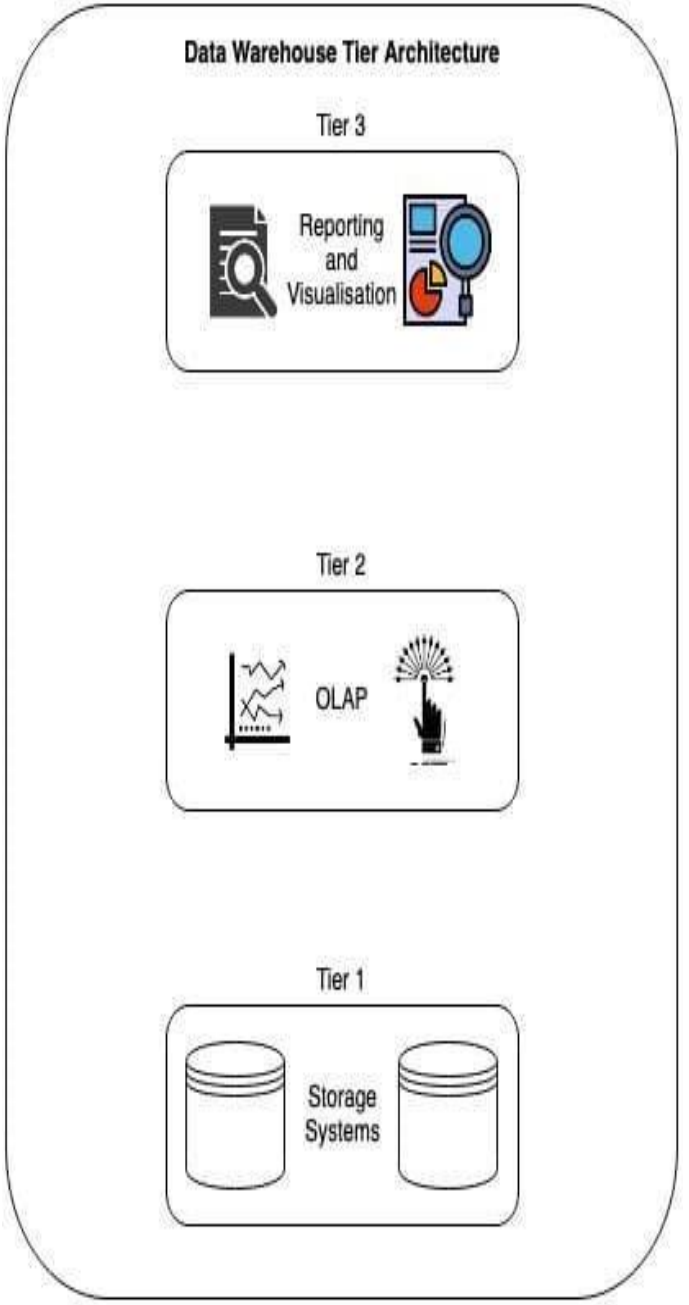


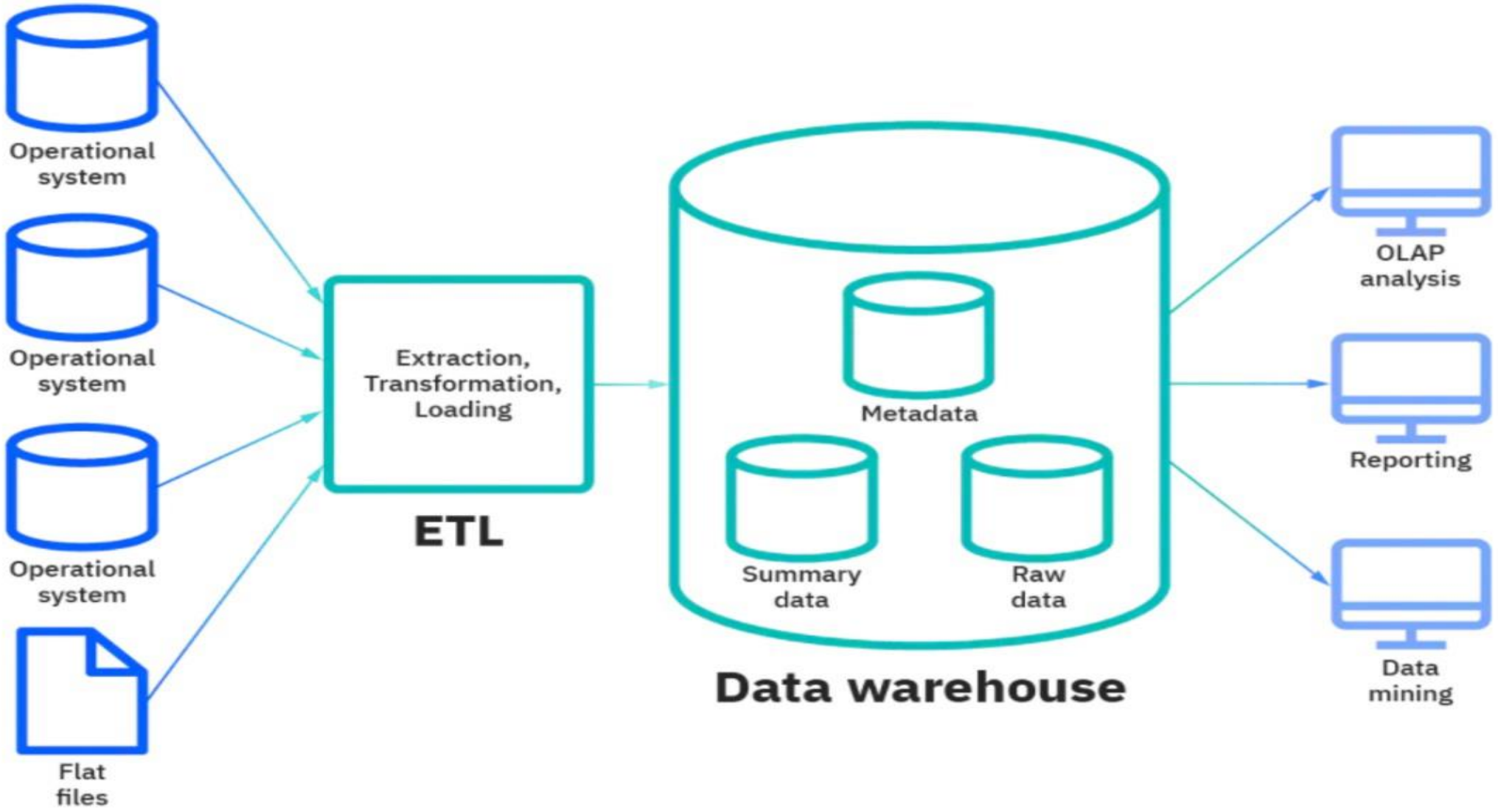
Data Warehousing

- Data warehousing is the process of collecting, cleaning, integrating, and storing large volumes of historical data from multiple sources into a central repository for analysis, reporting, and business intelligence (BI)
- Data mining attempts to depict meaningful patterns through a dependency on the data that is compiled in the data warehouse

Architecture-DW



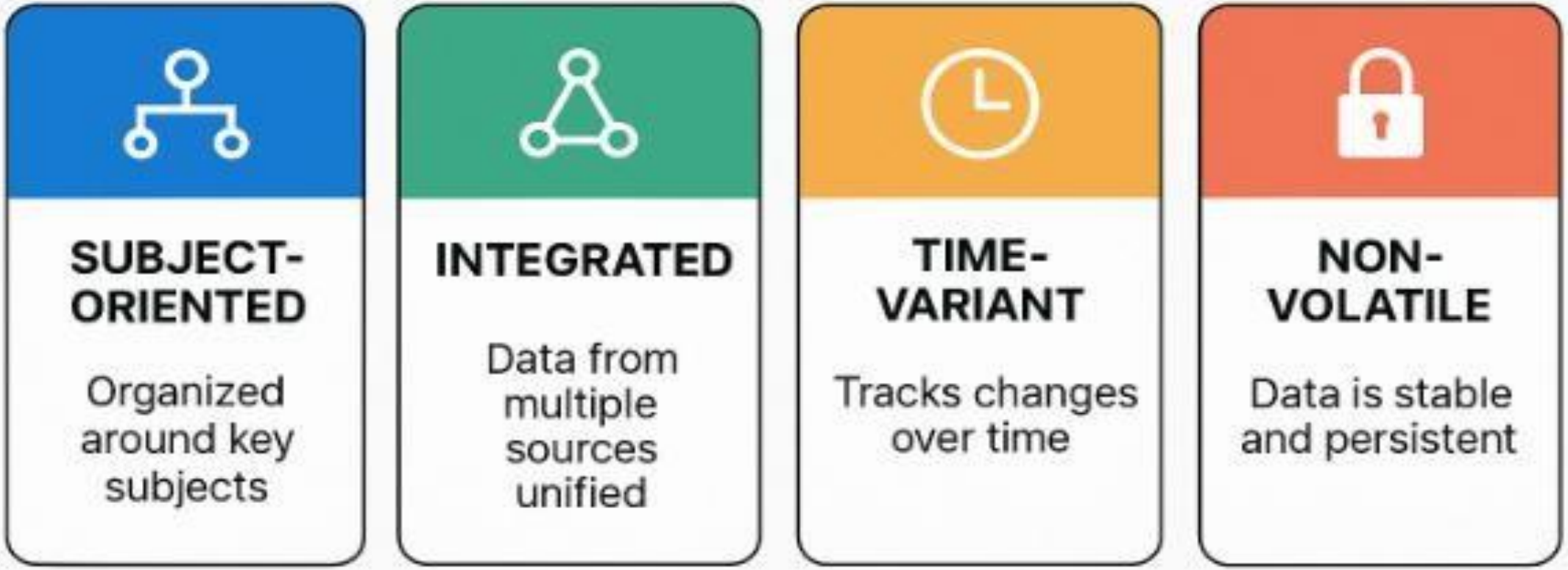



Operational system
Operational system
Operational system
Flat files
Extraction, Transformation, Loading
ETL
Metadata
Summary data
Raw data
Data warehouse
OLAP analysis
Reporting
Data mining

DATA WAREHOUSE:

- A data warehouse is where data can be collected for mining purposes, usually with large storage capacity.
- Source → Extract → Transform → Load → Target
- **Data warehouses collaborate data from several sources and ensure data accuracy, quality, and consistency.**
- System execution is boosted by differentiating the process of analytics from traditional databases.
- In a data warehouse, data is sorted into a formatted pattern by type and as needed.
- **Data warehouses store historical data** and **handle requests faster, helping in online analytical processing**, whereas a **database is used to store current transactions in a business process that is called online transaction processing.**

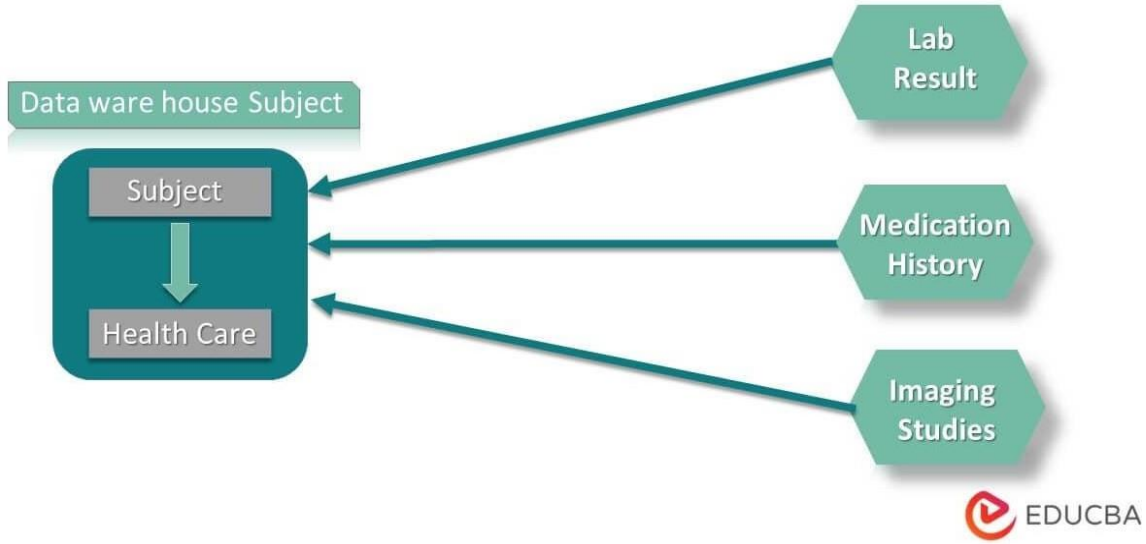
The Four Pillars of Data Warehousing



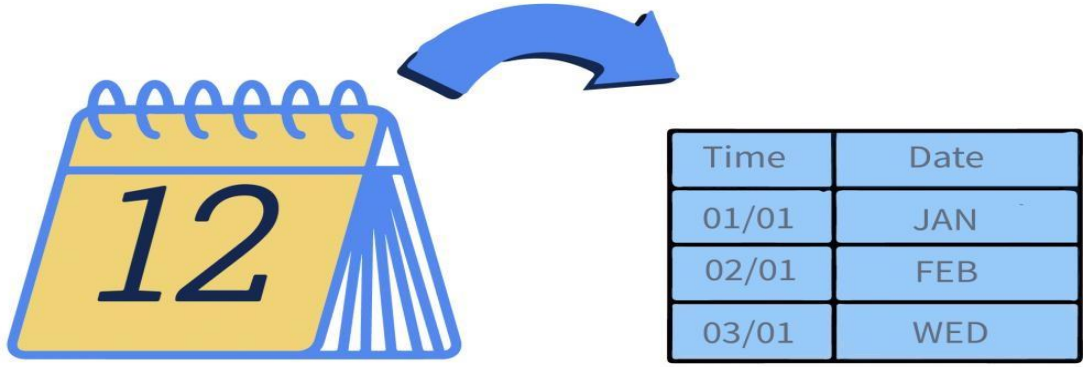
Subject-Oriented



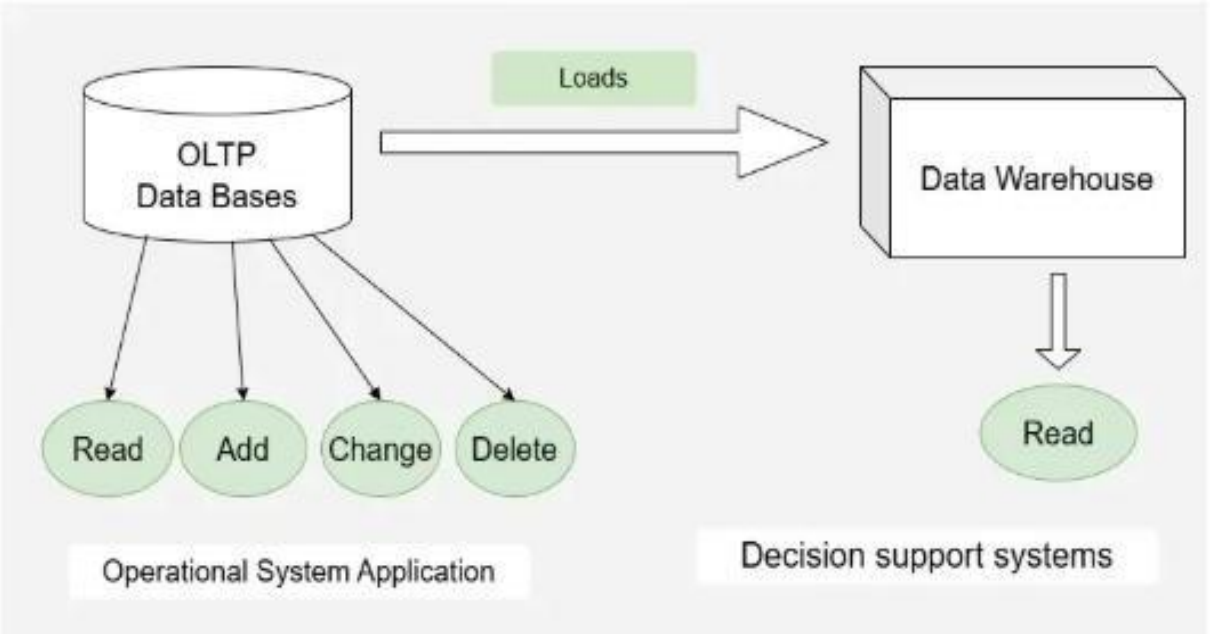
Integrated Data



Time-Variant



Non-Volatile



FEATURES OF DATA WAREHOUSES:

Subject Oriented:

- It provides you with important data about **a specific subject like suppliers, products, promotion, customers, etc**. Data warehousing usually handles the analysis and modeling of data that assist any organization to make data-driven decisions.

Integrated:

- Different heterogeneous sources are put together to build a data warehouse, **such as level documents or social databases.**

Time-Variant:

- The data collected in a **data warehouse is identified with a specific period.**

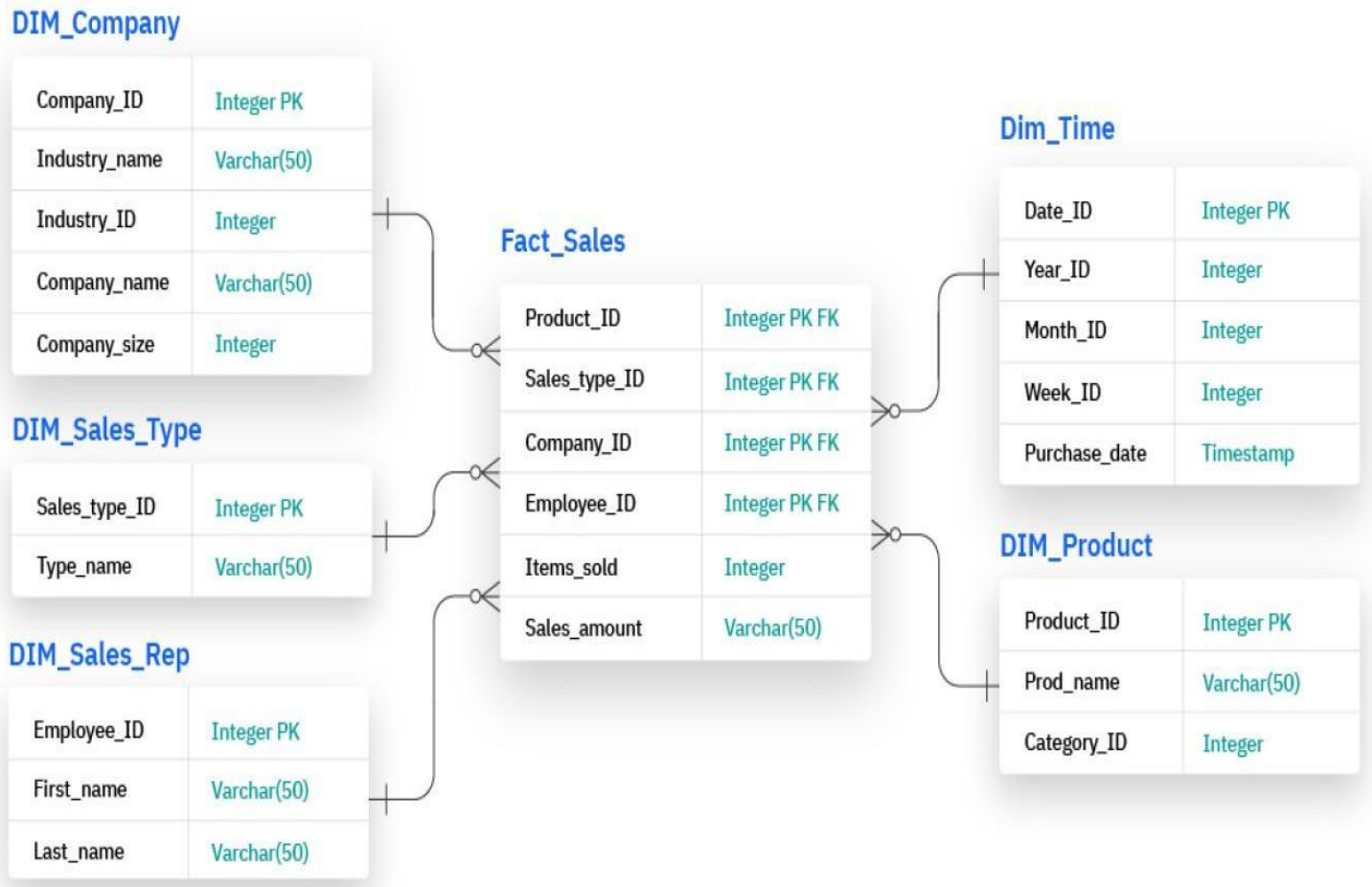
Nonvolatile:

- **This means the earlier data is not deleted when new data is added to the data warehouse.**
  - **The operational database and data warehouse are kept separate and thus continuous changes in the operational database are not shown in the data warehouse.**

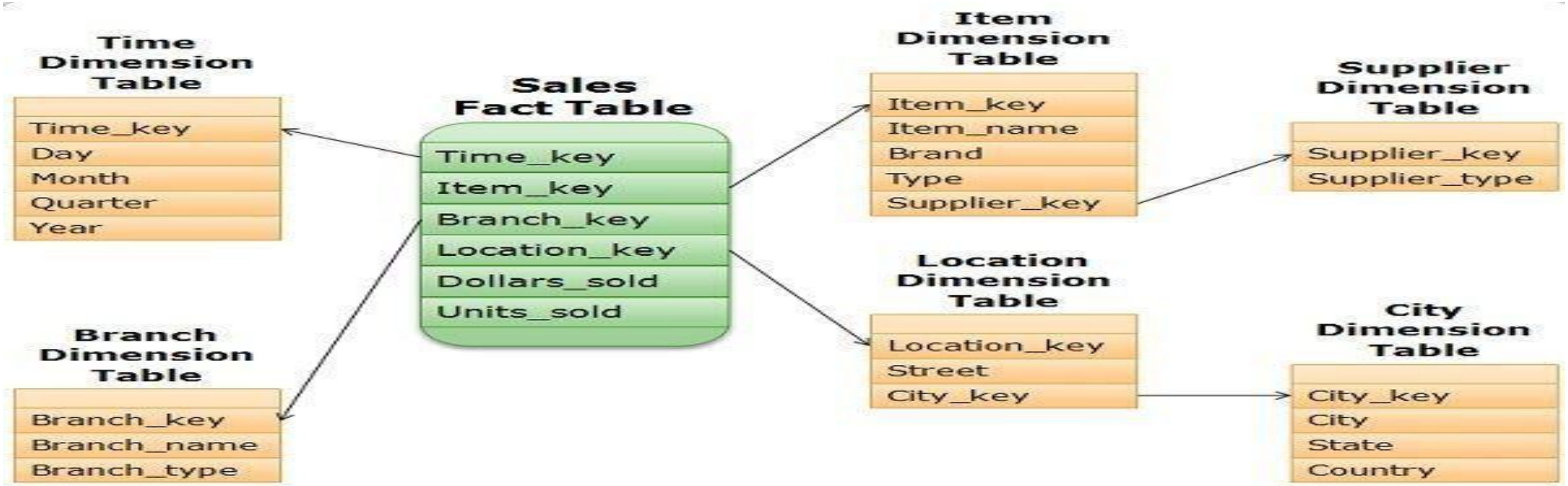
Star Schema-DW

- Star schemas consist of **a single, central fact table** surrounded by dimension tables. In a diagram, the fact table appears in the middle of a star pattern.

- The star schema is considered the simplest and most common type of schema, offering users faster querying speeds.



Snowflake Schema



Snowflake schema is the kind of the star schema which includes the hierarchical form of dimensional tables. In this schema, there is a fact table comprise of various dimension and sub-dimension table connected across through primary and foreign key to the fact table.

OLAP vs OLTP

- **OLAP** systems are **designed for high-speed**, **complex queries** and **multidimensional analysis** on large volumes of data.
- They use **“cubes”** (array-based multidimensional data structures) to enable faster, more flexible analysis across multiple dimensions.
- Common use cases include data mining, financial analysis, budgeting and forecast planning.
- **Online transaction processing (OLTP)** systems capture and update large volumes of real-time transactions from many users.
- In contrast, OLAP systems analyze data that has already been captured.

APPLICATIONS OF DATA WAREHOUSES:

- ***Data warehouses help analysts or senior executives analyze, organize, and use data for decision-making.***

Examples:

- Consumer goods
- Banking services
- Financial services
- Manufacturing
- Retail sectors

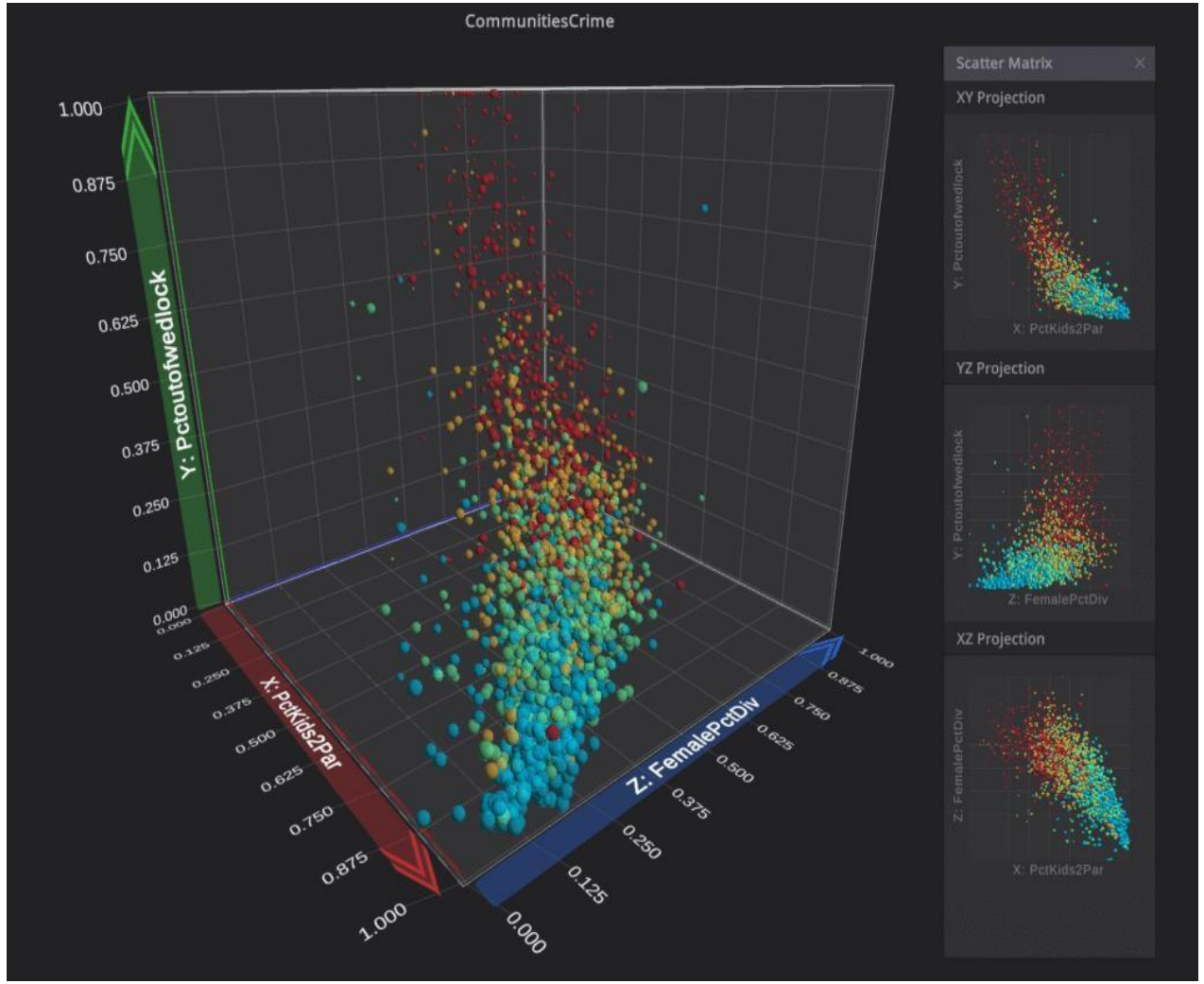
Components of Data Warehouse

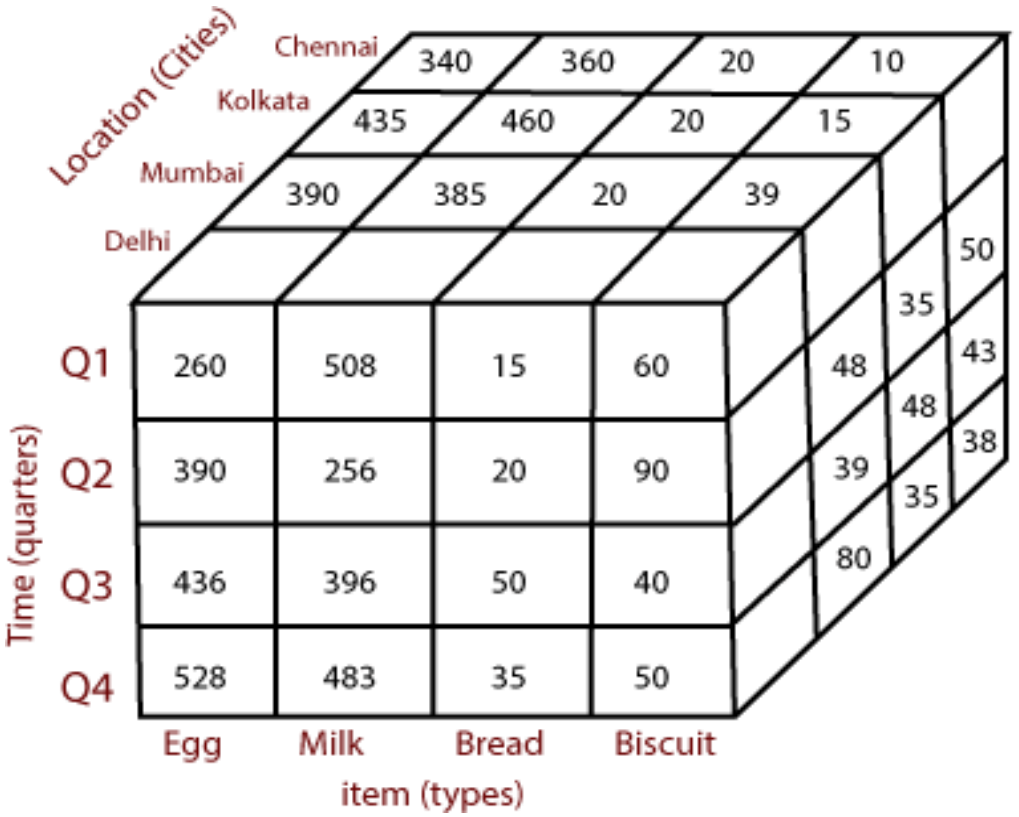
- **Data Sources**: These are the **various operational systems, databases, and external data** feeds that provide raw data to be stored in the warehouse.
- **ETL (Extract, Transform, Load) Process**: The ETL process is **responsible for extracting data from different sources**, **transforming it into a suitable format**, and **loading it into the data warehouse.**
- **Data Warehouse Database**: This is **the central repository *where cleaned and transformed data is stored***. It is typically organized in a **multidimensional format** for efficient querying and reporting.
- **Metadata**: Metadata describes the structure, source, and usage of data within the warehouse, making it easier for users and systems to understand and work with the data.

Cont..

- **Data Marts**: These are smaller, more focused data repositories derived from the data warehouse, designed to meet the needs of specific business departments or functions.
- **OLAP (Online Analytical Processing) Tools**: **OLAP tools** allow users to analyze data in multiple dimensions, providing deeper insights and supporting complex analytical queries.
- **End-User Access Tools**: These are reporting and analysis tools, such as dashboards or **Business Intelligence (BI) tools**, that enable business users to query the data warehouse and generate reports.

Multidimensional Data





Need for Data Warehousing

- **Handling Large Data Volumes**: **Traditional databases store limited data (MBs to GBs)**, while **data warehouses are built to handle huge datasets (up to TBs)**, making it easier to store and analyze long-term historical data.
- **Enhanced Analytics**: Databases handle transactions; data warehouses are optimized for complex analysis and historical insights.
- **Centralized Data Storage**: A data warehouse combines data from multiple sources, giving a single, unified view for better decision-making.
- **Trend Analysis**: By storing historical data, a data warehouse allows businesses to analyze trends over time, enabling them to make strategic decisions based on past performance and predict future outcomes.
- **Business Intelligence Support:** Data warehouses work with BI tools to give quick access to insights, helping in data-driven decisions and improving efficiency.

Types of Data Warehouses

- **Enterprise Data Warehouse (EDW)**: A centralized warehouse that stores data from across the organization for analysis and reporting.
- **Operational Data Store (ODS):** Stores real-time operational data used for day-to-day operations, not for deep analytics.
- **Data Mart**: A subset of a data warehouse, focusing on a specific business area or department.
- **Cloud Data Warehouse**: A data warehouse hosted in the cloud, offering scalability and flexibility.
- **Big Data Warehouse**: Designed to store vast amounts of unstructured and structured data for big data analysis.
- **Virtual Data Warehouse**: Provides access to data from multiple sources without physically storing it.
- **Hybrid Data Warehouse**: Combines on-premises and cloud-based storage to offer flexibility.
- **Real-time Data Warehouse**: Designed to handle real-time data streaming and analysis for immediate insights.

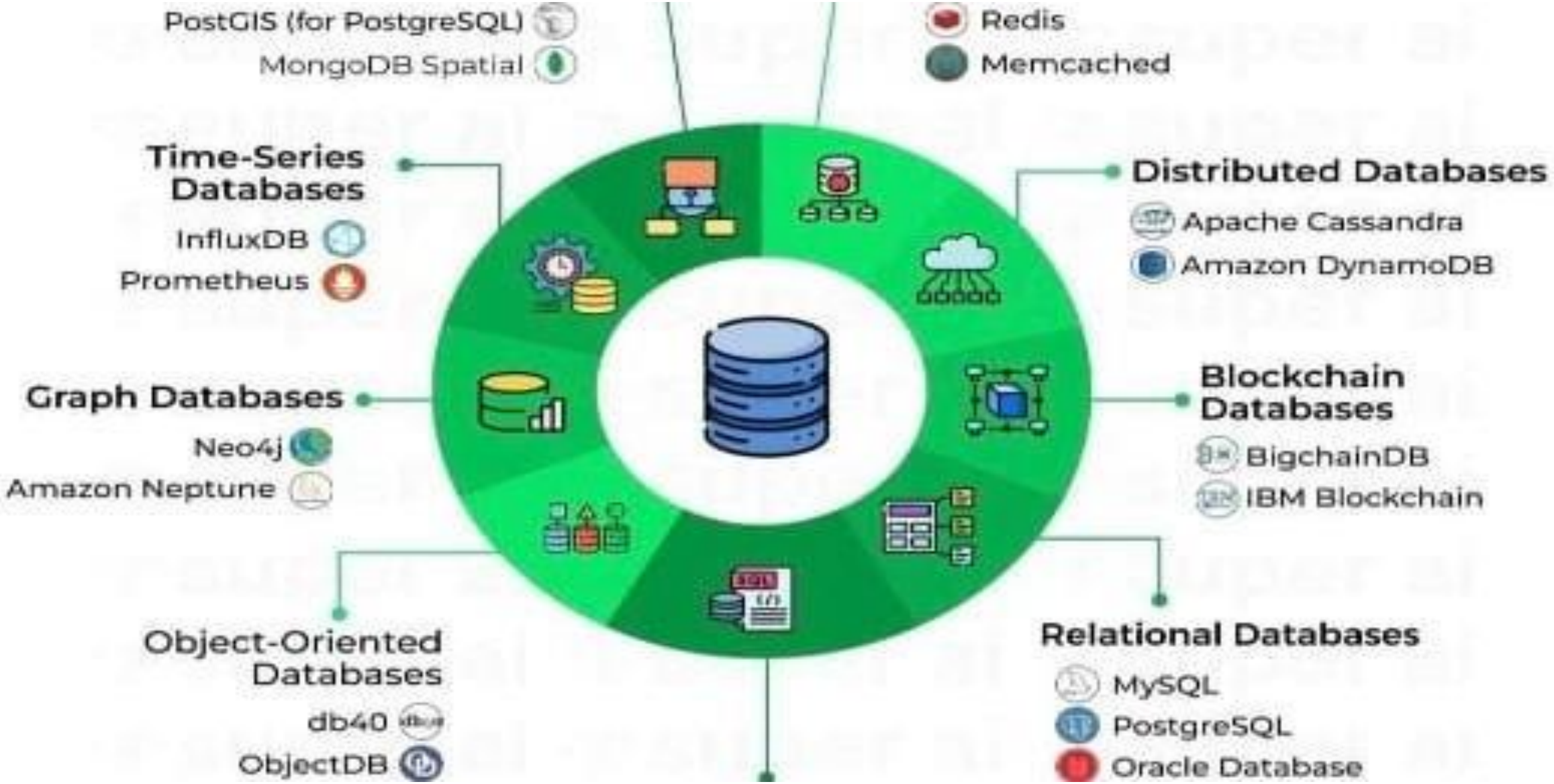
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- Consumer goods
- Banking services
- Financial services
- Manufacturing
- Retail sectors

Advantage and Disadvantage of Data Warehousing

Advantage	Disadvantage
Better Decisions: Centralized data supports faster, smarter decisions.	High Cost: Setup requires major investment.
Business Intelligence: Enables strong operational insights.	Complexity: Needs skilled professionals to manage.
High Data Quality: Ensures consistency and reliability.	Time-Consuming: Long setup and integration time.
Scalable: Handles large and growing datasets.	Integration Issues: Combining data from sources can be challenging.

Types of Databases



	Database	Data Lake	Data Warehouse
Data	Structured	Raw & Unstructured	Structured
Data Processing	Schema-on-write	Schema-on-read	Schema-on-write
Scalability	Varies	High	High
Cost	0 - $	$$	$$$
Users	Anyone	Data Scientists	Business Users
Use Cases	Real-time data processing, high transactional throughput, strong data consistency	Exploratory analysis, machine learning, data mining, or data science research	Reporting, analytics, and business intelligence

Data Mining

- **Data mining** focuses on analyzing data to discover patterns, trends, and insights, while data warehousing focuses on storing and managing data in a centralized location.

DATA MINING:

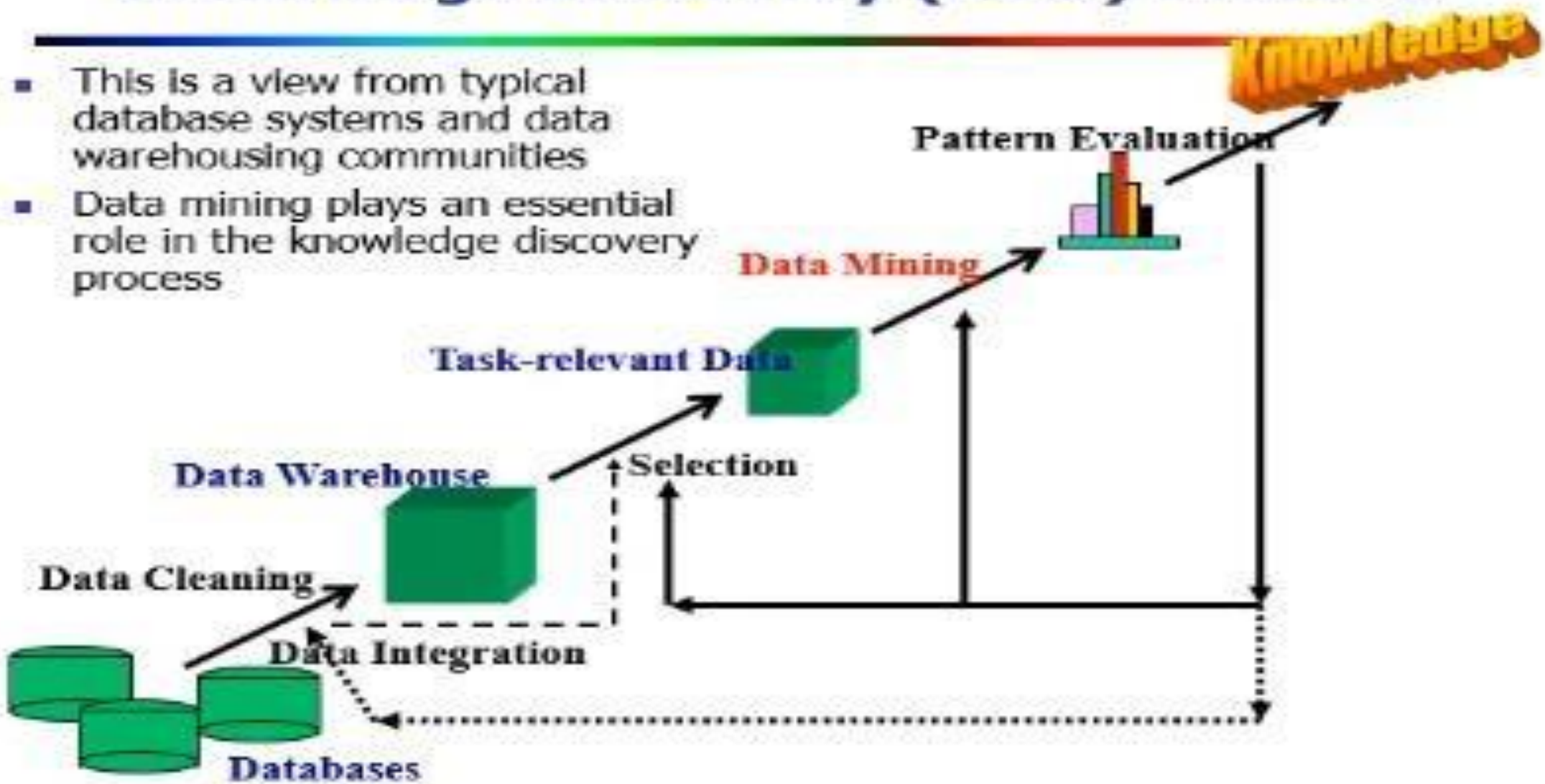
- **Data is extracted and analyzed to fetch useful information.**
- ***In data mining hidden patterns are researched from the dataset to predict future behavior***.
- Data mining is used to indicate and discover relationships through the data.
- **Data mining uses statistics, artificial intelligence, machine learning systems, and some databases to find hidden patterns in the data.**
- ***It supports business-related queries that are time-consuming to resolve.***

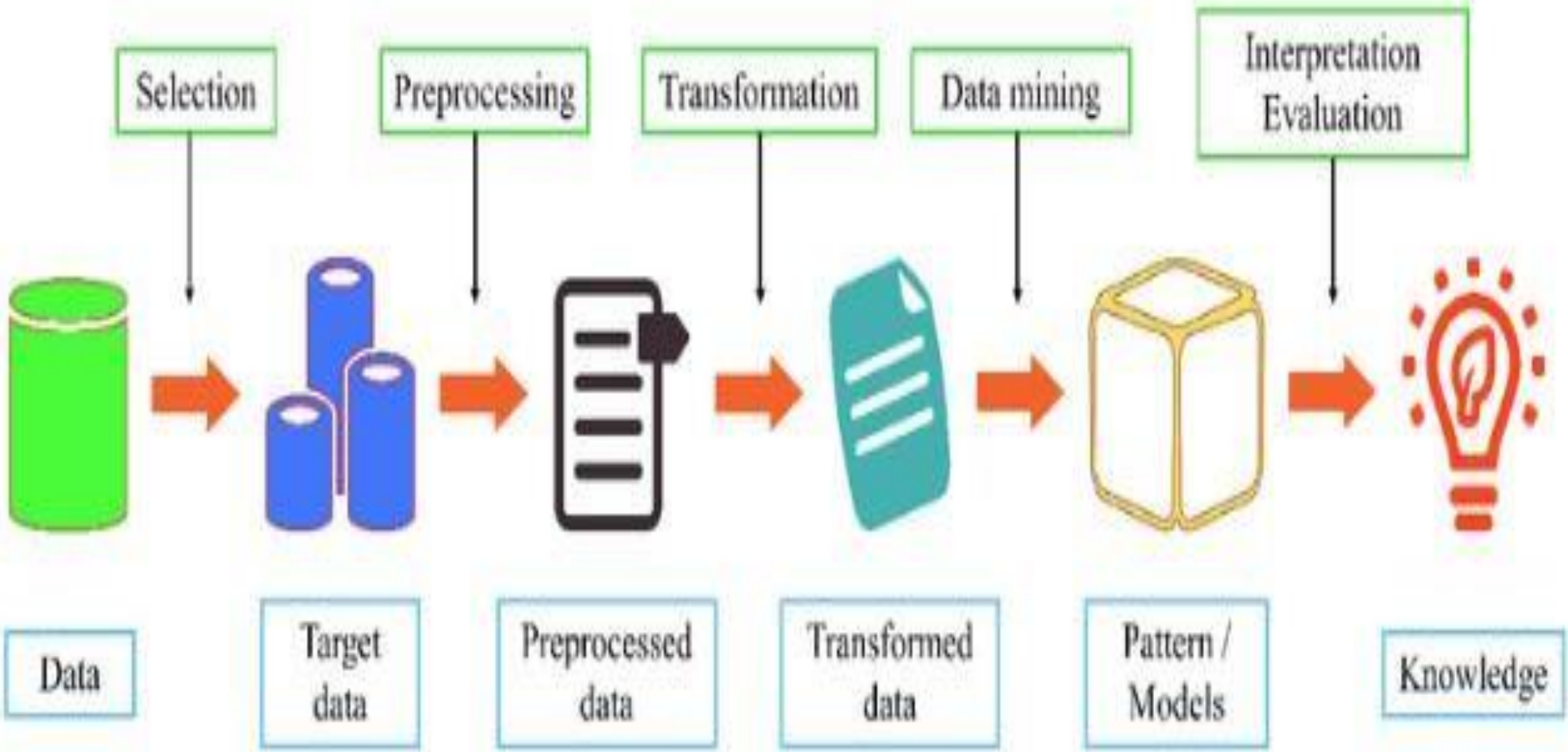
Data Mining

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Knowledge Discovery (KDD) Process

- This is a view from typical database systems and data warehousing communities
- Data mining plays an essential role in the knowledge discovery process




Selection
Preprocessing
Transformation
Data mining
Interpretation
Evaluation
Data
Target data
Preprocessed data
Transformed data
Pattern / Models
Knowledge

FEATURES OF DATA MINING:

- It is **good with large databases and datasets**
- It **predicts future results**
- It **creates actionable insights**
- It **utilizes the automated discovery of patterns**

ADVANTAGES OF DATA MINING:

Fraud Detection:

- It is used to **find which insurance claims, phone calls, debit or credit purchases are fraud.**

Trend Analysis:

- Existing marketplace trends are analyzed,which provides **a strategic benefit as it helps in reduction of costs, as in manufacturing per demand**.

Market Analysis:

- It can predict the market and therefore help to make business decisions.

For example: it can identify a target market for a retailer, or certain types of products desired by types of customers.

DATA MINING TECHNIQUES:

Classification:

- It is used to **fetch the appropriate information from the dataset and to segregate different classes that are present in the dataset**.

Below are the classification models.

- K-nearest neighbors
- Support Vector Machine
- Gaussian Naïve Bayes, etc.

FEATURES OF DATA MINING:

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- It creates actionable insights
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DATA MINING TECHNIQUES:

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- Gaussian Naïve Bayes, etc.

Algorithm of Data Mining

Clustering:

- It is used to find similarities in data by putting related data together and helping to identify different variations in the dataset.
- It helps to find hidden patterns.
- An example of clustering is text mining, medical diagnostics, etc.

Association Rules:

- They are used to identify a connection of two or more items.
- For example, if-then scenarios of items that are frequently purchased in tandem in a grocery store can calculate the proportion of items that are bought by customers together.
- Lift, confidence, and support are techniques used in association rules.

Outlier Detection:

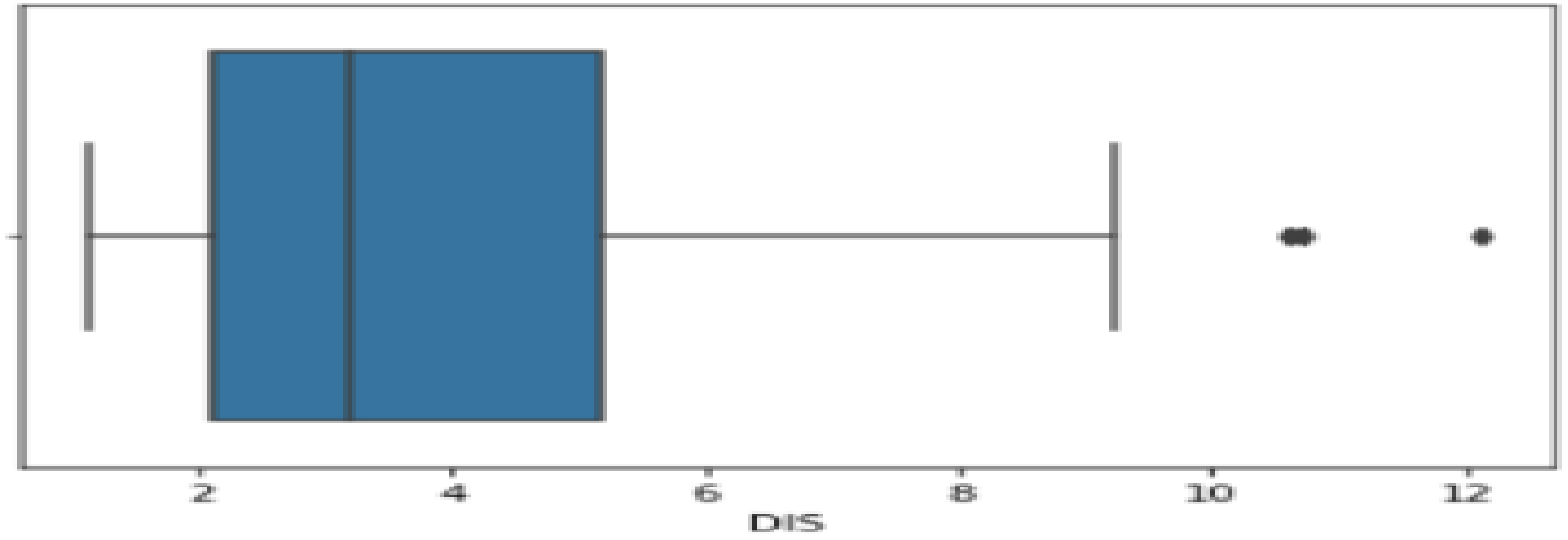
- It is used to identify patterns that do not match the normal behavior in the data, as the outlier deviates from the rest of the data points. It helps in fraud detection, intrusion, etc. **Boxplot and z-score are ways to detect outliers.**

Box Plot Analysis:

Sample code for boxplot:

```
1  Import seaborn as sns
2  sns.boxplot(df_data['DIS'])
```

Output:



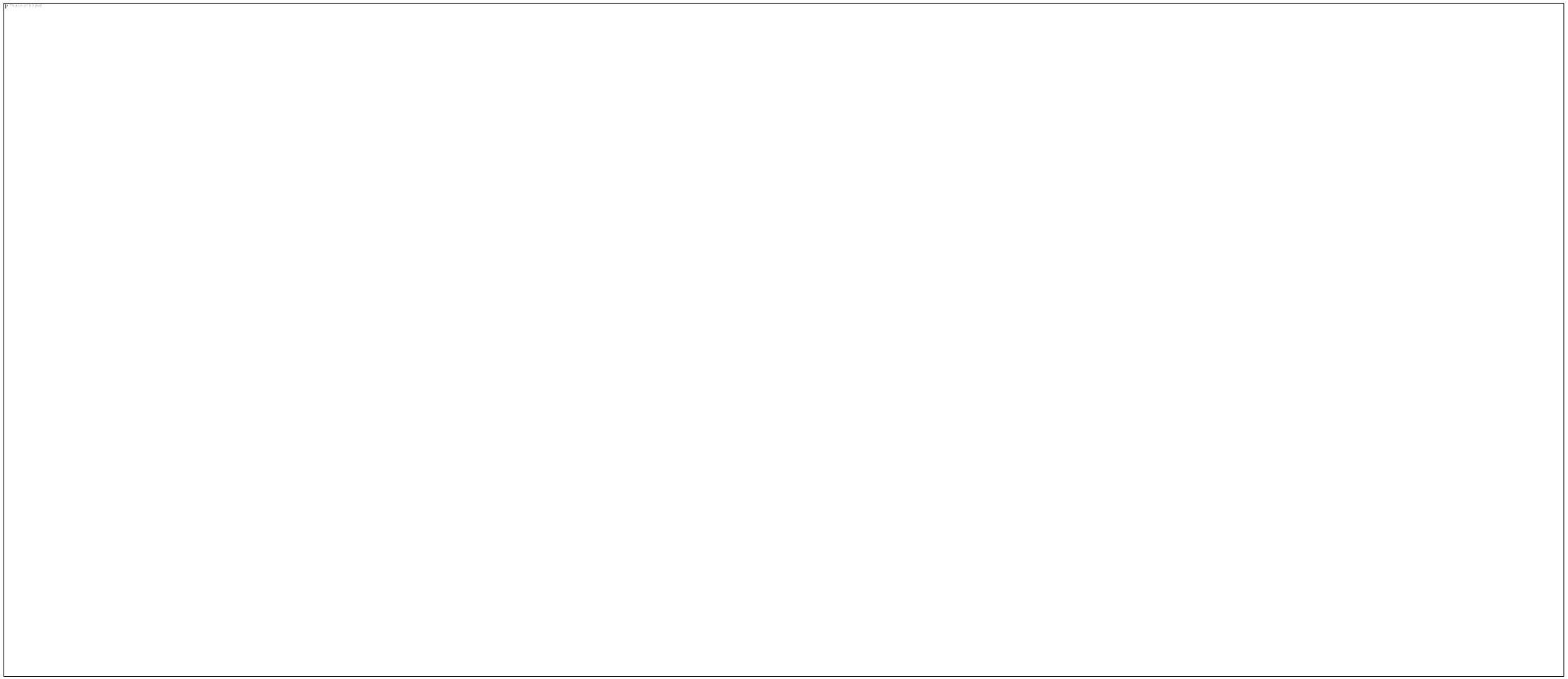
Difference Between Data Mining and Data Warehousing

Basis of Comparison	Data Warehousing	Data Mining
Definition	A data warehouse is a database system that is designed for analytical analysis instead of transactional work.	Data mining is the process of analyzing data patterns.
Process	Data is stored periodically.	Data is analyzed regularly.
Purpose	Data warehousing is the process of extracting and storing data to allow easier reporting.	Data mining is the use of pattern recognition logic to identify patterns.
Managing Authorities	Data warehousing is solely carried out by engineers.	Data mining is carried out by business users with the help of engineers.
Data Handling	Data warehousing is the process of pooling all relevant data together.	Data mining is considered as a process of extracting data from large data sets.
Functionality	Subject-oriented, integrated, time-varying and non-volatile constitute data warehouses.	AI, statistics, databases, and machine learning systems are all used in data mining technologies.

Research Goals in Data Science Project:



Research Goals in Data Science



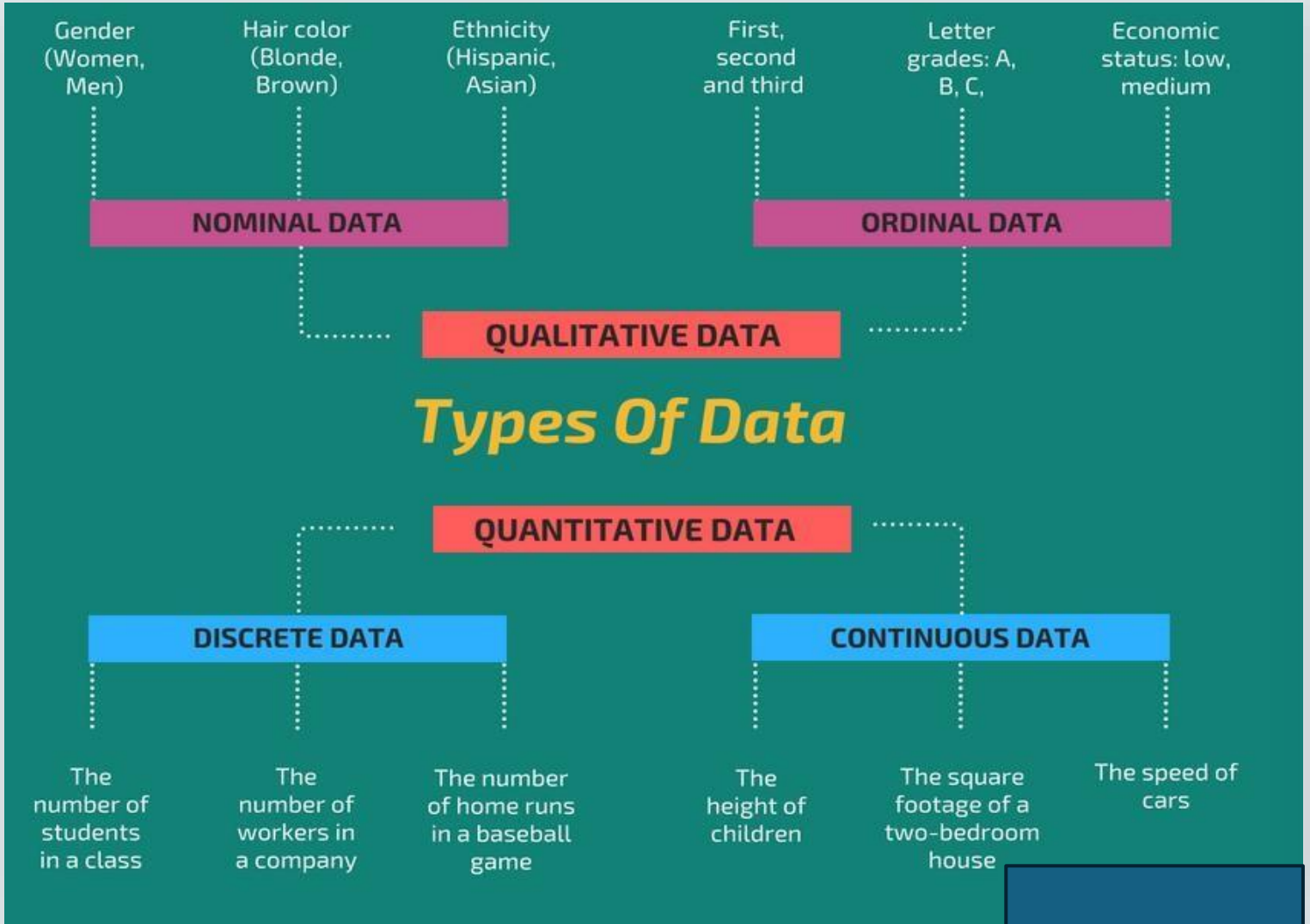
Any Questions?
Unit-1 End

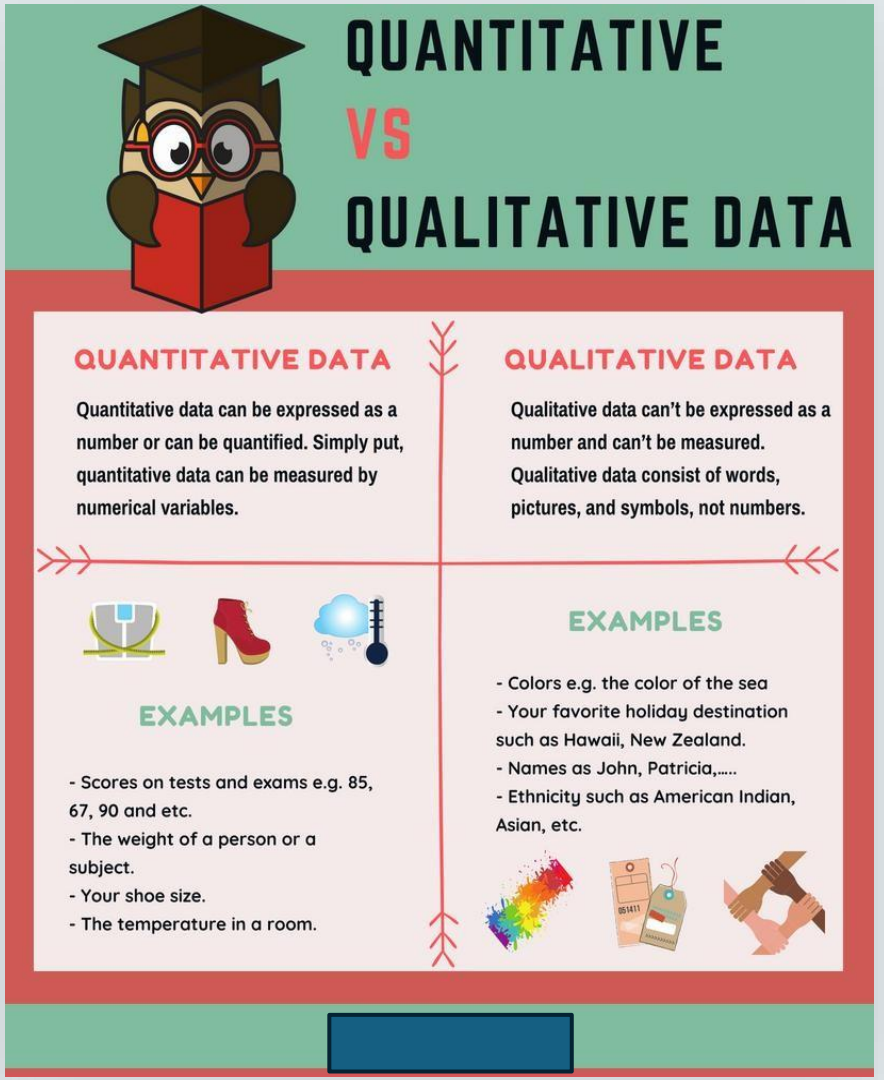
Unit 2: Describing Data

- Describing Data : Types of data and variables
- Descriptive statistics: tables, graphs, averages, Measures of variability
- Normal distributions and z-scores, measure of dispersion,
- Data distribution and shape, Frequency Distributions
- Tabulation

Types of Data

Done





Variable

- Variable is a term derived from the word '**vary**', meaning '**to change**' or '**to differ**'. Therefore, variables are the characteristics that change or differ.
- A **variable** refers to the quantity or characteristic whose value varies from one investigation to another.
- The difference in the value of a ***variable may be with respect to a place, item, individual, or time, and each value within this range is known as Variate*.**

- ***For example,*** *Quantity is a variable as different commodities have different quantities.*
- **Similarly, Price, Age, Weight, Height, Income, Production, etc., are also variables.**
- *Different variables are measured in different units, such as* ***quantity is measured in units or tonnes****, etc.,* ***height in centimeters or inches****,* ***age in years,*** *etc.*

Types of Variables:

1. **Discrete Variable (Discontinuous Variable)**
2. **Continuous Variable.**

1. Discrete Variable (Discontinuous Variable)

- ***Discrete variables are the variables that take only exact values and not fractional values.*** In simple terms, the variables which are expressed in terms of complete numbers are known as discrete variables.
- **For example,** the number of teachers in a school is a discrete variable, as they cannot be expressed in fractions

Team	No. of Workers
Marketing	10
Sales	15
Accounting	8
Project	14
Technical	20

2. Continuous Variable

- **Continuous variables are the variables that can take every possible value**, whether it is integral or fractional, in a specified given range.
- **For example,** the height of an individual can be in decimals or in exact numbers within the specified range.

Weight (in kg)	No. of Students
20-25	20
25-30	5
30-35	10
35-40	14
40-45	2

Raw Data

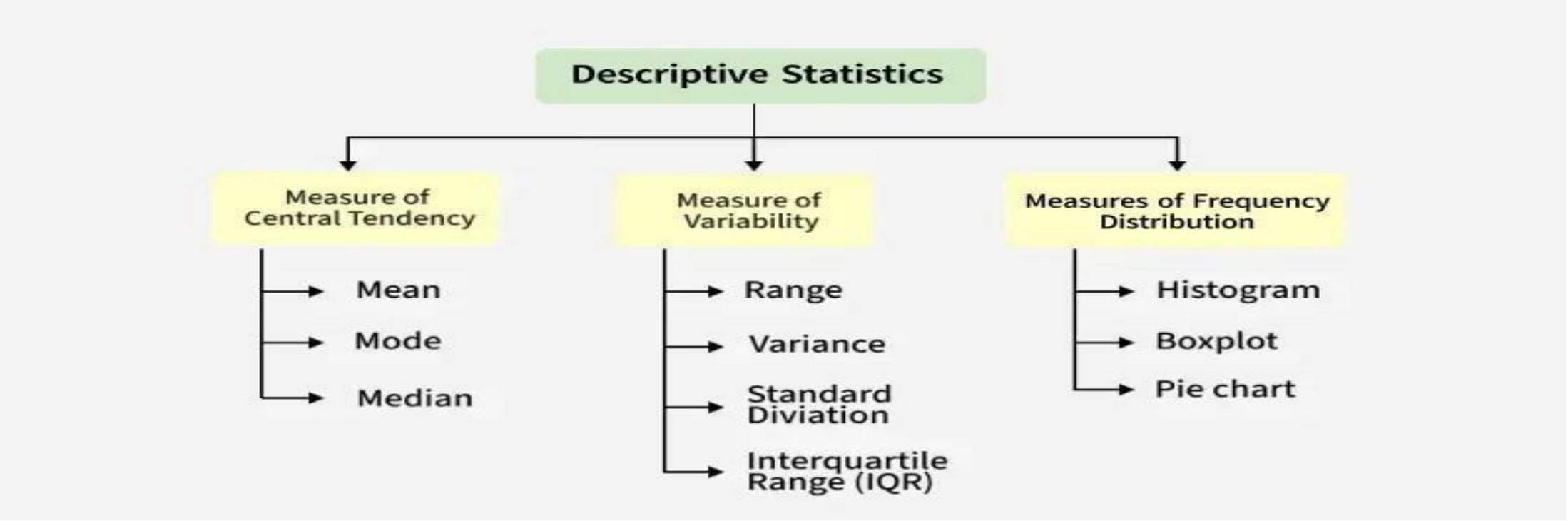
- Raw data is the mass of data in its crude form. In other words, the unorganized mass of different items, which are yet to be organized by an investigator is known as **raw data.**
- **For example,** the raw data of the weight of 40 students in a class can be expressed as follows:

25	38	30	55	32	32	47	30	58	44
40	55	35	63	25	55	38	70	43	69
26	65	69	42	57	40	42	64	28	55
55	27	44	40	36	35	44	30	70	51

The data presented in the above table is raw data.

Descriptive statistics

- Statistics is the foundation of data science.
- Descriptive statistics are simple tools that help us ***understand and summarize data.***
- They show the basic features of a dataset, like the average, highest and lowest values and how spread out the numbers

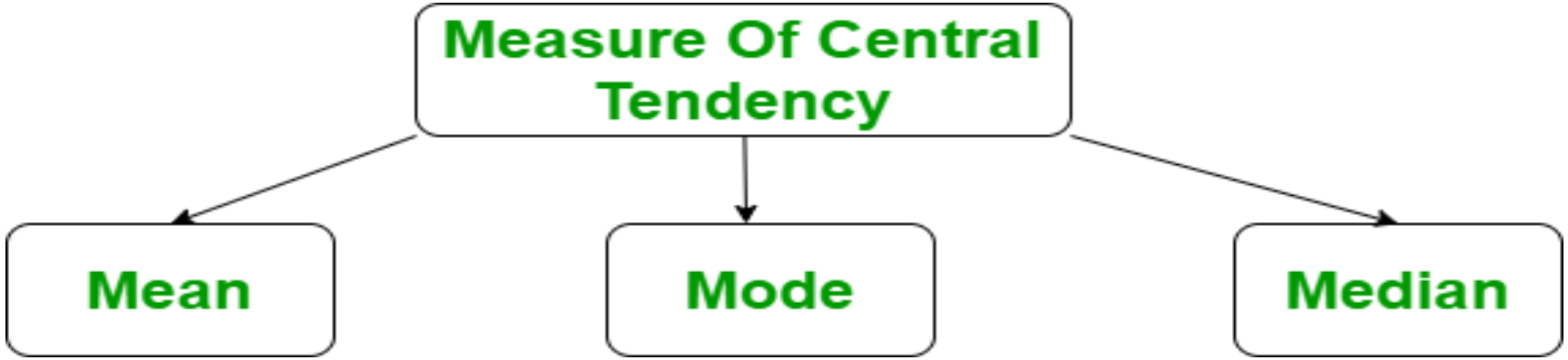


Types of Descriptive Statistics

1. **Where the data centers** (Measures of Central Tendency)
2. **How spread out the data is** (Measure of Variability)
3. **How the data is distributed** (Measures of Frequency Distribution)

1. Measures of Central Tendency

- Statistical values that **describe the central position within a dataset.**



$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Total number of all values}}$$

$$\text{Median} = \text{Middle value (when the data are arranged in order)}$$

$$\text{Mode} = \text{Most common value}$$

Mean:

- **Mean**: is the sum of observations divided by the total number of observations.
- It is also defined as average which is the sum divided by count.

$$\bar{x} = \frac{\sum x}{n}$$

where,

- x = Observations
- n = number of terms

Python Code

```
import numpy as np

# Sample Data
arr = [5, 6, 11]

# Mean
mean = np.mean(arr)

print("Mean = ", mean)
```

Output

```
Mean =  7.333333333333333
```

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \cdots + X_n}{n} = \frac{\Sigma X}{n}$$

where n represents the total number of values in the sample.

For a population, the Greek letter μ (mu) is used for the mean.

$$\mu = \frac{X_1 + X_2 + X_3 + \cdots + X_N}{N} = \frac{\Sigma X}{N}$$

where N represents the total number of values in the population.

Example 1:

3 4 3 2 4 6 7 6 4

$$\bar{x} = \frac{3+4+3+2+4+6+7+6+4}{9}$$

$$= \frac{39}{9}$$

$$= 4.33$$

$$\bar{X} = \frac{\Sigma X}{n}$$

$\bar{x}$ = mean

ΣX = Sum of all X's

Mode:

- Mode: The **most frequently occurring value in the dataset.** It's useful for categorical data and in cases where knowing the most common choice is crucial

```
import scipy.stats as stats

# sample Data
arr = [1, 2, 2, 3]

# Mode
mode = stats.mode(arr)
print("Mode = ", mode)
```

Output:

Mode = ModeResult(mode=array([2]), count=array([2]))

Median:

- Median: The median is the middle value in a sorted dataset.
- If the number of values is odd, it's the center value, if even, it's the average of the two middle values. It's often better than the mean for skewed data.

```
import numpy as np

# sample Data
arr = [1, 2, 3, 4]

# Median
median = np.median(arr)

print("Median = ", median)
```

Output

```
Median =  2.5
```

Example 3 4 3 2 4 6 7 6 4

1. Arrange data in an array

2 3 3 4 **4** 4 6 6 7

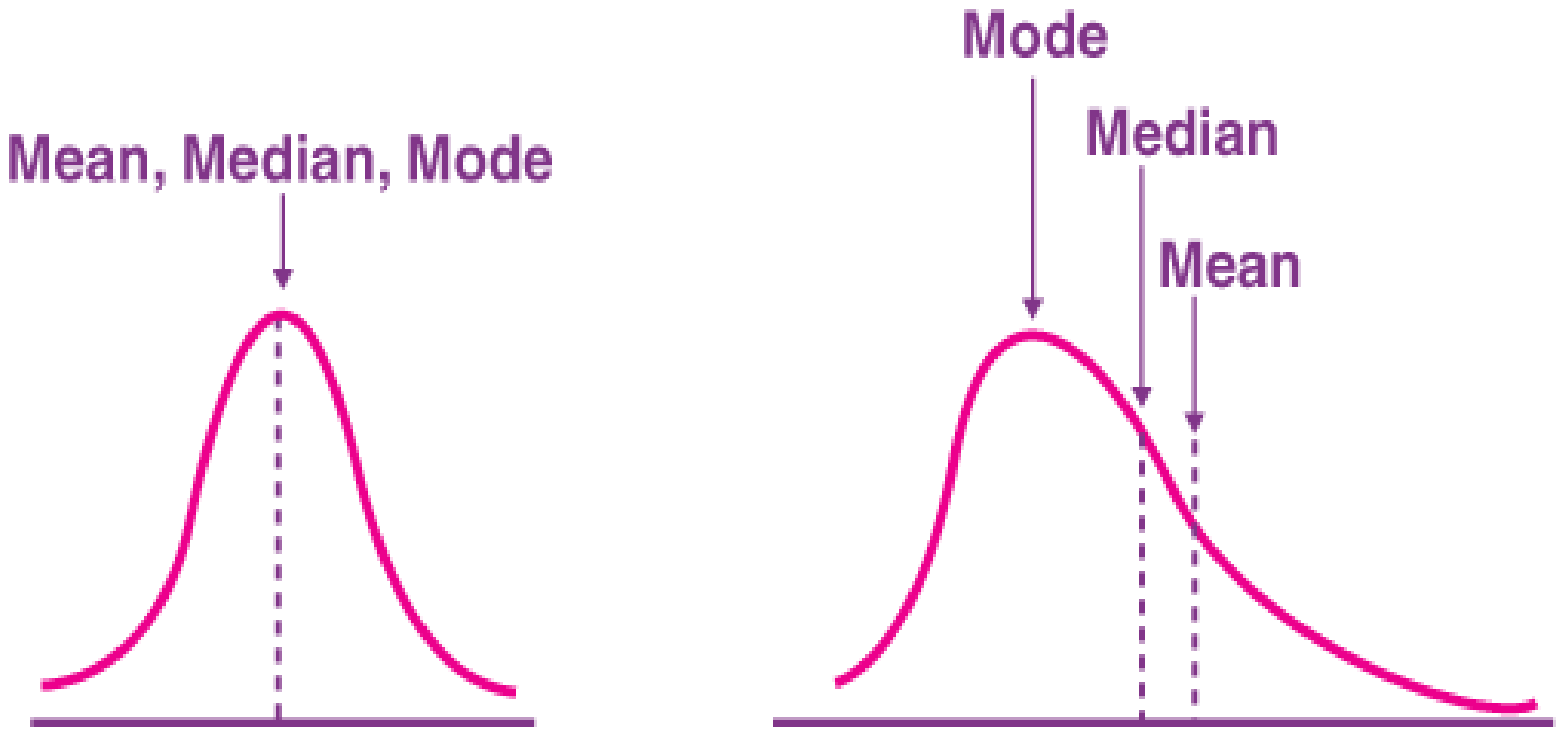
2.calculate the median location (ML)

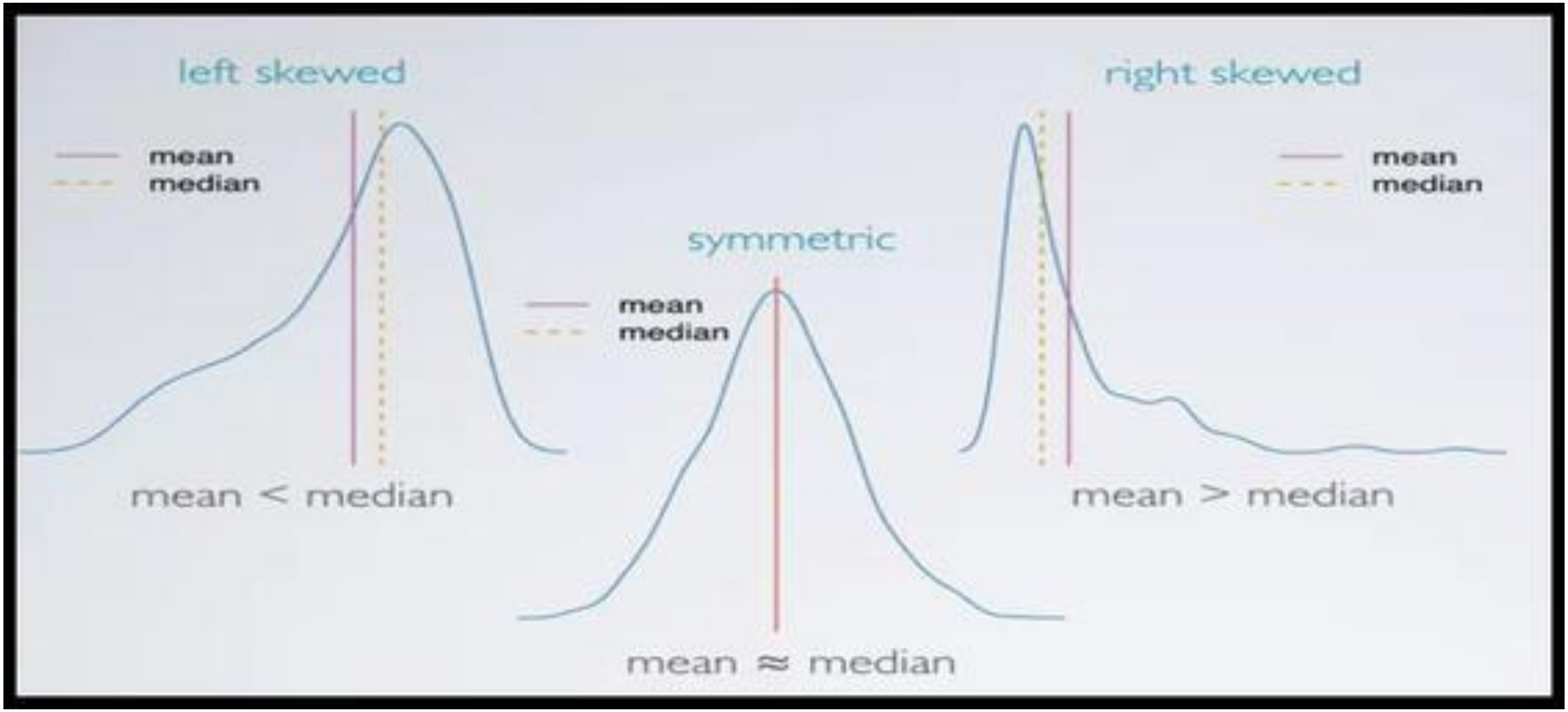
= (9+1) / 2 = 5

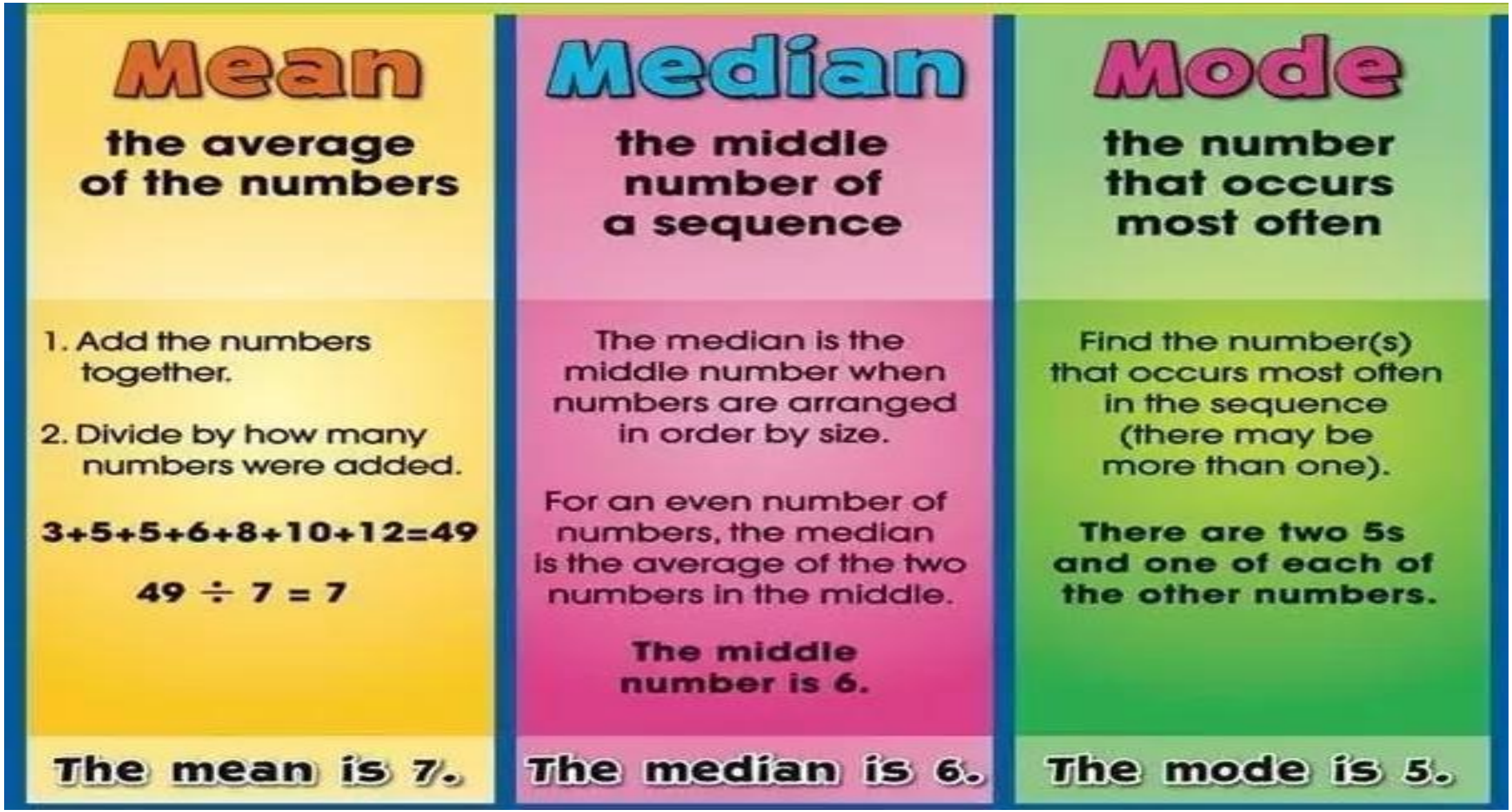
$$ML = \frac{n+1}{2}$$

3. Locate the median value

Measures of Central Tendency, Mean, Median & Mode




left skewed
mean
median
mean < median
symmetric
mean
median
mean ≈ median
right skewed
mean
median
mean > median


Mean
the average of the numbers
1. Add the numbers together.
2. Divide by how many numbers were added.
3+5+5+6+8+10+12=49
49 ÷ 7 = 7
The mean is 7.
Median
the middle number of a sequence
The median is the middle number when numbers are arranged in order by size.
For an even number of numbers, the median is the average of the two numbers in the middle.
The middle number is 6.
The median is 6.
Mode
the number that occurs most often
Find the number(s) that occurs most often in the sequence (there may be more than one).
There are two 5s and one of each of the other numbers.
The mode is 5.

2. Measure of Variability

- Measures of variability, also called measures of **dispersion**, help us spot the spread or distribution of observations in a dataset.
- They identifying outliers, assessing model assumptions and understanding data variability in relation to its mean.

The key measures of variability include:

1. Range : describes the difference between the largest and smallest data point in our data set. The bigger the range, the more the spread of data and vice versa. While easy to compute **range is sensitive to outliers**. This measure can provide a quick sense of the data spread but should be complemented with other statistics.

- **Range = Largest data value - smallest data value**

Python Code:

```
import numpy as np
# Sample Data
arr = [1, 2, 3, 4, 5]

# Finding Max
Maximum = max(arr)
# Finding Min
Minimum = min(arr)

# Difference Of Max and Min
Range = Maximum - Minimum
print("Maximum = {}, Minimum = {} and Range = {}".format(
    Maximum, Minimum, Range)
```

Output

```
Maximum = 5, Minimum = 1 and Range = 4
```

2. Variance:

Variance: is defined as an **average squared deviation from the mean**.

Variance is defined as "***The measure of how far the set of data is dispersed from their mean value***".

Variance is represented with the symbol σ^2.

In other words, we can also say that the variance is the average of the squared difference from the mean.

.

$$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$$

where,

- **x** -> Observation under consideration
- **N** -> number of terms
- **mu** -> Mean

Cont..

- **Variance and Standard Deviation** are the important measures used in Mathematics and statistics to find meaning from a large set of data.

- **Variance** is the measure of ***how the data points vary according to the mean***, while ***standard deviation is the measure of the central tendency of the distribution of the data.***

- *The* ***major difference between variance and standard deviation*** *is in their* ***units of measurement.***
- *Standard deviation is measured in a unit similar to the units of the mean of data,* ***whereas the variance is measured in squared units***

Code

- import statistics

- # sample data
- arr = [1, 2, 3, 4, 5]
- # variance
- print("Var = ", (statistics.variance(arr)))

Output

```
Var =  2.5
```

3. Standard deviation:

- **Standard deviation**: Standard deviation is widely used to measure the **extent of variation in data**. It's especially important when assessing model performance (e.g., residuals) or comparing datasets with different means.
- It is defined as the square root of the variance. It is calculated by finding the mean, then **subtracting each number from the mean which is also known as the average and squaring the result**. Adding all the values and then dividing by the no of terms followed by the square root.

residuals : what is left or remains after something is gone

$$\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}}$$

where,

- x = Observation under consideration
- N = number of terms
- mu = Mean

Code

- import statistics
- arr = [1, 2, 3, 4, 5]
- print("Std = ", (statistics.stdev(arr)))

Output

```
Std =  1.5811388300841898
```

Variance

Here is the formula for sample and population variance and standard deviation. There is slight difference observe them carefully.

For samples:

$$\text{variance} = s^2 = \frac{\Sigma (x - \bar{x})^2}{n - 1}$$

$$\text{standard deviation} = s = \sqrt{s^2}$$

Calculating Formula

$$s^2 = \frac{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}{n - 1}$$

For populations:

$$\text{variance} = \sigma^2 = \frac{\Sigma (x - \bar{x})^2}{n}$$

$$\text{standard deviation} = \sigma = \sqrt{\sigma^2}$$

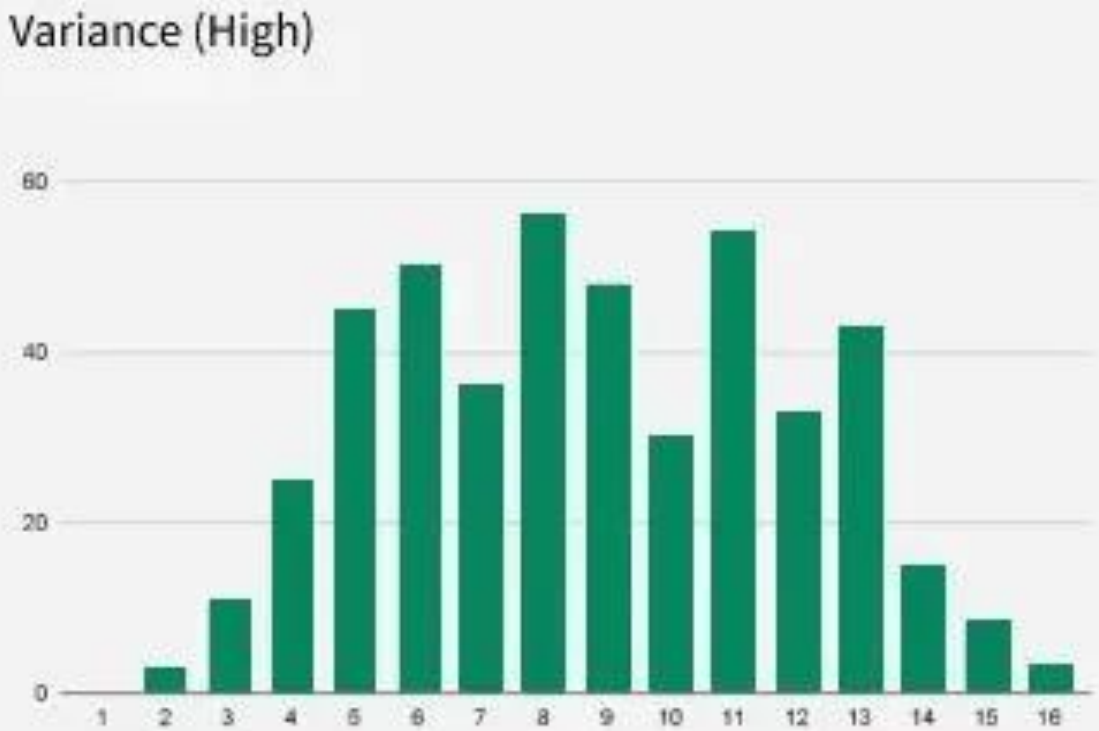
Calculating Formula

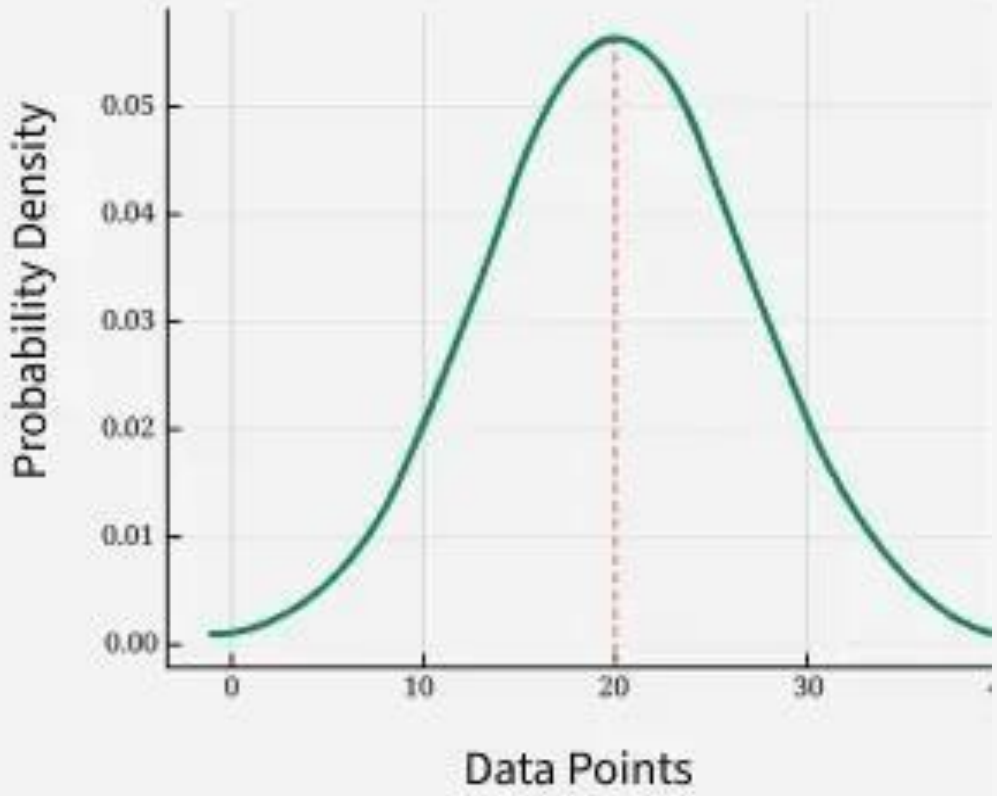
$$\sigma^2 = \frac{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}{n}$$

Where

- X is individual one value
- N is size of population
- $\bar{x}$ is the mean of population

Variance & Standard Deviation





Cont..

- **Variance** measures how far a set of numbers is spread out from their average or mean value.
- It is calculated by taking the average of the squared differences between each number and the mean of the dataset.
- **A higher variance indicates that the numbers are more spread out from their mean value.**
- **Standard deviation** is the square root of the variance and provides a measure of how much the data deviates from the mean.
- It is expressed in the same units as the data and is a more intuitive measure of the spread of the data since it is on the same scale.

- Both variance and standard deviation are important measures in many areas of statistics**, including hypothesis testing, quality control, and data analysis.**

Example

- **How to calculate variance step by step**
- Calculate the mean $\bar{x}$.
- Subtract the mean from each observation. $X - \bar{x}$
- Square each of the resulting observations. $(X - \bar{x})^2$
- Add these squared results together.
- Divide this total by the number of observations n (in case of population) to get variance $\mathbf{S^2}$. If you are calculating sample variance then divide by n-1.
- Use the positive square root to get standard deviation **S**.

Intuition

- **If variance is high, that means you have larger variability in your dataset**.
- In the other way, we can say more values are spread out around your mean value.
- Standard deviation represents the average distance of an observation from the mean
- The larger the standard deviation, larger the variability of the data.

Properties of Variance

- It is always non-negative since each term in the variance sum is squared and therefore the result is either positive or zero.
- Variance always has squared units.
- For example, the variance of a set of weights estimated in kilograms will be given in kg squared.
- Since the population variance is squared, we cannot compare it directly with the mean or the data themselves.

Standard Deviation

- The **Standard Deviation is a measure of how spread out numbers are.**
- Its symbol is **σ** (the greek letter sigma) **for population standard deviation** and S for sample standard deviation.
- It is the square root of the Variance.

Properties of Standard Deviation

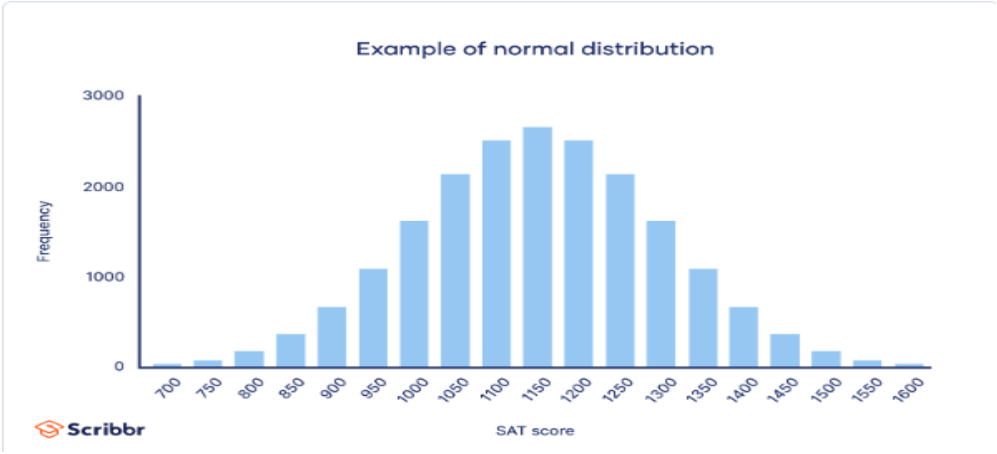
- It describes the square root of the mean of the squares of all values in a data set and is also called the root-mean-square deviation.
- The smallest value of the standard deviation is 0 since it cannot be negative.
- **When the data values of a group are similar, then the standard deviation will be very low or close to zero**. But when the data values vary with each other, then the standard variation is high or far from zero

Population vs. Sample Variance and Standard Deviation

- The primary task of inferential statistics (or estimating or forecasting) is making an opinion about something by using only an incomplete sample of data.
- In statistics, it is very important to distinguish between population and sample.
- A population is defined as all members (e.g. occurrences, prices, annual returns) of a specified group. **Population is the whole group.**
- A sample is a part of a population that is used to describe the characteristics (e.g. mean or standard deviation) of the whole population.
- **The size of a sample can be less than 1%, or 10%, or 60% of the population, but it is never the whole population**.
- As both sample and population are not same thing therefore slight difference is there in their formula.

Normal distribution

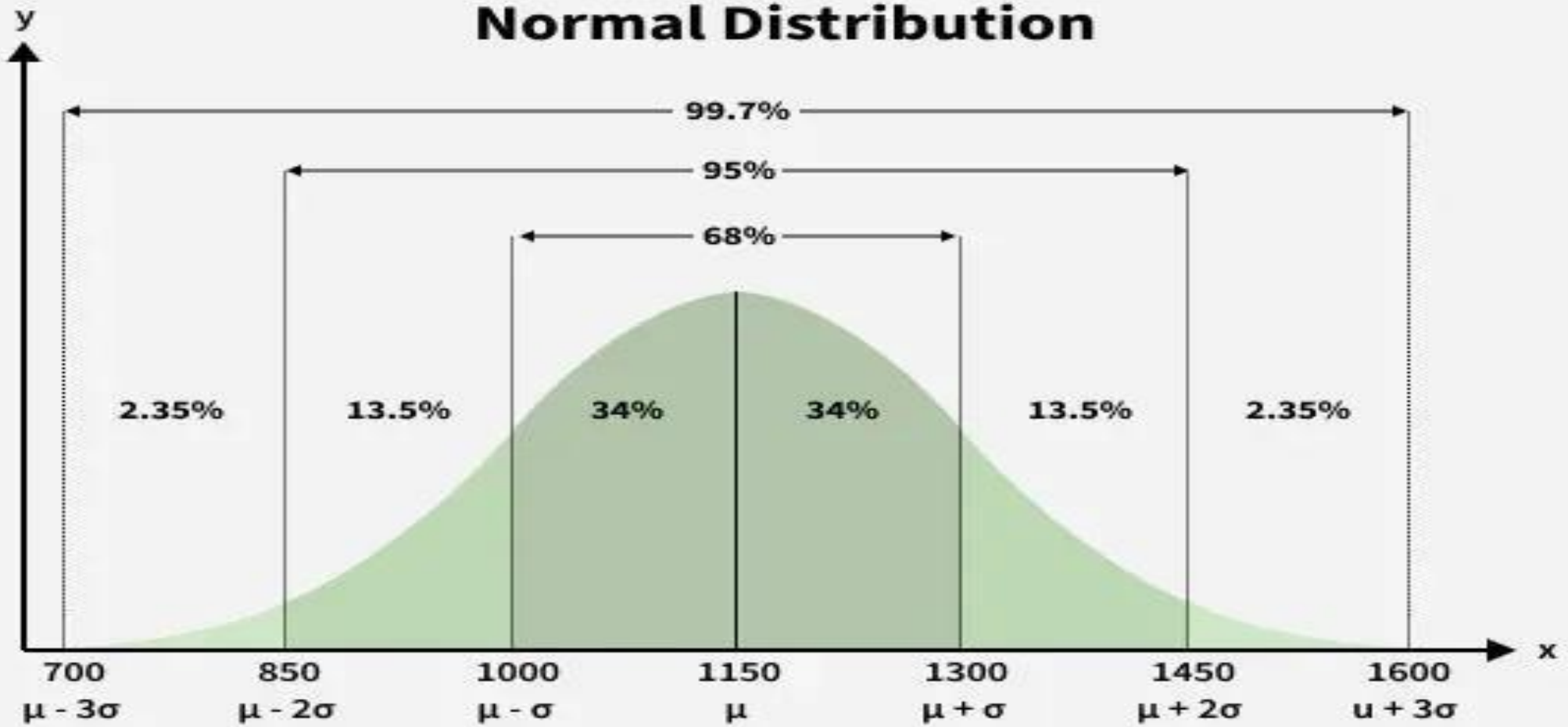
- ***Normal distribution*** *is a continuous probability distribution that is symmetric about the mean,* ***depicting that data near the mean are more frequent in occurrence than data far from the mean.***
- the **Normal Distribution,** also called the **Gaussian Distribution,** is the most significant continuous probability distribution. Sometimes it is also called a **bell curve**.
- A large number of random variables are either nearly or exactly represented by the normal distribution



Cont..

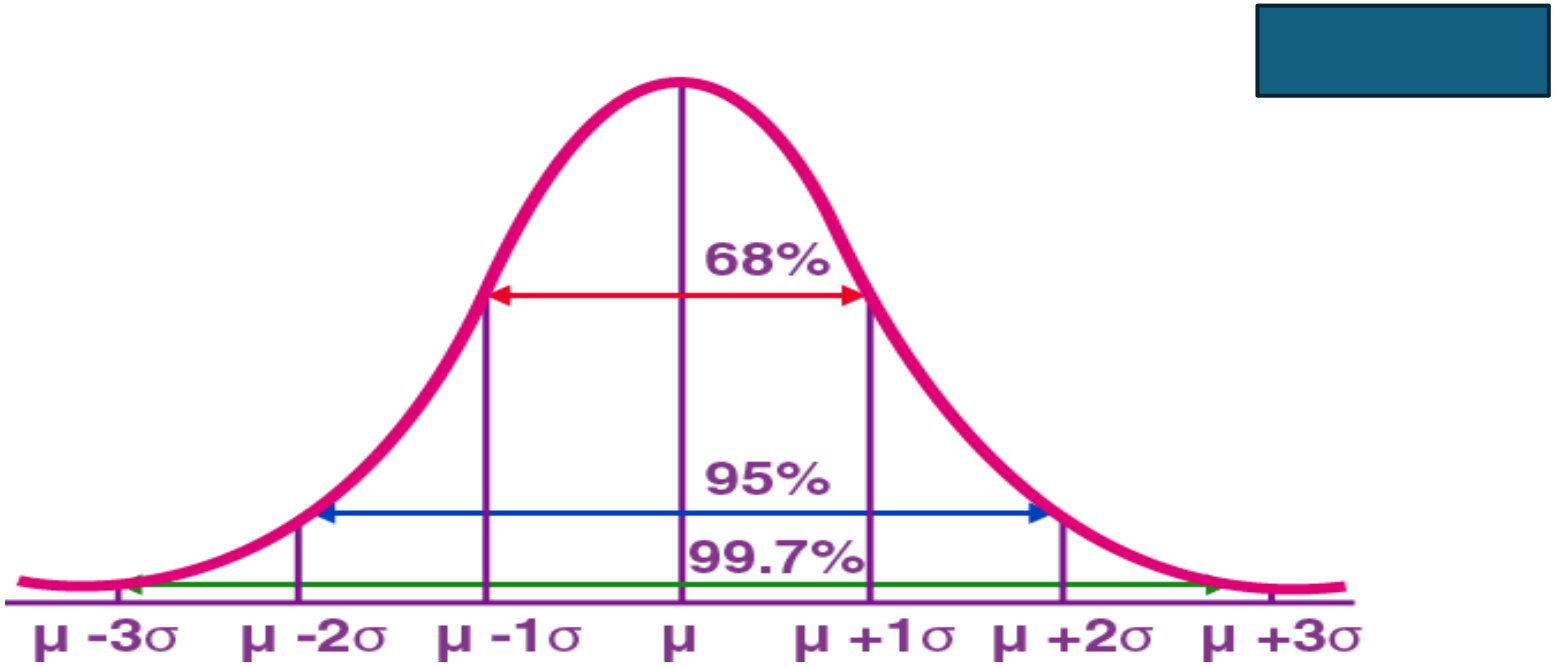
- We know that the **mean helps to determine the line of symmetry of a graph**, whereas the **standard deviation helps to know how far the data are spread out.**
- **If the standard deviation is smaller**, the data are somewhat close to each other and the **graph becomes narrower**.
- **If the standard deviation is larger**, the data are dispersed more, and the **graph becomes wider.**

Chart: Normal Distribution

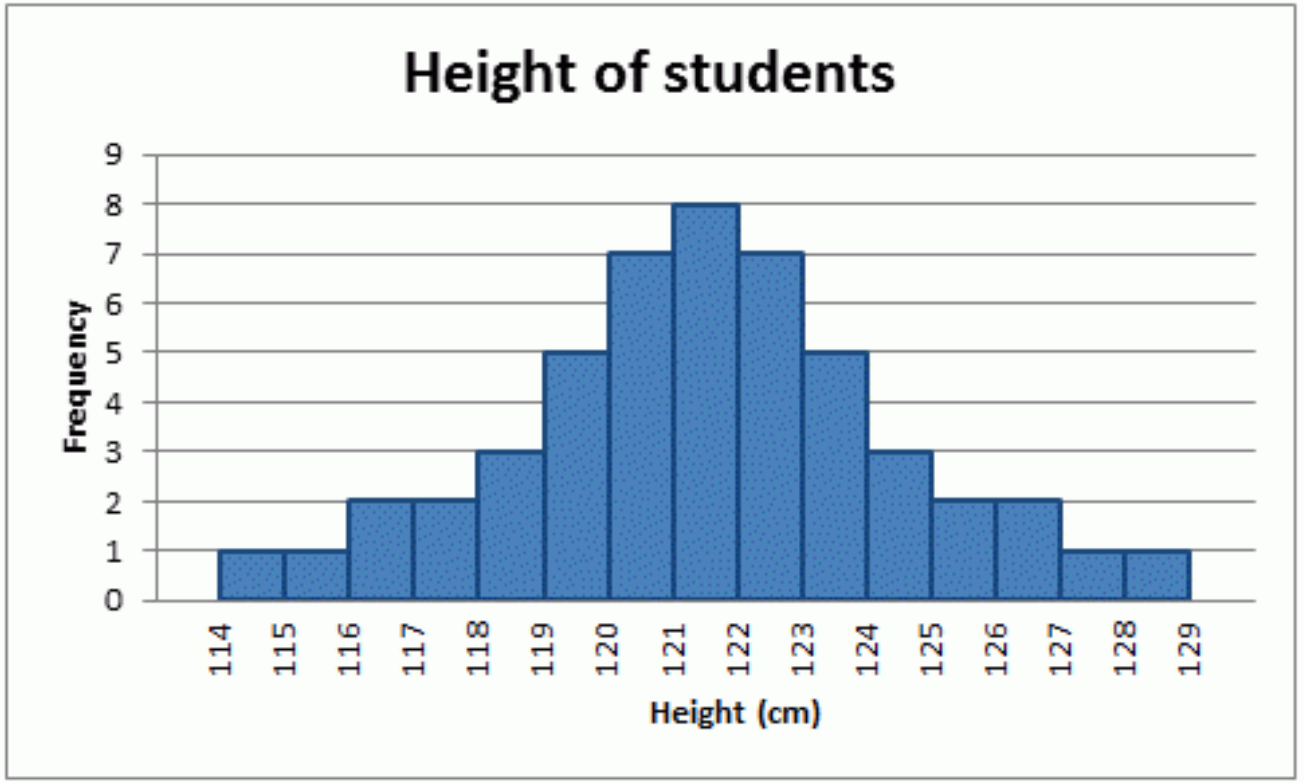


Using 1 standard deviation, the Empirical Rule states

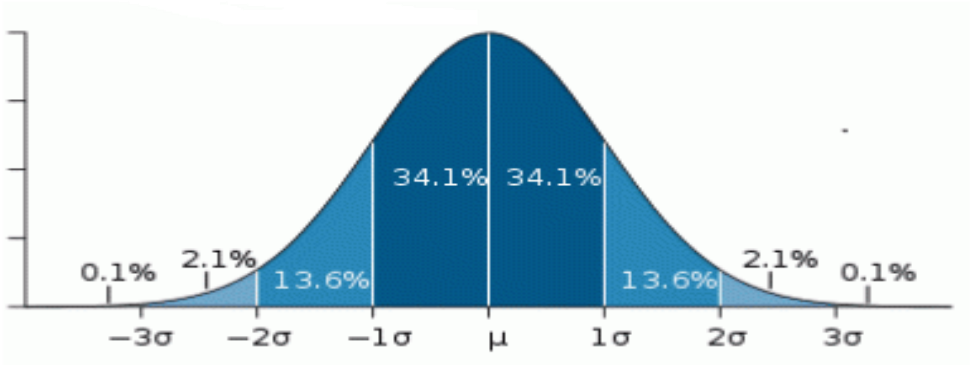
- Approximately 68% of the data falls within **one standard deviation of the mean.** (i.e., Between Mean- one Standard Deviation and Mean + one standard deviation)
- Approximately 95% of the data falls within **two standard deviations of the mean**. (i.e., Between Mean- two Standard Deviation and Mean + two standard deviations)
- Approximately 99.7% of the data fall within **three standard deviations of the mean**. (i.e., Between Mean- three Standard Deviation and Mean + three standard deviations)



Normal distribution: Height of students



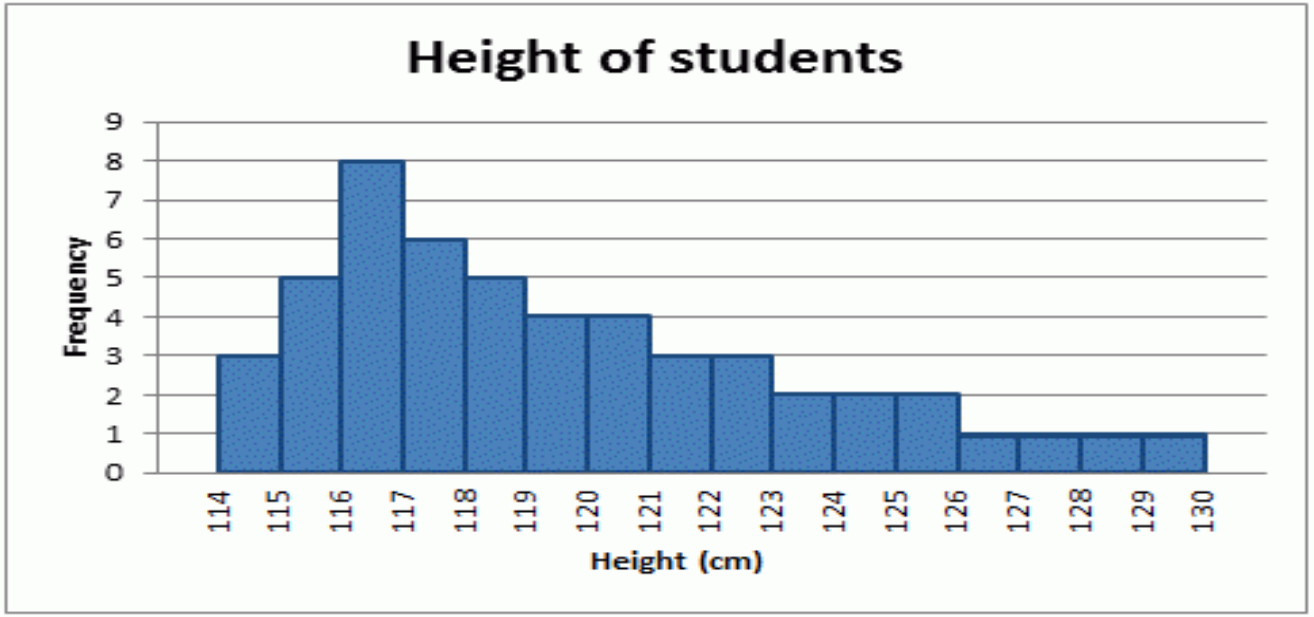
Bell curve example



Key features of the normal distribution:

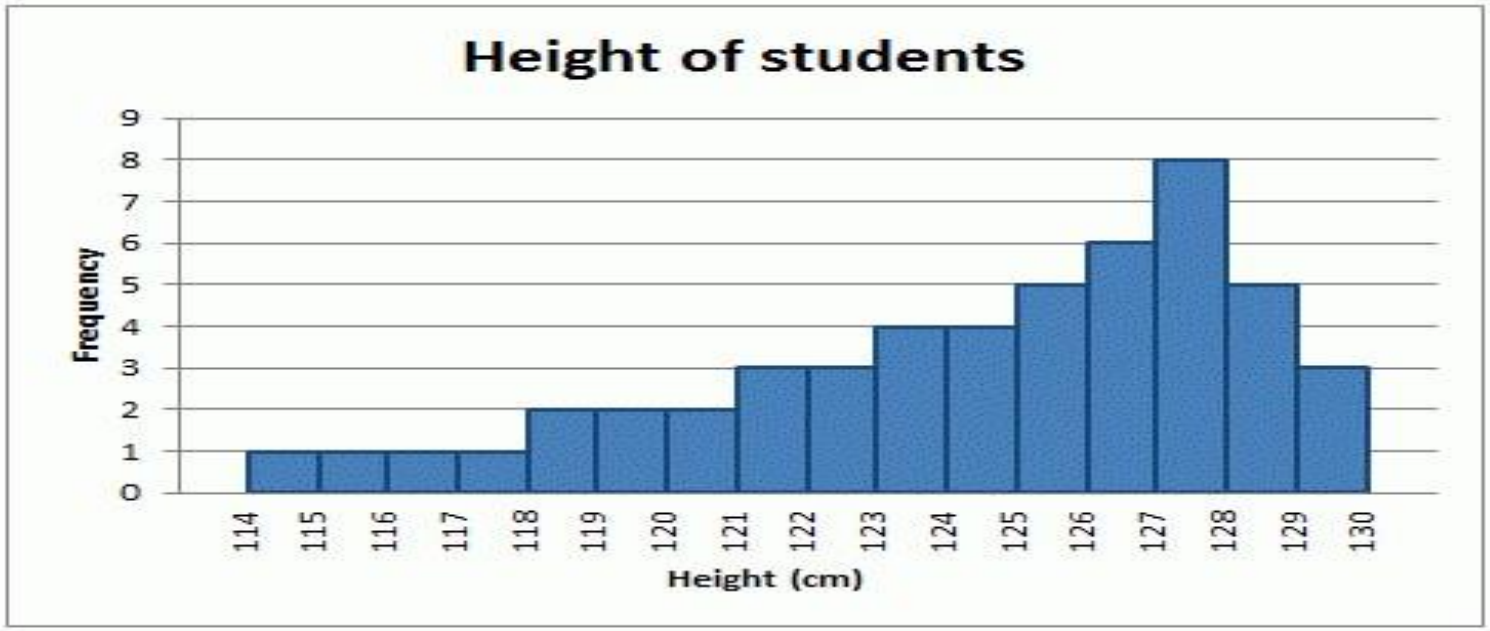
- symmetrical shape
- mode, median and mean are the same and are together in the centre of the curve
- there can only be one mode (i.e. there is only one value which is most frequently observed)
- most of the data are clustered around the centre, while the more extreme values on either side of the centre become less rare as the distance from the centre increases (i.e. About 68% of values lie within one standard deviation (σ) away from the mean; about 95% of the values lie within two standard deviations; and about 99.7% are within three standard deviations. This is known as the empirical rule or the 3-sigma rule.

Positively skewed distribution: Height of students



A distribution is said to be negatively skewed when the tail on the left side of the histogram is longer than the right side. Most of the values tend to cluster toward the right side of the x-axis (i.e. the larger values), with increasingly less values on the left side of the x-axis (i.e. the smaller values).

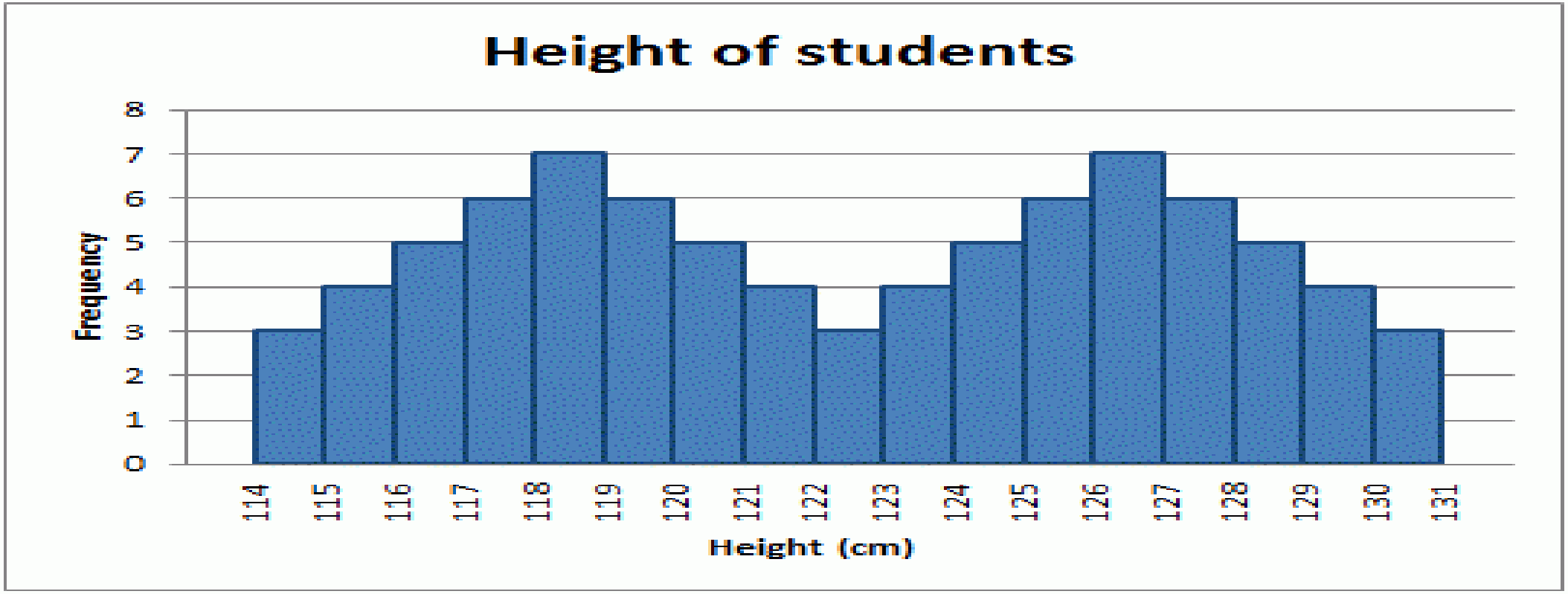
Negatively skewed distribution: Height of students



Key features of the skewed distribution:

- asymmetrical shape
- mean and median have different values and do not all lie at the centre of the curve
- there can be more than one mode
- the distribution of the data tends towards the high or low end of the dataset

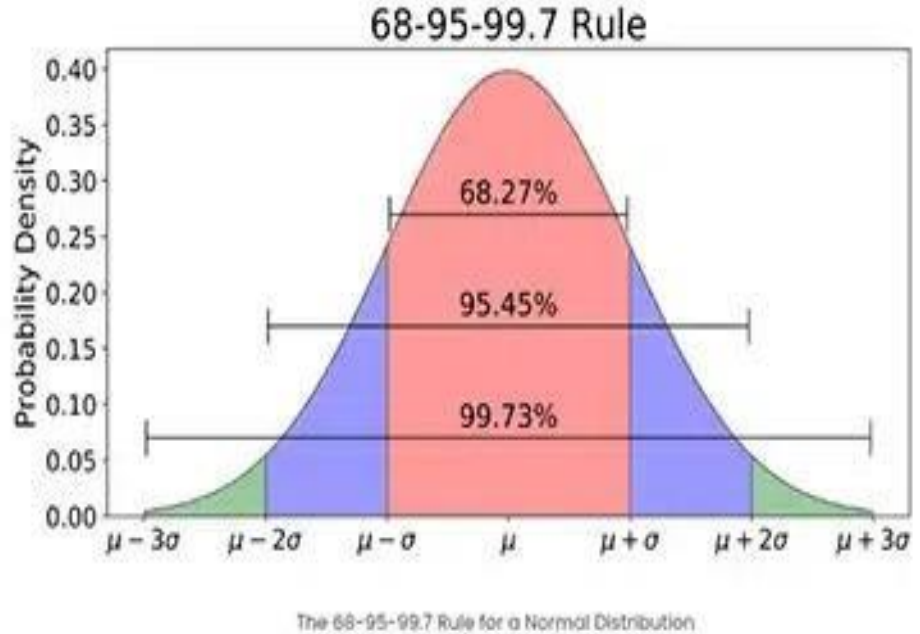
Bi-modal distribution: Height of students



Z-SCORE

In Statistics

Z-Score in Statistics is a measurement of how many standard deviations a data point is from the mean of a distribution.



The 68-95-99.7 Rule for a Normal Distribution

$$z = (x - \mu)/\sigma$$

What is Z-Score?

- **Z-score,** also known as the **standard score**, **tells us the deviation of a data point from the mean** by expressing it in terms of standard deviations above or below the mean.

- It gives us an idea of how far a data point is from the mean.
- Hence, the Z-Score is measured in terms of standard deviation from the mean.

- For example, a **Z-score of 2 indicates the value is 2 standard deviations away from the mean**.

Definition of Z-Score

- ***Z-score is a statistical measure that describes a value's position relative to the mean of a group of values.***
- ***It is expressed in terms of standard deviations from the mean. The Z-score indicates how many standard deviations an element is from the mean.***

Z-Score Formula

To calculate the z- score for any given data we need the value of the element along with the mean and standard deviation.

A z-score can be calculated using the following Z- score formula.

$$z = (X - \mu) / \sigma$$

where,

- z = Z-Score
- X = Value of Element
- μ = Population Mean
- σ = Population Standard Deviation

Example 1: You take the GATE examination and score 500. The mean score for the GATE is 390 and the standard deviation is 45. How well did you score on the test compared to the average test taker?

Solution:

- *Raw score/observed value = X = 500*
- *Mean score = μ = 390*
- *Standard deviation = σ = 45*
- *By applying the z-score formula ,*
- *z = (X - μ) / σ*
- *z = (500 - 390) / 45*
- *z = 110 / 45 = 2.44*

This means that your z-score is 2.44.

- *Since the Z-Score is positive 2.44, we will make use of the positive Z-Table.*
- *Now let's take a look at Z Table (CC-BY) to know how well you scored compared to the other test-takers.*
- *Follow the instruction below to find the probability from the table.*
- *Here,* ***z-score = 2.44,*** *which* ***i****ndicates that the data point is 2.44 standard deviations above the mean.*
- *Firstly, map the first two digits 2.4 on the Y-axis.*
- *Then along the X-axis, map 0.04*
- *Join both axes. The intersection of the two will provide you the cumulative probability associated with the Z-score value you're looking for*

[This probability represents the area under the standard normal curve to the left of the Z-score]

Z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.9608	0.96164
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926
1.9	0.97128	0.97193	0.97257	0.9732	0.97381	0.97441	0.975	0.97558
2	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.9803	0.98077
2.1	0.98214	0.98257	0.983	0.98341	0.98382	0.98422	0.98461	0.985
2.2	0.9861	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.9884
2.3	0.98928	0.98956	0.98983	0.9901	0.99036	0.99061	0.99086	0.99111
2.4	0.9918	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324
2.5	0.99379	0.99396	0.99413	0.9943	0.99446	0.99461	0.99477	0.99492
2.6	0.99534	0.99547	0.9956	0.99573	0.99585	0.99598	0.99609	0.99621
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.9972



- *As a result, final value which is **0.99266**.*
- *Now, we need to compare how our original score of 500 on the GATE examination compares to the average score of the batch. To do that we need to convert the cumulative probability associated with the Z-score into a percentage value.*
- ***0.99266 × 100 = 99.266%***
- *Finally, you can say that you have performed well than almost **99%** of other test-takers.*

Example 1: Z-score with negative result

- *Mean (μ) = 100*
- *Standard deviation (σ) = 15*
- *Data point (x) = 85*

Solution:

- *Z- score formula : $Z = (x - \mu) / \sigma$*
- *$Z = (85 - 100) / 15$*
- *$Z = -15 / 15$*
- *$Z = -1$*

This means the data point is 1 standard deviation below the mean.

Example 2: Basic Z-score calculation

- *Mean (μ) = 70*
- *Standard deviation (σ) = 5*
- *Data point (x) = 80*
- *Z-score formula: $Z = (x - \mu) / \sigma$*

Solution:

- *$Z = (80 - 70) / 5$*
- *$Z = 10 / 5$*
- *$Z = 2$*

This means the data point is 2 standard deviations above the mean.

Example 4:

Finding the corresponding height for a given a Z-score

- **Mean height (μ) = 170 cm, Standard deviation (σ) = 8 cm, Z-score = -1.5**

Use the Z-score formula and solve for x:

- *-1.5 = (x - 170) / 8*
- *-12 = x - 170*
- *x = 158 cm*

The height corresponding to a Z-score of -1.5 is 158 cm.

Class Test

Example 2: A student appeared for two tests. He secured 80 in the first and 75 in the second.
The mean and deviation for the first were 70 and 15 respectively, while for the second it was 64 and 12 respectively.

What conclusion can you make on comparing the student's performance for both exams?

Solution:

First test:

- Marks secured, $x1 = 80$
- Standard deviation, $\sigma 1 = 15$
- Mean, $\mu 1 = 70$
- z score = $80 - 7015 = 0.667$

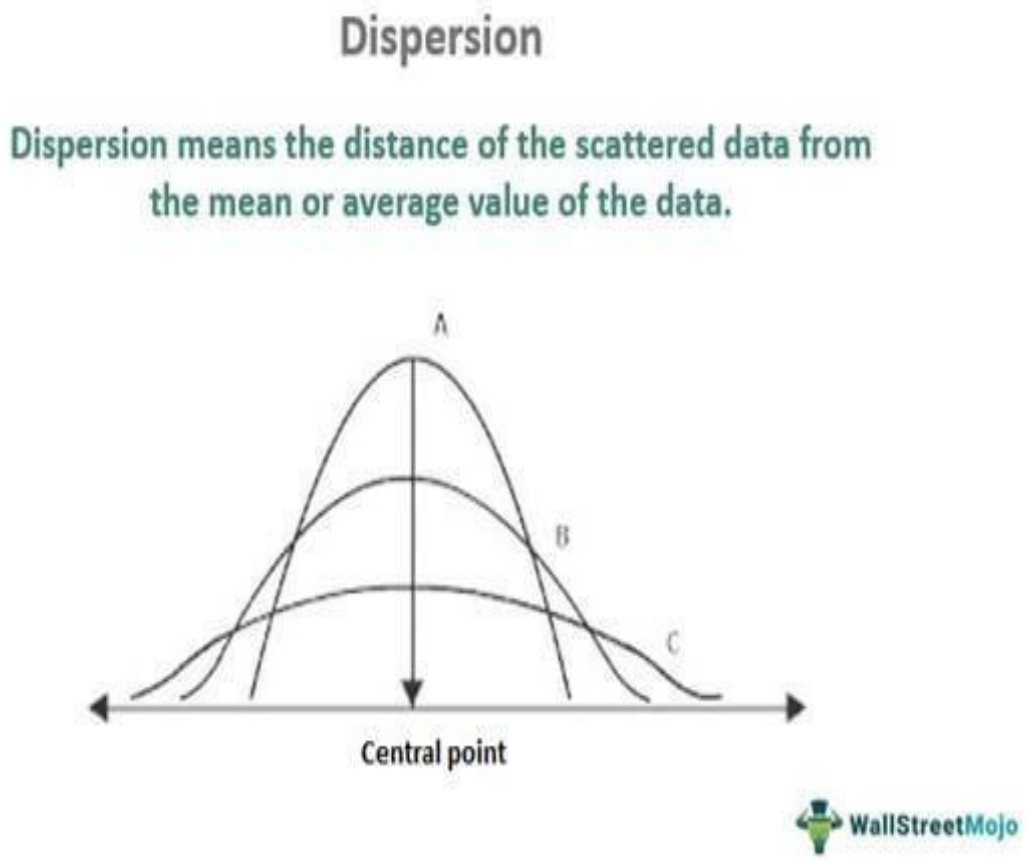
Second test:

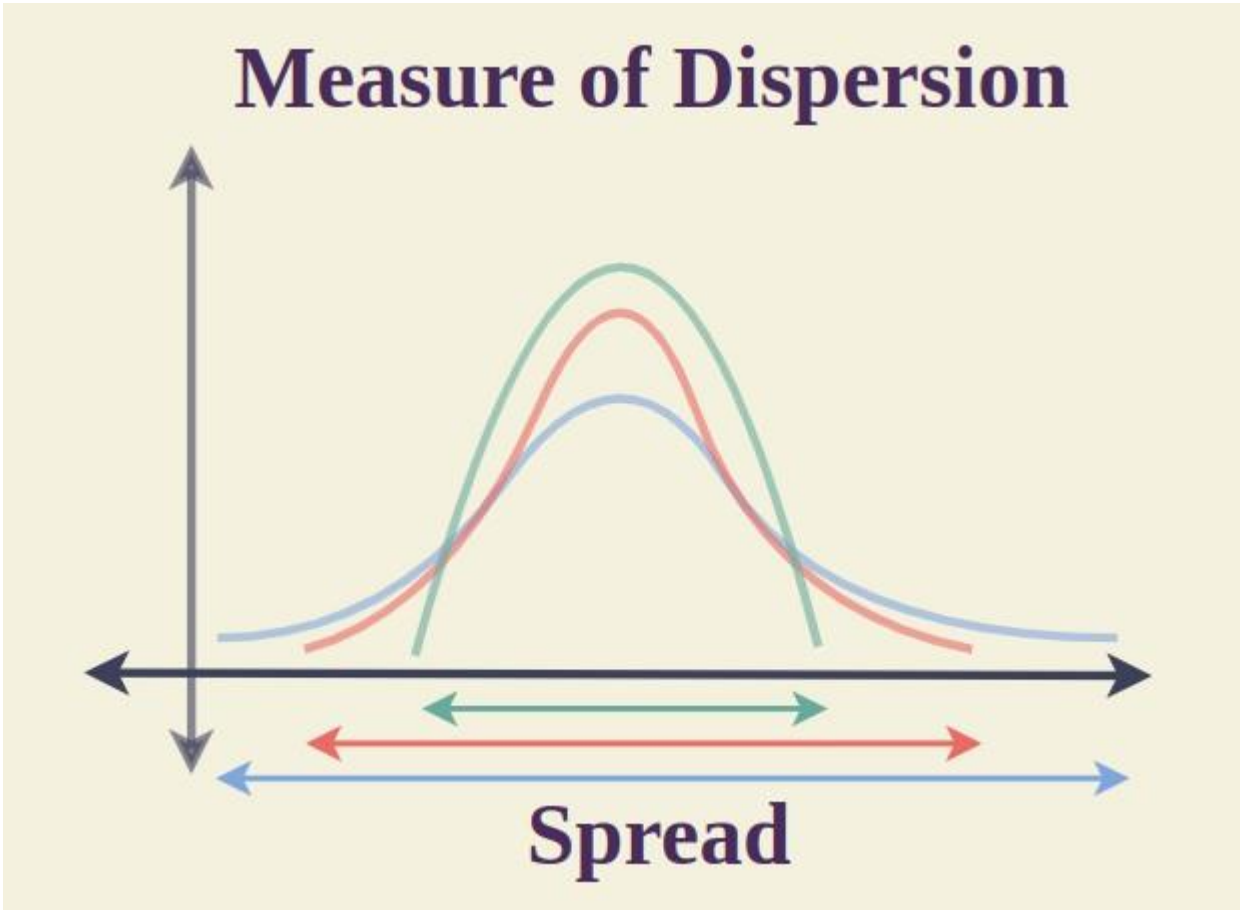
- Marks secured, $x2 = 75$
- Standard deviation, $\sigma 2 = 12$
- Mean, $\mu 2 = 64$
- z score = $75 - 6412 = 0.9167$

Since the z score is more for the second test, the student performed better in the second test.
Answer: The student performed better in the second test.

Measure of Dispersion

- **Measures of Dispersion/Spread** are used to represent the scattering of data. These are the numbers that show the various aspects of the data spread across multiple parameters.



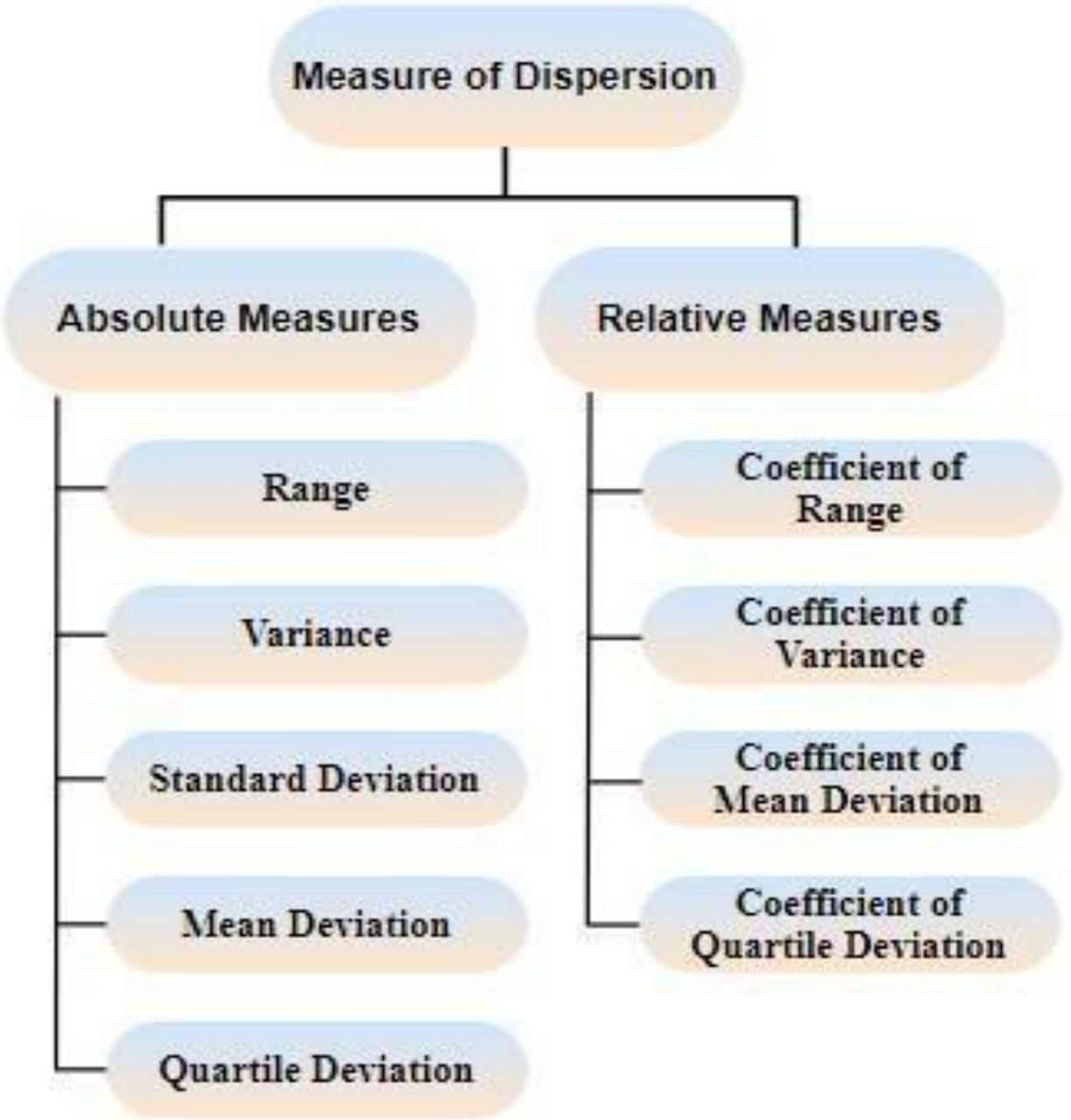


Dispersion in Statistics

- Dispersion in statistics is **a way to describe how spread out or scattered the data is around an average value.**
- It helps to understand if the data points are close together or far apart.
- Dispersion shows the variability or consistency in a set of data.
- There are different measures of dispersion**, like range, variance, and standard deviation.**

Types of Measures of Dispersion

1. Absolute Measure of Dispersion
2. Relative Measure of Dispersion



Absolute Measures of Dispersion

Absolute Measures of Dispersion	Related Formulas
Formulae of Measures of Dispersion	
Range	H - S where H is the Largest Value and S is the Smallest Value
Variance	Population Variance, $\sigma^2 = \Sigma(x_i-\mu)^2/n$ Sample Variance, $S^2 = \Sigma(x_i-\mu)^2/(n-1)$ where μ is the mean and n is the number of observations
Standard Deviation	S.D. = $\sqrt{(\sigma^2)}$
Mean Deviation	$\mu = (x - a)/n$ where, **a** is the central value(mean, median, mode) and **n** is the number of observation
Quartile Deviation	$(Q_3 - Q_1)/2$ where, Q_3 = Third Quartile and Q_1 = First Quartile

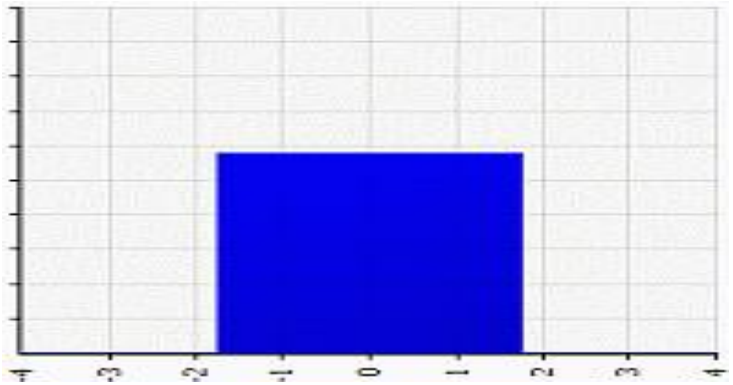
Coefficient of Dispersion

Relative Measures of Dispersion	Related Formulas
Coefficient of Range	(H - S)/(H + S)
Coefficient of Variation	(SD/Mean)×100
Coefficient of Mean Deviation	(Mean Deviation)/μ where, μ is the central point for which the mean is calculated
Coefficient of Quartile Deviation	$(Q_3 - Q_1)/(Q_3 + Q_1)$

Data Distribution and Shape

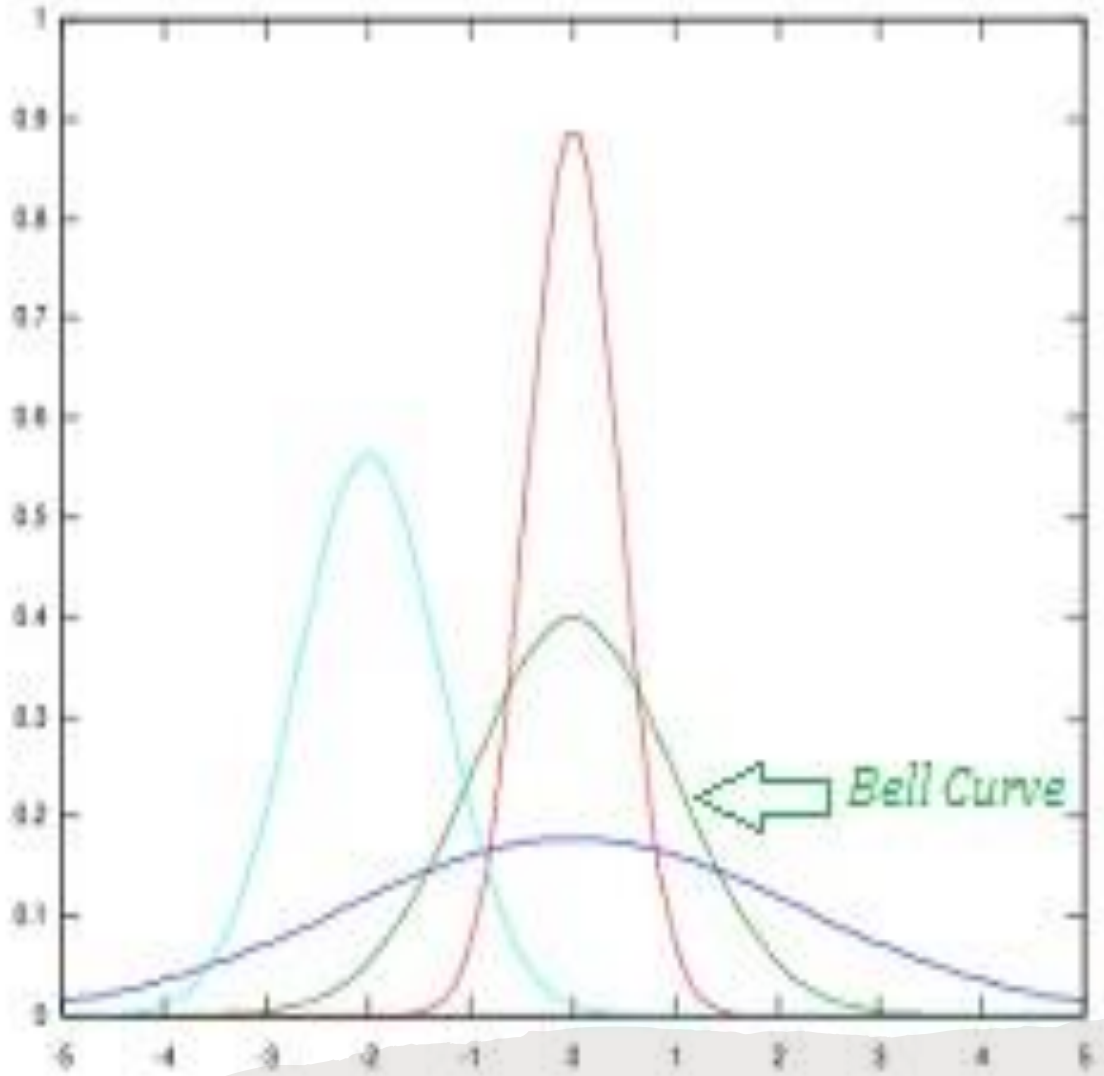
- **The shape of the distribution refers to the graph of the data.**
- The **data can be graphed using a histogram, which is like a bar graph.**
- **If the height of all the bars is the same, the shape is called uniform**, and the data is distributed evenly across the range

If a data set has no clear peaks (i.e. the whole graph looks flat), it's called a "*uniform distribution.*"



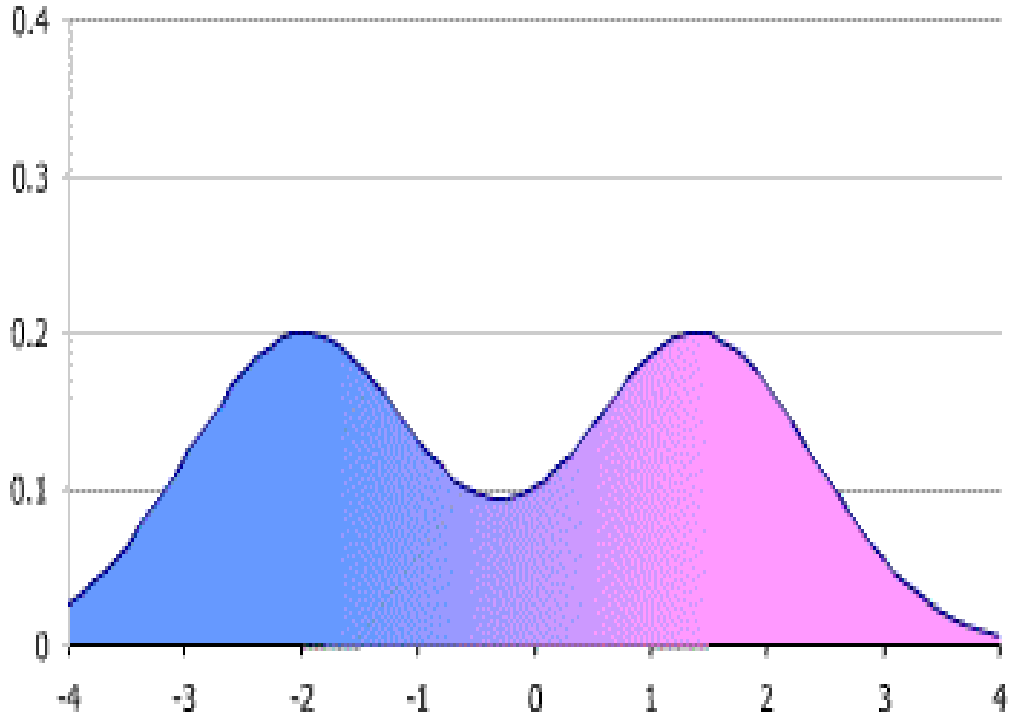
Number of peaks: Unimodal

- A **unimodal** distribution has **one** mode.
- A single peak can take on many shapes **(e.g. very tall and thin or very squat and fat).**
- **One of the most common types of unimodal distributions is the *normal distribution*,** sometimes called the "bell curve" because its shape looks like a bell.

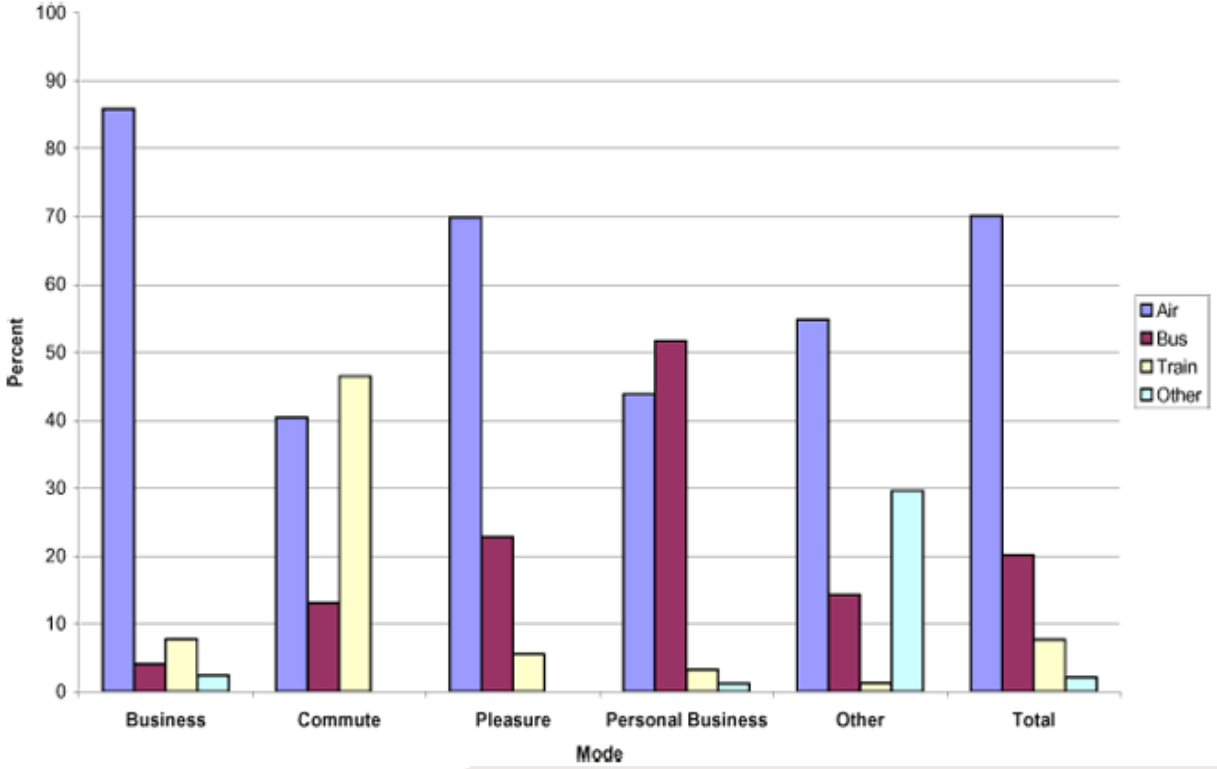


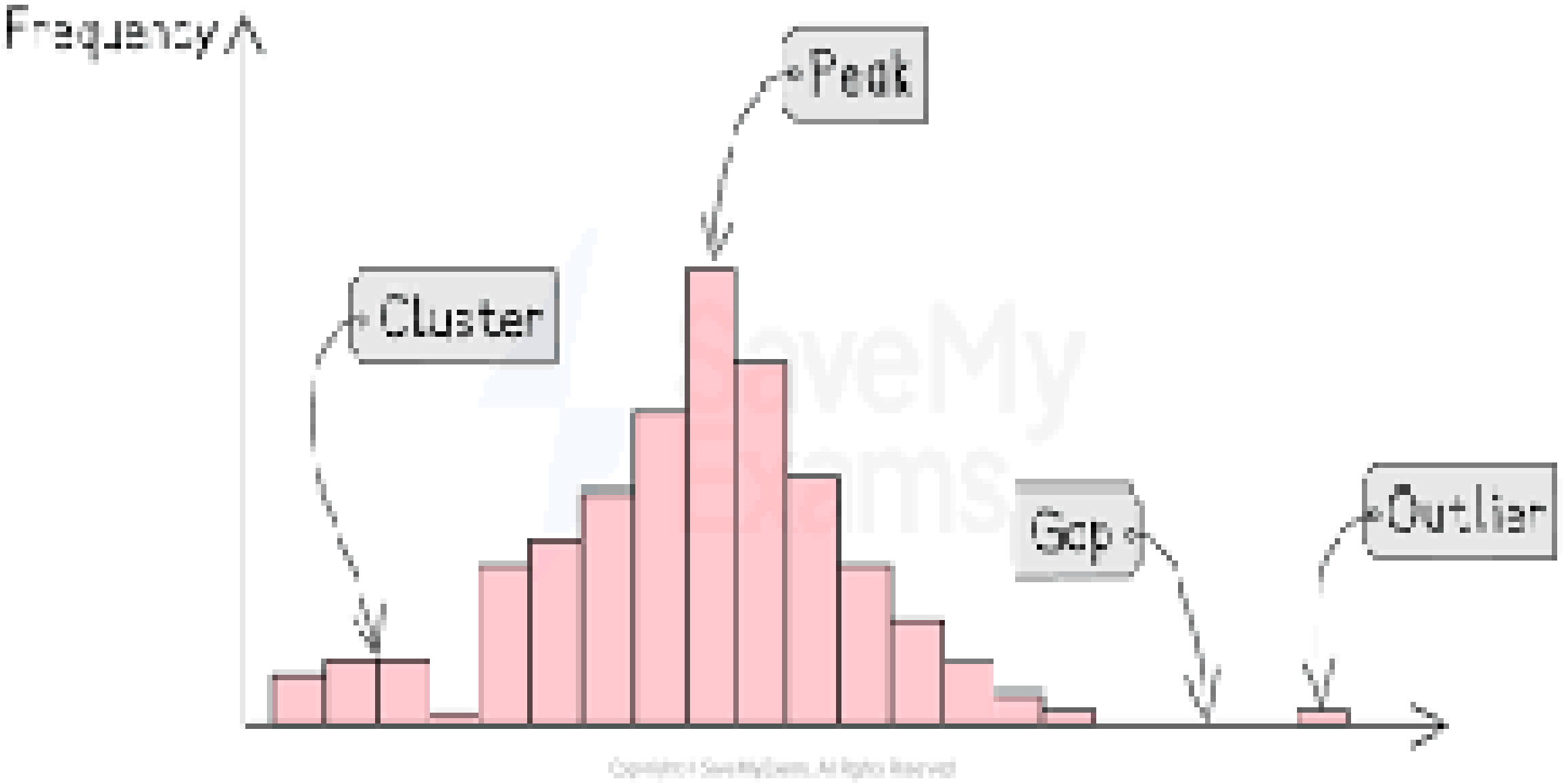
Bio Modal and Multi Modal:

- A **bimodal** distribution has **two** modes.



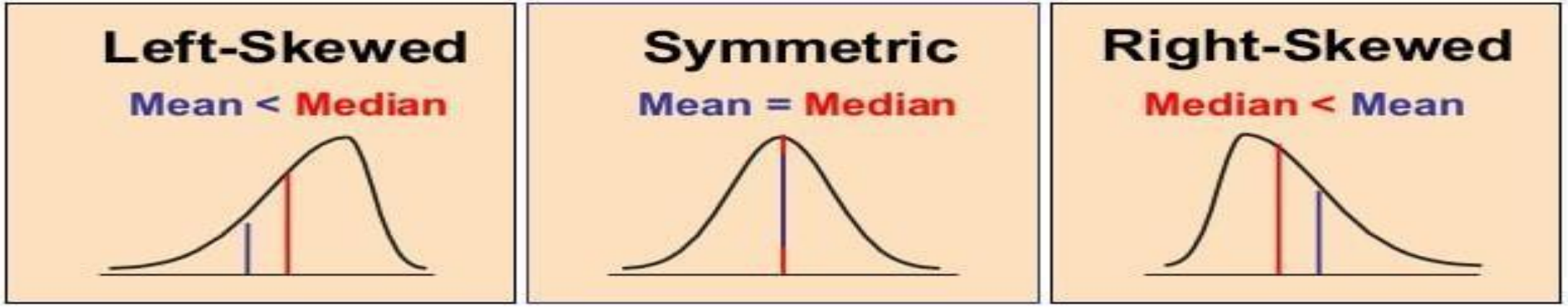
A **multimodal** distribution has **three or more** modes.




Frequency
Peak
Cluster
Gap
Outlier

Shape of a Distribution

- Describes how data are distributed
- Measures of shape
  - Symmetric or skewed

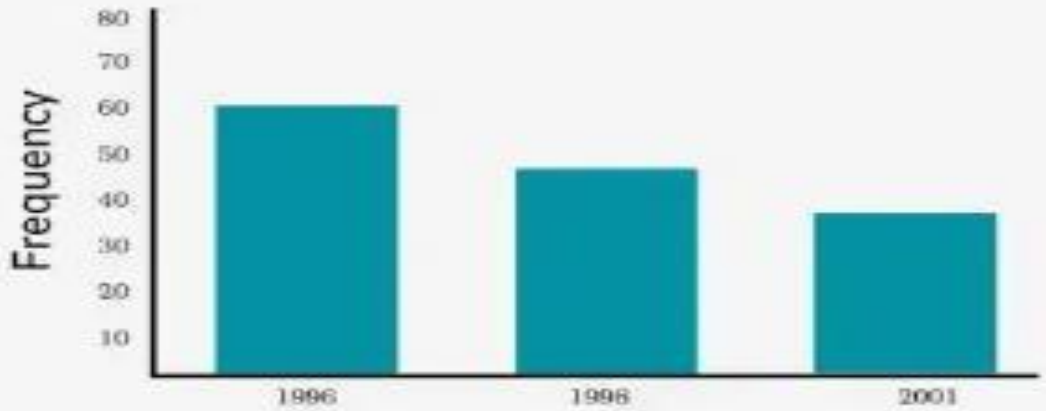


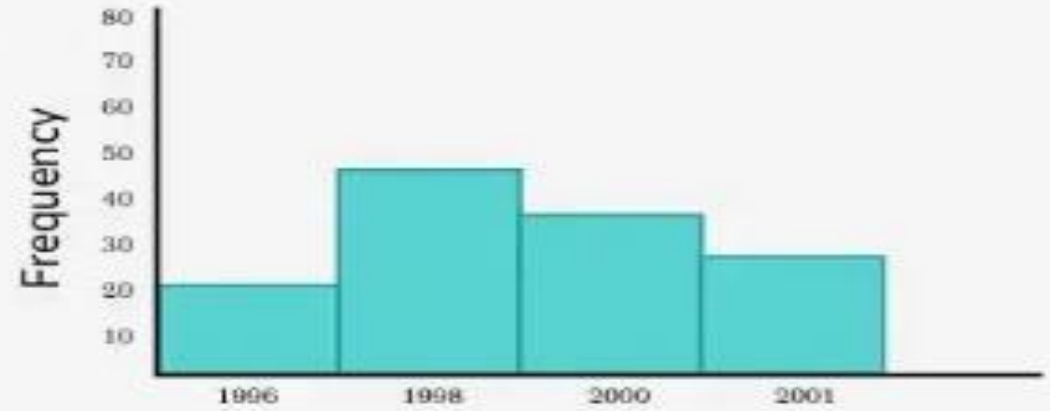
Frequency Distribution

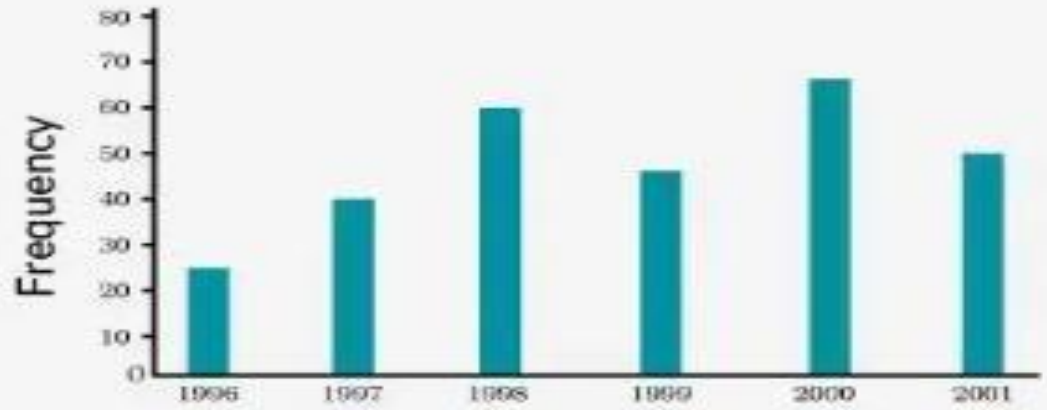
- A **frequency distribution** is a way **to organize data and see how often each value appears.**
- It shows **how many times each value or range of values occurs in a dataset.**
- This helps us **understand patterns**, like which values are common and which are rare.
- **Frequency distributions are often shown in tables or graphs, and make it easier to analyze and conclude data.**

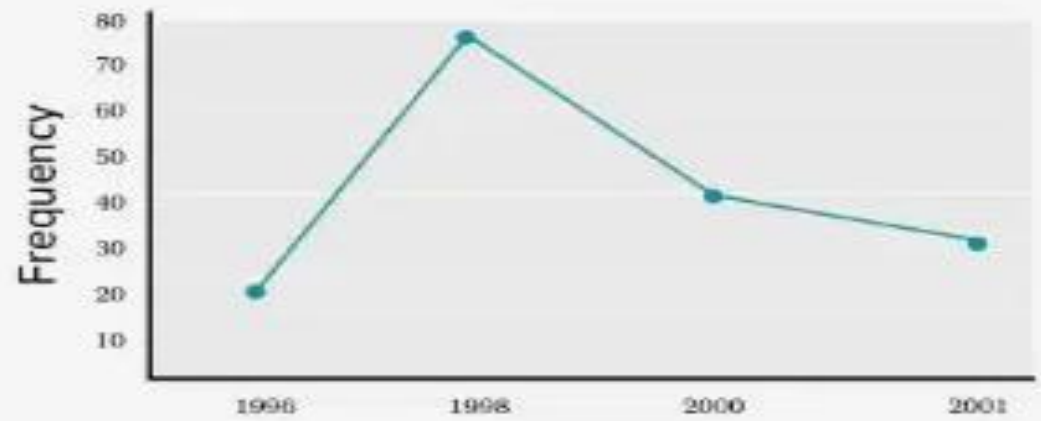
FREQUENCY DISTRIBUTION

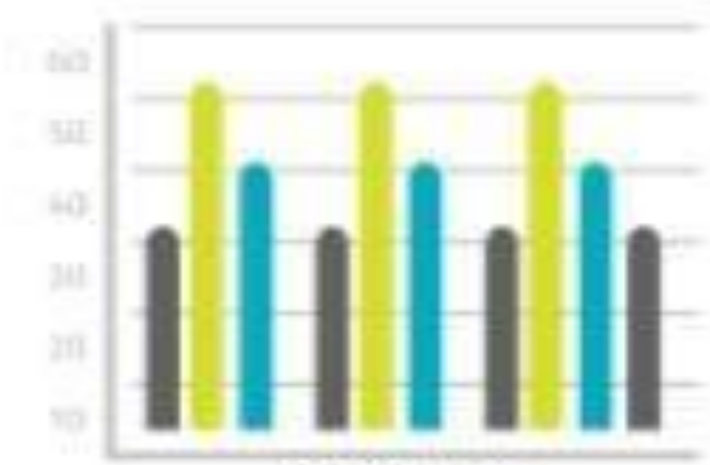
Frequency Distribution is a tool in statistics that helps us organize the data and also helps us reach meaningful conclusions.



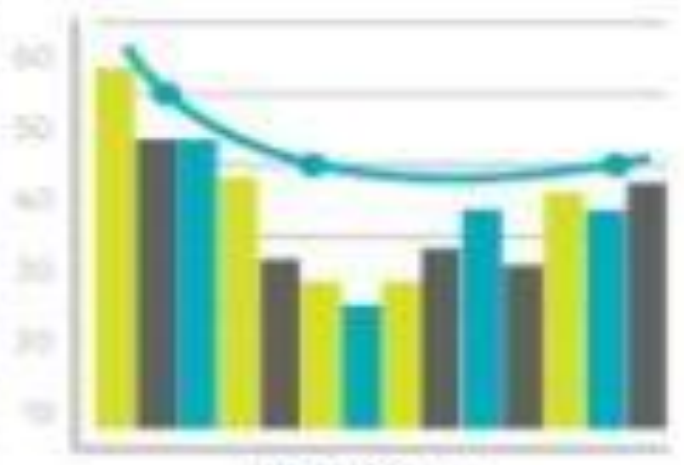




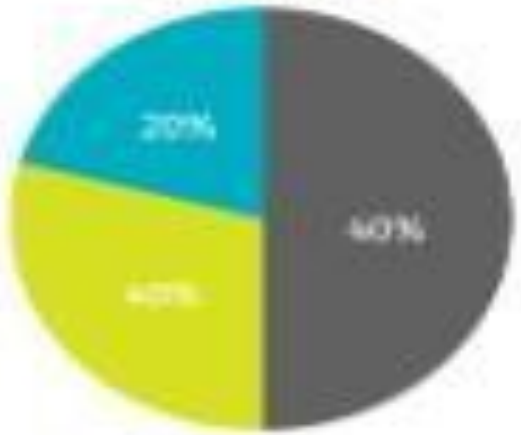




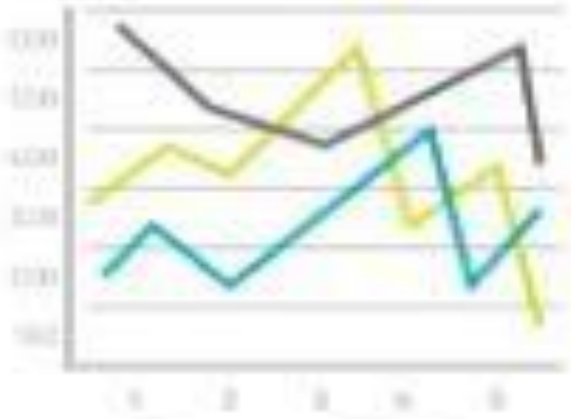
Bar Graph



Histogram



Pie Chart



Frequency Polygon

Tabulation

- In statistics, tabulation is the systematic organization of collected, classified data into rows and columns to facilitate comparison, analysis, and interpretation.

- It presents data in a concise, visual format to simplify complex information, identify patterns and trends, and support the drawing of valid conclusions.

- This organized presentation helps investigators understand data with ease and is a logical step after classifying raw data.

FORMAT OF TABLE

Table Number: .....

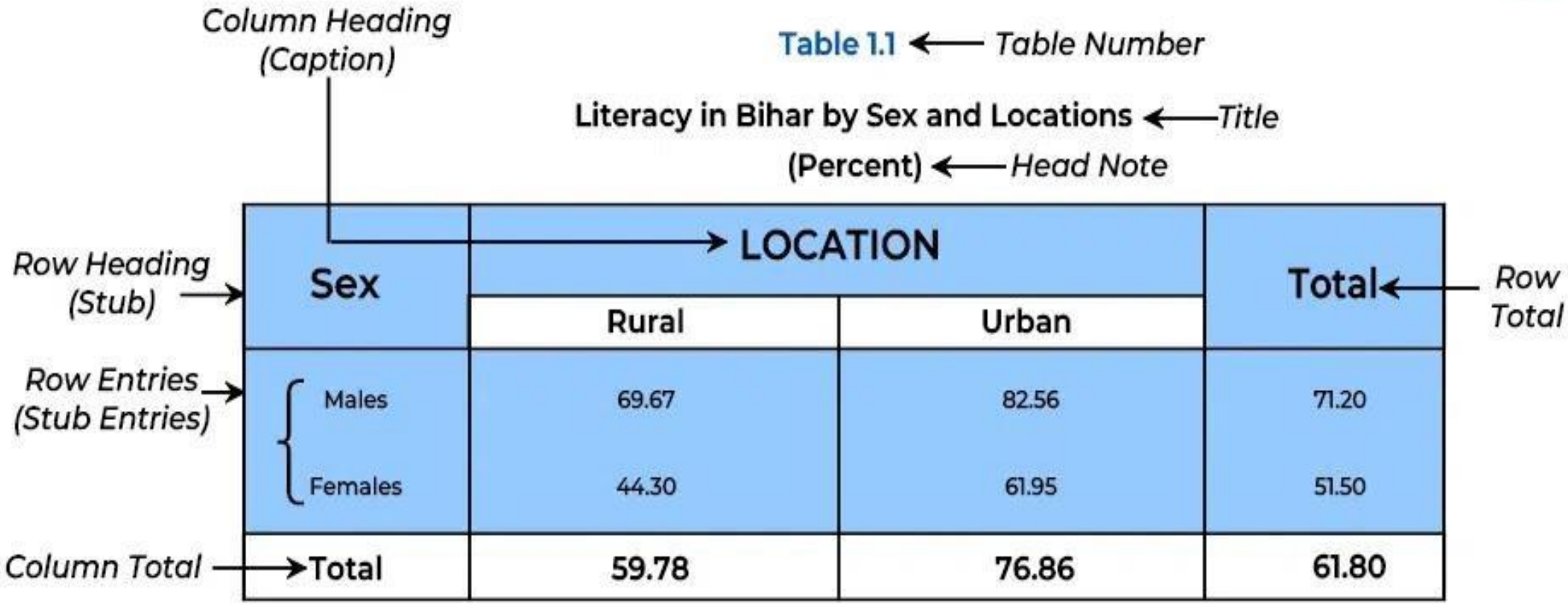
Title: .....

(Head Note, if any)

Stub (Row Heading)	Caption (Column Heading)				Total (Rows)
	Sub-head		Sub-head		
	Column-head	Column-head	Column-head	Column-head	
Stub Entries (Row Entries)	BODY				
Total (Columns)					

Source Note:

Footnote:


Column Heading (Caption)
Table 1.1 ← Table Number
Literacy in Bihar by Sex and Locations ← Title
(Percent) ← Head Note
Row Heading (Stub)
Sex
LOCATION
Rural
Urban
Total
Row Total
Row Entries (Stub Entries)
Males
69.67
82.56
71.20
Females
44.30
61.95
51.50
Column Total
Total
59.78
76.86
61.80
Source Note → Source Note: Census of India 2011, Provisional Population Totals
Footnote → Note: Figures are rounded to two digits after decimal.

End of the Unit-2

Thank You

Unit-3-**Relationships Descriptions**

- Correlation and scatter plots
- Computational formula for correlation coefficient/Persons
- Regression analysis
- Regression lines
- least squares method
- Standard error of estimate
- **Interpretation of R^2** (coefficient of determination)
- Multiple regression equations
- Regression towards the mean

Correlation

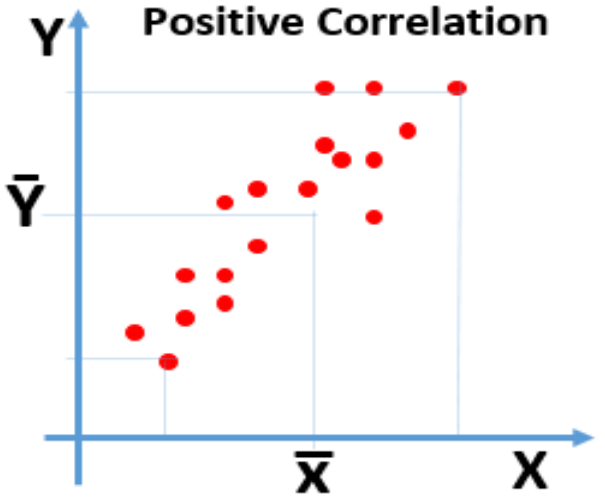
- **Correlation is a statistical measure that describes the extent of a linear relationship between two variables, ranging from -1 to 1.**

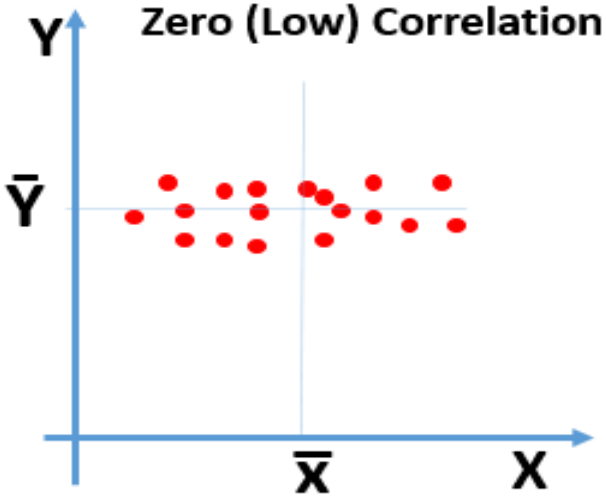
- A **positive correlation** indicates variables move in the same direction, a **negative correlation** shows opposite directions, and a **zero correlation** means no clear linear relationship

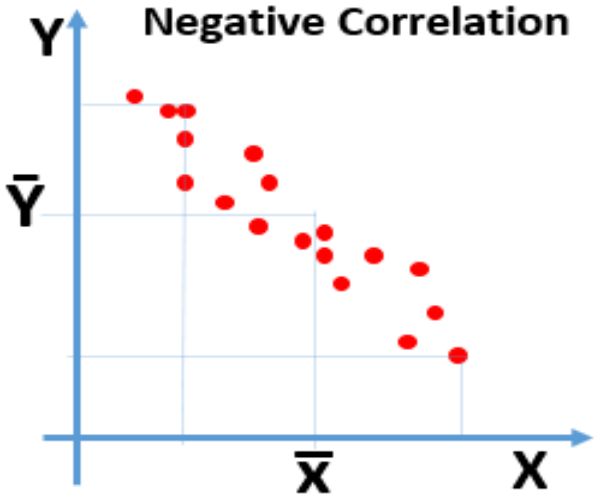
Analysis of Two Continuous Variables

Scatter plot and **Correlation** are a great way of analyzing two continuous variables.

- **Correlation** measures the strength of linear relationship between Two Continuous Variables
- **Scatter Plot** visually shows the relation between the Two Continuous Variables using X – Y axis coordinate system.





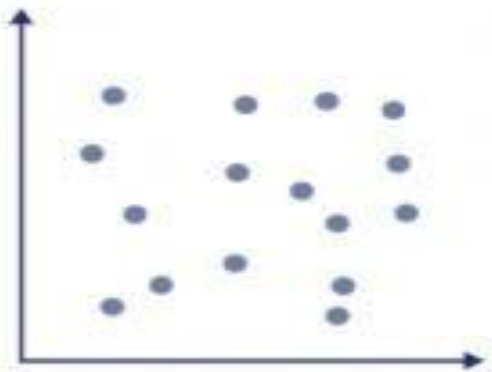


Correlation coefficient value	Correlation type	Meaning
1	Perfect positive correlation	When one variable changes, the other variables change in the same direction.
0	Zero correlation	There is no relationship between the variables.
-1	Perfect negative correlation	When one variable changes, the other variables change in the opposite direction.

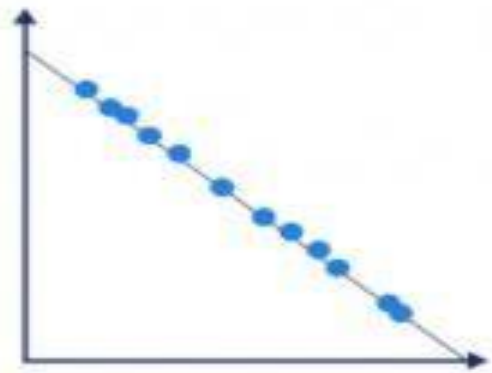
Perfect positive correlation



Zero correlation



Perfect negative correlation



Scatter Diagram

- A scatter diagram is a diagram that **shows the values of two variables X and Y,** along with the way in which these two variables relate to each other.
- The values of variable **X are given along the horizontal axis**, with the **values of the variable Y given on the vertical axis**.

Types of Correlations

- **Positive Correlation** – when the values of the two variables move in the same direction so that an increase/decrease in the value of one variable is followed by an increase/decrease in the value of the other variable.

- **Negative Correlation** – when the values of the two variables move in the opposite direction so that an increase/decrease in the value of one variable is followed by decrease/increase in the value of the other variable.

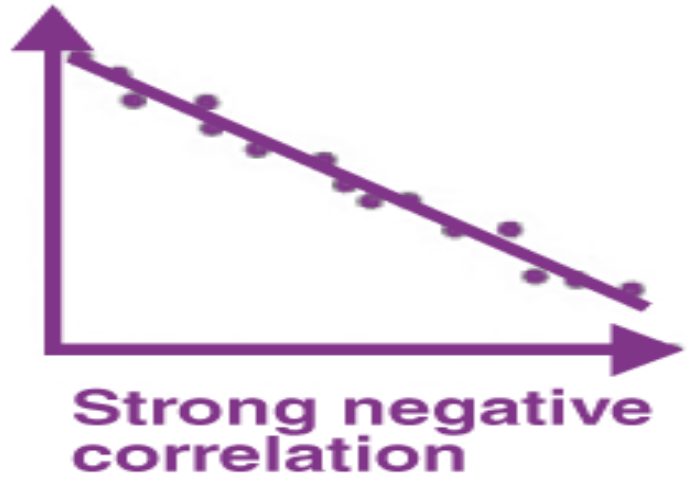
- **No Correlation** – when there is no linear dependence or no relation between the two variables.



Strong positive correlation



Weak positive correlation



Strong negative correlation



Weak negative correlation



Moderate negative correlation



No correlation

What is a Scatter Diagram?

- A simple and attractive method of measuring correlation by diagrammatically representing bivariate distribution for determination of the nature of the correlation between the variables is known as the **Scatter Diagram Method**.

- This method gives the investigator/analyst a visual idea of the nature of the association between the two variables.

- **It is the simplest method of studying the relationship between two variables as there is no need to calculate any numerical value.**

How to draw a Scatter Diagram?

- Plot the values of the given variables (say X and Y) along the X-axis and Y-axis, respectively.

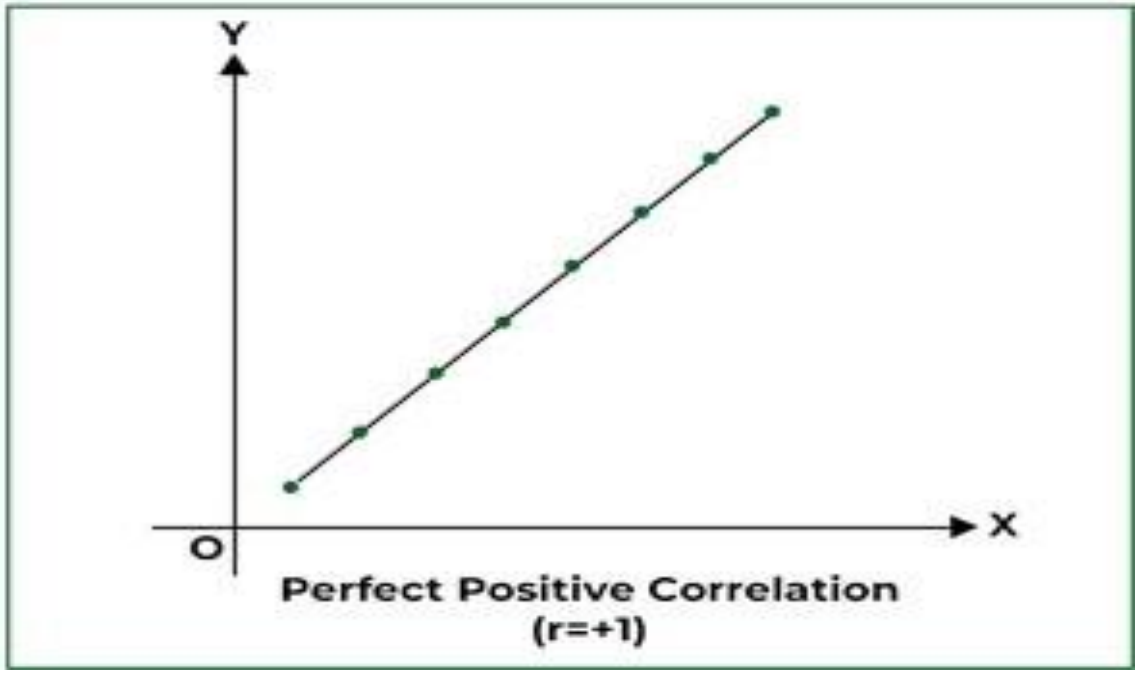
- Show these plotted values on the graph by dots. Each of these dots represents a pair of values.

- After observing the pattern of dots, one can know the presence or absence of correlation and its type

The scatter diagram can be interpreted in the following ways:

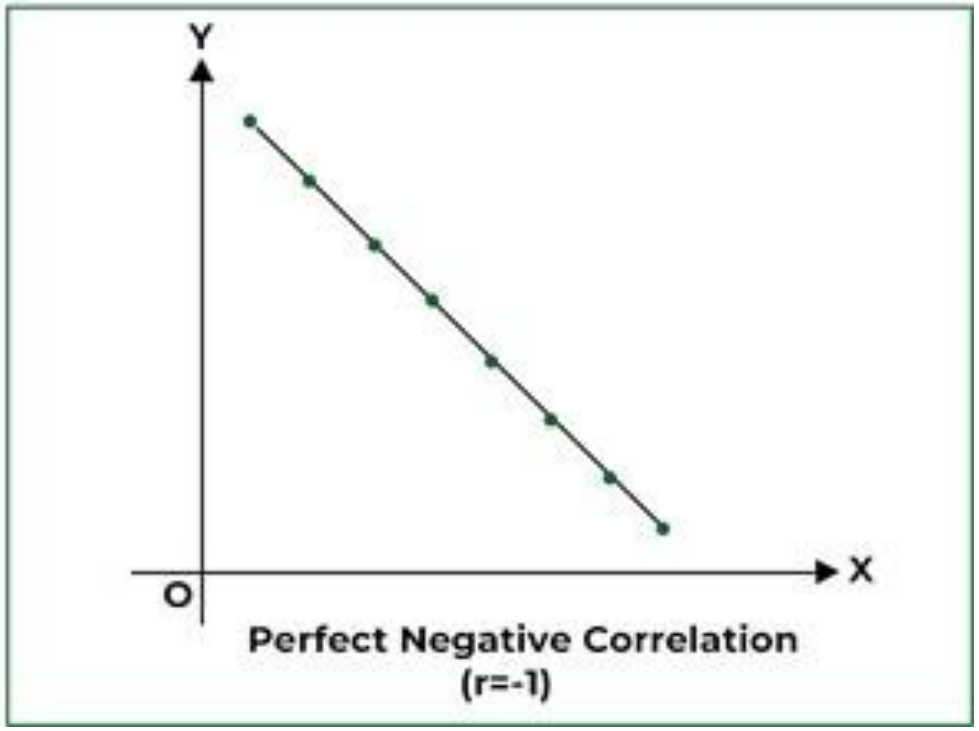
1. **Perfect Positive Correlation**

If the points of the scatter diagram fall on a straight line and have a positive(upward) slope, then the correlation is said to be perfectly positive; i.e., r = +1.



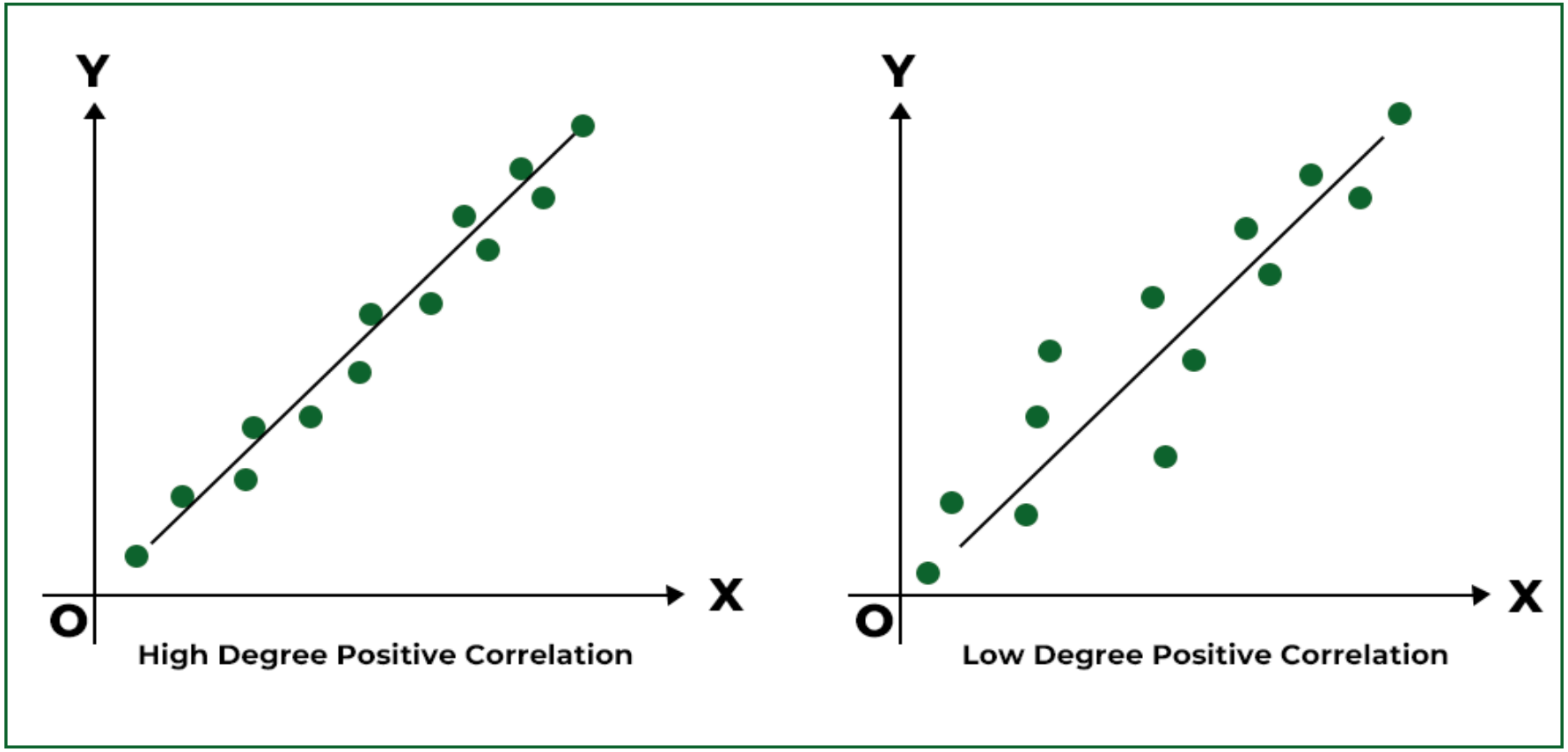
2. Perfect Negative Correlation

- **If the points of the scatter diagram fall on a straight line(opposite side)** and have a negative(downward) slope, then the correlation is said to be perfectly negative; i.e., r = -1.



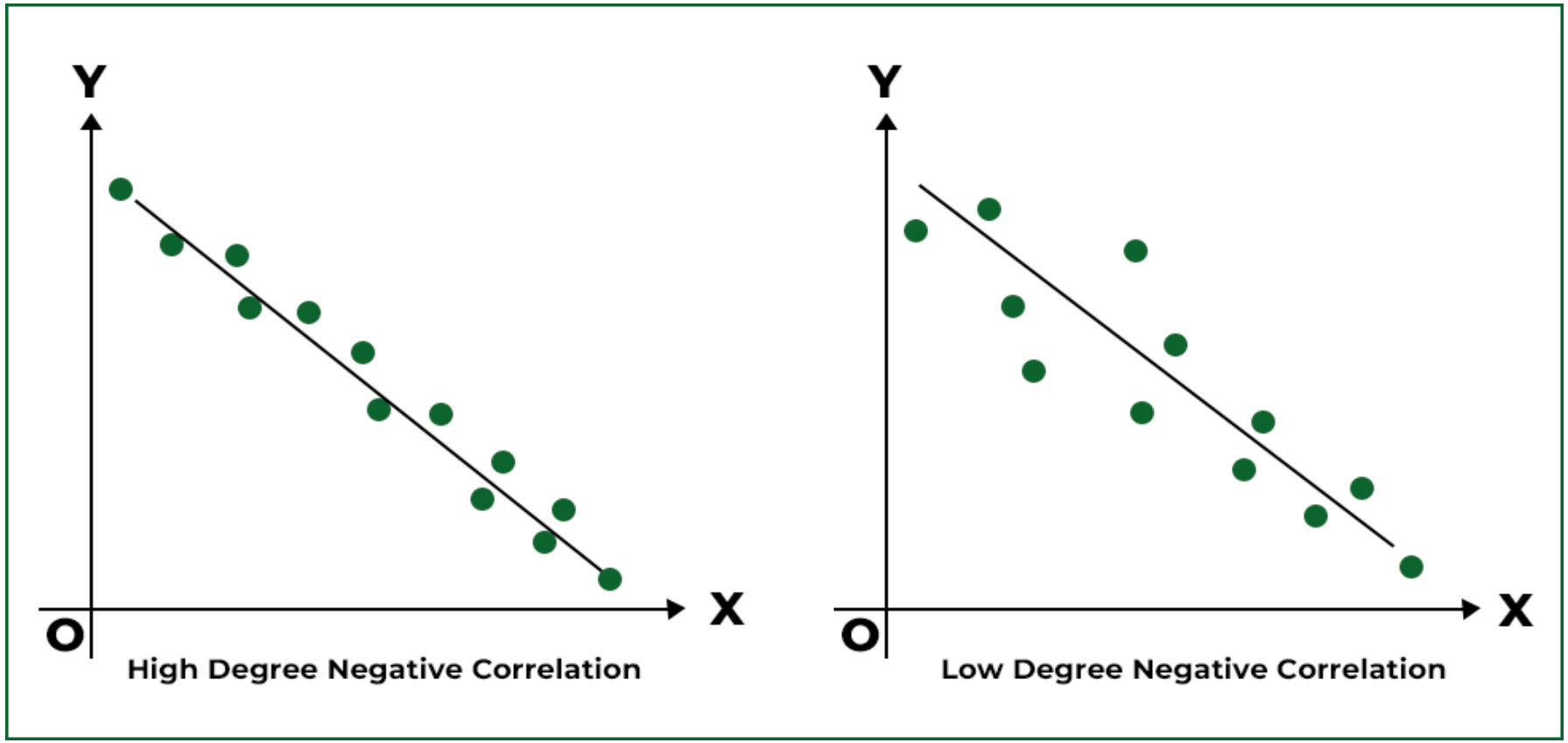
3. Positive Correlation

- **When the points of the scatter diagram cluster around a straight line (upward slope from left to right),** then the correlation is said to be positive.



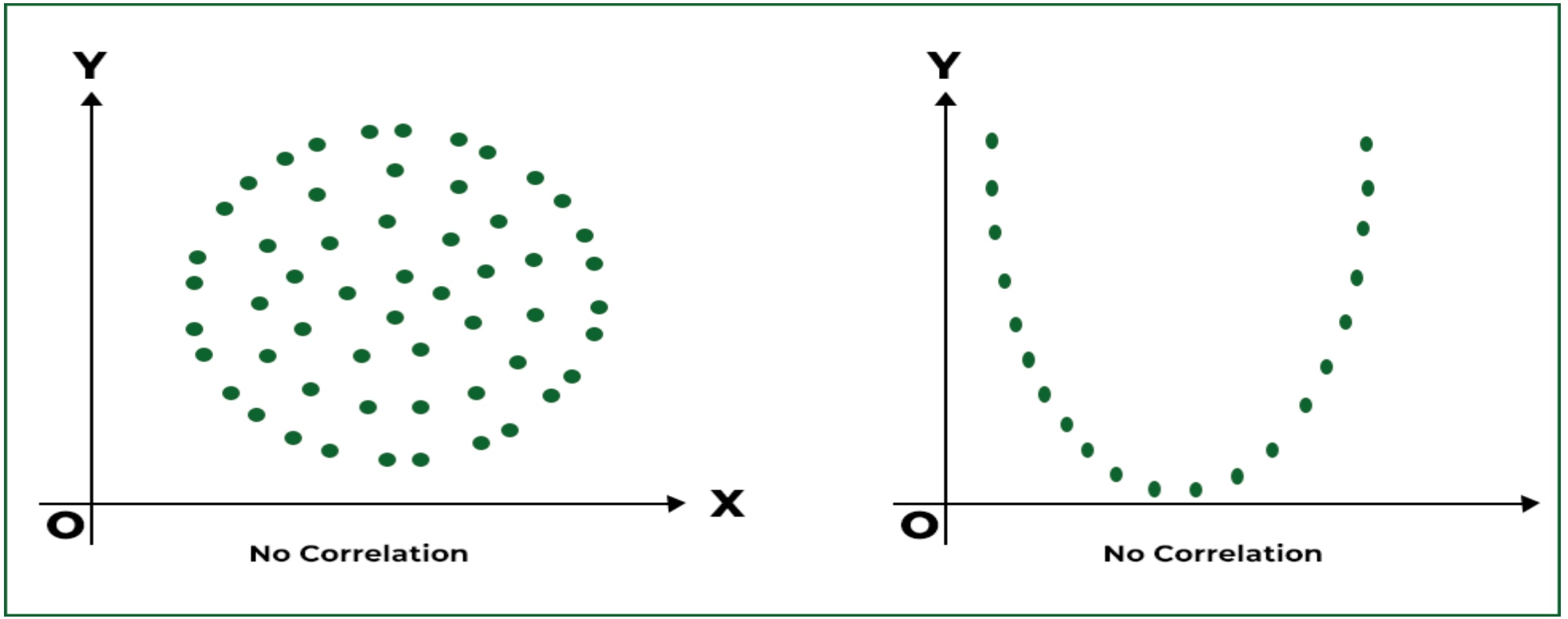
4. Negative Correlation

- **When the points of the scatter diagram cluster around a straight line (downward/negative slope),** then the correlation is said to be negative.
-

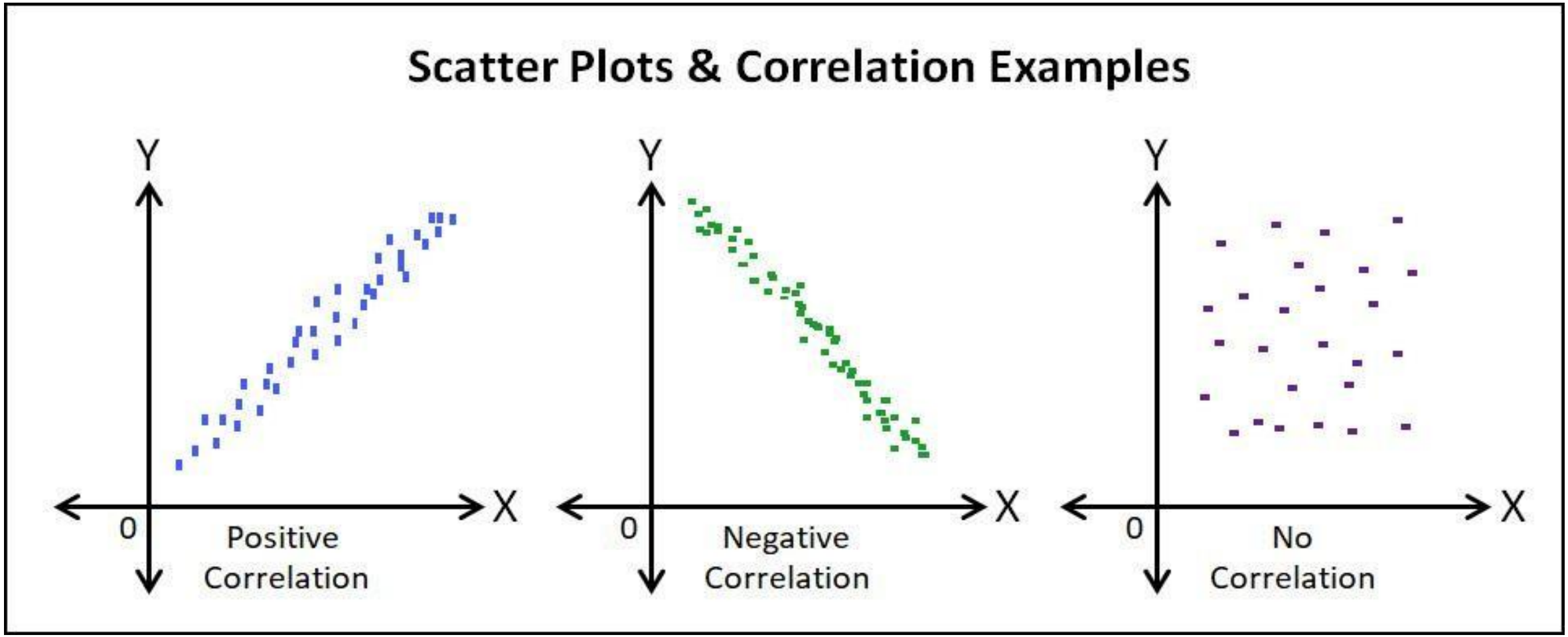


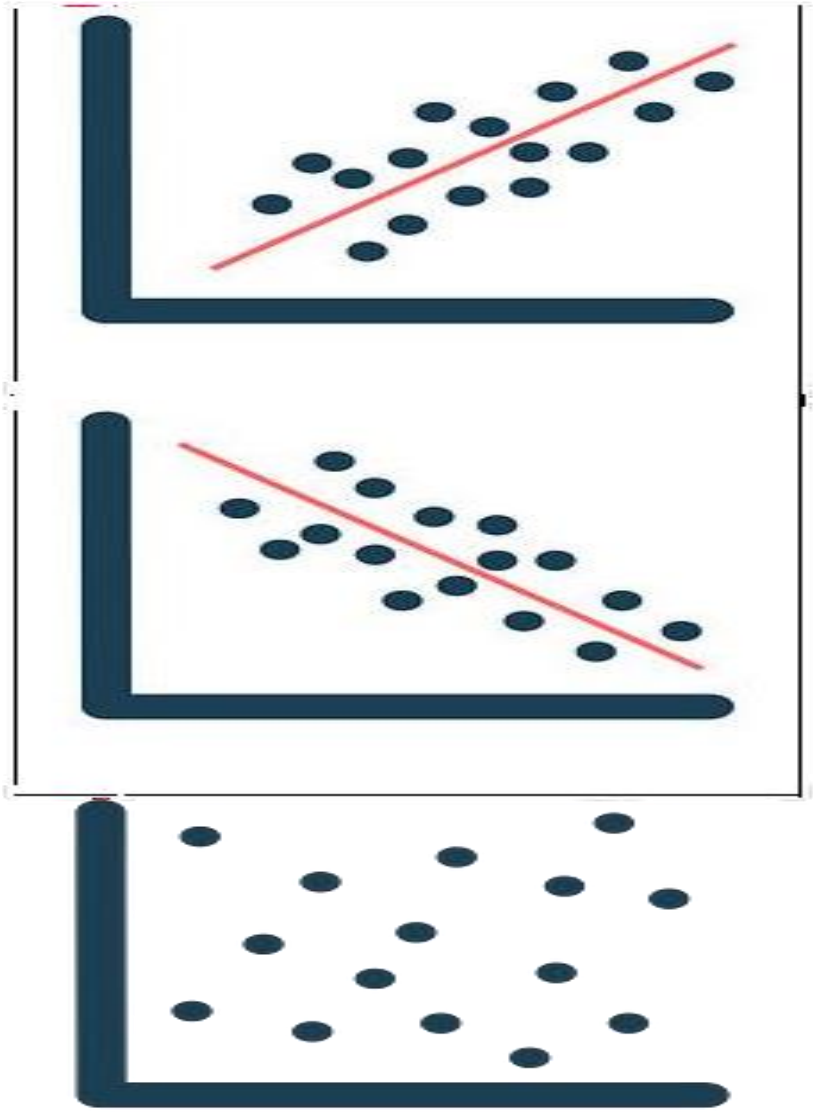
5. No Correlation

- **When the points of the scatter diagram are scattered in a haphazard manner,** then there is zero or no correlation.



Examples:


Scatter Plots & Correlation Examples
Y
X
0
Positive Correlation
Y
X
0
Negative Correlation
Y
X
0
No Correlation



Positive correlation

As one variable increases so does the other variable.

Negative correlation

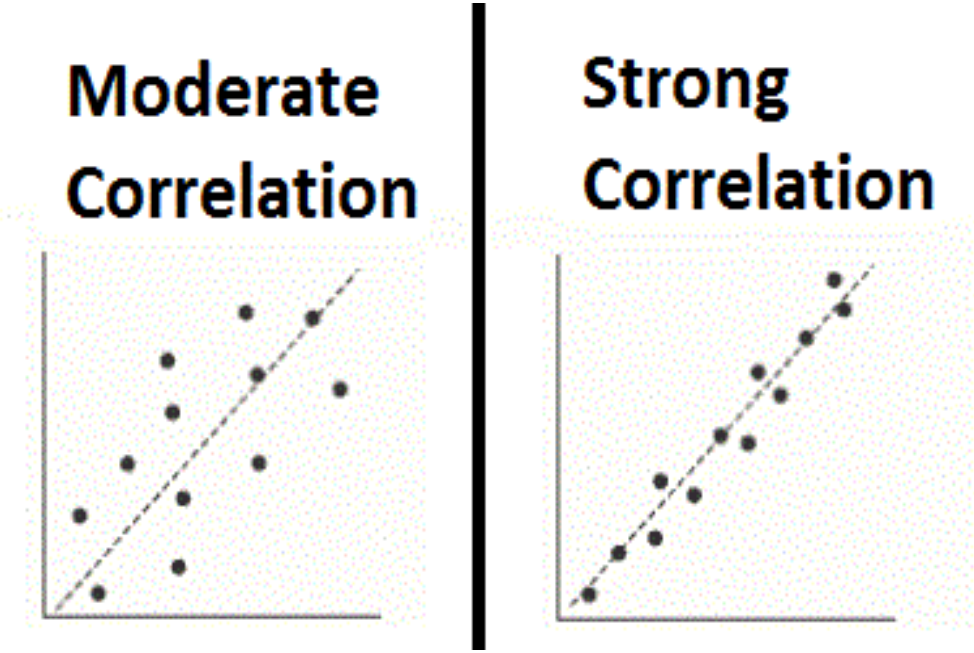
As one variable increases the other variable decreases.

No correlation

There is no relationship between the two variables.

Line of Best Fit & The Strength of Correlation (R)

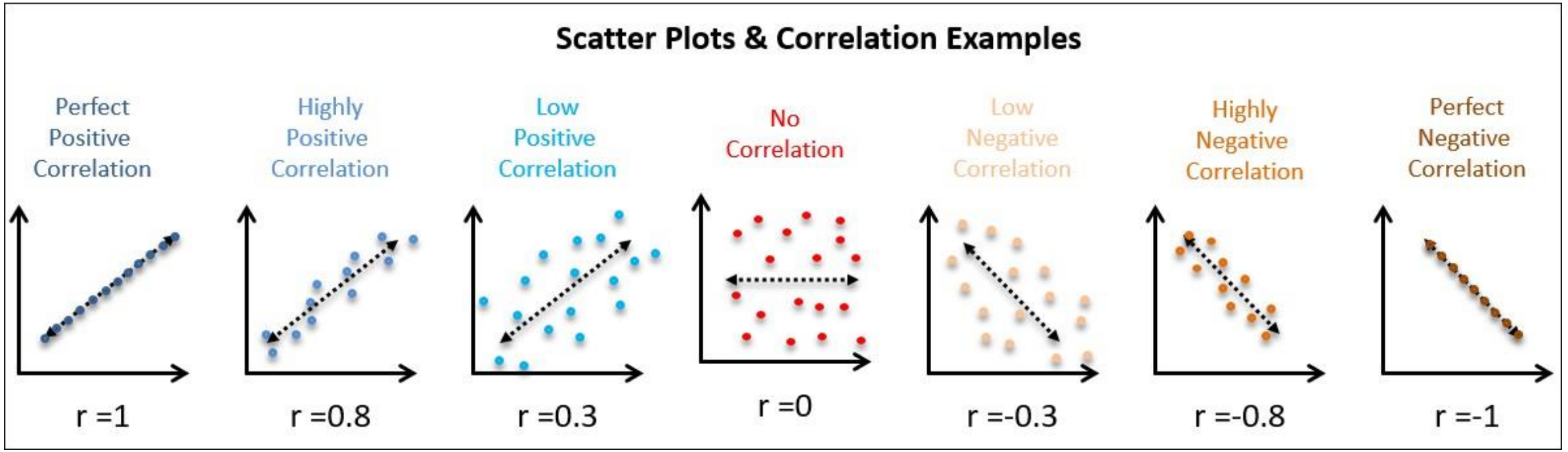
- Both Plots share the same Line of Best Fit, however **the data points on the right hand side of the graph are much closer to the trend line.**
- This indicates that the level of correlation between **the two variables is much stronger than the data on the left.**
- A Low Correlation can also be interpreted to mean that there are likely other variables or factors that contribute to “Observed” or Dependent Variable.



Pearson's Linear Correlation Coefficient (R/r)

- The R Value, which is also called Pearson's Correlation Coefficient is a measure of the strength and direction of the *linear* relationship between two variables.

- R Values range from +1 to -1, with +1 being a perfect positive correlation, -1 being a perfectly negative correlation and 0 being absolutely no measurable correlation.



Pearson Correlation Coefficient

- R = The Correlation Coefficient
- R^2 = The Coefficient of Determination
- *So What is R^2 and what does it represent?*
- R^2 represents the % of the total variation in *y*, that can be explained by the linear relationship between *x* & *y*.

- Let's say you measure r and it turns out to be 0.866. Then you calculate R^2 = 0.75 by just squaring your R value.
- **You can then interpret your data to mean that 75% of the total variation in *y* can be explained by the linear relationship between *x* & *y*.**

THE COEFFICIENT OF DETERMINATION | R SQUARE

- The coefficient of Determination is the direct indicator of how good our model is in terms of performance whether it is accuracy, Precision or Recall.

- **The Coefficient of Determination is the measure of the variance in response variable 'y' that can be predicted using predictor variable 'x'**. It is the most common way to measure the strength of the model.

- The coefficient of determination (R^{Square}) for a linear regression model with one independent variable can be calculated as below:

$$R\ Square = \{ (1 / N) * \Sigma [(x__i - xbar) * (y__i - ybar)] / (\sigma_x * \sigma_y) \}^{\wedge 2}$$

Cont..

where N is the number of observations used to fit the model

- Σ is the summation symbol
- x_i is the x value for observation i
- xbar is the mean of x values
- y_i is the y value for observation i
- ybar is the mean of y value
- σ_x is the standard deviation of x
- σ_y is the standard deviation of y.

There is one more way to calculate R Square value as follows:

Coefficient of Determination (R Square)

$$R^2 = \frac{SSR}{SST}$$

$$SSR = \sum_i (\hat{y}_i - \bar{y})^2$$

$$SST = \sum_i (y_i - \bar{y})^2$$

Where,

- SSR is Sum of Squared Regression also known as variation explained by the model
- SST is Total variation in the data also known as sum of squared total
- y_i is the y value for observation i
- y_bar is the mean of y value
- y_bar_hat is predicted value of y for observation i

R Squared Formula

The coefficient of determination which is represented by R^2 is determined using the following formula:

$$R^2 = 1 - (RSS/TSS)$$

Where,

- R^2 *represents the requrired R Squared value,*
- ***RSS*** *represents the residual sum of squares, and*
- ***TSS*** *represents the total sum of squares.*

If we are not provided with the residual sum of squares (RSS), it can be calculated as follows:

$$\text{RSS} = \Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Where,

- y_i *is the* i^{th} *observation, and*
- $\hat{y}_i$ *is the estimated value of* y_i.

The coefficient of determination can also be calculated using another formula which is given by:

$$R^2 = r^2$$

Where **r** represents the correlation coefficient and is calculated using the following formula:

$$\mathbf{r} = \frac{\mathbf{n\Sigma(xy) - \Sigma x \Sigma y}}{\sqrt{[\mathbf{n\Sigma x^2 - (\Sigma x)^2}][\mathbf{n\Sigma y^2 - (\Sigma y)^2}]}}$$

Where,

- ***n*** *is the total observations,*
- ***x*** *is the first variable, and*
- ***y*** *is the second variable.*

Example: Calculating R^2 using the correlation coefficient

You are studying the relationship between heart rate and age in children, and you find that the two variables have a negative Pearson correlation:

$$r = -0.28$$

This value can be used to calculate the coefficient of determination (R^2) using Formula 1:

$$R^2 = (r)^2$$

$$R^2 = (-0.28)^2$$

$$R^2 = 0.08$$

Cont..

- **Pearson Correlation Coefficient (PCC)** is used for measuring the strength and direction of a linear relationship between two variables.

- It is important in fields like data science, finance, healthcare, and social sciences, where understanding relationships between different factors is important.

- Quantifying the degree to which two variables are related helps analysts, researchers, and decision-makers to find patterns that make predictions and inform decisions.
- **Value of PCC(denoted by r) ranges from -1 to 1:**

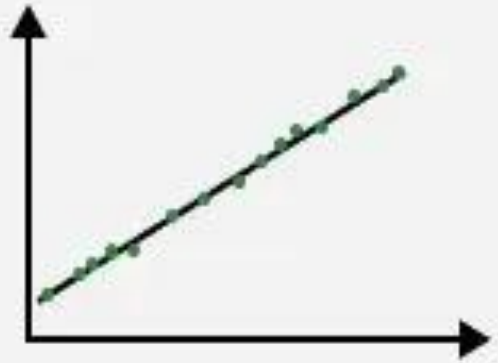
Cont..

- 1 shows a perfect positive correlation where both variables increase or decrease together at a constant rate.

- -1 shows a perfect negative correlation where one variable increases as the other decreases proportionally.

- 0 shows no linear relationship means changes in one variable do not predict changes in the other.

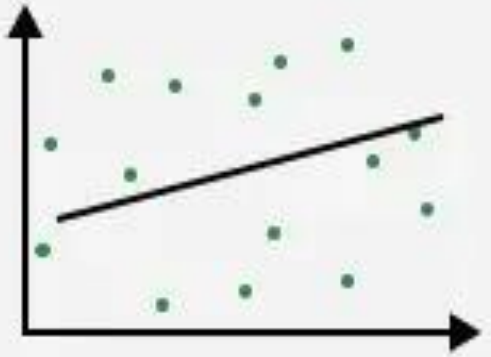
Pearson Correlation Coefficient



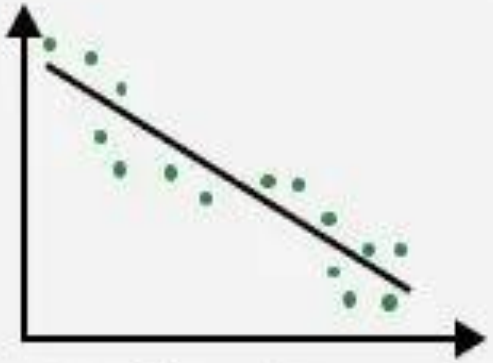
1. Strong Positive Correlation



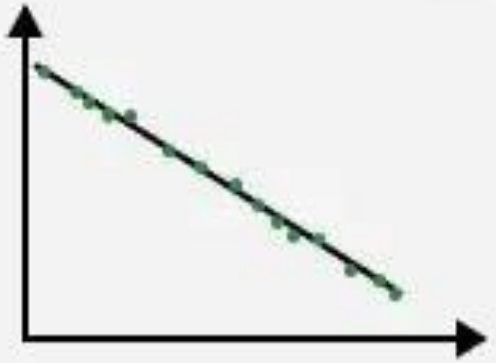
2. Medium Positive Correlation



3. Weak / No Correlation



4. Medium Negative Correlation



5. Strong Negative Correlation

Pearson's Correlation Coefficient Formula and Interpretation

The formula for Pearson's correlation coefficient is shown below:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Where:

- n = number of data points
- x and y = the two variables being compared
- $\sum$ denotes summation

This formula calculates the covariance of the variables and normalizes it by their standard deviations. It provides a scale-independent measure means it is unaffected by the unit of measurement.

Pearson Correlation Coefficient Table

Pearson Correlation Coefficient Table

Pearson Correlation Coefficient (r) Range	Type of Correlation	Interpretation
$0 < r \leq 1$	Positive	Both variables increase or decrease together.
$r = 0$	None	No relationship exists between the changes in both variables.
$-1 \leq r < 0$	Negative	As one variable increases, the other decreases.

Example:

- r = 0.85 suggests a strong positive correlation such as more study time leading to better test scores.
- r = -0.75 shows a strong negative correlation like the inverse relationship between outdoor temperature and heating costs.

Pearson correlation coefficient (*r*) value	Strength	Direction
Greater than .5	Strong	Positive
Between .3 and .5	Moderate	Positive
Between 0 and .3	Weak	Positive
0	None	None
Between 0 and –.3	Weak	Negative
Between –.3 and –.5	Moderate	Negative
Less than –.5	Strong	Negative

Class Test-1,
Marks:02

Question1:
Last Digit of Roll No: 0-2-4-6-8

- Calculate the correlation coefficient for the following data by the help of Pearson's correlation coefficient formula:
- **X = 21, 31, 25, 40, 47, 38 and Y = 70, 55, 60, 78, 66, 80**

Question1:
Last Digit of Roll No: 1-3-5-7-9

Question: Marks obtained by 5 students in algebra and trigonometry as given below:

Algebra	15	16	12	10	8
Trigonometry	18	11	10	20	17

Calculate the Pearson correlation coefficient.

Solution

Solution:

Given variables are, X = 21, 31, 25, 40, 47, 38, and Y = 70, 55, 60, 78, 66, 80

To find the correlation coefficient of the following variables Firstly a table is to be constructed as follows, to get the values required in the formula also add all the values in the columns to get the values used in the formula.

X
21
31
25
40
47
38
$\sum x = 202$

$\sum xy = 13937$, $\sum x = 202$, $\sum y = 409$, $\sum x^2 = 7280$, $\sum y^2 = 28265$

Put n =6 all the values in the Pearson's correlation coefficient formula:-

$$R = n(\sum xy) - (\sum x)(\sum y) / \sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2}$$

$R = 6(13937) - (202)(409) / \sqrt{[6(7280) - (202)^2][6(28265) - (409)^2]}$

$R = 1004 / \sqrt{[2876][2909]}$

$R = 1004 / 2892.452938$

$R = -0.3471$

The correlation coefficient is -0.3471

Solution:

Construct the following table:

x	y	x^2	y^2	xy
15	18	225	324	270
16	11	256	121	176
12	10	144	100	120
10	20	100	400	200
8	17	64	289	136
$\sum x$ = 61	$\sum y$ = 76	$\sum x^2$ = 789	$\sum y^2$ = 1234	$\sum xy$ = 902

Formula for Pearson correlation coefficient is given by:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

$$r = \frac{5 \times 902 - 61 \times 76}{\sqrt{[5 \times 789 - (61)^2][5 \times 1234 - (76)^2]}}$$

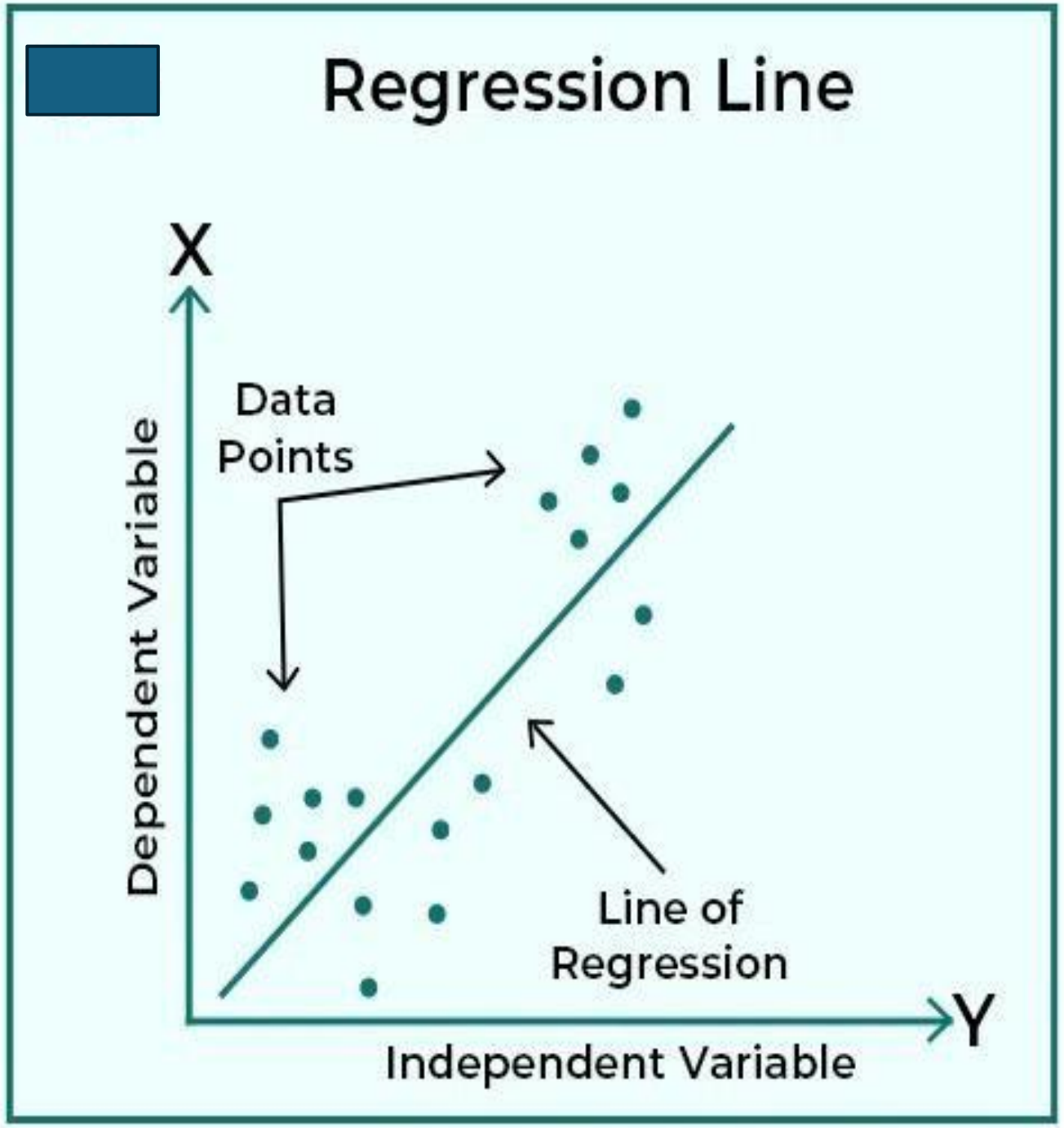
$$r = -0.424$$

Regression Analysis

- **Regression analysis** is a way to find trends in data set.
- Regression analysis will provide you with an equation for a graph so that you can make predictions about your data. This process is called a regression analysis.
- A regression line is a straight line that describes how a response variable y changes as an explanatory variable x changes
- For example, if you've been putting on weight over the last few years, it can predict how much you'll weigh in ten years time if you continue to put on weight at the same rate.
- It will also give a slew of statistics (including a p-value and a correlation coefficient) to tell you how accurate your model is in Machine Learning.

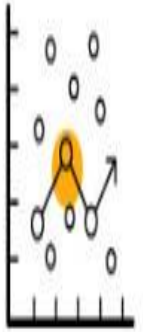
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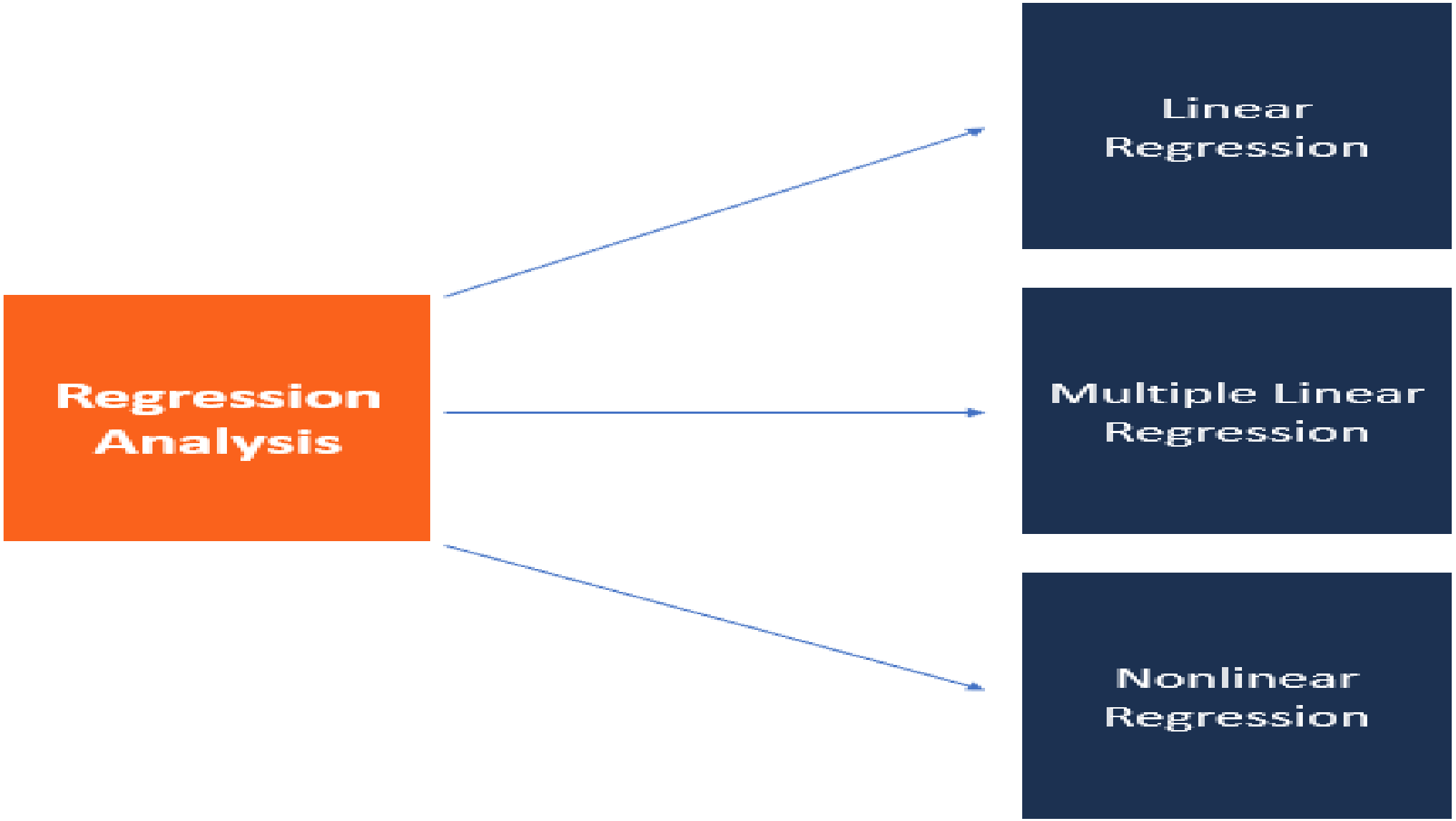
- **Regression analysis is one of the most widely used statistical techniques, particularly in fields such as economics, finance, healthcare, and engineering.**
- **It allows us to understand the relationships between different variables and predict outcomes based on input data.**
- Regression analysis is essentially a way to estimate the relationships among variables.
- At its core, the goal is **to predict the dependent variable (also known as the response or outcome variable)** based **on one or more independent variables (also known as predictors or explanatory variables).**



What is Regression analysis?

A set of statistical processes for **estimating the relationships between a dependent variable** ('outcome' or 'response') **and one or more independent variables** ('predictors', 'covariates', 'explanatory variables' or 'features').

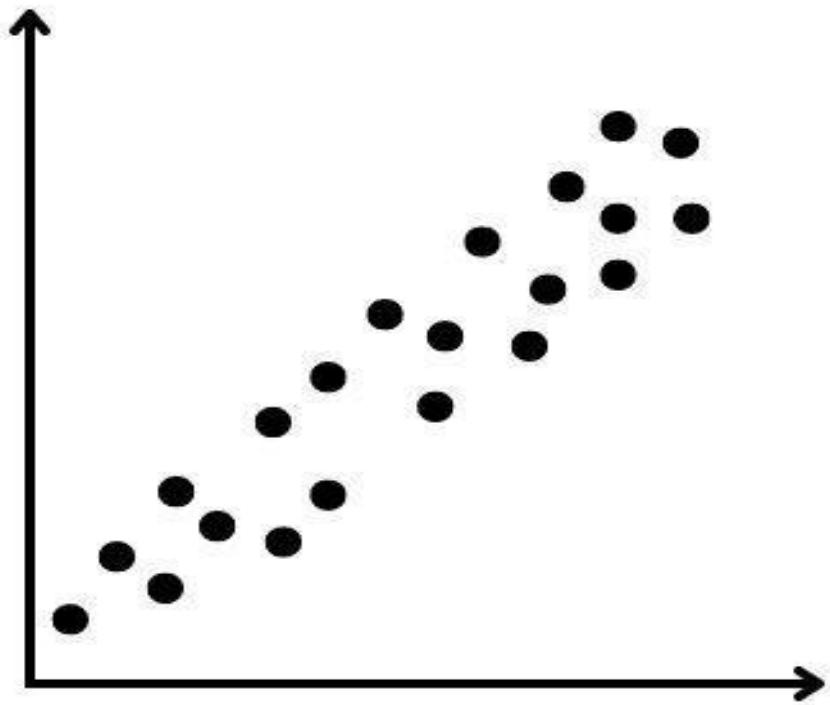



Regression Analysis
Linear Regression
Multiple Linear Regression
Nonlinear Regression

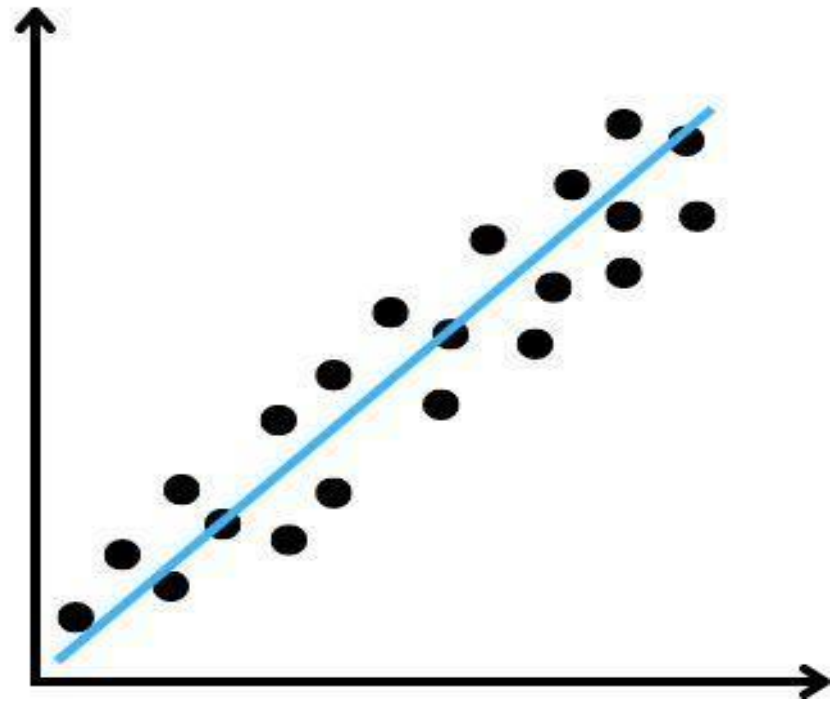
Regression Analysis

aicorr.com

Correlation



Linear Regression



Regression Line

- **Regression Line** is defined as a statistical concept that facilitates and predicts **the relationship between two or more variables**.

- A **regression line is a straight line that reflects the best-fit connection** in a dataset **between independent and dependent variables.**

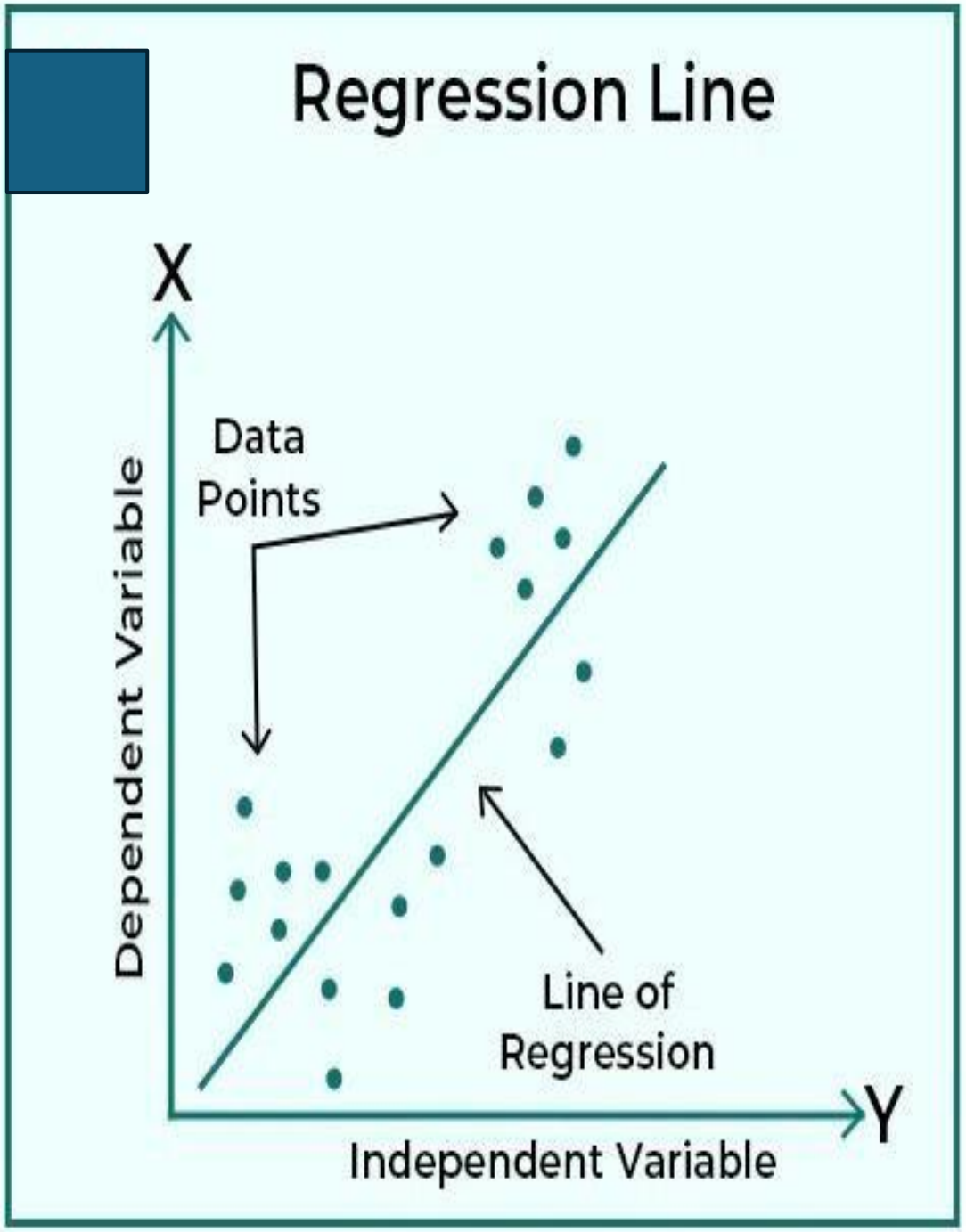
- The **independent variable is generally shown on the X-axis** and **the dependent variable is shown on the Y-axis.**

- The main purpose of developing a regression line is to predict or estimate the value of the dependent variable based on the values of one or more independent variables.

Linear Regression Line Formula:

- The linear regression line formula is y = a + bX (or **y = mx + b**), where **'y' is the dependent variable,**
- **'x' is the independent variable**,
- **'a' (or 'm') is the y-intercept**, and **'b' (or 'b') is the slope of the line.**

- The slope (b) indicates how much 'y' changes for every one-unit change in 'x', and the intercept (a) is the predicted value of 'y' when 'x' is zero.



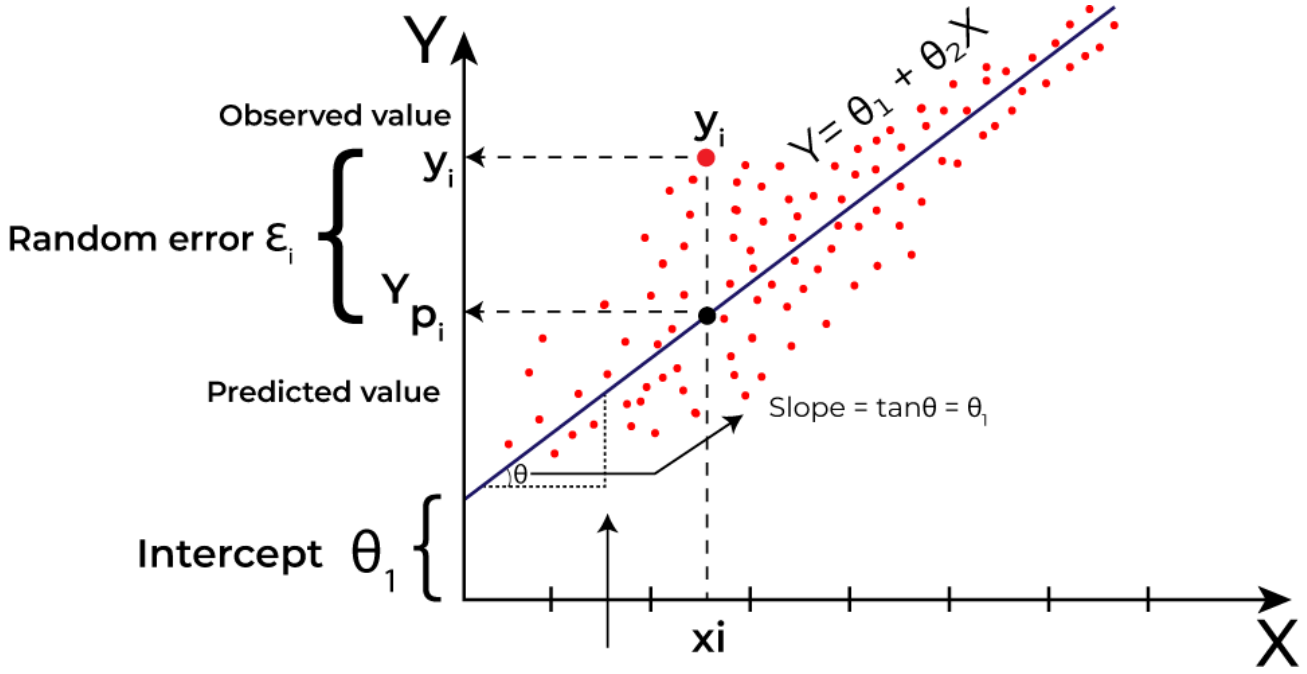
Equation of Regression Line

$$Y = a + bX + \varepsilon$$
$$Y = mX + C$$

- Y is the dependent variable
- X is the independent variable
- a is the y-intercept, which represents the value of Y when X is 0.
- b is the slope, which represents the change in Y for a unit change in X
- ε is residual error.

Goal of the Best-Fit Line

- The goal of linear regression is to find a straight line that minimizes the error (the difference) between the observed data points and the predicted values.
- This line helps us predict the dependent variable for new, unseen data.



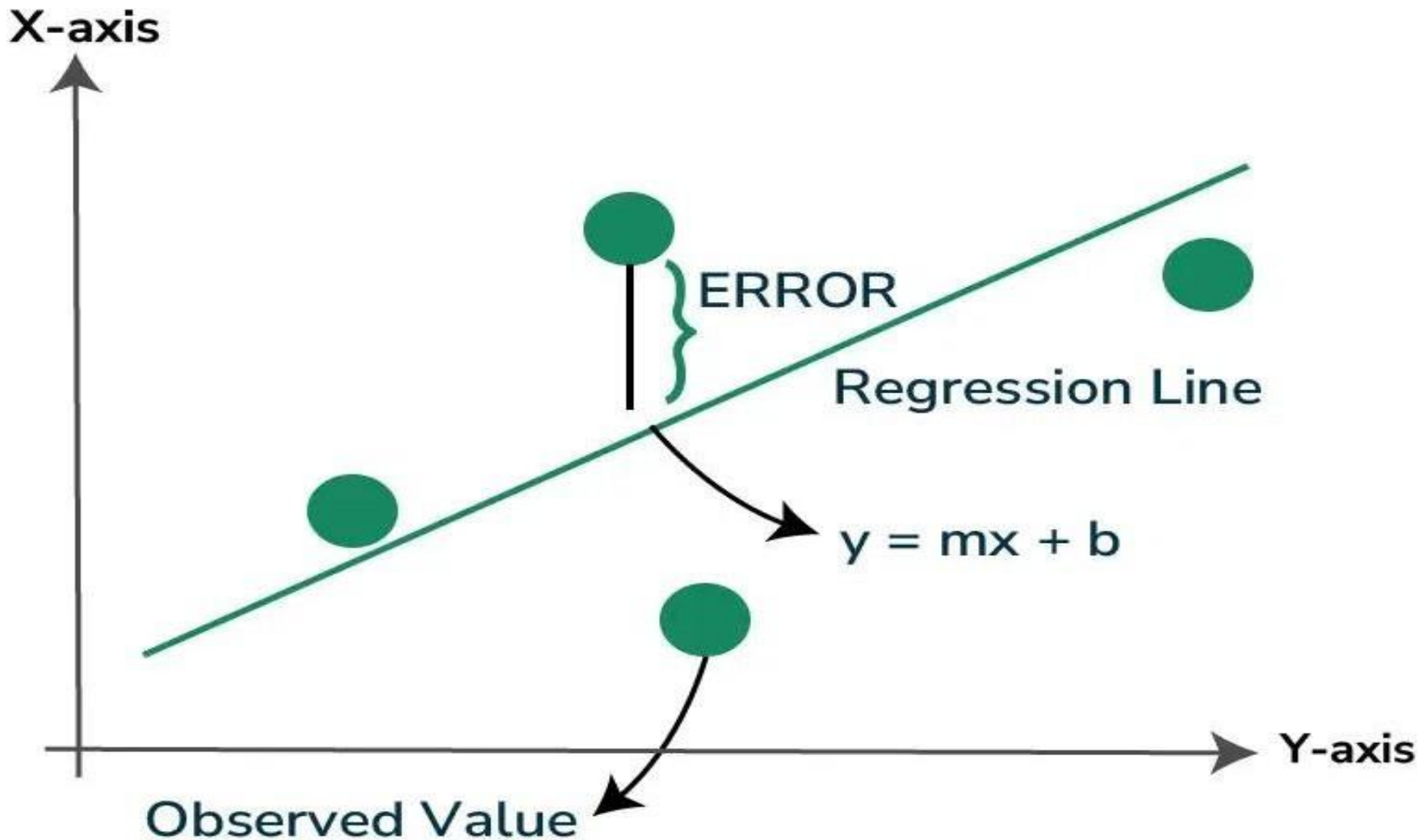
Least Square Method-LSM

- **The Least Square method** is a popular mathematical approach **used in data fitting, regression analysis, and predictive modeling.**

- **It helps find the best-fit line** or curve that minimizes the sum of squared differences between the observed data points and the predicted values.

- **This technique is widely used in statistics, machine learning, and engineering applications.**

Least Square Method



Least Square Method Formula

- **The Least Square Method formula finds the best-fitting line through a set of data points.**
- For a simple linear regression, which is a line of the form $y=mx+c$, where y is the dependent variable,
- x is the independent variable,
- a is the slope of the line, and b is the y-intercept, the formulas to calculate the slope (m) and intercept (c) of the line are derived from the following equations:

- ***Slope (m) Formula:*** $m = n(\sum xy) - (\sum x)(\sum y) \,/\, n(\sum x^2) - (\sum x)^2$

- ***Intercept (c) Formula:*** $c = (\sum y) - a(\sum x) \,/\, n$

Slope and Intercept:

- **Slope (m):** The slope of the best-fit line indicates how much the dependent variable (y) changes with each unit change in the independent variable (x).
- For example if the slope is 5, it means that for every 1-unit increase in x, the value of y increases by 5 units.

- **Intercept (b):** The intercept represents the predicted value of y when x = 0. It’s the point where the line crosses the y-axis.

Cont..

- Where:
- n is the number of data points,
- $\sum xy$ is the sum of the product of each pair of x and y values,
- $\sum x$ is the sum of all x values,
- $\sum y$ is the sum of all y values,
- $\sum x^2$ is the sum of the squares of the x values.

Minimizing the Error: The Least Squares Method

- To find the best-fit line, we use a method called **Least Squares**.
- The idea behind this method is to minimize the sum of squared differences between the actual values (data points) and the predicted values from the line.
- These differences are called residuals.

The formula for residuals is:

$$Residual = y_i - \hat{y}_i$$

Where:

- y_i is the actual observed value
- $\hat{y}_i$ is the predicted value from the line for that x_i

The least squares method minimizes the sum of the squared residuals:

$$Sum of squared errors (SSE) = \Sigma(y_i - \hat{y}_i)^2$$

Example

- ***Slope (m) Formula:*** $m = n(\sum xy) - (\sum x)(\sum y) \,/\, n(\sum x^2) - (\sum x)^2$
- ***Intercept (c) Formula:*** $c = (\sum y) - a(\sum x) \,/\, n$
- **Problem 1: Find the line of best fit for the following data points using the Least Square method: (x,y) = (1,3), (2,4), (4,8), (6,10), (8,15).**

Solution:

Here, we have x as the independent variable and y as the dependent variable. First, we calculate the means of x and y values denoted by X and Y respectively.

$X = (1+2+4+6+8)/5 = 4.2$
$Y = (3+4+8+10+15)/5 = 8$

	x_i	y_i	$X - x_i$	$Y - y_i$	$(X-x_i)*(Y-y_i)$	$(X - x_i)^2$
	1	3	3.2	5	16	10.24
	2	4	2.2	4	8.8	4.84
	4	8	0.2	0	0	0.04
	6	10	-1.8	-2	3.6	3.24
	8	15	-3.8	-7	26.6	14.44
Sum (Σ)			0	0	55	32.8

The slope of the line of best fit can be calculated from the formula as follows:
$m = (\Sigma (X - x_i)*(Y - y_i)) / \Sigma (X - x_i)^2$
$m = 55/32.8 = 1.68$ *(rounded upto 2 decimal places)*

Now, the intercept will be calculated from the formula as follows:
$c = Y - mX$
$c = 8 - 1.68*4.2 = 0.94$

Thus, the equation of the line of best fit becomes, $y = 1.68x + 0.94$.

Why Linear Regression is Important?

- **Simplicity and interpretability:** It's a relatively easy concept to understand and apply. The resulting simple linear regression model is a straightforward equation that shows how one variable affects another.
- **Prediction:** Linear regression allows you to predict future values based on existing data. For instance, you can use it to predict sales based on marketing spend or house prices based on square footage.
- **Foundation for other techniques:** It serves as a building block for many other data science and machine learning methods. Even complex algorithms often rely on linear regression as a starting point or for comparison purposes.
- **Widespread applicability:** Linear regression can be used in various fields, from finance and economics to science and social sciences. It's a versatile tool for uncovering relationships between variables in many real-world scenarios.

Class Test-3

- **Problem 2: Find the line of best fit for the following data of heights and weights of students of a school using the Least Squares method:**
- **Height (in centimeters): [160, 162, 164, 166, 168]**
- **Weight (in kilograms): [52, 55, 57, 60, 61]**

Solution:

Here, we denote Height as x (independent variable) and Weight as y (dependent variable). Now, we calculate the means of x and y values denoted by X and Y respectively.

$X = (160 + 162 + 164 + 166 + 168) / 5 = 164$
$Y = (52 + 55 + 57 + 60 + 61) / 5 = 57$

	x_i	y_i	$X - x_i$	$Y - y_i$	$(X-x_i)*(Y-y_i)$	$(X - x_i)^2$
	160	52	4	5	20	16
	162	55	2	2	4	4
	164	57	0	0	0	0
	166	60	-2	-3	6	4
	168	61	-4	-4	16	16
Sum (Σ)			0	0	46	40

Now, the slope of the line of best fit can be calculated from the formula as follows:
$m = (\Sigma (X - x_i)\times(Y - y_i)) / \Sigma(X - x_i)^2$
$m = 46/40 = 1.15$

Now, the intercept will be calculated from the formula as follows:
$c = Y - mX$
$c = 57 - 1.15*164 = -131.6$

Thus, the equation of the line of best fit becomes, $y = 1.15x - 131.6$

Evaluation Metrics for Linear Regression

The most used metrics are:

1. Coefficient of Determination or R-squared (R2)

2. Root Mean Squared Error (RSME) and Residual Standard Error (RSE)

Coefficient of Determination or R-Squared (R2)

- R-squared is a number that explains the amount of variation that is explained/captured by the developed model.
- It always ranges between 0 & 1 .
- Overall, the higher the value of R-squared, the better the model fits the data.

Mathematically it can be represented as,

$$R^2 = 1 - (RSS/TSS)$$

- **Residual sum of Squares (RSS)** is defined as the sum of squares of the residual for each data point in the plot/data. It is the measure of the difference between the expected and the actual observed output.

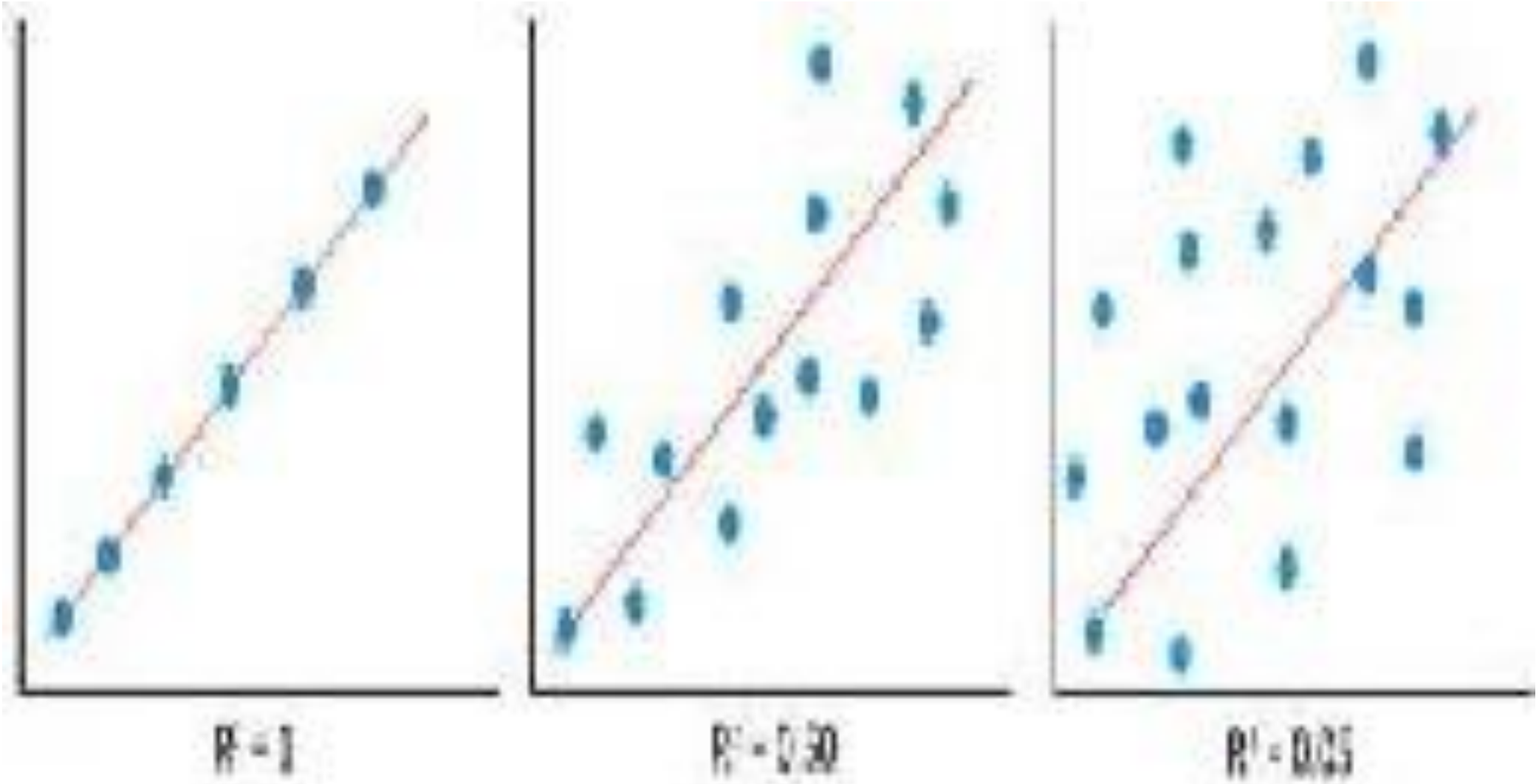
$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$$

- **Total Sum of Squares (TSS)** is defined as the sum of errors of the data points from the mean of the response variable. Mathematically TSS is,
-

$$TSS = \sum (y_i - \bar{y}_i)^2$$

where y hat is the mean of the sample data points.

Cont..



Root Mean Squared Error

- The Root Mean Squared Error is the square root of the variance of the residuals.
- It specifies the absolute fit of the model to the data i.e. how close the observed data points are to the predicted values.

$$RMSE = \sqrt{\frac{RSS}{n}} = \sqrt{\sum_{i=1}^{n} \left(y_i^{Actual} - y_i^{Predicted} \right)^2 / n}$$

Importance of Regression Line

- **1. Error Analysis:** Regression analysis provides a way to assess the goodness of fit of a model. By examining residuals (the differences between observed and predicted values), one can identify patterns and trends in the errors, which further helps in the improvement of models.
- **2. Variable Selection:** Regression analysis helps in the selection of relevant variables. While having a large dataset with many potential predictors, regression analysis can provide guidance in identifying which variables have a significant impact on the outcome, enabling more efficient and parsimonious models.
- **3. Quality Control:** In manufacturing and quality control processes, regression analysis can be used to monitor and control product quality. By understanding the relationship between input variables and product quality, manufacturers can make adjustments to maintain or improve quality standards.
- **4. Forecasting:** Regression models can be used for time series analysis and forecasting. This is valuable in industries like retail, where understanding historical sales data can help in predicting future sales, optimising inventory levels, and planning for seasonal demand.
- **5. Risk Assessment:** In finance and insurance, regression analysis is crucial for assessing and managing risk. It can help identify factors affecting investment returns, loan defaults, or insurance claims, aiding in risk assessment and pricing.
- **6. Policy Evaluation:** In social sciences and public policy, regression analysis is employed to evaluate the impact of policy changes or interventions. By examining the relationship between policy variables and relevant outcomes, researchers can assess the effectiveness of different policies and inform decision-makers.

Types of Regression Lines

- **Linear Regression Line:** Linear regression line is utilised when there is a linear relationship between the reliant variable and at least one free variables. The condition of a straightforward linear relapse line is typically; $Y = a + bX + \varepsilon$, where Y is the reliant variable, X is the free variable, a is the y-intercept, b is the slope, and ε is error.
- **Logistic Regression Line**: **Logistic regression is used when the dependent variable is discrete. It models the probability of a binary outcome using a logistic function**. The equation is typically expressed as the log-odds of the probability.
- **Polynomial Regression Line:** Polynomial regression is used when the relationship between the dependent and independent variables is best represented by a polynomial equation. The equation is $Y = aX^2 + bX + c$, or even higher-order polynomial equations.
- **Non-Linear Regression Line:** For situations where the relationships between variables is not linear, non-linear regression lines must be used to defined the relationship.
- **Multiple Regression Line:** This involves multiple independant variables to predict a dependant variable. It is an extension of linear regression.
- **Time Series Regression Line:** This approach is used to deal with time-series data, and models how the dependent variable changes over time.

Applications of Regression Line

- **1.Economics:** Regression analysis is used in economics to anticipate economic trends, evaluate consumer behaviour, and **identify factors influencing economic variables such as GDP, inflation, and unemployment.**

- **2. Finance:** Regression analysis is used in portfolio management **to estimate risk and return of investments.** It aids in the prediction of stock prices, bond yields, and other financial measures.

- **3. Medicine:** Regression analysis is used in the medical field to investigate the link between variables such as **dosage and patient response, as well as to predict patient outcomes based on a variety of criteria.**

- **4. Marketing:** Regression analysis is used by marketers **to understand the impact of advertising, pricing, and other marketing** initiatives on sales and customer behavior.

- **5. Environmental Science:** Regression analysis is used by researchers to model the link between **environmental parameters (such as temperature and pollution levels) and their impact on ecosystems.**

The Standard Error of Estimate

When a $\hat{y}$-value is predicted from an x-value, the prediction is a point estimate.

An interval can also be constructed.

The **standard error of estimate s_e** is the standard deviation of the observed y_i-values about the predicted $\hat{y}$-value for a given x_i-value. It is given by

$$s_e = \sqrt{\frac{\Sigma(y_i - \hat{y}_i)^2}{n-2}}$$

where n is the number of ordered pairs in the data set.

The closer the observed y-values are to the predicted y-values, the smaller the standard error of estimate will be.

Note: two parameters (the slope and the intercept) of the regression line are estimated from the data

Standard Error of Estimate

- **Standard Error** is the measure of the variability of a sample statistic used to estimate the variability of a population.

- Standard Error is important in dealing with sample statistics, such as sample mean, sample proportion, etc.
- Sample Error Formula is used to determine the accuracy of a sample that reflects a population.

- The standard error formula is the discrepancy between the sample mean and the population mean.

Standard Error(SE) Formula

- The SE formula is used to determine the reliability of a sampling that represents a population.
- The sample mean that differs from the provided population and is expressed as:
- $SE = S/\sqrt{(n)}$

where,

- ***S*** *is Standard Deviation of Data*
- ***n*** *is Number of Observations*

Standard Error of Mean (SEM)

- Standard Error of Mean(SEM) is also known by the name **Standard Deviation of Mean**, is the standard deviation of the measure of sample mean of the population.

- Standard Error is used when we have to find the population mean from the sample of data and we take two samples to get two different mean in that case we use Standard Error Mean.

- Standard Error Mean formula is the ratio of standard deviation to the root of sample size,
- $SEM = S/\sqrt{(n)}$

where,

- *S is Standard Deviation*
- *n is Number of Observations*

Standard Error of Estimate (SEE)

- Standard Error Estimate is use to find the accuracy of prediction of any event. Its abbreviation is SEE.

- Standard Error Estimate (SEE) is also called the Sum of Sqaures Error. SEE is the square root of average squared deviation.

Standard Error of Estimate(SEE) formula:

- $SEE = \sqrt{[\Sigma(x_i - \mu)/(n - 2)]}$

where,

- x_i *is values of Data*
- μ *is Mean Value of Data*
- n *is Sample Size*

Example 1: Find the standard Error Mean for the sample data: 1, 2, 3, 4, 5.

Mean of Given Data = (1+2+3+4+5)/5

- $\mu = 15/5$
- $\mu = 3$

Standard Deviation $= \sqrt{((1 - 3)^2 + (2 - 3)^2 + (3 - 3)^2 + (4 - 3)^2 + (5 - 3)^2)/(5 - 1)}$

- $\sigma = \sqrt{((4 + 1 + 0 + 1 + 4)/4)}$
- $\sigma = \sqrt{(10/4)}$
- $\sigma = 1.5$

SEM = S/√(n)

- *SE = 1.5/√5*
- *SE = 0.67*

Example 2: Find the standard error for the sample data: 2, 3, 4, 5, 6.

- *Mean of Given Data?*
- *Standard Deviation?*

Solution:

Mean of Given Data = (2+3+4+5+6)/5

- $\mu = 20/5$
- $\mu = 4$

Standard Deviation(σ) = $\sqrt{((2 - 4)^2 + (3 - 4)^2 + (4 - 4)^2 + (5 - 4)^2 + (6 - 4)^2)/(5 - 1)}$

- $\sigma = \sqrt{((4 + 1 + 0 + 1 + 4)/4)}$
- $\sigma = \sqrt{(10/4)}$
- $\sigma = 1.58$
- $SE = 1.58/\sqrt{5}$
- $SE = 0.706$

Example 3: Find the standard error for the sample data: 5, 8, 10, 12.

Solution:

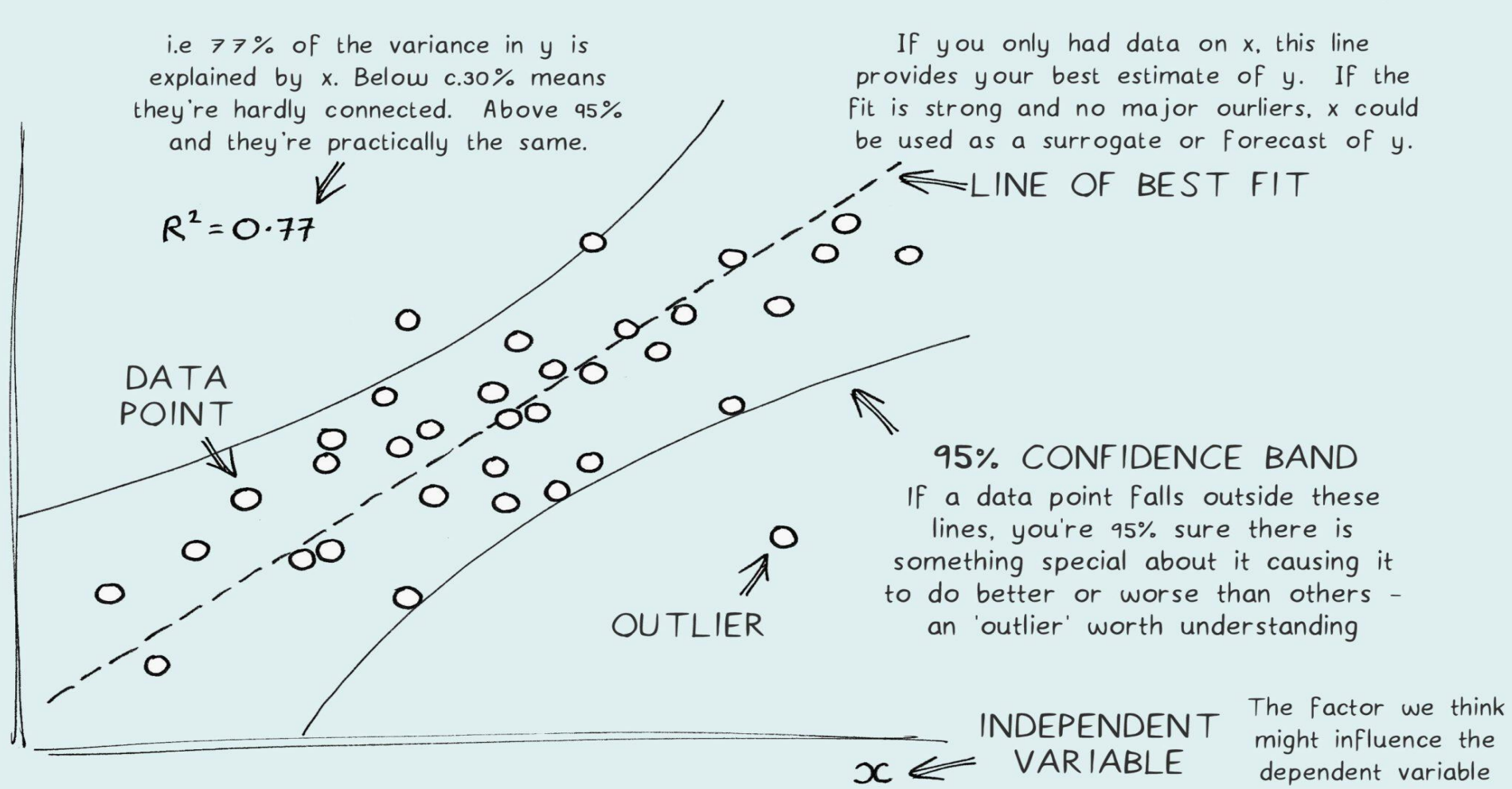
Mean of Given Data(μ) = (5+8+10+12)/4

- *μ = 8.75*

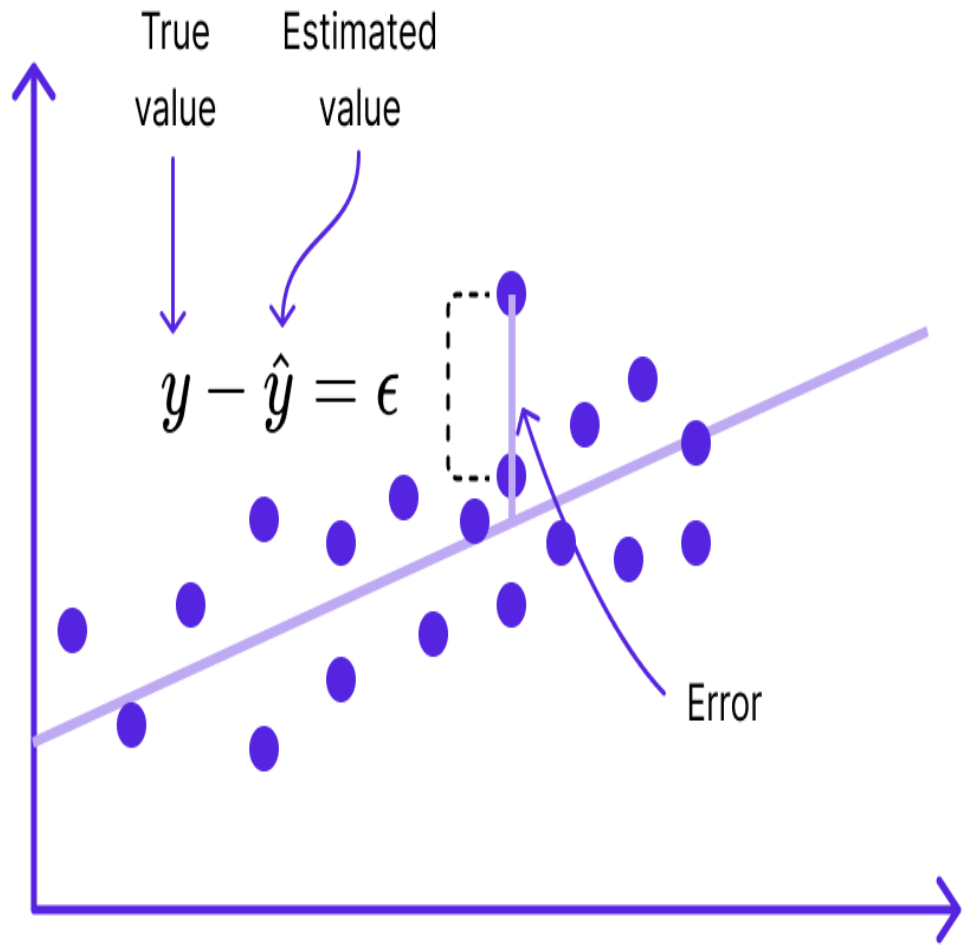
Standard Deviation = $\sqrt{((5 - 8.75)^2 + (8 - 8.75)^2 + (10 - 8.75)^2 + (12 - 8.75)^2)/(4 - 1)}$

- *$\sigma = \sqrt{(26.75/3)}$*
- *σ = 2.98*
- *SE = 2.98/$\sqrt{8.75}$*
- *SE = 1.0074*

LINEAR REGRESSION



Simple Linear Regression Model



Residual / Error:

Actual value - Predicted value

Sum of Residuals / Errors:

Sum(Actual value - Predicted values)

Square of Sum of Residuals / Errors:

$(\text{Sum}(\text{Actual value} - \text{Predicted values}))^2$

Exponent

hours	score
1	65
2	70
3	75
4	80
5	85
6	90

Student score dataset

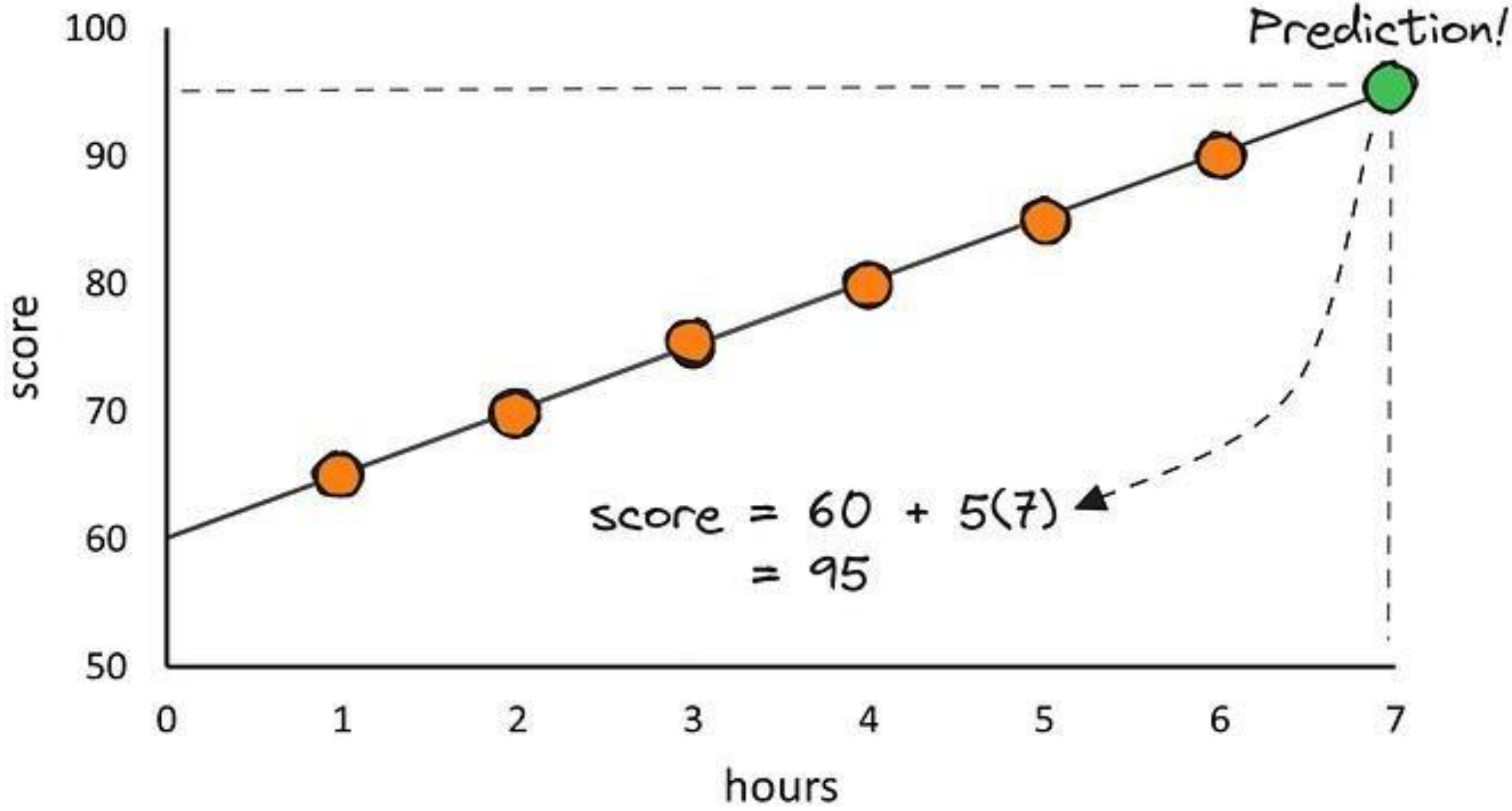
•**y** → the **response/target variable** that we are trying to predict from the explanatory variables. In the given problem, **this is the "score".**

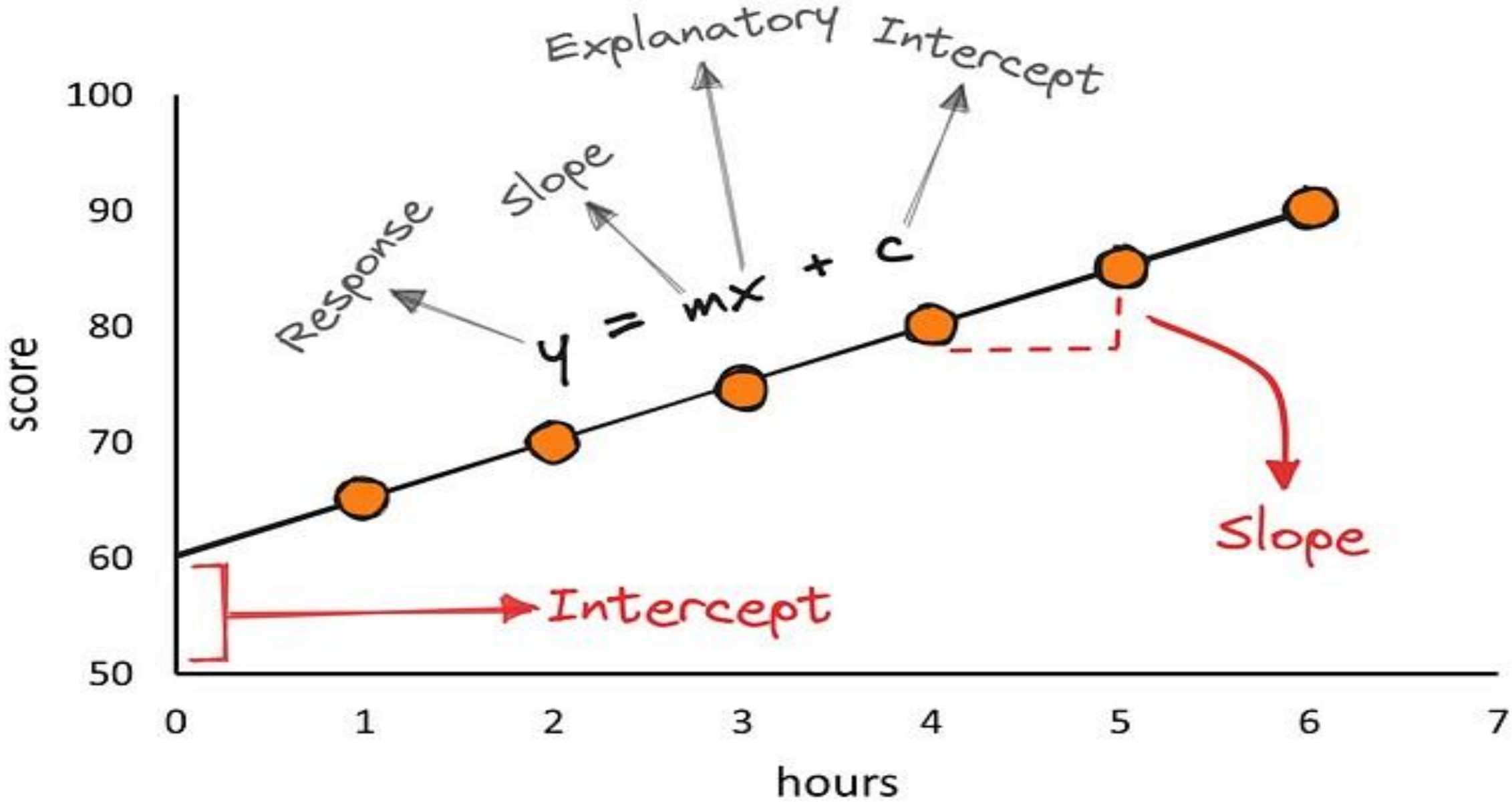
•**x** → the explanatory variable we are using to make a prediction. In the problem above, **this is "hours".**

•**m** → Slope is the steepness of the regression line that tells us **how much we expect the y value to increase when we increase the x value by one unit**. In machine learning, the slope is also known as **weight**.

•**c** → The intercept tells us **where the regression line crosses the vertical y-axis**. In machine learning, the intercept is also known as bias.

score = 60 + 5*(study duration in hours)


Prediction!
100
90
80
70
60
50
score
score = 60 + 5(7)
= 95
0
1
2
3
4
5
6
7
hours


Explanatory
Intercept
Response
Slope
y = mx + c
100
90
80
70
60
50
score
0
1
2
3
4
5
6
7
hours
Slope
Intercept

Equation of the Best-Fit Line

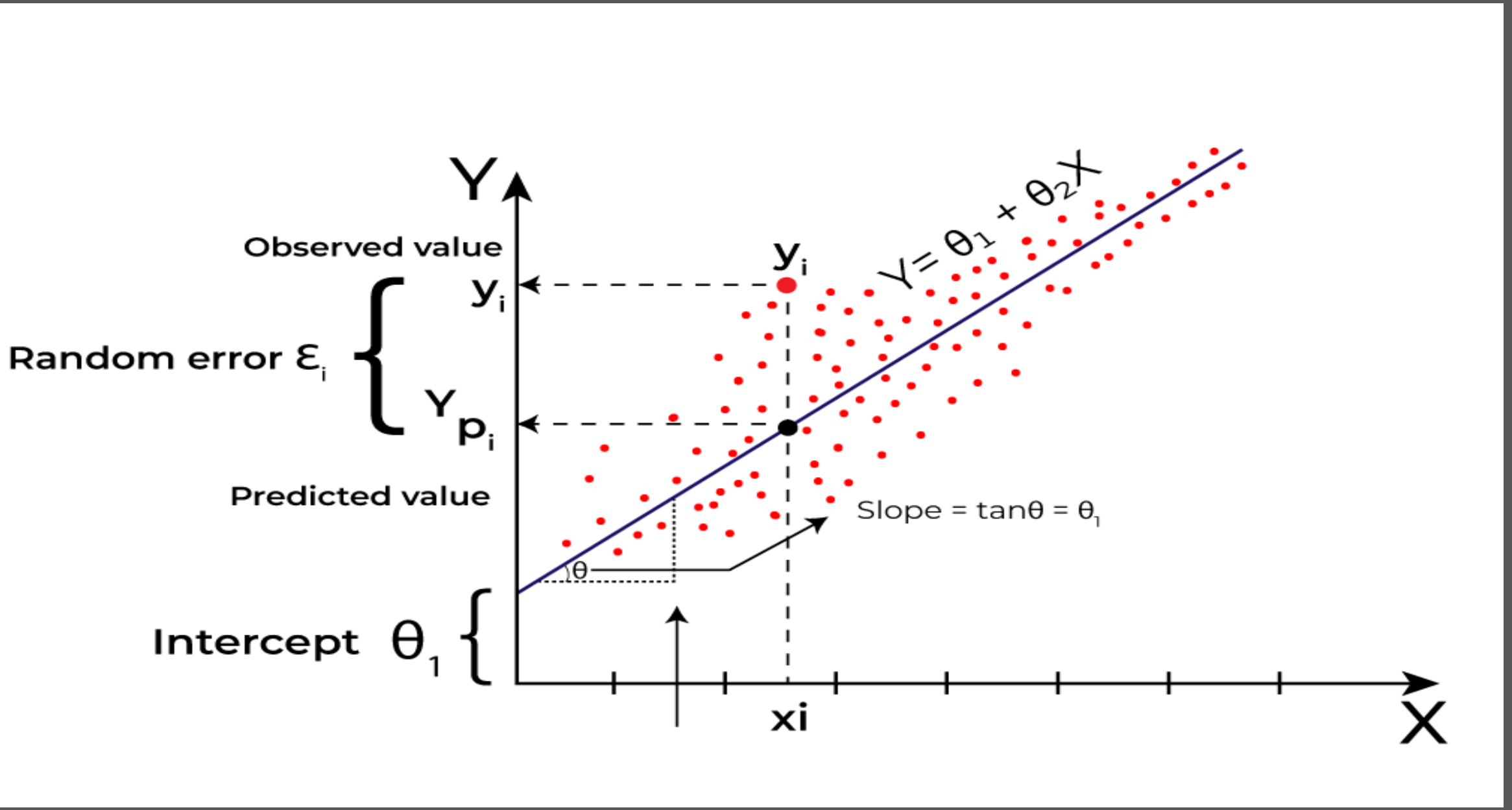
- For simple linear regression (with one independent variable), the best-fit line is represented by the equation:

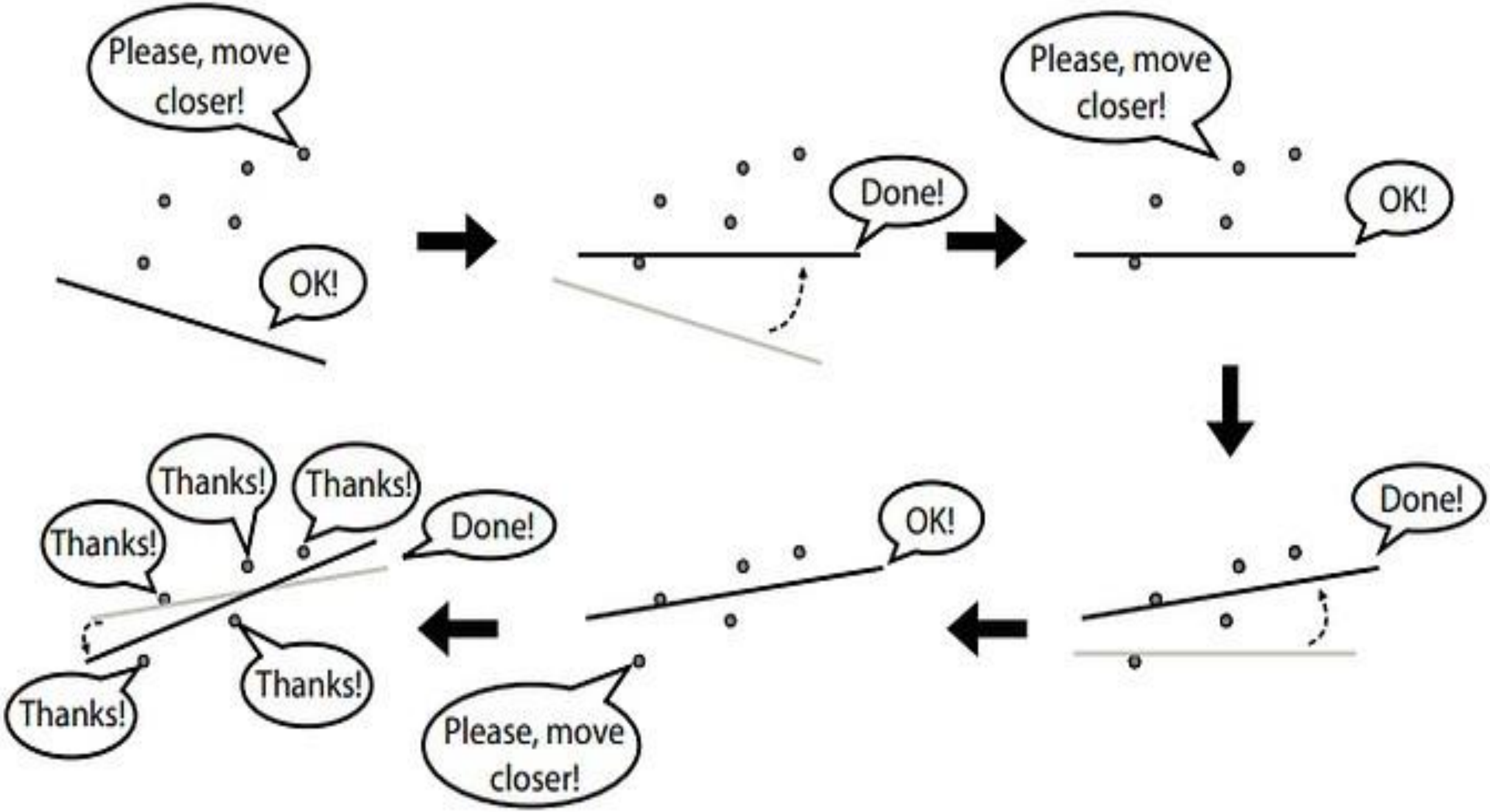
$$y = mx + b$$

Where:

- **y** is the predicted value (dependent variable)
- **x** is the input (independent variable)
- **m** is the slope of the line (how much y changes when x changes)
- **b** is the intercept (the value of y when x = 0)

- The best-fit line will be the one that optimizes the values of m (slope) and b (intercept) so that the predicted y values are as close as possible to the actual data points.


Y
Observed value
y_i
Random error ε_i
Y_{p_i}
Predicted value
Intercept θ_1
$Y = \theta_1 + \theta_2 X$
y_i
Slope = tanθ = θ_1
θ
xi
X

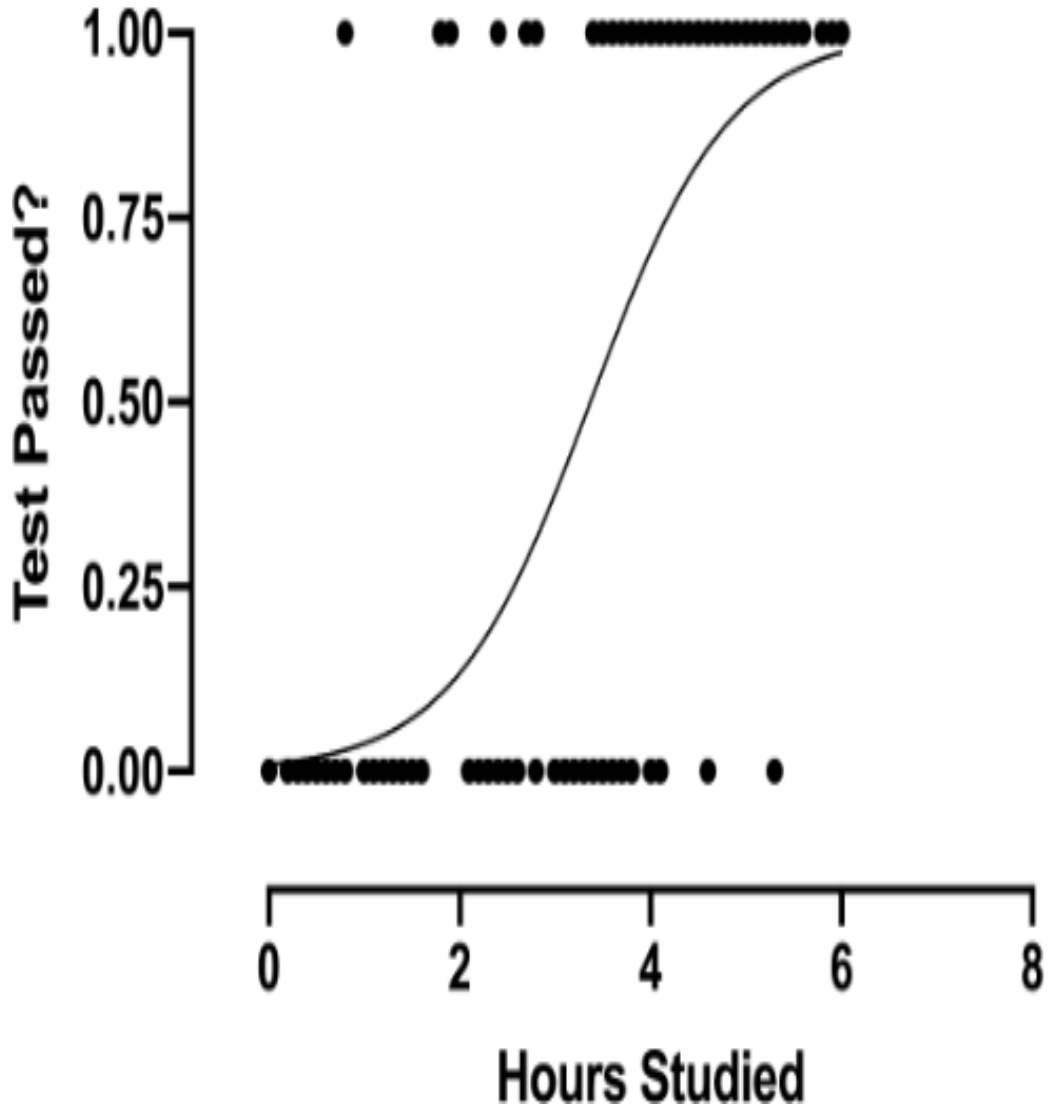

Please, move closer!
OK!
Done!
Please, move closer!
OK!
Done!
OK!
Please, move closer!
Thanks!
Thanks!
Thanks!
Thanks!
Thanks!
Done!

Logistic Regression

- When the dependent variable is discrete, the logistic regression technique is applicable.

- In other words, this technique is used to compute the probability of mutually exclusive occurrences such as pass/fail, true/false, 0/1, and so forth.

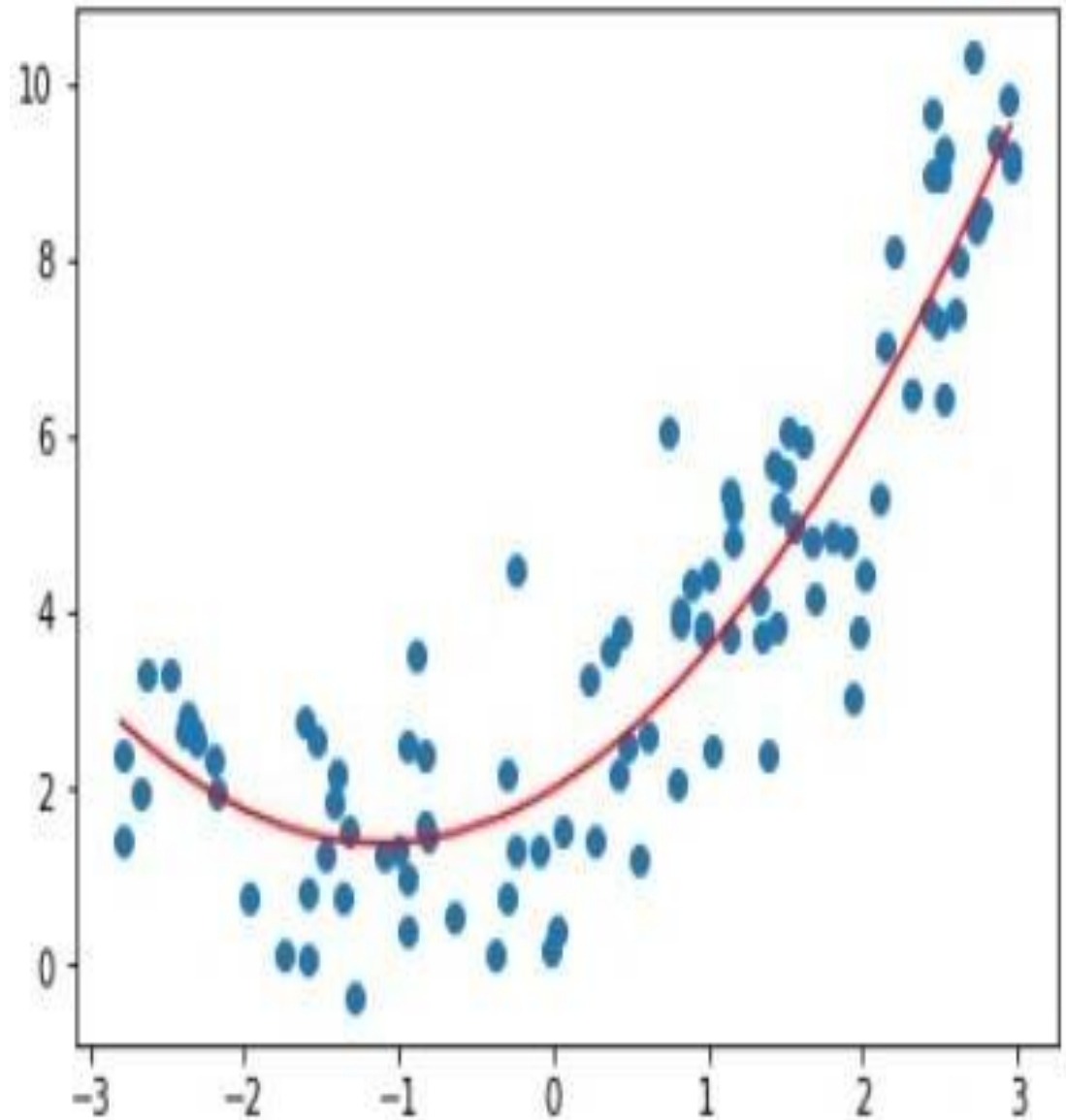
- Thus, the target variable can take on only one of two values, and a sigmoid curve represents its connection to the independent variable, and probability has a value between 0 and 1.


1.00
0.75
0.50
0.25
0.00
Test Passed?
0
2
4
6
8
Hours Studied

Polynomial Regression

- Polynomial regression analysis represents a non-linear relationship between dependent and independent variables.

- This technique is a variant of the multiple linear regression model, but the best fit line is curved rather than straight.


10
8
6
4
2
0
−3
−2
−1
0
1
2
3

Multiple Linear Regression

- **Multiple linear regression** is a technique to understand the relationship between a *single* dependent variable and *multiple* independent variables.
- The formulation for multiple linear regression is also similar to simple linear regression with
- the small change that instead of having one beta variable, you will now have betas for all the variables used.

The formula is given as:

- $Y = B_0 + B_1X_1 + B_2X_2 + \ldots + B_pX_p + \boldsymbol{\varepsilon}$

THANK YOU

Unit-4

Python **Libraries for Data Wrangling**

Course Code	Course Title	L	T	P	Credits
BT-AIDS-205A	Foundations of Data Science	3	-	-	3
		CIE	SEE		Total
		40	60		100

Course Outcomes

CO1	Understand the fundamentals and processes of data science.
CO2	Learn to describe data effectively within the data science workflow.
CO3	Explore relationships between different data variables.
CO4	Utilize Python libraries for data wrangling tasks.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	-	-	-	-	-	-	-	1	3	2
CO2	3	3	3	2	-	-	-	-	-	-	-	1	3	2
CO3	2	1	2	1	-	-	-	-	-	-	-	1	3	2
CO4	3	1	3	1	-	-	-	-	-	-	-	2	2	3

COURSE CIRRICULAM

Course Outline

Unit: I	Introduction To Data Science	Contact Hours:12
Benefits and uses of Data Science, Facets of data, Overview of the Data Science Process, Defining research goals, Data retrieval and preparation, Exploratory Data Analysis (EDA), Model building and application development Data Mining and Data Warehousing		
Unit: II	**Describing Data**	**Contact Hours:10**
Types of data and variables, Descriptive statistics: tables, graphs, averages, Measures of variability, Normal distributions and z-scores, measure of dispersion, Data distribution and shape, Frequency Distributions and Tabulation		
Unit: III	**Relationships Descriptions**	**Contact Hours:14**
Correlation and scatter plots, Computational formula for correlation coefficient, Regression analysis: regression lines, least squares method, Standard error of estimate, Interpretation of R^2 (coefficient of determination), Multiple regression equations, Regression towards the mean		
Unit: IV	**Python Libraries for Data Wrangling**	**Contact Hours:08**
Basics of NumPy arrays, Aggregations and computations on arrays, Comparisons, masks, and Boolean logic, Fancy indexing and structured arrays, Data manipulation with Pandas: indexing, selection, operations, handling missing data Hierarchical indexing, combining datasets, Aggregation, grouping, and pivot tables		

Unit-4

Python **Libraries for Data Wrangling**





Python **Libraries for Data Wrangling**

- Data Wrangling is the ***process of gathering, collecting, and transforming Raw data into another format for better understanding, decision-making, accessing, and analysis in less time.***
- Data Wrangling is also known as Data Munging.

- Data wrangling involves processing the data in ***various formats like - merging, grouping, concatenating etc. for the purpose of analysing or getting them ready to be used with another set of data.***

- Python has built-in features to apply these wrangling methods to various data sets to achieve the analytical goal.

Importance Of Data Wrangling

- Books selling Website want to show top-selling books of different domains, according to user preference.
- **For example,** if a new user searches for motivational books, then they want to show those motivational books which sell the most or have a high rating, etc.
- But on their website, there are plenty of raw data from different users. Here the concept of Data Munging or Data Wrangling is used.
- **Data wrangling is not by the System itself.**
- This process is done by Data Scientists.
- Data Scientist will wrangle data in such a way that they will sort the motivational books that are sold more or have high ratings or user buy this book with these package of Books, etc.On the basis of that, the new user will make a choice.

Data Wrangling in Python

- Data Wrangling is essential to learn for Data Science and Data Analysis.
- **Pandas Framework** of Python is used for Data Wrangling.
- It is used for **processes like data sorting or filtration, Data grouping, etc.**

Data wrangling in Python :

- **Data exploration: T**he data is studied, analyzed, and understood by visualizing representations of data.
- **Dealing with missing values:** Most of the datasets having a vast amount of data contain missing values of *NaN*, *they are needed to be taken* care of by replacing them with mean, mode, the most frequent value of the column, or simply by dropping the row having a *NaN* value.

Cont..

- **Reshaping data:** In this process, **data is manipulated according to the requirements,** where new data can be added or pre-existing data can be modified.

- **Filtering data:** Some times datasets are comprised of unwanted rows or columns which are required to be removed or filtered

Data Wrangling in Python

- Data Wrangling is a crucial topic for Data Science and Data Analysis. Pandas Framework of Python is used for Data Wrangling. Pandas is an open-source library in Python specifically developed for Data Analysis and Data Science. It is used for processes like data sorting or filtration, Data grouping, etc.

Data wrangling in Python deals with the below functionalities:

- **Data exploration:** In this process, the data is studied, analyzed, and understood by visualizing representations of data.
- **Dealing with missing values:** Most of the datasets having a vast amount of data contain missing values of *NaN*, *they are needed to be taken* care of by replacing them with mean, mode, the most frequent value of the column, or simply by dropping the row having a *NaN* value.
- **Reshaping data:** In this process, data is manipulated according to the requirements, where new data can be added or pre-existing data can be modified.
- **Filtering data:** Some times datasets are comprised of unwanted rows or columns which are required to be removed or filtered
- **Other:** After dealing with the raw dataset with the above functionalities we get an efficient dataset as per our requirements and then it can be used for a required purpose like data analyzing, machine learning, data visualization, model training etc.

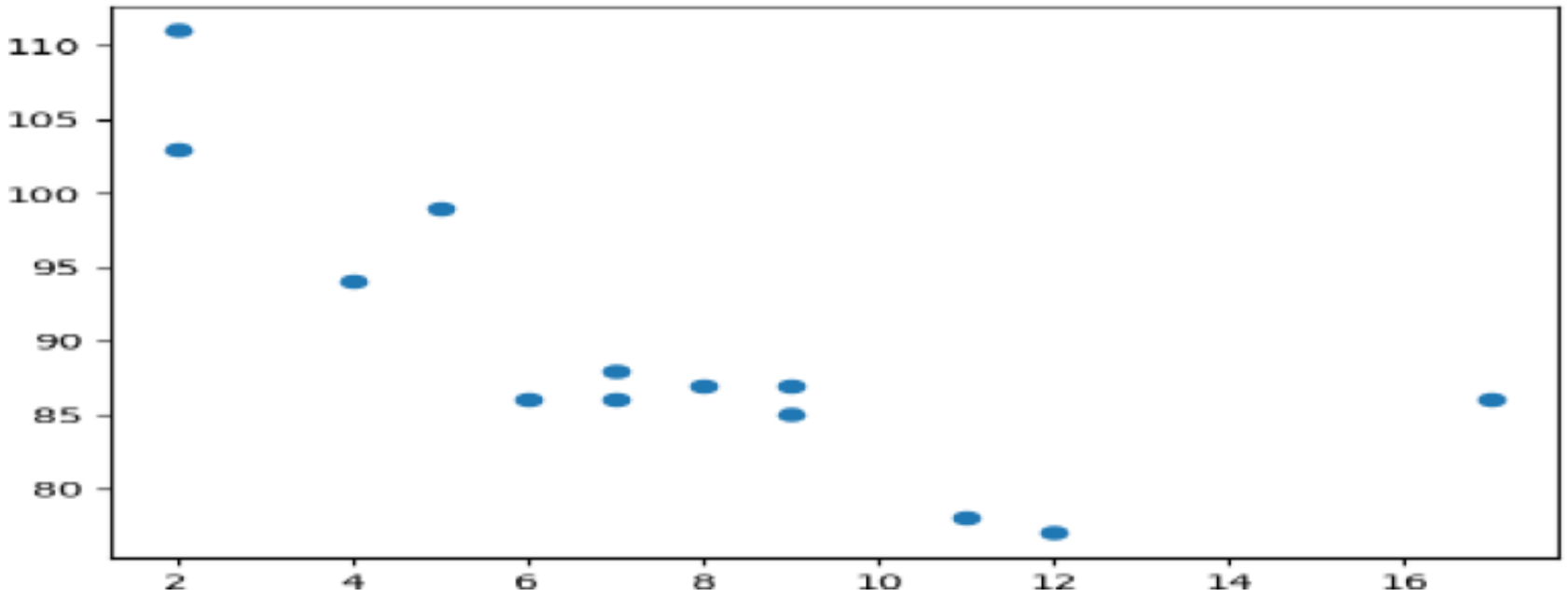
Scatter plots Using Python: Home Assignment

- The scatter() function plots one dot for each observation. It needs two arrays of the same length, one for the values of the x-axis, and one for values on the y-axis:

Example : simple scatter plot:

- import matplotlib.pyplot as plt
- import numpy as np
- x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
- y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
- plt.scatter(x, y)
- plt.show()

Result:



Histogram using Python: Home Assignment

- A histogram is a graph showing *frequency* distributions.
- It is a graph showing the number of observations within each given interval

Create Histogram:

- import matplotlib.pyplot as plt
- import numpy as np
- x = np.random.normal(170, 10, 250)
- plt.hist(x)
- plt.show()

Example1: *we load the data into a dataframe, and then we visualize the data in a tabular format.*

```
# Import pandas package
import pandas as pd

# Assign data
data = {'Name': ['Jai', 'Princi', 'Gaurav',
                 'Anuj', 'Ravi', 'Natasha', 'Riya'],
        'Age': [17, 17, 18, 17, 18, 17, 17],
        'Gender': ['M', 'F', 'M', 'M', 'M', 'F', 'F'],
        'Marks': [90, 76, 'NaN', 74, 65, 'NaN', 71]}

# Convert into DataFrame
df = pd.DataFrame(data)

# Display data
df
```

Output:

	Name	Age	Gender	Marks
0	Jai	17	M	90
1	Princi	17	F	76
2	Gaurav	18	M	NaN
3	Anuj	17	M	74
4	Ravi	18	M	65
5	Natasha	17	F	NaN
6	Riya	17	F	71

defining the dataframe and displaying in tabular format

Example 2: Dealing with missing values in Python

```
# Compute average
c = avg = 0
for ele in df['Marks']:
    if str(ele).isnumeric():
        c += 1
        avg += ele
avg /= c

# Replace missing values
df = df.replace(to_replace="NaN",
                value=avg)

# Display data
df
```

There are *NaN* values present in the *MARKS* column which is a missing value in the dataframe that is going to be taken care of in data wrangling by replacing them with the column mean.

Output:

	Name	Age	Gender	Marks
0	Jai	17	M	90.0
1	Princi	17	F	76.0
2	Gaurav	18	M	75.2
3	Anuj	17	M	74.0
4	Ravi	18	M	65.0
5	Natasha	17	F	75.2
6	Riya	17	F	71.0

replacing Nan values with average

Example3: Data Replacing in Data Wrangling

- In the *GENDER* column, we can replace the Gender column data by categorizing them into different numbers.

```
# Categorize gender
df['Gender'] = df['Gender'].map({'M': 0,
                                 'F': 1, }).astype(float)

# Display data
df
```

Output:

	Name	Age	Gender	Marks
0	Jai	17	0.0	90.0
1	Princi	17	1.0	76.0
2	Gaurav	18	0.0	75.2
3	Anuj	17	0.0	74.0
4	Ravi	18	0.0	65.0
5	Natasha	17	1.0	75.2
6	Riya	17	1.0	71.0

Data encoding for gender variable in data wrangling

Filtering data in Data Wrangling

suppose there is a requirement for the details regarding name, gender, and marks of the top-scoring students.

Here we need to remove some using the pandas slicing method in data wrangling from unwanted data.

```
# Filter top scoring students
df = df[df['Marks'] >= 75].copy()

# Remove age column from filtered DataFrame
df.drop('Age', axis=1, inplace=True)

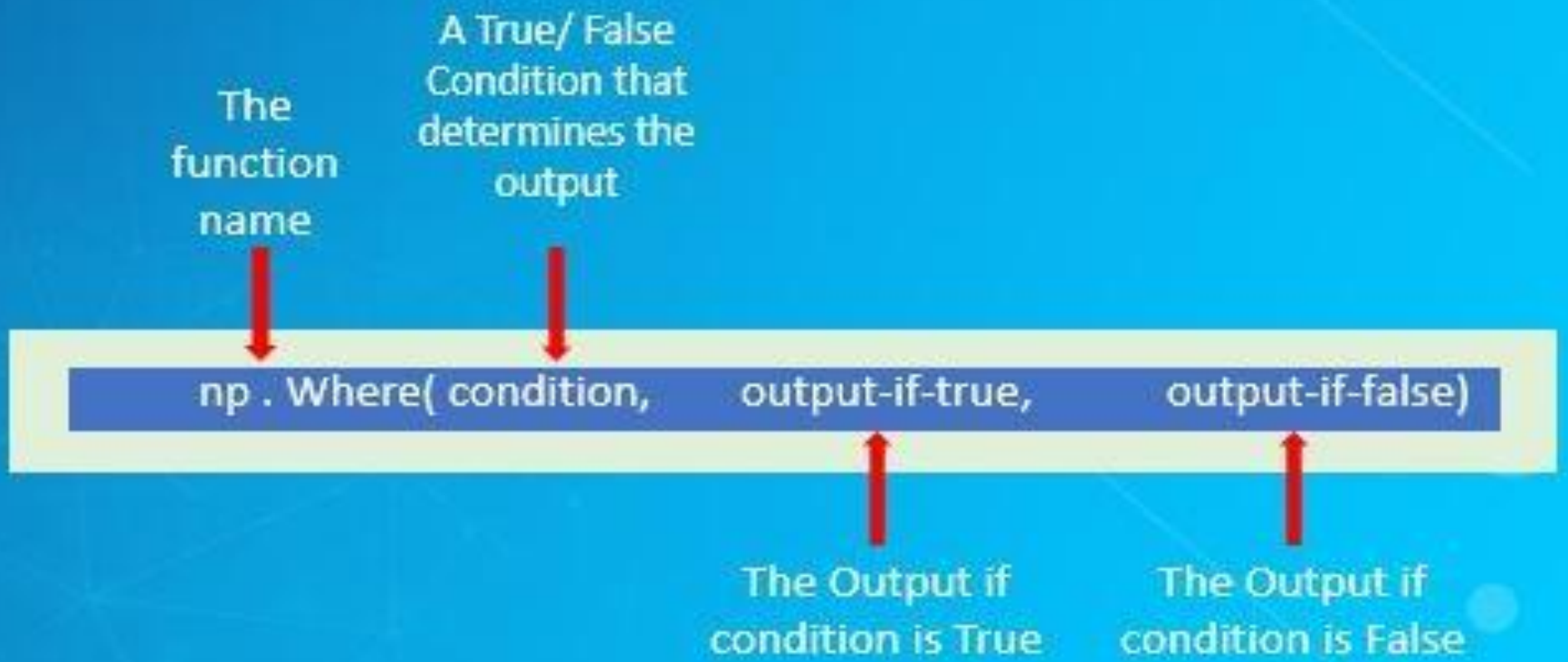
# Display data
df
```

Output:

	Name	Gender	Marks
0	Jai	0.0	90.0
1	Princi	1.0	76.0
2	Gaurav	0.0	75.2
5	Natasha	1.0	75.2

Dropping column and filtering rows

numPy.where()


The function name
A True/ False Condition that determines the output
np . Where(condition, output-if-true, output-if-false)
The Output if condition is True
The Output if condition is False





Python - numpy.where()



- The numpy.where() function can be used to select elements from arrays depending on a condition
- This will choose elements from an array X if the condition holds. Otherwise, it will choose from another array Y.
- Numpy broadcasts the condition array, X and Y before applying this function on our target array.

```
op = numpy.where(condition, X, Y)
```

What is NumPy?

NumPy is a Python library designed for efficient array manipulation and numerical computations.

Faster than Python lists and optimized for handling large datasets.

Supports vectorized operations enabling faster execution of mathematical operations over arrays.



Basics of NumPy Arrays

What is NumPy?

- NumPy is a Python library **used for working with arrays**.
- It also has functions for working in domain of linear algebra, Fourier transform, and matrices.
- NumPy was created in 2005 by Travis Oliphant.
- It is an open source project and you can use it freely.
- NumPy stands for Numerical Python.

Why Use NumPy? Compare to List

Why NumPy?

- NumPy arrays are stored at one continuous place in memory unlike lists, so processes can access and manipulate them very efficiently. This behavior is called locality of reference in computer science.
- This is the main reason why Numpy is faster than lists. Also it is optimized to work with latest CPU architectures.

Numpy Fundamentals

Numpy provides efficient numerical computing with features like Array Manipulation, mathematical Operation, Linear Algebra and Random number Generation.

01

ARRAY MANIPULATION

Create and reshape arrays

02

MATHEMATICAL FUNCTIONS

Perform element wise Operations

03

LINEAR ALGEBRA

Perform matrix operations

04

RANDOM NUMBER GENERATION

Generate Random Data

Numpy Array Manipulation

Create and reshape array easily using Numpy

Creation

np.array ([1,2,3,4,5,6])



Reshape

Original =np.array ([1,2,3,4,5,6])

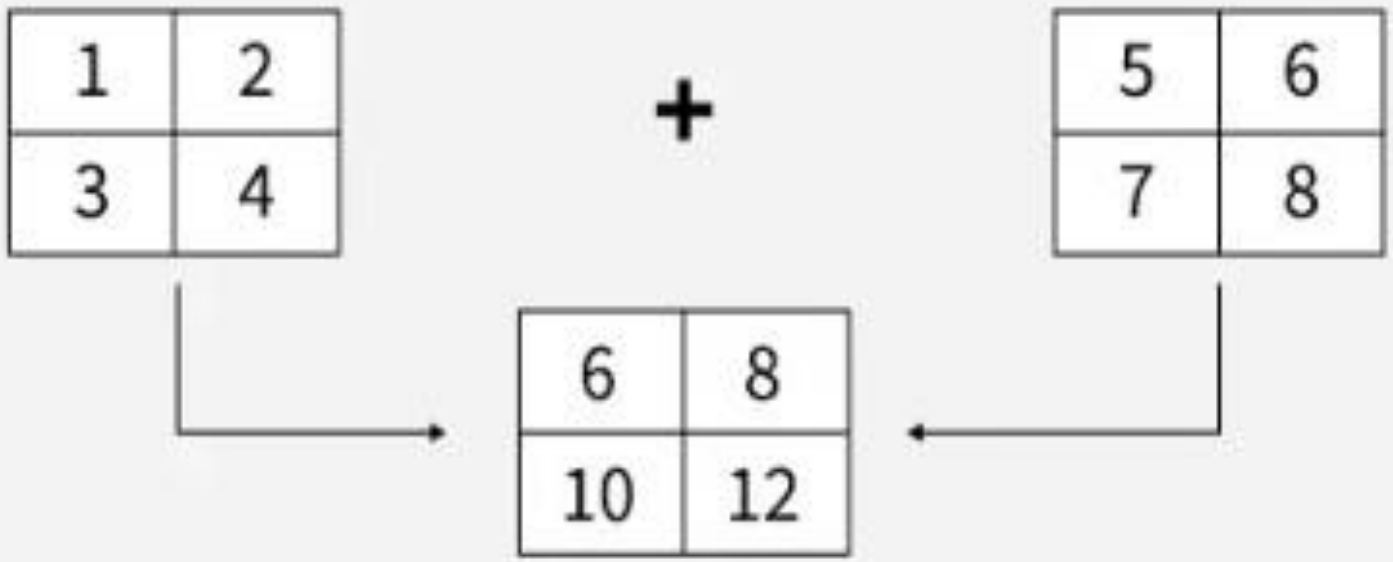


Mathematical Operations in Numpy

Perform element-wise arithmetic operations using Numpy

a = np.array ([[1,2], [3,4]])

b = np.array ([[5,6], [7,8]])



Numpy Linear Algebra





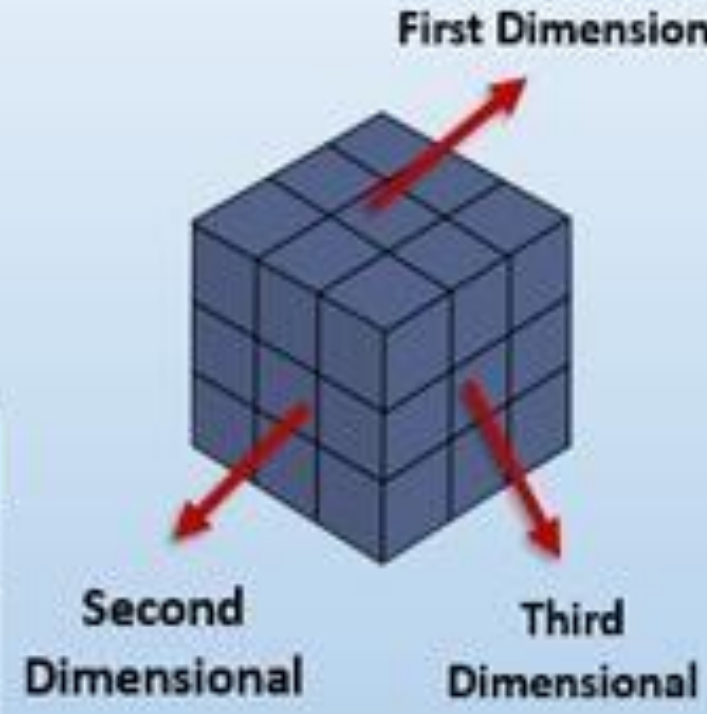
NumPy Ndarray

10	15	13	8	25

1D-Array

	Column 0	Column 1	Column 2
Row 0	X[0][0]	X[0][1]	X[0][2]
Row 1	X[1][0]	X[1][1]	X[1][2]
Row 2	X[2][0]	X[2][1]	X[2][2]

2D-Array



3D-Array

NumPy - Features

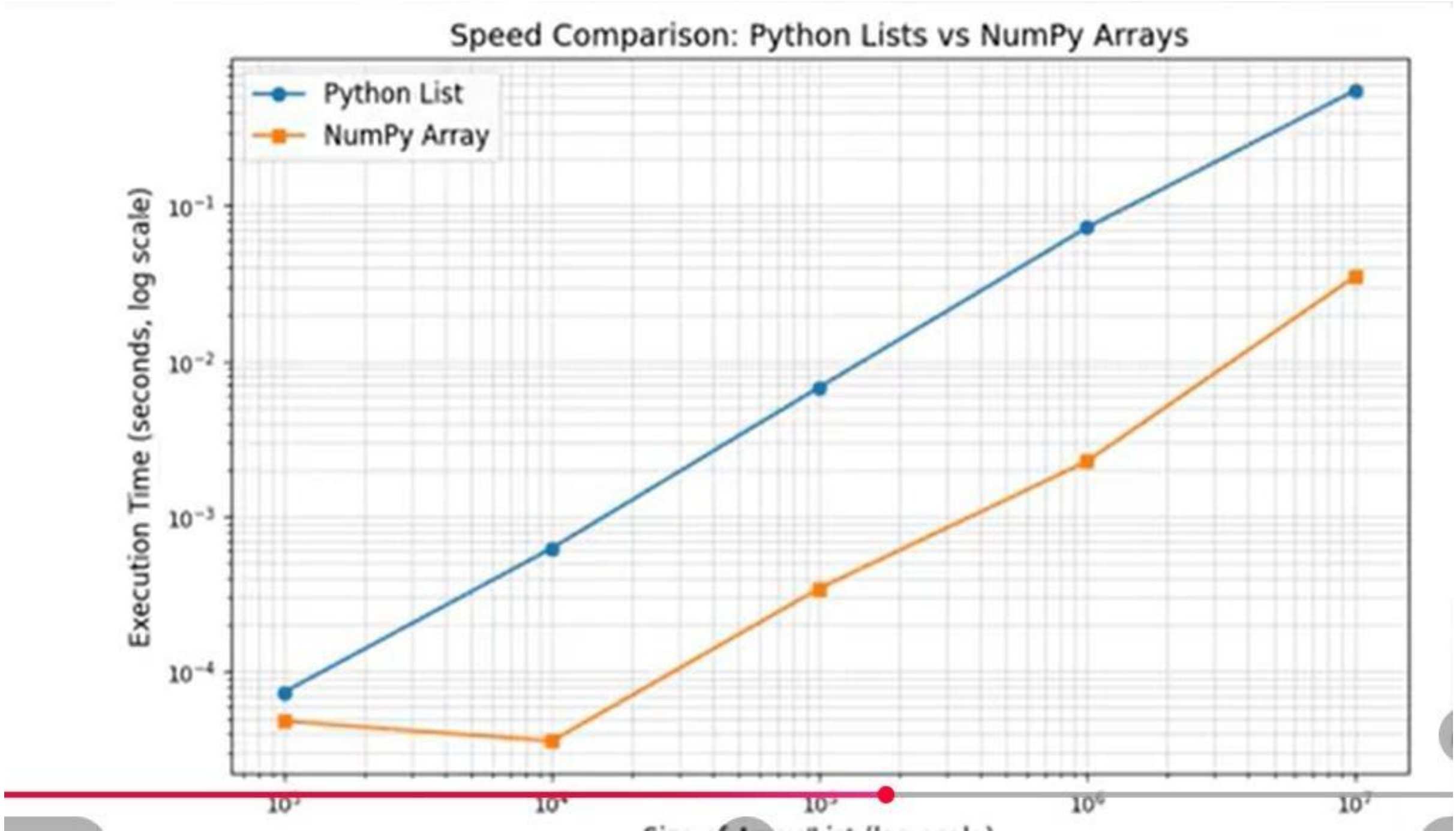
- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- **Tools for integrating C/C++ and Fortran code**
- **Useful linear algebra, Fourier transform, and random number capabilities**

Why Numpy faster than list?

```
# Plot the results
plt.figure(figsize=(10, 6))
plt.plot(sizes, python_times, label="Python List", marker='o')
plt.plot(sizes, numpy_times, label="NumPy Array", marker='s')

# Add labels and title
plt.xscale("log")
plt.yscale("log")
plt.xlabel("Size of Array/List (log scale)", fontsize=12)
plt.ylabel("Execution Time (seconds, log scale)", fontsize=12)
plt.title("Speed Comparison: Python Lists vs NumPy Arrays", fontsize=14)
plt.legend(fontsize=12)
plt.grid(True, which="both", linestyle="--", linewidth=0.5)

# Show the plot
plt.show()
```


Speed Comparison: Python Lists vs NumPy Arrays
Python List
NumPy Array
Execution Time (seconds, log scale)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{5}
10^{6}
10^{7}

In [1]:
```
import pandas as pd

dataset = pd.read_csv("C:/Users/Admin/Desktop/Blog/Missing values/data.csv")
dataset
```

Out[1]:

	Height	Weight	Country	Place	Number of days	Some column
0	12.0	35.0	India	Bengaluru	1.0	NaN
1	NaN	36.0	US	New York	2.0	NaN
2	13.0	32.0	UK	London	NaN	NaN
3	15.0	NaN	France	Paris	4.0	NaN
4	16.0	39.0	US	California	5.0	12.0
5	NaN	NaN	NaN	Mumbai	NaN	NaN
6	NaN	NaN	NaN	NaN	6.0	NaN

Python List Operation	NumPy Array Operation	Description
`len(list)`	`array.shape`	Get number of elements
`list.append(item)`	N/A	Add an item (NumPy arrays are fixed-size)
`for item in list`	Vectorized operations	Apply operation to each item
`list1 + list2`	`array1 + array2`	Concatenate lists vs. element-wise addition
`sum(list)`	`np.sum(array)`	Sum of all elements

Handling Missing Data

- Managing missing values helps avoid errors and incorrect conclusions in data analysis.
- Without complete data, our conclusions may be flawed.
- By filling in missing information, we make sure our findings are accurate and trustworthy.

Identifying missing values:

How?

Identifying missing values:

```
# Check for missing values in the DataFrame
print(df.isnull().sum())
```

Dropping or filling missing values:

```
# Drop rows with missing values
df.dropna()

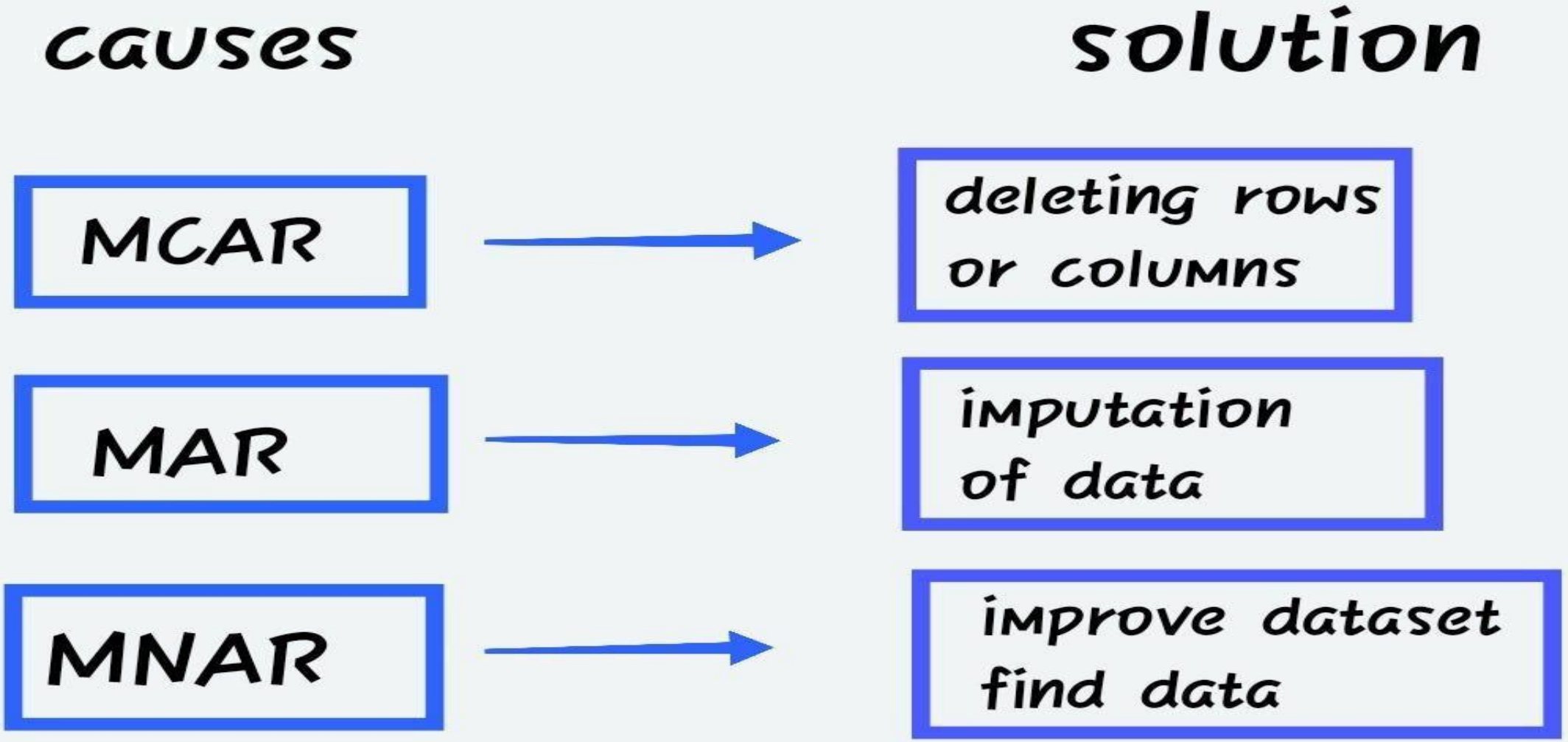

# Fill missing values with a specific value
df.fillna(0)

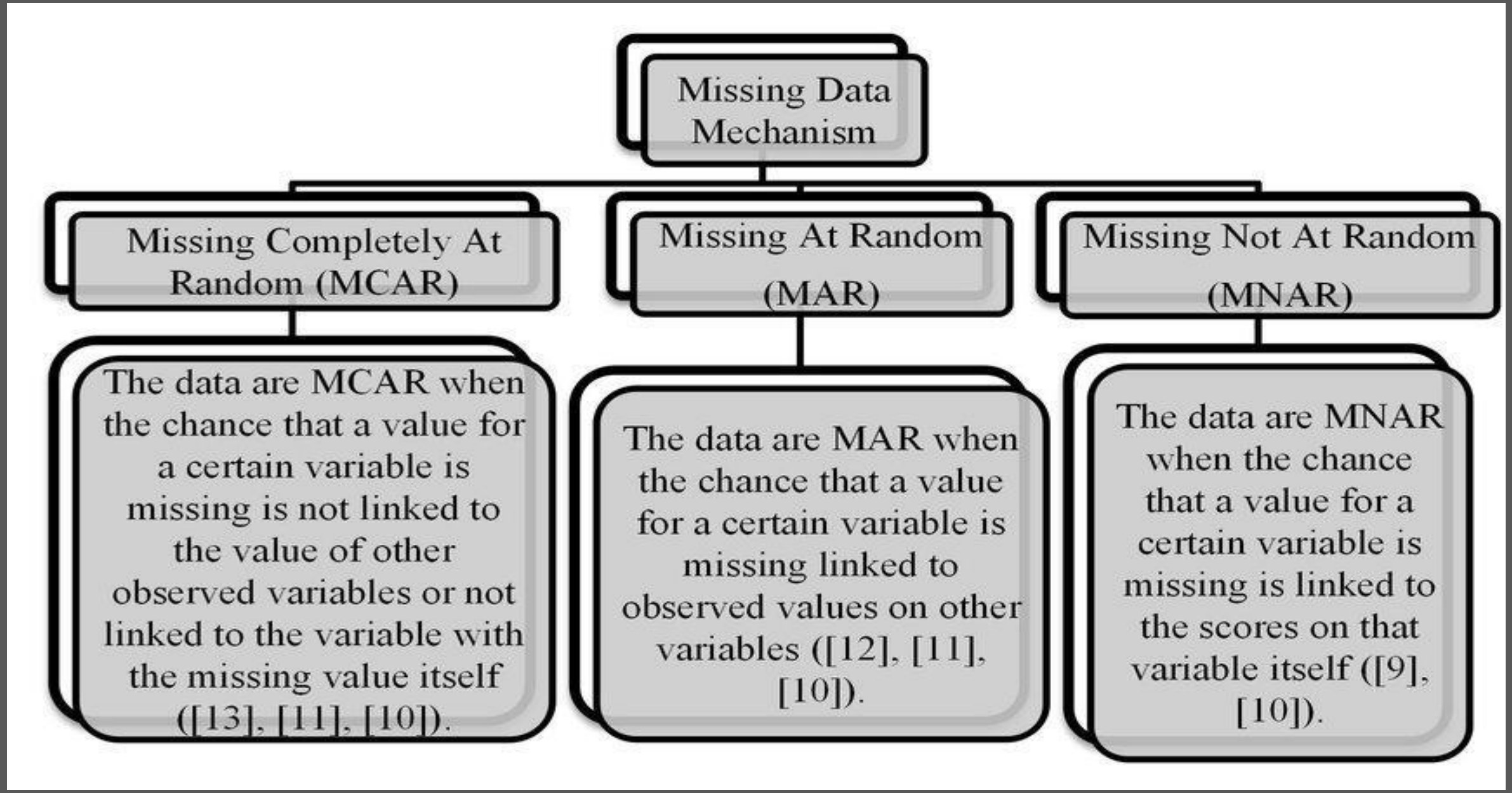

# Fill missing values using a function
df.fillna(df.mean())
```

Types of Missing Values



Figure 1 - Different Types of Missing Values in Datasets


causes
solution
MCAR
deleting rows or columns
MAR
imputation of data
MNAR
improve dataset find data


Missing Data Mechanism
Missing Completely At Random (MCAR)
Missing At Random (MAR)
Missing Not At Random (MNAR)
The data are MCAR when the chance that a value for a certain variable is missing is not linked to the value of other observed variables or not linked to the variable with the missing value itself ([13], [11], [10]).
The data are MAR when the chance that a value for a certain variable is missing linked to observed values on other variables ([12], [11], [10]).
The data are MNAR when the chance that a value for a certain variable is missing is linked to the scores on that variable itself ([9], [10]).

MCAR

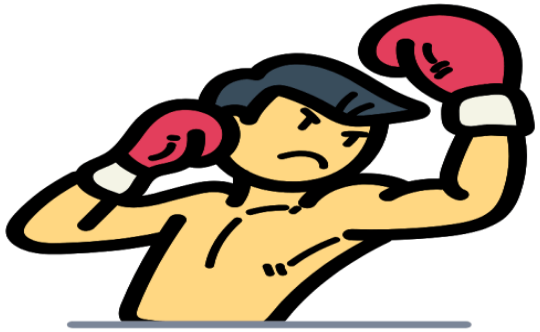


Oops, I dropped that one sample...

EXAMPLE

SAMPLE ID	SAMPLE ATT 1	SAMPLE ATT 2
1	0.51	100.1
2	0.34	100.3
3	0.44	99.5
4	0.21	99.9
5	0.40	99.7

MAR



I'm a man! I don't talk about my feeling!

EXAMPLE

AGE	SEX	HOW YOU FEEL
47	F	Angry
35	F	Happy
32	M ➤	Alone
56	M ➤	Angry
45	F	Alone

MNAR



Urgh, I'm too rich to know how much I got.

EXAMPLE

AGE	SEX	INCOME
47	M	45k
35	F	120k
32	M	2.1M
56	F	78k
45	M	1.34M

Missing Completely At Random (MCAR)

- **Example:** In a survey about library books, some overdue book values in the dataset are missing due to human error in recording.

Missing At Random (MAR)

- **Example:** In a survey, ‘Age’ values might be missing for those who did not disclose their ‘Gender’.
- Here, the missingness of ‘Age’ depends on ‘Gender’, but the missing ‘Age’ values are random among those who did not disclose their ‘Gender’.

Missing Not At Random (MNAR)

- **Example:** In a survey about library books, people with more overdue books might be less likely to respond to the survey. Thus, the number of overdue books is missing and depends on the number of books overdue.

Why Do Missing Values Occur?

- **Data Entry Errors**: Sometimes, it's just human error. Someone might forget to input a value or accidentally delete one.
- **Sensor Malfunctions**: In IoT or scientific experiments, a faulty sensor might fail to record data at certain times.
- **Survey Non-Response**: In surveys, respondents might skip questions they're uncomfortable answering or don't understand.
- **Merged Datasets**: When combining data from multiple sources, some entries might not have corresponding values in all datasets.
- **Data Corruption**: During data transfer or storage, some values might get corrupted and become unreadable.
- **Intentional Omissions**: Some data might be intentionally left out due to privacy concerns or irrelevance.
- **Sampling Issues**: The data collection method might systematically miss certain types of data.
- **Time-Sensitive Data**: In time series data, values might be missing for periods when data wasn't collected (e.g., weekends, holidays).

How to Handle Missing Data?

- Missing data is a common headache in any field that deals with datasets.

- It can arise for various reasons, from human error during data collection to limitations of data gathering methods.

- Here are two main approaches:

1. **Deletion**
2. **Imputation**

Deletion:

- **Deletion:** This involves removing rows or columns with missing values.

- This is a straightforward method, but it can be problematic if a significant portion of your data is missing.

- Discarding too much data can affect the reliability of your conclusions.

Imputation:

- **Imputation:** This replaces missing values with estimates. There are various imputation techniques, each with its strengths and weaknesses. Here are some common ones:

  - **Mean/Median/Mode Imputation:** Replace missing entries with the average (mean), middle value (median), or most frequent value (mode) of the corresponding column. This is a quick and easy approach, but it can introduce bias if the missing data is not randomly distributed.

  - **K-Nearest Neighbors (KNN Imputation):** This method finds the closest data points (neighbors) based on available features and uses their values to estimate the missing value. KNN is useful when you have a lot of data and the missing values are scattered.

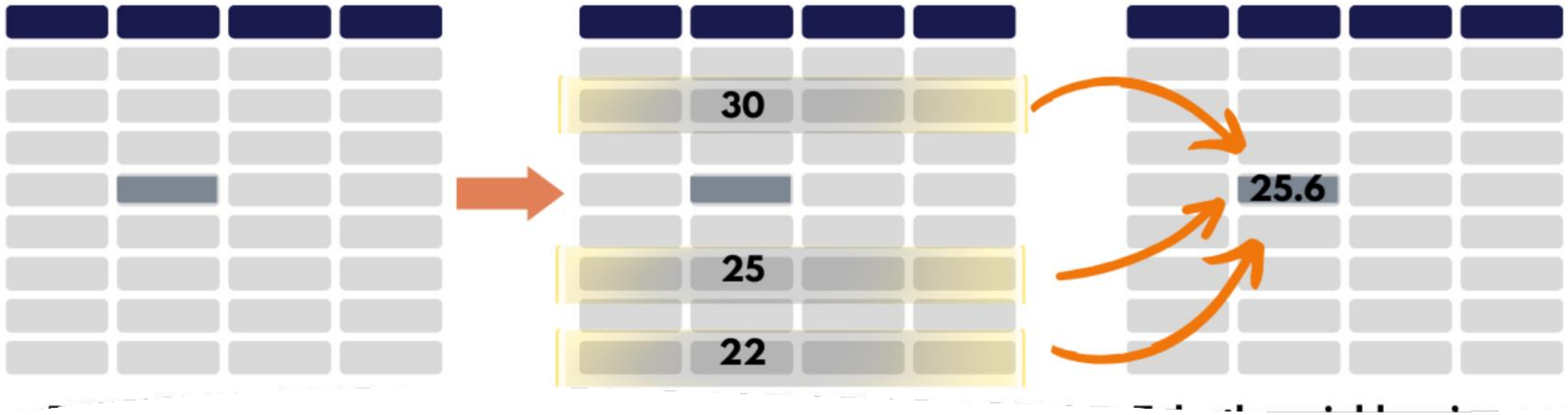
  - **Model-based Imputation:** This involves creating a statistical model to predict the missing values based on other features in the data. This can be a powerful technique, but it requires more expertise and can be computationally expensive.pen_spark.

K-Nearest Neighbors (KNN Imputation):

- Nearest neighbor imputation algorithms efficiently fill in missing data by replacing each missing value with a value derived from similar cases within the entire dataset.

- Imputation with K-Nearest Neighbors (KNN) estimates missing values in a dataset by considering the values of the closest data points, determined by a distance metric like Euclidean distance.

- The missing value is then assigned the average of these nearest neighbors' values, weighted by their proximity.



- Consider the following dataset with 5 variables and 11 observations.
- We aim to impute the missing value in the 5th row of variable 2.
- First, we identify the row's 3 closest neighbors (highlighted by the squared boxes) using a KNN algorithm.
- Then, we calculate the average of the values for variable 2 from these neighbors.

Calculation:

- The imputed value is calculated as (Value1 x w1 + Value2 x w2 + Value3 x w3) / 3, where w1, w2, and w3 are weights proportional to the distance of each neighbor from the data point being imputed.

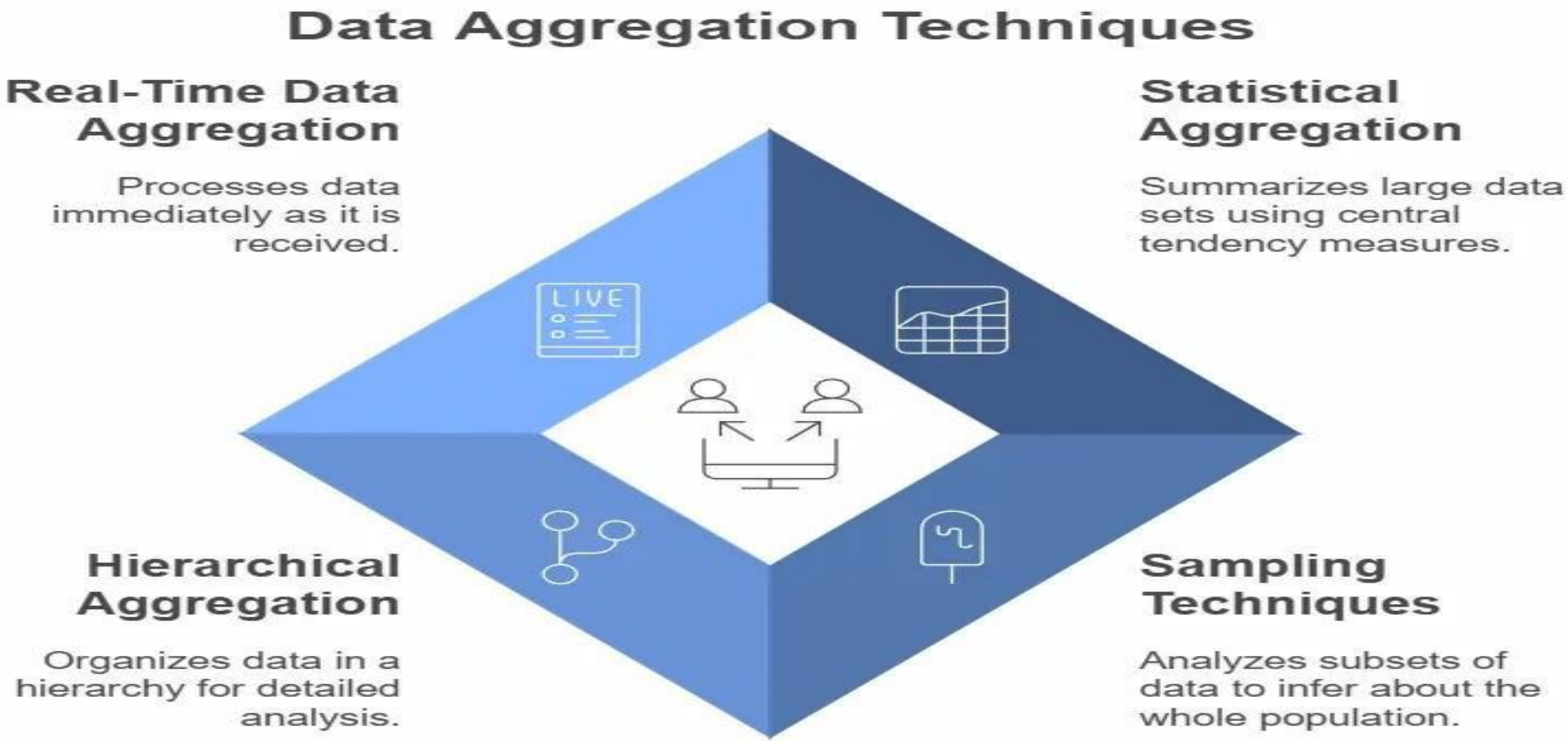
Why Do We Need to Care About Handling Missing Data?

- It is important to handle the missing values appropriately.
- **Many machine learning algorithms fail if the dataset contains missing values**. However, algorithms like K-nearest and Naive Bayes support data with missing values.
- You may end up building a biased machine learning model, leading to incorrect results if the missing values are not handled properly.
- Missing data can lead to a lack of precision in the statistical analysis.

Handling Missing Data and Hierarchical Indexing

- **NaN (Not a Number)**: In many programming languages and data analysis tools, missing values are represented as NaN. This is the default for libraries like **Pandas** in Python.
- **NULL or None**: In databases and some programming languages, missing values are often represented as NULL or None. For instance, in **SQL databases**, a missing value is typically recorded as NULL.
- **Empty Strings**: Sometimes, missing values are denoted by empty strings (""). This is common in text-based data or CSV files where a field might be left blank.
- **Special Indicators**: Datasets might use specific indicators like -999, 9999, or other unlikely values to signify missing data. This is often seen in older datasets or specific industries where such conventions were established.
- **Blanks or Spaces**: In some cases, particularly in fixed-width text files, missing values might be represented by spaces or blank fields.

Aggregations and Computations on Arrays



Cont..

- **Statistical Aggregation:** This is a technique that **summarises huge bulks of data** using measures of central tendency, such as mean, median, and mode. It is like the average temperature over one week; it gives a general sense of the trend without being confused by the daily ups and downs.

  Sampling Techniques: Sampling allows a **subset of data can be analysed instead of the entire population.** Think of it like a taste test, just one small scoop can give you the flavour of the whole cake! One of the most common sampling techniques is called stratified sampling, in which the population is divided into subgroups before sampling.

  Hierarchical Aggregation: This technique is about **organising data in a hierarchy**, enabling zooming down and up through the levels of detail. You know, like a family tree: it's broad at the top, and you go on finding the branches!

  Real-Time Data Aggregation: In this approach, the **data is aggregated and processed as soon as it comes in.** Just like receiving live updates of an event; you catch everything right on the spot!

Example of Data Set:

Dataset

	02	SAT SUN						
1	08-01	0	0	25.1	99	0		0.14
2	08-02	1	0	26.4		0		
3	08-03	2	0		96	0		0.21
4	08-04	3	0	24.1	68	0		0.68
5	08-05	4		24.7	98	0		0.2
6	08-06	5	0	26.5	98	0		0.32
7	08-07	6	0	27.6	78	0		0.72
8	08-08	0	0	28.2		0		0.61
9	08-09	1	0	27.1	70	1		
10	08-10	2	1	26.7	75			0.54
11	08-11	3	0			0		
12	08-12	4		24.3	77	1		0.67
13	08-13	5	0	23.1	77	1		0.66
14	08-14	6	0	22.4	89	1		0.38
15	08-15	0	0		80	1		0.46
16	08-16	1	0	26.5	88	0		
17	08-17	2	0	28.6	76	0		0.52
18	08-18	3	0					
19	08-19	4	0	27.0	73	1		0.62
20	08-20	5	0	26.9	73	0		0.81

```python
import pandas as pd
import numpy as np

# Create the dataset as a dictionary
data = {
    'Date': ['08-01', '08-02', '08-03', '08-04', '08-05', '08-06', '08-0
             '08-11', '08-12', '08-13', '08-14', '08-15', '08-16', '08-1
    'Weekday': [0, 1, 2, 3, 4, 5, 6, 0, 1, 2, 3, 4, 5, 6, 0, 1, 2, 3, 4,
    'Holiday': [0.0, 0.0, 0.0, 0.0, np.nan, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0
    'Temp': [25.1, 26.4, np.nan, 24.1, 24.7, 26.5, 27.6, 28.2, 27.1, 26.
    'Humidity': [99.0, np.nan, 96.0, 68.0, 98.0, 98.0, 78.0, np.nan, 70.
    'Wind': [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, np.nan, 0.0, 0
    'Outlook': ['rainy', 'sunny', 'rainy', 'overcast', 'rainy', np.nan,
    'Crowdedness': [0.14, np.nan, 0.21, 0.68, 0.20, 0.32, 0.72, 0.61, np
}

# Create a DataFrame from the dictionary
df = pd.DataFrame(data)

# Display basic information about the dataset
print(df.info())

# Display the first few rows of the dataset
print(df.head())

# Display the count of missing values in each column
print(df.isnull().sum())
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 8 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Date         20 non-null     object
 1   Weekday      20 non-null     int64
 2   Holiday      19 non-null     float64
 3   Temp         16 non-null     float64
 4   Humidity     17 non-null     float64
 5   Wind         19 non-null     float64
 6   Outlook      17 non-null     object
 7   Crowdedness  15 non-null     float64
dtypes: float64(5), int64(1), object(2)
memory usage: 1.3+ KB
```

	Date	Weekday	Holiday	Temp	Humidity	Wind	Outlook	Crowdedness
0	08-01	0	0.0	25.1	99.0	0.0	rainy	0.14
1	08-02	1	0.0	26.4	NaN	0.0	sunny	NaN
2	08-03	2	0.0	NaN	96.0	0.0	rainy	0.21
3	08-04	3	0.0	24.1	68.0	0.0	overcast	0.68
4	08-05	4	NaN	24.7	98.0	0.0	rainy	0.20

```
Date           0
Weekday        0
Holiday        1
Temp           4
Humidity       3
Wind           1
Outlook        3
Crowdedness    5
dtype: int64
```

Comparisons, Mask and Boolean Logic

- Boolean masks to examine and manipulate values within NumPy arrays.
- Masking comes up when you want to extract, modify, count, or otherwise manipulate values in an array based on some criterion:
- For example, you might wish to count all values greater than a certain value, or perhaps remove all outliers that are above some threshold.

- In NumPy, Boolean masking is often the most efficient way to accomplish these types of tasks

Operator	Equivalent ufunc
==	np.equal
!=	np.not_equal
<	np.less
<=	np.less_equal
>	np.greater
>=	np.greater_equal

Boolean Masks

Array + Condition → Boolean Array (mask)

8	2	1	6
1	2	8	9
7	6	4	3

\+ “ < 5 “ →

F	T	T	F
T	T	F	F
F	F	T	T

0	1	1	0
1	1	0	0
0	0	1	1

Fancy indexing with integer arrays

- Indexing with array of integers
- Appears naturally to select specific values from their indices

In [35]:
```
x = np.arange(6)
indices = [1, 2, -1, 1]
print(x)
print(x[indices])
```

```
[0 1 2 3 4 5]
[1 2 5 1]
```

Fancing Indexing and Structured Arrays

- Fancy indexing is like the simple indexing we've already seen, but we pass arrays of indices in place of single scalars.
- This allows us to very quickly access and modify complicated subsets of an array's values.

- **Exploring Fancy Indexing:** Fancy indexing is conceptually simple: it means passing an array of indices to access multiple array elements at once.

Types of fancy indexing

- Indexing / accessing more values
- Array of indices
- In multi-dimensional
- Standard indexing

Example

```
import numpy as np
rand = np.random.RandomState(42)
x = rand.randint(100, size=10)
print(x)
```

```
[51 92 14 71 60 20 82 86 74 74]
```

Indexing / accessing more values

Suppose we want to access three different elements. We could do it like this:

```
[x[3], x[7], x[2]]
```

```
[71, 86, 14]
```

Array of indices

We can pass a single list or array of indices to obtain the same result.

ind = [3, 7, 4]

x[ind]

array([71, 86, 60])

In multi-dimensional

Fancy indexing also works in multiple dimensions. Consider the following array.

```
X = np.arange(12).reshape((3, 4))
X
```

```
array([[ 0, 1, 2, 3],
       [ 4, 5, 6, 7],
       [ 8, 9, 10, 11]])
```

Standard indexing

Like with standard indexing, the first index refers to the row, and the second to the column.

```
row = np.array([0, 1, 2])
col = np.array([2, 1, 3])
X[row, col]
array ([ 2, 5, 11])
```

```
Example array
print(X)
[[ 0  1  2  3]
 [ 4  5  6  7]
 [ 8  9 10 11]]
```

- Combine fancy and simple indices

```
X[2, [2, 0, 1]]
array([10, 8, 9])
```

- Combine fancy indexing with slicing

```
X[1:, [2, 0, 1]]
array([[ 6, 4, 5], [10, 8, 9]])
```

- Combine fancy indexing with masking

```
mask = np.array([1, 0, 1, 0], dtype=bool)
```

Data Manipulation with Pandas: Output?

```
# Importing the pandas library
import pandas as pd

# creating a dataframe object
student_register = pd.DataFrame()

# assigning values to the
# rows and columns of the dataframe
student_register['Name']  =  ['Abhijit','Smriti',
                             'Akash', 'Roshni']
student_register['Age'] = [20, 19, 20, 14]
student_register['Student'] = [False, True,
                               True, False]

print(student_register)
```

Output:

```
      Name   Age   Student
0   Abhijit    20     False
1    Smriti    19      True
2     Akash    20      True
3    Roshni    14     False
```

Statement: As you can see, the dataframe object has four rows [0, 1, 2, 3] and three columns[“Name”, “Age”, “Student”] respectively. The column which contains the index values i.e. [0, 1, 2, 3] is known as the **index column**, which is a default part in pandas datagram. We can change that as per our requirement too because Python is powerful.

Data Frame: A DataFrame is a two-dimensional labeled data structure resembling a spreadsheet. Here's an example of creating a DataFrame with student information:

```
# Create a dictionary with student data
student_data = {
     "Name": ["Alex", "Ben", "Clara", "Deric", "Eva", "Frank"],
     "Age": [20, 21, 19, 22, 23, 20],
     "Major": ["CS", "Math", "Physics", "Economics", "Biology", "Chemistry"],
     "Score": [85, 92, 78, 95, 88, 90],
     "Hometown": ["York", "Angeles", "Chicago", "Houston", "Boston", "Francisco"]
     "Graduation Year": [2022, 2023, 2022, 2023, 2024, 2023],
     "Scholarship": [False, True, False, True, False, False]
}

# Create a DataFrame from the dictionary
df = pd.DataFrame(student_data)

# Print the DataFrame
print(df)
```

```
#Output:

     Name  Age        Major  Score    Hometown  Graduation Year  Scholarship
0    Alex   20          CS     85        York             2022        False
1     Ben   21        Math     92     Angeles             2023         True
2   Clara   19     Physics     78     Chicago             2022        False
3   Deric   22   Economics     95     Houston             2023         True
4     Eva   23     Biology     88      Boston             2024        False
5   Frank   20   Chemistry     90   Francisco             2023        False
```

Series: A Series is a **one-dimensional** labelled array.

```
import pandas as pd

# Create a Series with student names as labels and scores as data
student_scores = pd.Series([85, 92, 78, 95], index=["Alex", "Ben", "Clara", "Der

# Print the Series
print(student_scores)
```

```
#Output:

Alex     85
Ben      92
Clara    78
Deric    95
dtype: int64
```

Data Exploration

- Before performing data manipulation, it’s essential to explore and understand your dataset.
- Pandas provides several handy functions for this purpose.

- **head():** This function returns the first few rows in the DataFrame. By default, it returns the first 5 rows.
- Similarly, **tail()** returns the last few rows in the DataFrame.

```python
# Display the first few rows of the DataFrame
print(df.head())
```

```
#Output:

      Name  Age       Major  Score Hometown  Graduation Year  Scholarship
0    Alex   20         CS     85      York              2022        False
1     Ben   21        Math    92   Angeles              2023         True
2   Clara   19     Physics     78   Chicago              2022        False
3   Deric   22   Economics     95   Houston              2023         True
4     Eva   23     Biology     88    Boston              2024        False
```

describe():

- describe(): This function provides a **statistical summary** of the numerical columns in the DataFrame.

```
# Display a summary of the DataFrame
print(df.describe())
```

```
#Output:

              Age       Score  Graduation Year
count    6.000000    6.000000         6.000000
mean    20.833333   88.000000      2022.833333
std      1.471960    5.966574         0.752773
min     19.000000   78.000000      2022.000000
25%     20.000000   85.750000      2022.250000
50%     20.500000   89.000000      2023.000000
75%     21.750000   91.500000      2023.000000
max     23.000000   95.000000      2024.000000
```

info():

This function displays a concise summary of a DataFrame, including data types, non-null values, and memory usage.

```
# Display information about the DataFrame
print(df.info())
```

```
# Output:

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 7 columns):
 #    Column           Non-Null Count   Dtype
---   ------           --------------   -----
 0    Name             6 non-null       object
 1    Age              6 non-null       int64
 2    Major            6 non-null       object
 3    Score            6 non-null       int64
 4    Hometown         6 non-null       object
 5    Graduation Year  6 non-null       int64
 6    Scholarship      6 non-null       bool
dtypes: bool(1), int64(3), object(3)
memory usage: 422.0+ bytes
```

df.shape :

Used to get a tuple containing the DataFrame’s dimensions.

6 Rows and 7 Col

```
# Display DataFrame Dimensions
print(df.shape)

# Output:
(6, 7)
```

Data Selection and Filtering

- Pandas provides various methods to select and filter data from a DataFrame based on labels, integer positions, or boolean conditions.

Selecting Columns:

```
# Select a single column
df['Major']

# Select multiple columns
df[['Name', 'Age', 'Major']]
```

Filtering Rows:

```
# Filter rows based on a condition
df[df['Score'] > 90]

# Filter rows with multiple conditions
df[(df['Graduation Year'] == 2023) & (df['Scholarship'] == True)]
```

Data Aggregation and Grouping

- Data Aggregation is the process of **combining data in a manner that results in new information.**
- The most common aggregations are **sum, mean, maximum, minimum, etc.**

- Grouping is the process of categorizing data based on given conditions so that a group of data can be analyzed.

```
# Group the data by Major and calculate the mean of the scores
grouped_data = df.groupby('Scholarship')['Score'].mean()

print(grouped_data)
```

```
# Output:

Scholarship
False    85.25
True     93.50
Name: Score, dtype: float64
```

After grouping data, you can apply aggregate functions to each group. This enables you to summarize or calculate statistics for each group, providing comparisons between groups.

Merging and Joining Data

- **Merges:**
- In Pandas, merge is the process of combining two or more DataFrames based on common columns.

The different types of merges are:

- **Inner Merge**: Only the common values in both the left and right dataframes are kept.

- **Left Merge**: Combines all rows from the left DataFrame with matching rows from the right DataFrame, filling missing values with NaN.

- **Right Merge**: Combines all rows from the right DataFrame with matching rows from the left DataFrame, filling missing values with NaN.

- **Outer Merge**: All the values from both the left and right dataframes are kept.

```python
# Create a new dataframe
df_new = pd.DataFrame({'Name': ['Alex', 'Ben', 'Clara', 'Deric'],
'Courses': ['Algorithms', 'Calculus', 'Quantum Physics', 'Macroeconomics']})

# Left merge
left_merge = df.merge(df_new, on='Name', how="left")

print(left_merge)
```

```
# Output:

    Name  Age       Major  Score    Hometown  Graduation Year  Scholarship          
0   Alex   20         CS     85        York              2022         False         A
1    Ben   21        Math     92     Angeles              2023          True          
2  Clara   19     Physics     78     Chicago              2022         False   Quantu
3  Deric   22   Economics     95     Houston              2023          True    Macro
4    Eva   23     Biology     88      Boston              2024         False          
5  Frank   20   Chemistry     90  Francisco              2023         False          
```

Joins:

- This method is used to **join** rows from two or more DataFrames based on their index values.
- The join() method works similarly to merge() but only merges on the index values instead of specific columns.
- Thus, the join() method doesn’t need a common column.
- It’s handy for combining data that has a direct index-based relationship.

The method has four types of joins: Inner Join, Left Join, Right Join, and Outer Join.

```
# Create a new dataframe
df_new = pd.DataFrame({'Courses': ['Algorithms', 'Calculus', 'Quantum Physics',

# Inner join
inner_join = df.join(df_new, how="inner")

print(inner_join)
```

```
# Output:

    Name  Age      Major  Score Hometown  Graduation Year  Scholarship
0   Alex   20         CS     85     York             2022        False
1    Ben   21       Math     92  Angeles             2023         True
2  Clara   19    Physics     78  Chicago             2022        False
3  Deric   22  Economics     95  Houston             2023         True
```

Concatenation:

- **concat()** is used to append or concatenate rows or columns from one DataFrame to another. It doesn't consider common values or matching indexes.

- **Vertically** (axis=0): Appends rows from one DataFrame to another, stacking them vertically.
- **Horizontally** (axis=1): Appends columns from one DataFrame to another, stacking them horizontally.
- Assume we have another DataFrame (df_new) of student information with similar columns:

```
# Concatenate the two DataFrames vertically
pd.concat([df, df_new], axis=0, ignore_index=True)
```

Working with Categorical data

- Working with categorical data in Pandas involves handling variables that represent discrete, non-numerical values, such as gender, nationality, or status.

- Pandas offers a specific data type, **Categorical**, which provides more efficient storage and additional methods for working with categorical variables.

```python
# Create a dictionary with student data
student_data = {
    "Name": ["Alex", "Ben", "Clara", "Deric", "Eva", "Frank"],
    "Age": [20, 21, 19, 22, 23, 20],
    "Major": ["CS", "Math", "Physics", "Economics", "Biology", "Chemistry"],
    "Score": [85, 92, 78, 95, 88, 90],
    "Hometown": ["York", "Angeles", "Chicago", "Houston", "Boston", "Francisco"]
    "Graduation Year": [2022, 2023, 2022, 2023, 2024, 2023],
    "Scholarship": [False, True, False, True, False, False]
}

# Create a DataFrame from the dictionary
df = pd.DataFrame(student_data)

# Print the DataFrame
print(df)
```

```
#Output:

      Name   Age        Major   Score    Hometown   Graduation  Year   Scholarship
0     Alex    20           CS      85        York                2022         False
1      Ben    21         Math      92     Angeles                2023          True
2    Clara    19      Physics      78     Chicago                2022         False
3    Deric    22    Economics      95     Houston                2023          True
4      Eva    23      Biology      88      Boston                2024         False
5    Frank    20    Chemistry      90   Francisco               2023         False
```

```python
# Convert a column to Categorical
df['Major'] = df['Major'].astype('category')

print(df['Major'])
```

```
# Output:

0           CS
1         Math
2      Physics
3    Economics
4      Biology
5    Chemistry
Name: Major, dtype: category
Categories (6, object): ['Biology', 'CS', 'Chemistry', 'Economics', 'Math', 'Phy
```

Data Visualization

- With pandas, quickly create visual graphs like line plots, bar charts, and scatter plots from your data tables.

- These graphs help you understand patterns and relationships in your data easily, making it simpler to communicate your findings visually.

```
# Line Plot
df['Score'].plot()

# Histogram
df['Age'].hist()

# Scatter Plot
df.plot.scatter(x='Age', y='Score')

# Box Plot
df.boxplot(column='Score', by='Major')
```

To display the plots created with Pandas, you need to use `plt.show()` from the Matplotlib library.

Pivot Table: Tool

- A **pivot table is a data analysis tool** that summarizes, sorts, and analyzes large datasets by rearranging and "pivoting" rows and columns.

- It calculates aggregations like sums, averages, or counts to reveal patterns, trends, and comparisons within data without requiring complex formulas

Fancy Indexing in NumPy

- Fancy indexing in NumPy is a method to select multiple elements from an array using arrays or lists of specific index where, index is used to represent the position of element in the array.
- Instead of picking elements one by one, you can select multiple elements at once on your choice.

- ***Fancy indexing is the advance form of simple indexing. In simple indexing we access single elements or slices using integers, while fancy indexing access multiple elements using arrays or lists of integers. It returns a new array that is independent of the original one.***

Example: Simple Indexing

- Let us create a 1D array to access a single element using it's position. In the below code arr[3] access the element at index 3 (fourth position) of array which is 60.

- import numpy as np
- x = np.array([50, 90, 70, 60, 40, 100])
- print("By using simple indexing:" ,x[3])

Example1:

- Let us create a 1D array using arange() function that includes numbers from 20 to 30, then we will create a second NumPy array for indexing multiple elements at once.
- In the below code we have created a second array which contains indices to access multiple elements i.e. 3, 4, 6 and the elements in those positions are 23, 24, 26.

Cont..

- import numpy as np
- x = np.arange(20, 30)
- print(x)
- arr = np.array([3, 4, 6])
- print("The elements at 3, 4, 6 positions are :\n" , x[arr])

Following is the output of the above code –

[32 57 48 26 47 32 38 35 30 36]

Accessing multiple elements using python list: [57 32 48]

Converting 1D to 2D Array Using Fancy Indexing

- In this example, we used the arange() function to build a one-dimensional array containing numbers from 1 to 10.
- Here the elements we need to access is specified in 2D.
- In fancy indexing the shape of result is reflected by the shape of index array.

Following is the code –

- import numpy as np
- x = np.arange(1, 10)
- indices = np.array([[5, 3], [4, 5]])
- new_2D_arr = x[indices]
- print(new_2D_arr)

Cont..

Following is the output of the above code –

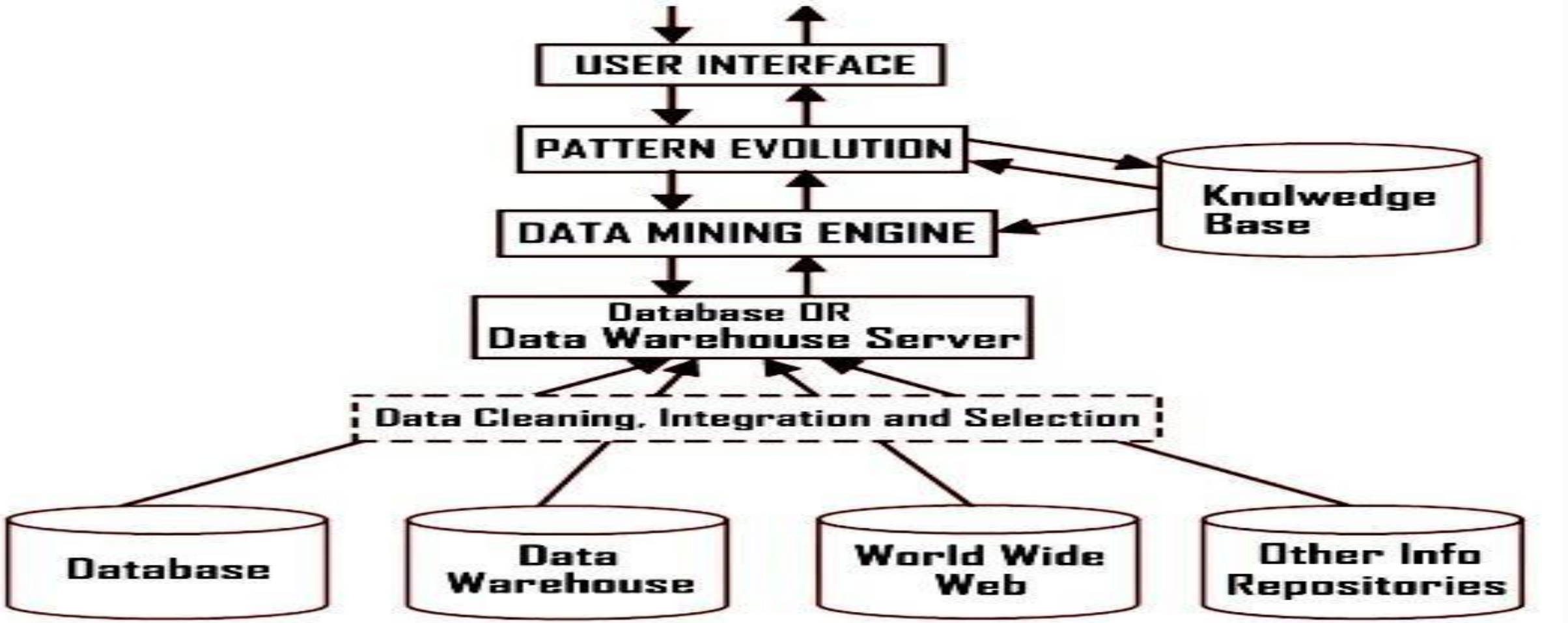
- [[6 4]
- [5 6]]

Data Mining

- Data mining is a very important process where potentially useful and previously unknown information is extracted from large volumes of data.
- There are several components involved in the data mining process.

- The major components of any data mining system are data source, **data warehouse server**, data mining engine, pattern evaluation module, graphical user interface and knowledge base.

Data Mining Architecture:



Data mining Engine

- The data mining engine is the **core component** of any data mining system.
- It consists of **several modules** for performing data mining tasks including **association**, **classification**, **characterization**, **clustering**, **prediction**, **time-series analysis** etc.

Pattern Evaluation Modules

- The pattern evaluation module is mainly responsible for the **measure of interestingness of the pattern** by using a **threshold value.** It interacts with the data mining engine to focus the search towards interesting patterns.

- **Graphical User Interface**

The graphical user interface module provides the **communication** between the **user** and the **data mining system**.

This module helps the user use the system easily and efficiently without knowing the real complexity behind the process.

- When the user specifies a query or a task, this module interacts with the data mining system and displays the result in an easily understandable manner.

Knowledge Base

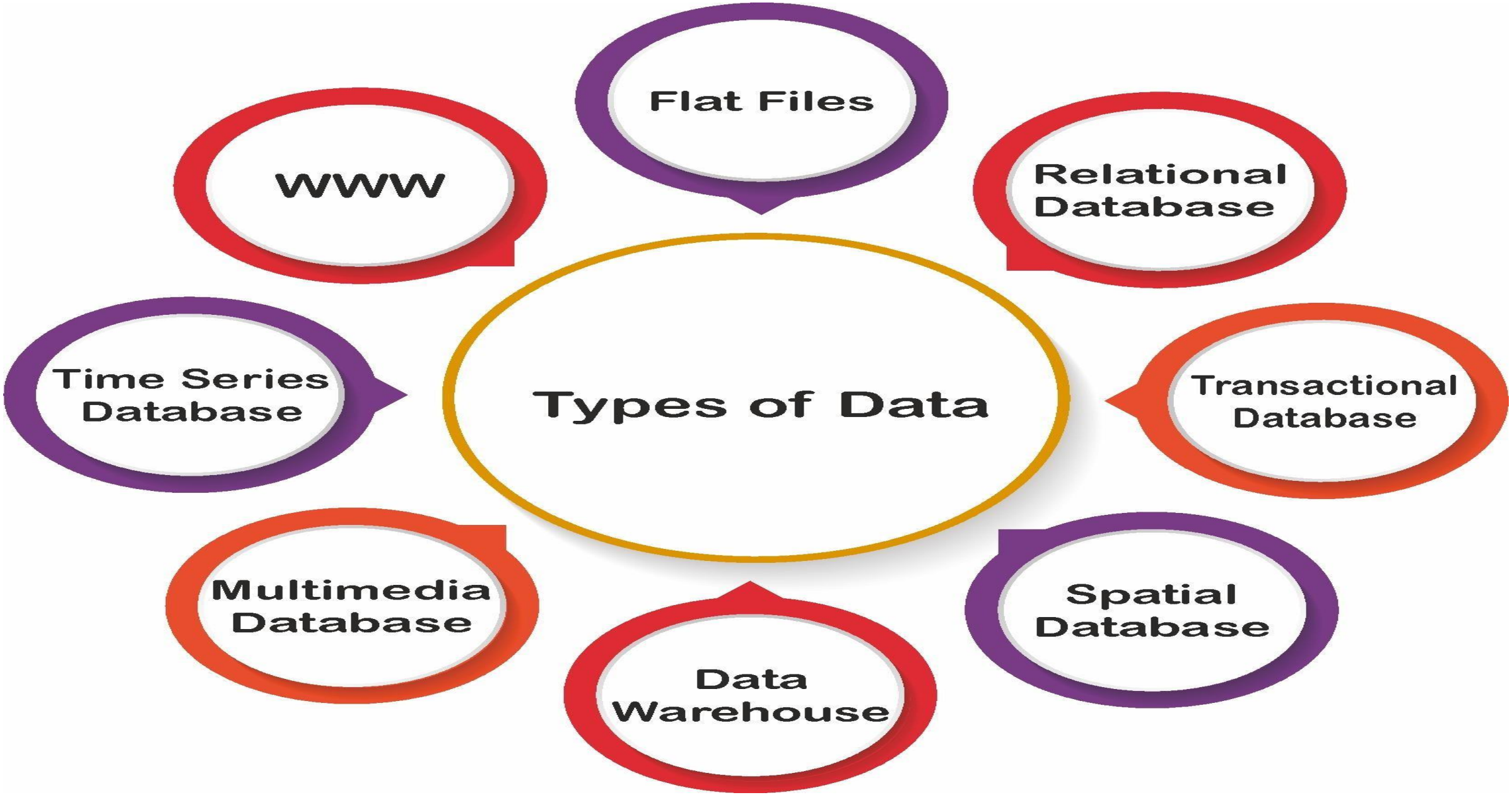
- The knowledge base is **helpful** in the **whole data mining process**.
- It might be useful for guiding the search or evaluating the interestingness of the result patterns.
- The knowledge base might even contain user beliefs and data from user experiences that can be useful in the process of data mining.

Types of Data:
What Kinds of Data Can Be Mined

Basic forms of data for mining

- Database Data (or) Relational database
- Data warehouse data
- Transactional data

- **other forms of data for mining**
- Multimedia Database
- Spatial Database
- World Wide Web
- Text data (Flat File)
- Time series database

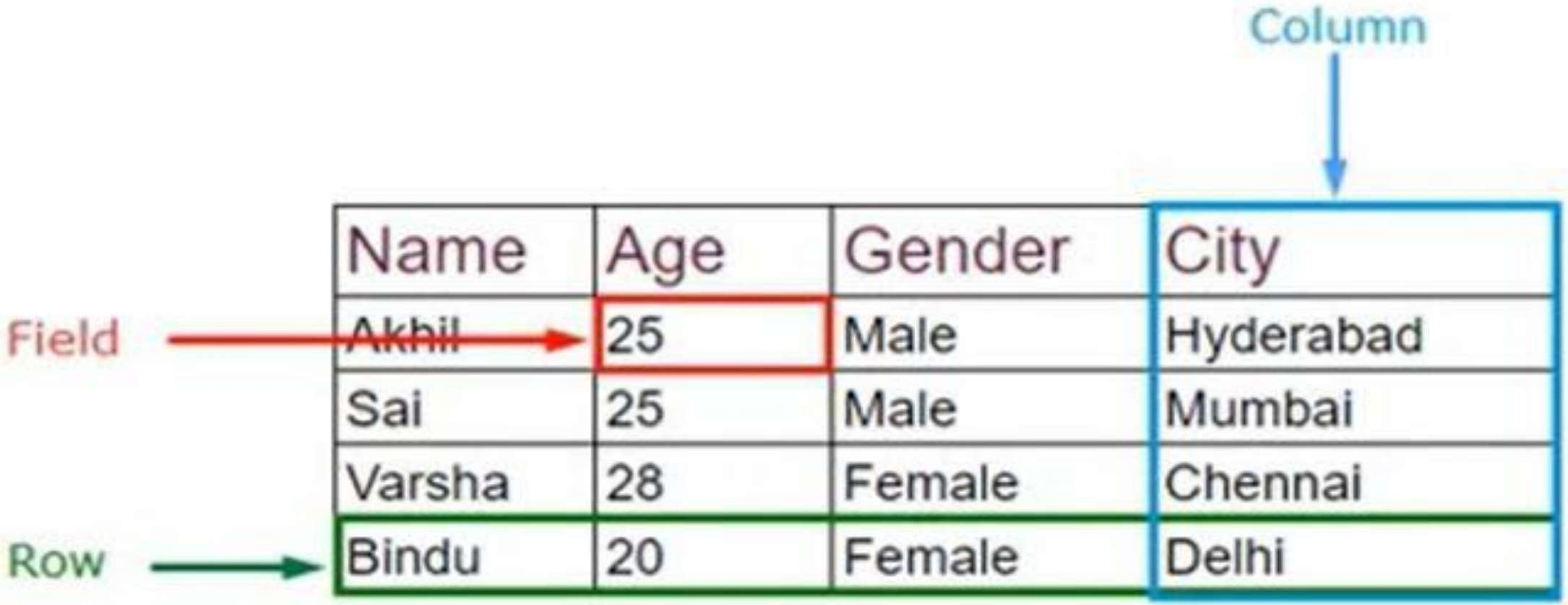

Types of Data
Flat Files
WWW
Relational Database
Time Series Database
Transactional Database
Multimedia Database
Data Warehouse
Spatial Database

Database Data (or) Relational database

- A database system, also called a database management system (DBMS), consists of a collection of interrelated data, known as a database, and a set of software programs to manage and access the data.

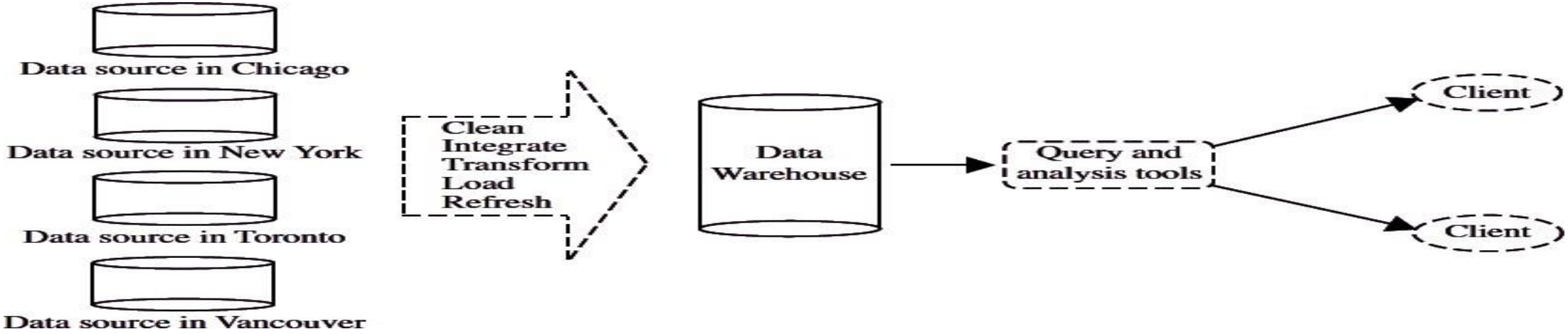
- A relational database: is a collection of tables, each of which is assigned a unique name, each table consists of a set of attributes (columns or fields) and usually stores a large set of tuples (records or rows).
- Each tuple in a relational table represents an object identified by a unique key and described by a set of attribute values.

Example:


Column
Field
Row
Name
Age
Gender
City
Akhil
25
Male
Hyderabad
Sai
25
Male
Mumbai
Varsha
28
Female
Chennai
Bindu
20
Female
Delhi

Data warehouse data

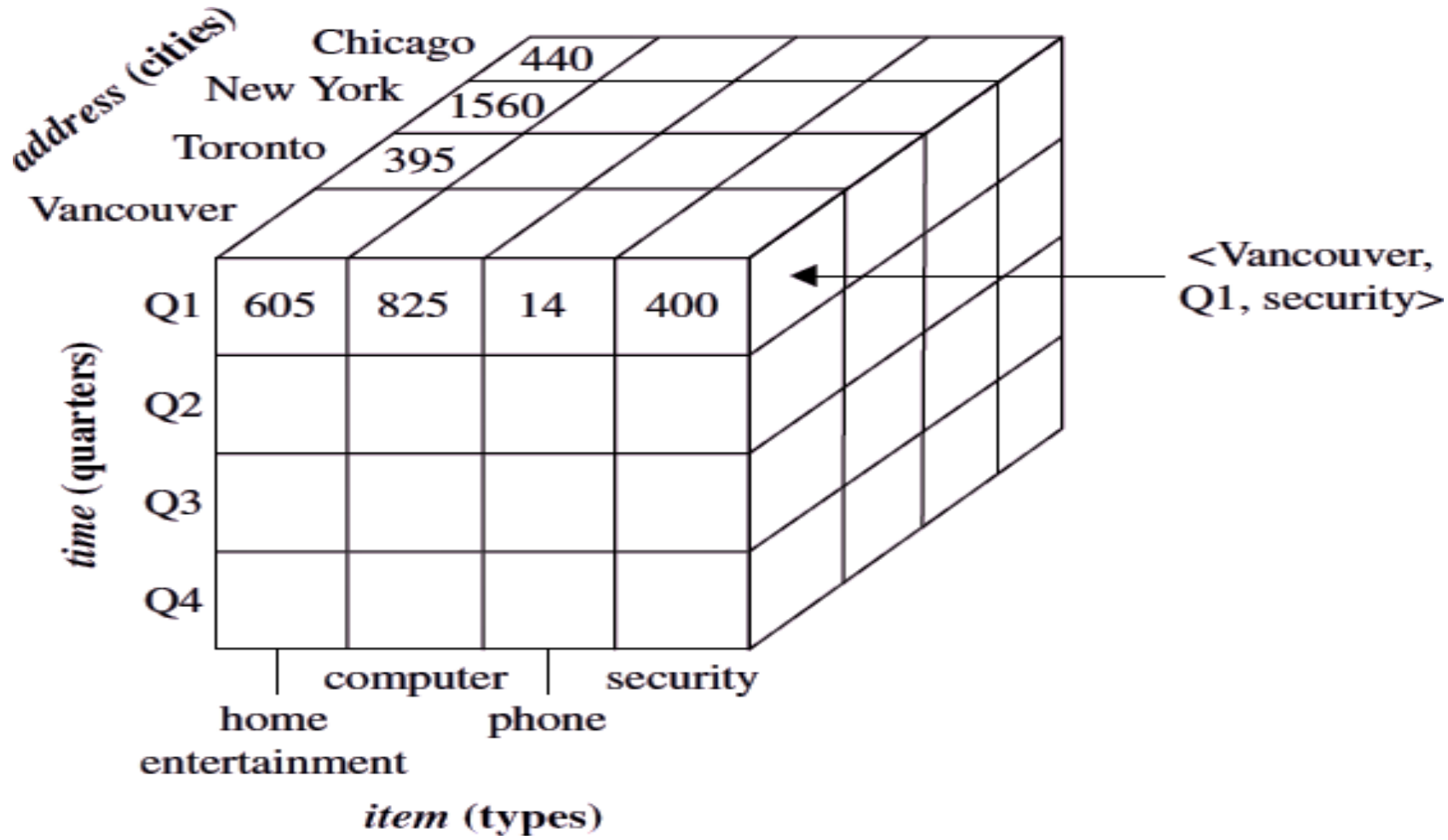
- A data warehouse is a repository of information collected from multiple sources, **stored under a unified schema, and usually residing at a single site.**
- Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing.



Cont..

- A data warehouse is defined as the collection of data integrated from multiple sources.
- Later this data can be mined for decision making.
- A data warehouse is usually **modelled by a multidimensional data structure**, called **a data cube**, in which each dimension corresponds to an attribute or a set of attributes in the schema, and each cell stores the value of some aggregate measure such as count or sum.
- A data cube provides a multidimensional view of data and **allows the precomputation and fast access of summarized data.**

Example:


address (cities)
Chicago
New York
Toronto
Vancouver
440
1560
395
Q1
Q2
Q3
Q4
605
825
14
400
time (quarters)
<Vancouver, Q1, security>
computer
security
home entertainment
phone
item (types)

Transactional data

- Transactional database is a collection of **data organized by time stamps, date etc to represent transaction in databases**. In general, each record in a transactional database captures a transaction, **such as a customer's purchase, a flight booking, or a user's clicks on a web page.**
- A transaction typically includes a unique transaction identity number (trans ID) and a list of the items making up the transaction, such as the items purchased in the transaction.

- This type of database has the **capability to roll back or undo operation when a transaction is not completed or committed**. And it follows ACID property of DBMS.

Example:

- TID Items
- T1 Bread, Coke, Milk
- T2 Popcorn, Bread
- T3 Popcorn, Coke, Egg, Milk
- T4 Popcorn, Bread, Egg, Milk
- T5 Coke, Egg, Milk

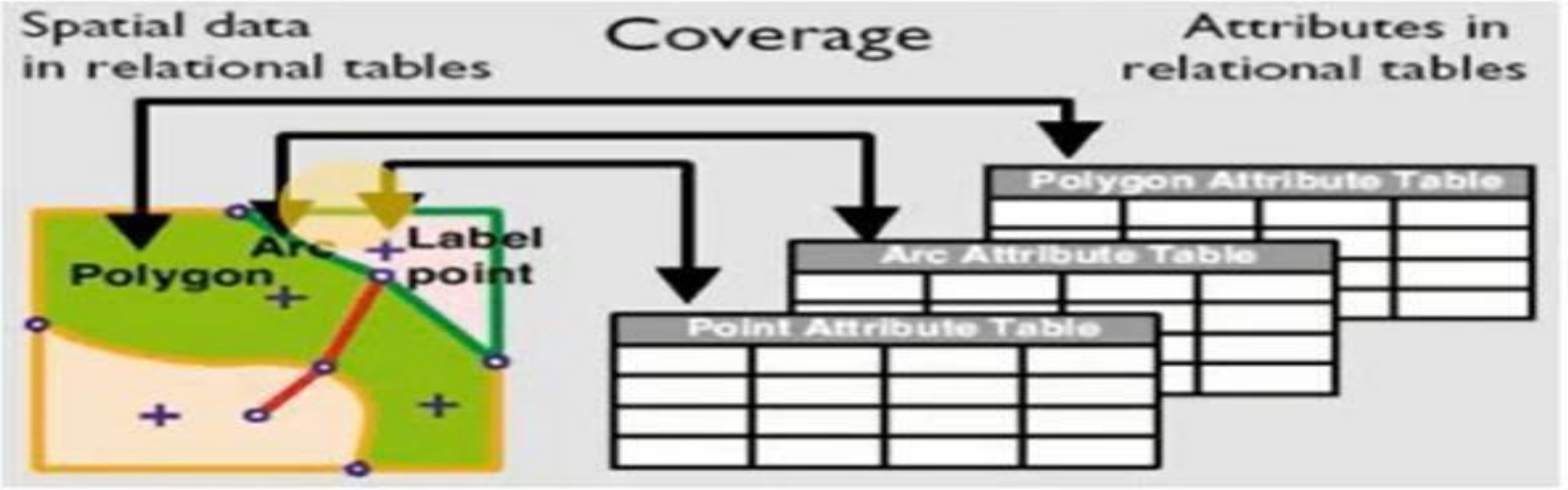
Multimedia database

- The multimedia databases are used to store multimedia data such as **images, animation, audio, video along with text.**
- This data is stored in the form of multiple file types like **.txt**(text), **.jpg**(images), **.swf**(videos), **.mp3**(audio) etc.



Spatial database

- A spatial database is a database that is **enhanced to store and access spatial data or data that defines a geometric space.**
- These data are often **associated with geographic locations** and features, or constructed features like cities. **Data on spatial databases are stored as coordinates, points, lines, polygons and topology**.



World Wide Web

- The World Wide Web is a collection of documents and resources such as audio, video, and text. It identifies all this by URLs of the web browsers which are linked through HTML pages.
- **Online shopping, job hunting, and research are some uses.**
- it is the most heterogeneous repository as it collects data from multiple resources.
- it is **dynamic in nature as Volume of data is continuously increasing and changing.**



Text data (Flat File)

- **Flat files are a type of structured data that are stored in a plain text format.**
- They are **called "flat" because they have no hierarchical structure, unlike a relational database table.**
- Flat files typically consist of rows and columns of data, with each row representing a single record and each column representing a field or attribute within that record.
- They can be stored in various formats such as CSV, tab-separated values (TSV) and fixed-width format.
- Flat files is defined as data files in text form or binary form with a structure that can be easily extracted by data mining algorithms.
- **Data stored in flat files have no relationship or path among themselves,** like **if a relational database is stored on flat file, then there will be no relations between the tables.**

Example:

*Untitled - Notepad

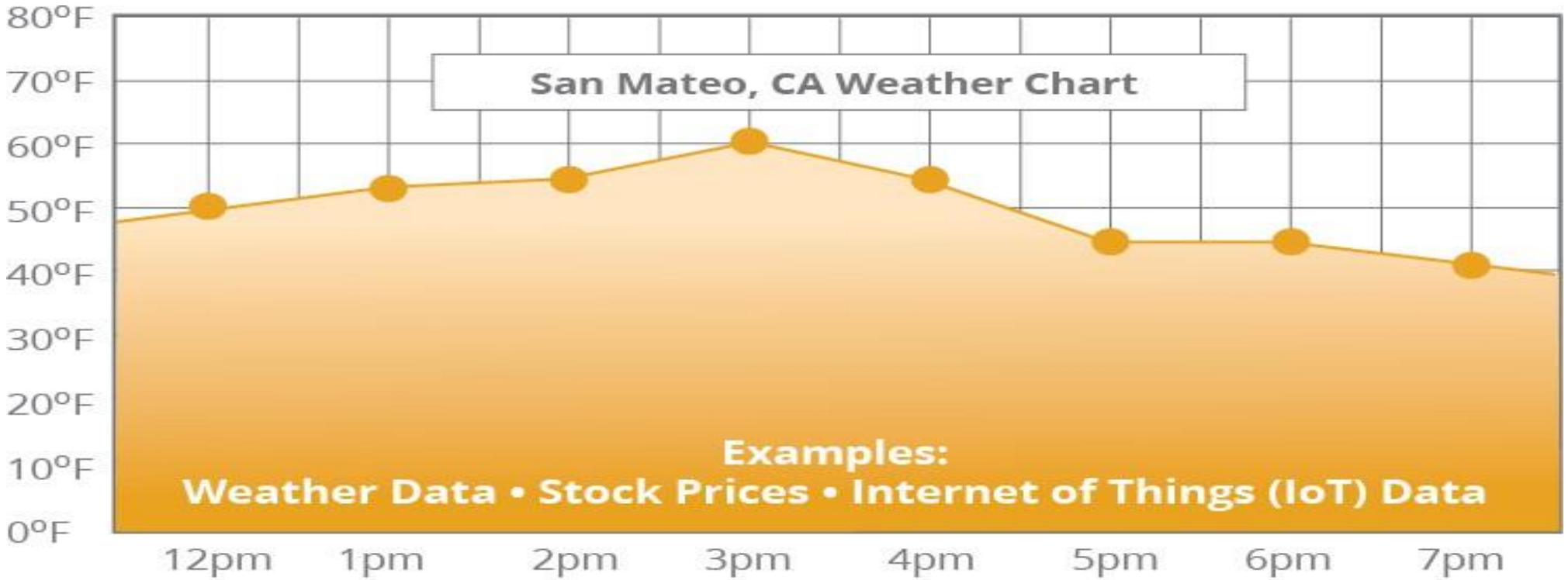
File Edit Format View Help

```
"OrderID", "CustomerID", "OrderDate"
"01","001", "06/06/2021"
"02","369", "06/06/2021"
"03","151", "06/06/2021"
"04","014", "06/06/2021"
"05","061", "06/06/2021"
"06","220", "06/06/2021"
```

Time series database

- **Time-series data is a sequence of data points collected over time intervals, allowing us to track changes over time.**
- Time-series data can track changes over milliseconds, days, or even years.
- A time series database (TSDB) is a database optimized for time-stamped or time series data.
- Time series data are simply measurements or events that are tracked, monitored, down-sampled, and aggregated over time.
- This could be server metrics, **application performance monitoring**, network data, **sensor data**, events, clicks, **trades in a market,** and many other types of analytics data.

Example:



A Time Series Database is a database that contains data for each point in time.

Data Mining Functionalities

- Data mining is important because there is so much data out there, and it's impossible for people to look through it all by themselves.
- Data mining uses various functionalities to analyze the data and find patterns, trends, and other information that would be hard for people to find on their own.
- Data mining functionalities are used to specify the kinds of patterns to be found in data mining tasks.
- In general, such data mining tasks can be classified into two categories: **descriptive** and **predictive**.

Descriptive data mining

- This kind of mining **focuses on transforming raw data into information that can be used in reports and analyses.**

- It provides **certain knowledge about the data, for instance, count, average**.
- **It gives information about what is happening inside the data without any previous idea**.

- It exhibits the common features in the data.
- To know the general properties of the data present in the database.

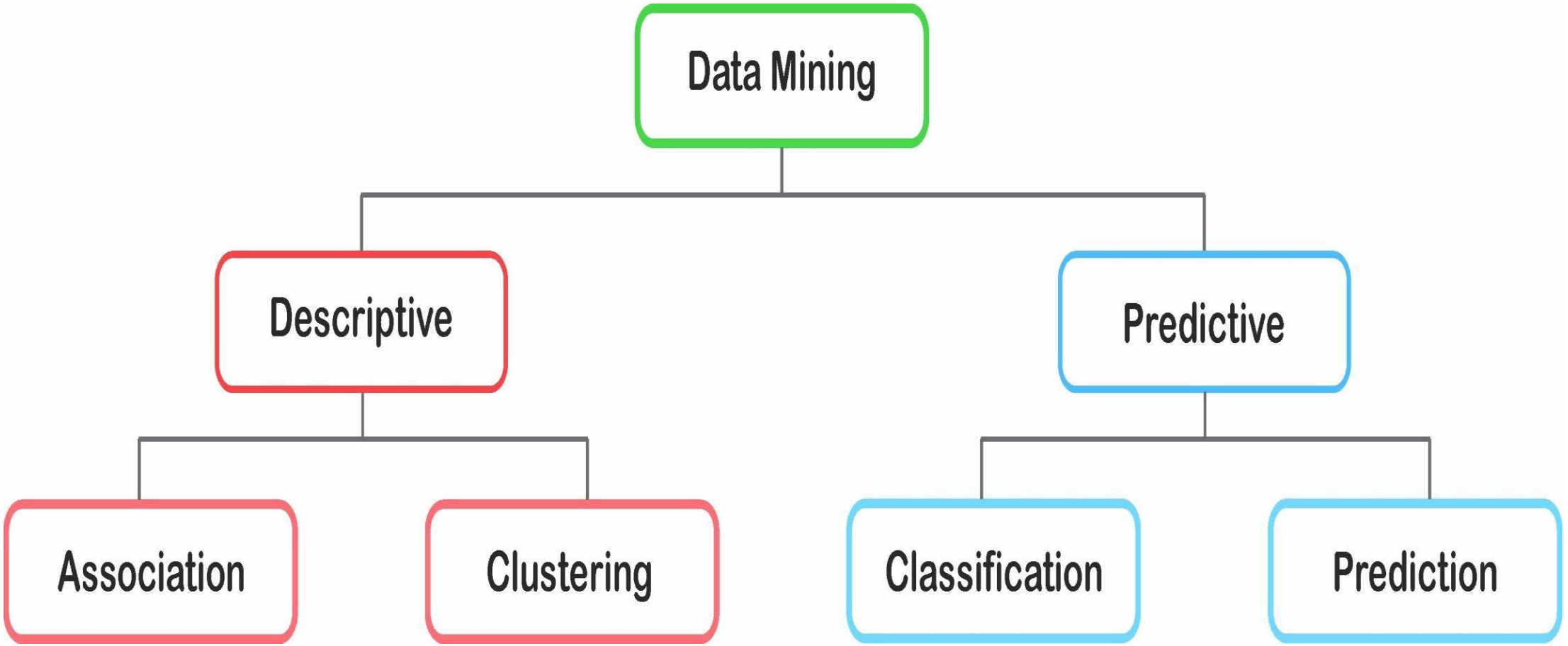
Predictive data mining

- These kind of mining tasks perform **inference on the current data in order to make predictions.**
- This **helps the developers in understanding the characteristics that are not explicitly available**.
- For instance, the prediction of business analysis in the next quarter with the performance of the previous quarters.
- In general, **the predictive analysis predicts or infers the characteristics with the previously available data.**

Data mining functionalities:

- Class/Concept Description (Characterization and Discrimination)
- Classification
- Prediction
- Association Analysis
- Cluster Analysis
- Outlier Analysis

Data mining functionalities chart:



Class/Concept Description: Characterization and Discrimination

- **Data is associated with classes or concepts.**
- **Class: A collection of things sharing a common attribute**

Example: Classes of items – computers and printers

- **Concept: An abstract or general idea derived from specific instances.**

Example: Concepts of customers – bigSpenders and budgetSpenders.

- It can be useful to describe individual classes and concepts in summarized, concise, and yet precise terms. Such descriptions of a class or a concept are called **class/concept descriptions**.
- These descriptions can be derived using **data characterization** and **data discrimination**, or both.

Data characterization

Data characterization is a summarization of the general characteristics or featuresof a target class of data.

Data summarization can be done based on statistical measures and plots.

The output of data characterization can be presented in various forms it includes pie charts, bar charts, curves, and multidimensional data cubes.

Example: A customer relationship manager at AllElectronics may order thefollowing data mining task: Summarize the characteristics of customers who spend morethan **$5000** a year at AllElectronics.

The result is a general profile of these customers,such as that they are **40 to 50 years old**, employed, and have **excellent credit ratings**

Data Discrimination

Data discrimination is one of the functionalities of data mining.
It compares the data between the two classes. Generally, it maps the target class with a predefined group or class. It compares and contrasts the characteristics of the class with the predefined class using a set of rules called discriminate rules.

Example: A customer relationship manager at AllElectronics may want to compare two groups of customers those who shop for computer products regularly(e.g., more than twice a month) and those who rarely shop for such products (e.g.,less than three times a year).

The resulting description provides a general comparativeprofile of these customers, such as that 80% of the customers who frequently purchasecomputer products are between 20 and 40 years old and have a university education,whereas 60% of the customers who infrequently buy such products are either seniors oryouths, and have no university degree.

Classification:

- Classification is a data mining technique that categorizes items in a collection based on some predefined properties.

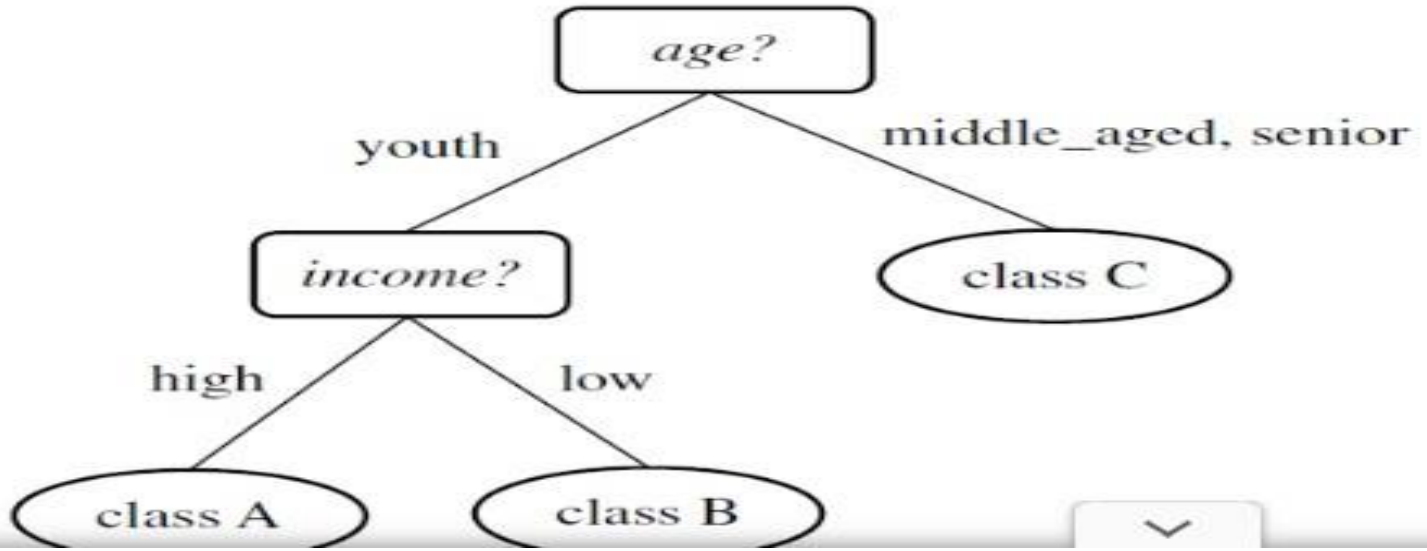
- It uses methods like IF-THEN, Decision trees or neural networks to predict a class or essentially classify a collection of items.

- Classification is a supervised learning technique used to categorize data into predefined classes or labels.

Example:

Example:

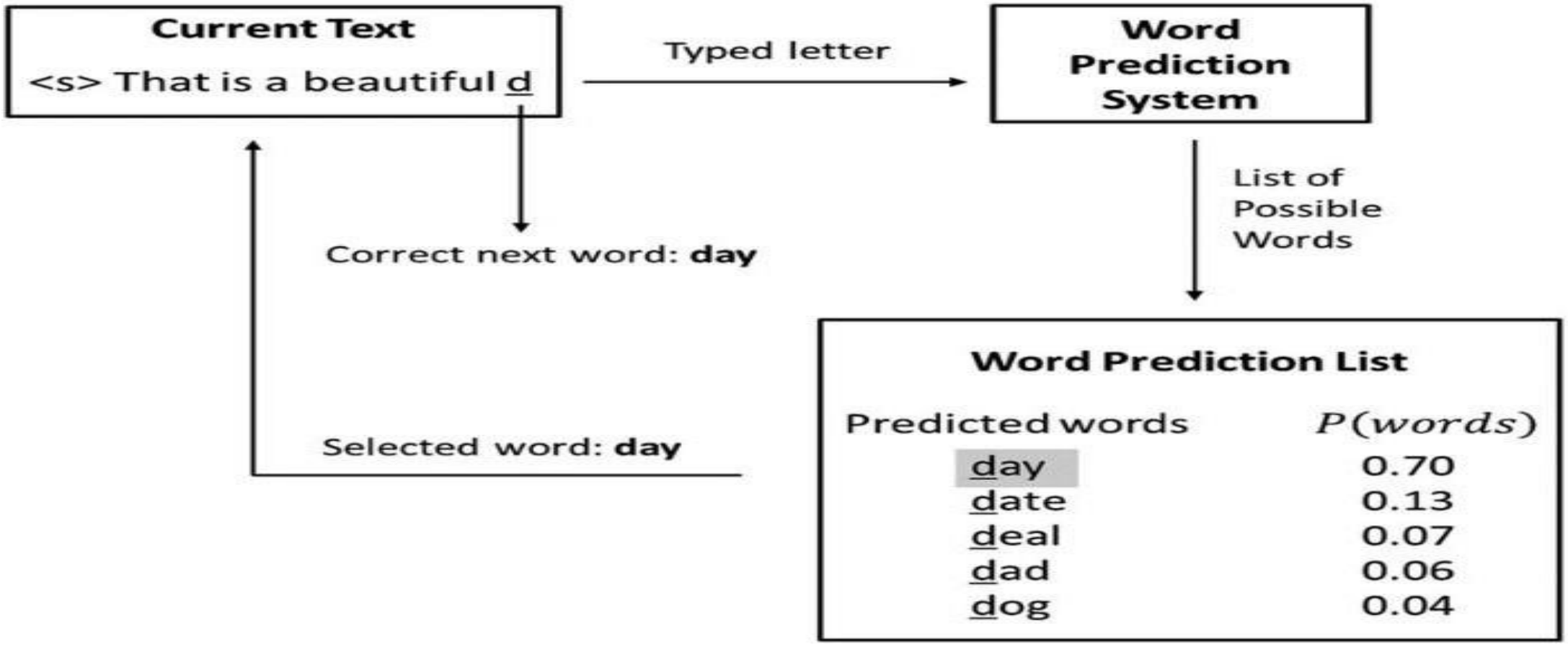
age(X, "youth") AND income(X, "high") → *class(X, "A")*
age(X, "youth") AND income(X, "low") → *class(X, "B")*
age(X, "middle_aged") → *class(X, "C")*
age(X, "senior") → *class(X, "C")*



Prediction

- Finding missing data in a database is very important for the accuracy of the analysis.
- Prediction is one of the data mining functionalities that help the analyst find the missing numeric values.
- If there is a missing class label, then this function is done using classification.
- It is very important in business intelligence and is very popular.
- One of the methods is to predict the missing or unavailable data using prediction analysis.

Example:


Current Text
<s> That is a beautiful d
Typed letter
Word Prediction System
List of Possible Words
Correct next word: day
Selected word: day
Word Prediction List
Predicted words
P(words)
day 0.70
date 0.13
deal 0.07
dad 0.06
dog 0.04

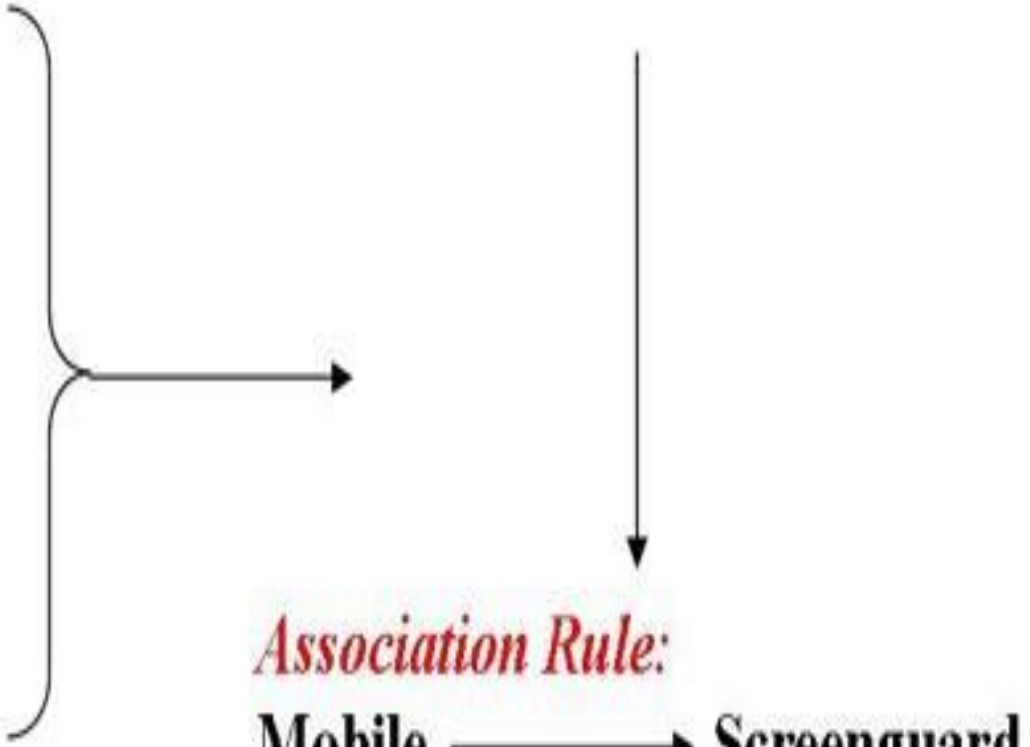
Association Analysis

- Association Analysis is a functionality of data mining. It relates two or more attributes of the data.
- It discovers the relationship between the data and the rules that are binding them.
- It is also known as Market Basket Analysis for its wide use in retail sales.
- The suggestion that Amazon shows on the bottom, “Customers who bought this also bought.” is a real-time example of association analysis.
- It relates two transactions of similar items and finds out the probability of the same happening again.
- This helps the companies improve their sales of various items.

Example:

Transaction Data:

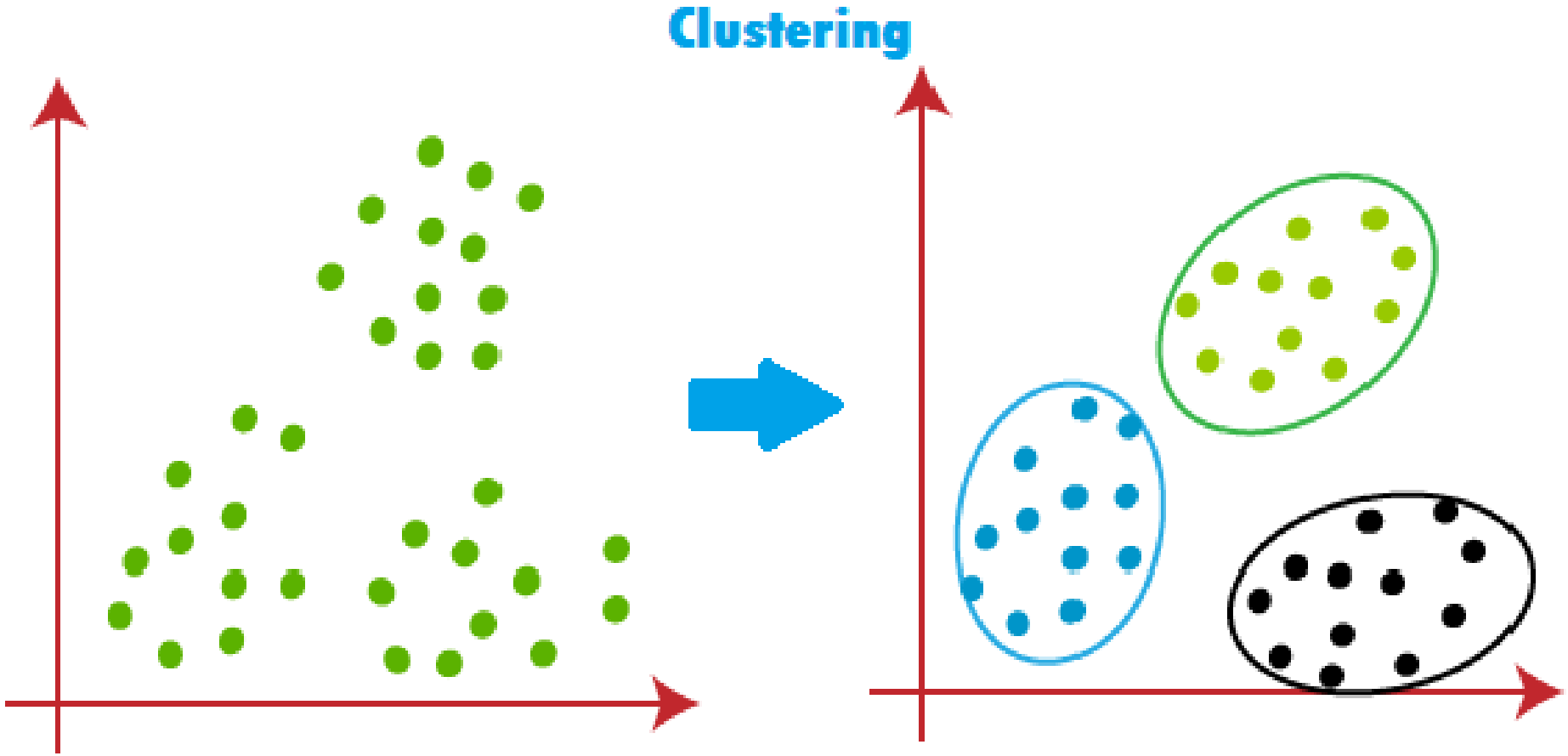
TID	Items
T1	Mobile, HeadPhones, Router
T2	Mobile, Screenguard, Cable
T3	Mobile, Screenguard, Backcover, HDD
T4	Mobile, Powerbank, Mouse
T5	Mobile, Screenguard, Pendrive



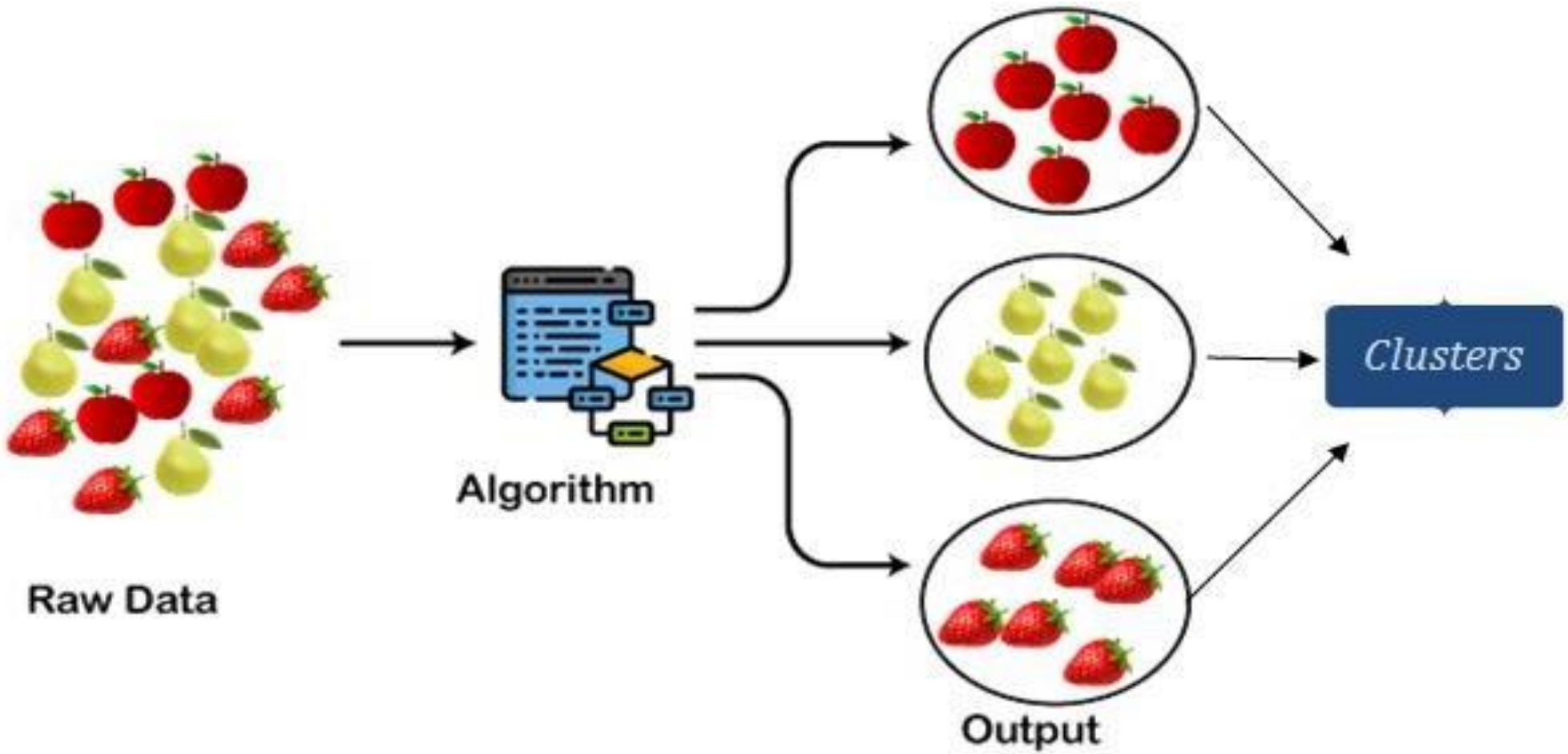
Cluster Analysis

- Clustering is an **unsupervised learning** technique **that group's similar data points together based on their features.**
- The goal is to identify underlying structures or patterns in the data. Some common clustering algorithms include K-means, hierarchical clustering, and DBSCAN.
- This data mining functionality is similar to classification. But in this case, the **class label is unknown.**
- **Similar objects are grouped in a cluster.**
- There are vast differences between one cluster and another.

Example1:



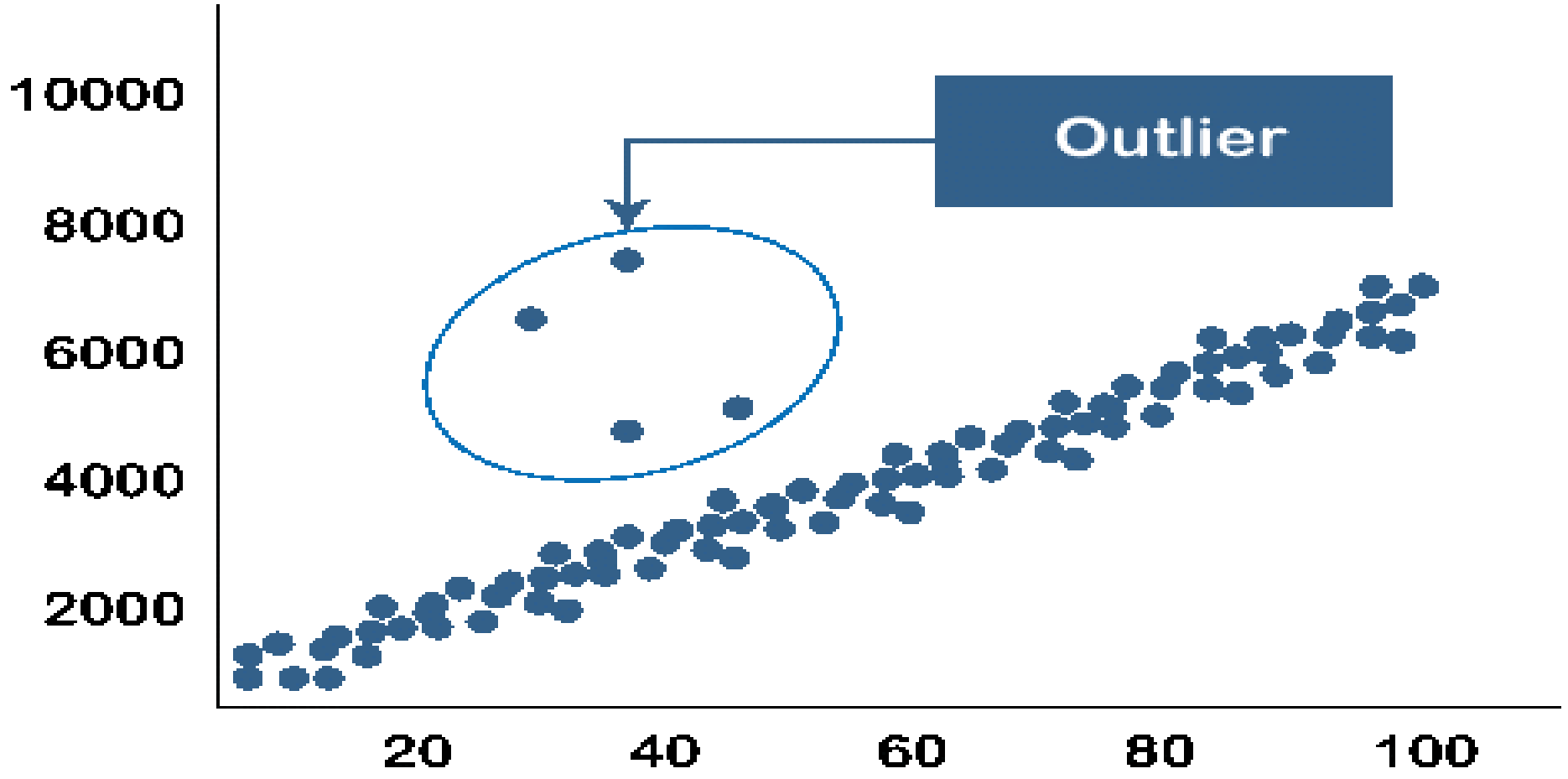
Example2: Clustering



Outlier Analysis

- **When data that cannot be grouped in any of the class appears, we use outlier analysis.**
- There will be occurrences of **data that will have different attributes/features to any of the other classes or clusters.** These outstanding **data are called outliers.**
- They are **usually considered noise or exceptions**, and the **analysis of these outliers is called outlier mining.**
- Outlier analysis is important to understand the **quality of data.**
- **If there are too many outliers, you cannot trust the data or draw patterns out of it.**

Example1:



Example2:

Average weight of first 4 kids = (30 + 35 + 40 + 50)/4 = 38.75 kg **Without Outliers**

Average weight of all kids = (30 + 35 + 40 + 50 + 300)/5 = 91 kg **With Outliers**



30 kg



35 kg



40 kg



50kg



300 kg

What is Data Preprocessing?

- Data preprocessing is a crucial step in data mining.
- It involves **transforming raw data into a clean, structured, and suitable format for mining.**
- Proper **data preprocessing helps improve the quality of the data**, enhances the **performance of algorithms, and ensures more accurate and reliable results.**

Why Preprocess the Data?

- In the real world, **many databases and data warehouses have noisy, missing, and inconsistent data due to their huge size.**
- **Low quality data leads to low quality data mining**.

Cont..

Noisy: Containing errors or outliers. **E.g., Salary = “-10”**

- Noisy data may come from Human or computer error at data entry.
- Errors in data transmission.

Missing: lacking certain attribute values or containing only aggregate data. **E.g., Occupation = “”**

- Missing (Incomplete) may data come from “Not applicable” data value when collected.
- Human/hardware/software problems.

Cont..

Inconsistent: Data inconsistency meaning is that different versions of the same data appear in different places.

For **example**, the ZIP code is saved in one table as **1234-567** numeric data format; while in another table it may be represented in **1234567.**

- Inconsistent data may come from Errors in data entry.
- Merging data from different sources with varying formats.
- Differences in the data collection process.
- Data preprocessing is used to improve the quality of data and mining results.
- The goal of data preprocessing is to enhance the accuracy, efficiency, and reliability of data mining algorithms.

Major Tasks in Data Preprocessing

- Data preprocessing is **an essential step in the knowledge discovery process**, because **quality decisions must be based on quality data.**
- Data Preprocessing involves **Data Cleaning, Data Integration, Data Reduction and Data Transformation.**

Steps in Data Preprocessing

1. Data Cleaning

- **Data cleaning is a process that "cleans" the data by filling in the missing values, smoothing noisy data, analyzing, and removing outliers, and removing inconsistencies in the data**.
- If users believe the data are dirty, they are unlikely to trust the results of any data mining that has been applied.
- Real-world data tend to be incomplete, noisy, and inconsistent.
- Data cleaning (or data cleansing) routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data.

Missing Values

- **Case Study:** Imagine that you need to analyze All Electronics sales and customer data. You note that many tuples have no recorded value for several attributes such as customer income. How can you go about filling in the missing values for this attribute?

There are several methods to fill the missing values.

Ignore the tuple: This is usually done when the class label is missing(classification). This method is not very effective, unless the tuple contains several attributes with missing values.

Fill in the missing value manually: In general, this approach is time consuming and may not be feasible given a large data set with many missing values.

Use a global constant to fill in the missing value: Replace all missing attribute values the same constant such as a label like “Unknown” or “- ∞ “.

Use the attribute mean or median to fill in the missing value: Replace all missing values in the attribute by the mean or median of that attribute values.

Noisy Data

- Noise is a random error or variance in a measured variable.Data smoothing techniques are used to eliminate noise and extract the useful patterns. The different techniques used for data smoothing are:
- Binning: Binning methods smooth a sorted data value by consulting its "neighbourhood," that is, the values around it. The sorted values are distributed into several "buckets," or bins. Because binning methods consult the neighbourhood of values, they perform local smoothing.

There are three kinds of binning. They are:

- **Smoothing by Bin Means:** In this method, each value in a bin is replaced by the mean value of the bin. For example, the mean of the values 4, 8, and 15 in Bin 1 is 9. Therefore, each original value in this bin is replaced by the value 9.
- **Smoothing by Bin Medians:** In this method, each value in a bin is replaced by the median value of the bin. For example, the median of the values 4, 8, and 15 in Bin 1 is 8. Therefore, each original value in this bin is replaced by the value 8.
- **Smoothing by Bin Boundaries:** In this method, the minimum and maximum values in each bin are identified as the bin boundaries. Each bin value is then replaced by the closest boundary value.
- For example, the middle value of the values 4, 8, and 15 in Bin 1is replaced with nearest boundary i.e., 4.

Example:

- Sorted data for price (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-frequency) bins:
Bin 1: 4, 8, 15
Bin 2: 21, 21, 24
Bin 3: 25, 28, 34

Smoothing by bin means:
Bin 1: 9, 9, 9
Bin 2: 22, 22, 22
Bin 3: 29, 29, 29

Smoothing by bin medians:
Bin 1: 8, 8, 8
Bin 2: 21, 21, 21
Bin 3: 28, 28, 28

Smoothing by bin boundaries:
Bin 1: 4, 4, 15
Bin 2: 21, 21, 24
Bin 3: 25, 25, 34

Cont..

- **Regression:** Data smoothing can also be done by regression, a technique that used to predict the numeric values in a given data set. **It analyses the relationship between a target variable (dependent) and its predictor variable (independent).**
  - Regression is a form of a supervised machine learning technique that tries to predict any continuous valued attribute.
  - Regression done in two ways; Linear regression involves finding the "best" line to fit two attributes (or variables) so that one attribute can be used to predict the other. Multiple linear regression is an extension of linear regression, where more than two attributes are involved and the data are fit to a multidimensional surface.

- **Clustering:**It supports in identifying the outliers. The similar values are organized into clusters and those values which fall outside the cluster are known as outliers.

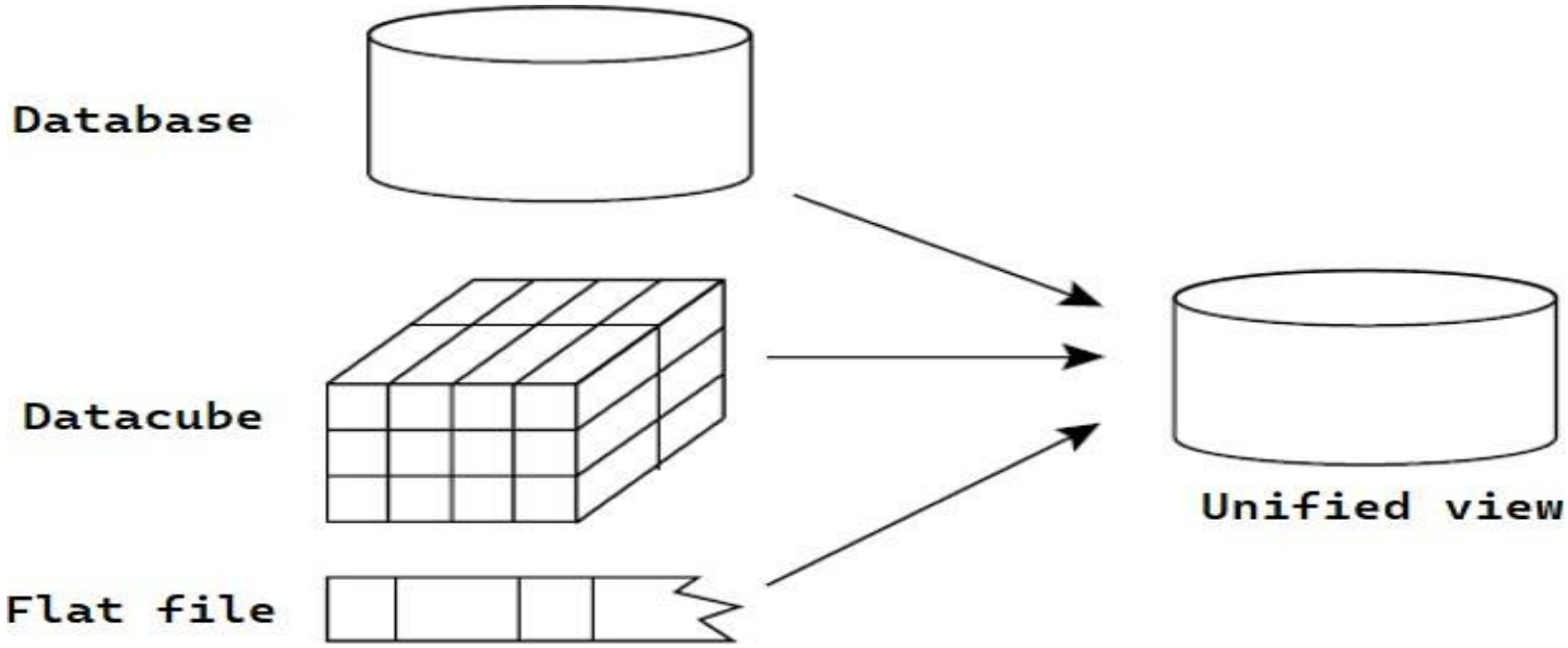
2. Data Integration

- Data integration is the process of combining data from multiple sources into a single, unified view.
- This process involves identifying and accessing the different data sources, mapping the data to a common format. Different data sources may include multiple data cubes, databases, or flat files.

- The goal of data integration is to make it easier to access and analyze data that is spread across multiple systems or platforms, in order to gain a more complete and accurate understanding of the data.

- Data integration strategy is typically described using a triple (G, S, M) approach, where G denotes the global schema, S denotes the schema of the heterogeneous data sources, and M represents the mapping between the queries of the source and global schema.

Example:



Example:

To understand the (G, S, M) approach, let us consider a data integration scenario that aims to combine employee data from two different HR databases, database A and database B. The global schema (G) would define the unified view of employee data, including attributes like EmployeeID, Name, Department, and Salary.

In the schema of heterogeneous sources, database A (S1) might have attributes like EmpID, FullName, Dept, and Pay, while database B's schema (S2) might have attributes like ID, EmployeeName, DepartmentName, and Wage.
The mappings (M) would then define how the attributes in S1 and S2 map to the attributes in G, allowing for the integration of employee data from both systems into the global schema.

Issues in Data Integration

- There are several issues that can arise when integrating data from multiple sources, including:
- **Data Quality:** Data from different sources may have varying levels of accuracy, completeness, and consistency, which can lead to data quality issues in the integrated data.

- **Data Semantics:** Integrating data from different sources can be challenging because the same data element may have different meanings across sources.

- **Data Heterogeneity:** Different sources may use different data formats, structures, or schemas, making it difficult to combine and analyze the data.

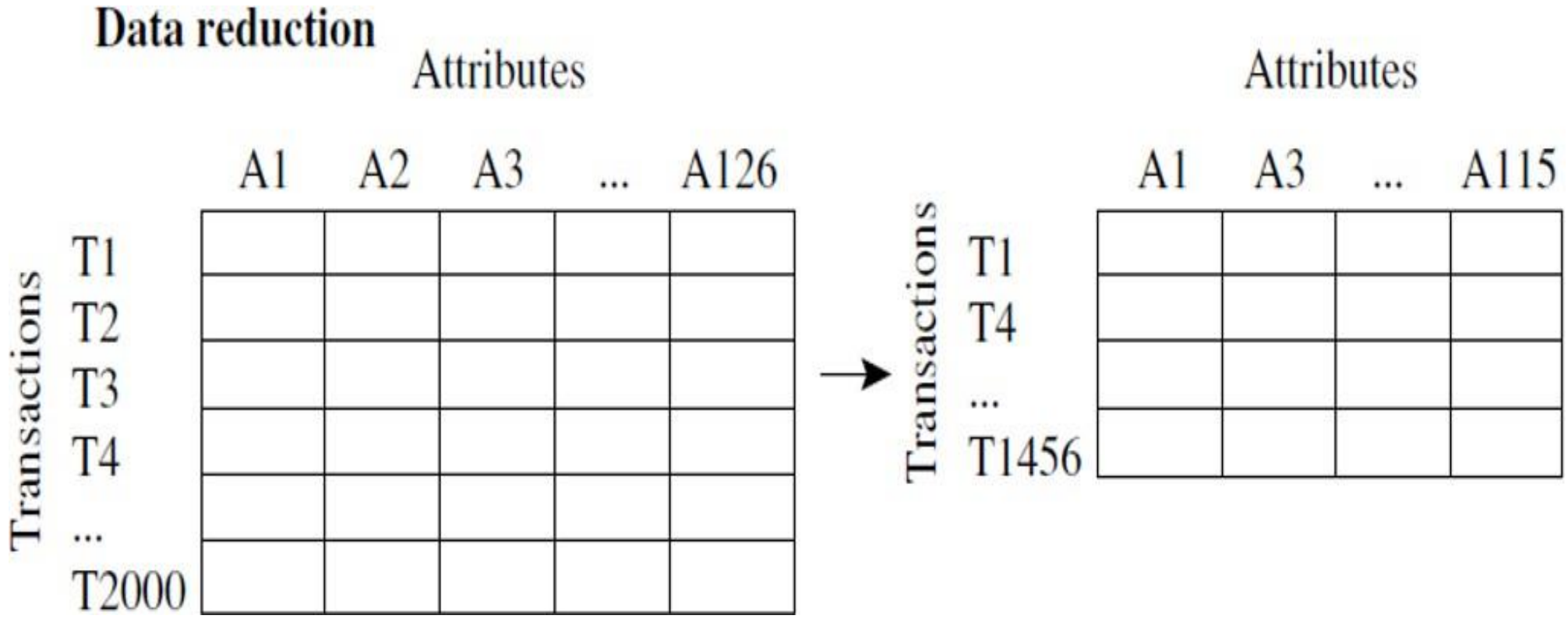
3. Data Reduction

- Imagine that you have selected data from the AllElectronics data warehouse for analysis.The data set will likely be huge! Complex data analysis and mining on huge amounts ofdata can take a long time, making such analysis impractical or infeasible.

- Data reduction techniques can be applied to obtain a reduced representation of thedata set that ismuch smaller in volume, yet closely maintains the integrity of the originaldata. That is, mining on the reduced data set should be more efficient yet produce thesame (or almost the same) analytical results.

- In simple words,Data reduction is a technique used in data mining to reduce the size of a dataset while still preserving the most important information. This can be beneficial in situations where the dataset is too large to be processed efficiently, or where the dataset contains a large amount of irrelevant or redundant information.

Example:


Data reduction
Attributes
A1
A2
A3
...
A126
Transactions
T1
T2
T3
T4
..
T2000
Attributes
A1
A3
...
A115
Transactions
T1
T4
..
T1456

Data reduction techniques:

- **Data Sampling:** This technique involves selecting a subset of the data to work with, rather than using the entire dataset. This can be useful for reducing the size of a dataset while still preserving the overall trends and patterns in the data.

- **Dimensionality Reduction:** This technique involves reducing the number of features in the dataset, either by removing features that are not relevant or by combining multiple features into a single feature.

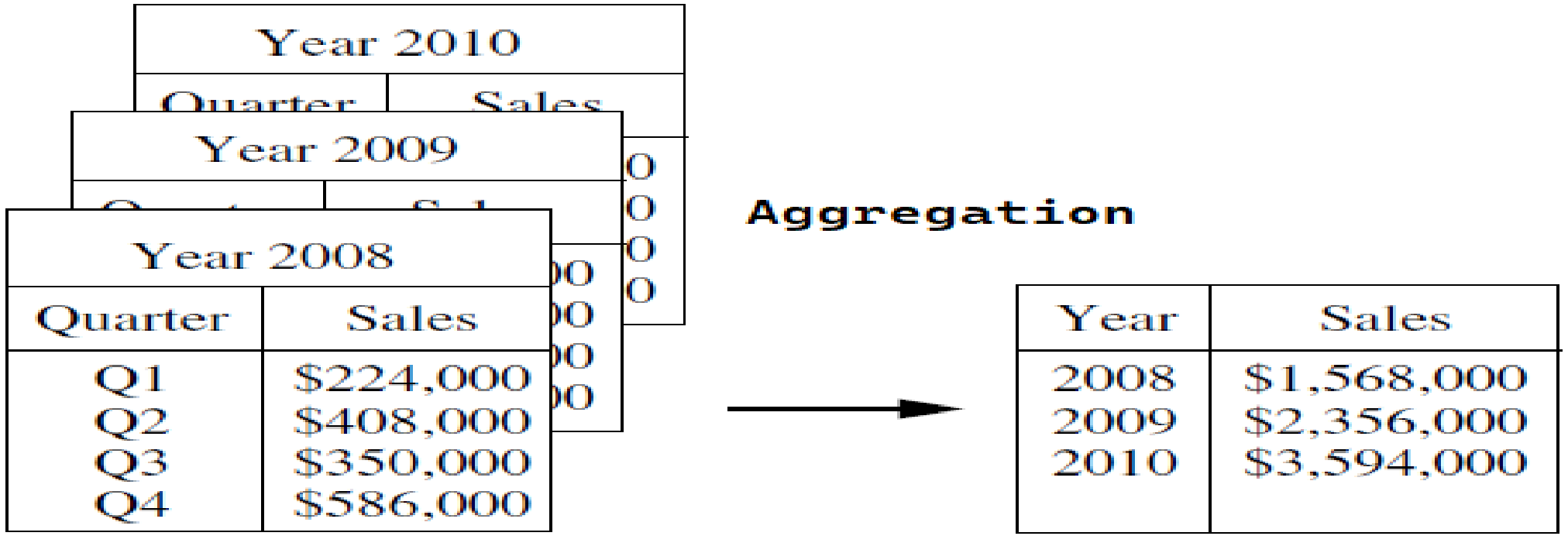
- **Data compression:**This is the process of altering, encoding, or transforming the structure of data in order to save space. By reducing duplication and encoding data in binary form, data compression creates a compact representation of information. And it involves the techniques such as lossy or lossless compression to reduce the size of a dataset.

4. Data Transformation

- Data transformation in data mining refers to the process of converting raw data into a format that is suitable for analysis and modelling.

- The goal of data transformation is to prepare the data for data mining so that it can be used to extract useful insights and knowledge.

Example:



Normalization

$-2, 32, 100, 59, 48 \longrightarrow -0.02, 0.32, 1.00, 0.59, 0.48$

Data transformation Steps:

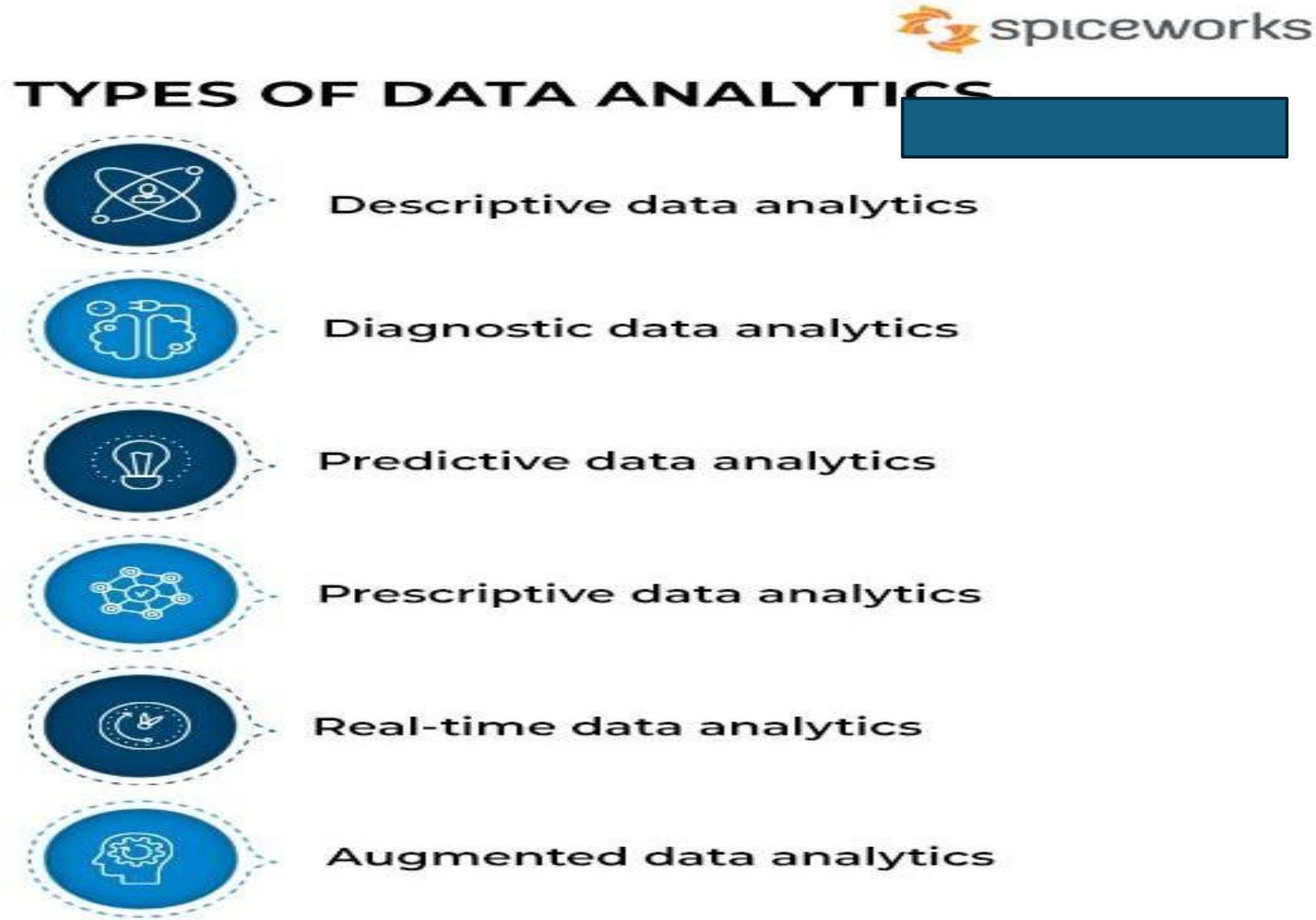
- **Smoothing:** It is a process that is used to remove noise from the dataset using techniques include binning,regression, and clustering.
- **Attribute construction (or feature construction):** In this, new attributes are constructedand added from the given set of attributes to help the mining process.
- **Aggregation:** In this, summary or aggregation operations are applied to the data. Forexample, the daily sales data may be aggregated to compute monthly and annualtotal amounts.
- **Data normalization:** This process involves converting all data variables into a small range.such as -1.0 to 1.0, or 0.0 to 1.0.
- **Generalization:** It converts low-level data attributes to high-level data attributes using concept hierarchy. For Example, Age initially in Numerical form (22,) is converted into categorical value (young, old).

Method Name	Irregularity	Output
Data Cleaning	Missing, Nosie, and Inconsistent data	Quality Data before Integration
Data Integration	Different data sources (data cubes, databases, or flat files)	Unified view
Data Reduction	Huge amounts of data can take a long time, making such analysis impractical or infeasible.	Reduce the size of a dataset and maintains the integrity.
Data Transformation	Raw data	Prepare the data for data mining

What Is Data Analytics?

- **Data analytics is defined as a set of processes, tools, and technologies that help manage qualitative and quantitative data to enable discovery, simplify organization, support governance, and generate insights for a business.**
- Data analytics is a discipline that **involves analyzing data sets to get information that would help solve problems in different sectors**. It employs several disciplines like computer programming, statistics, and mathematics, to give accurate data analysis.
- **The goal of data analytics can either be to describe, predict, or improve organizational performance.**
- They achieve this **using advanced data management techniques** like **data modeling, data mining, data transformation, etc., to describe, predict and solve present and future problems.**

Types of Data Analytics



1. **Descriptive** data analytics

- This type of data analytics **examines past data to explain what had happened**.
- It is the most **straightforward data analytics technique**.
- Depending on the scenario, some data analysts use descriptive analytics as a summary to support investigations and analysis from other types of analytics.
- This can be tagged as “best practice” because it explains the results from other analytics regarding historical data.
- **Companies use statistical analysis techniques to perform descriptive data analysis.**
- **This type of data analytics helps them compare past results, identify anomalies, distinguish strengths and weaknesses, etc.**
- **Companies use descriptive analysis to identify a problem.**

2. **Diagnostic** data analytics

- Diagnostic data analytics **examines past data to explain the cause of an anomaly.**
- This type of **analytics aims to answer "why did this happen?"** from a descriptive analytics result.
- The techniques used for **diagnostic data analytics are drill-down, data discovery, data mining, and correlations.**
- Data analysts use the data discovery technique to find sources that might help them deduce reasons from a result.
- Data mining involves getting information from a large set of raw data by automated processes.
- **Results from diagnostic analytics are obtained by finding correlations or patterns between different data.**
- Results from **this type of analytics help companies draft accurate solutions to problems instead of relying on guesswork.**

3. **Predictive** data analytics

- **Predictive data analytics involves using current or historical data to predict future actions.**
- **Individuals and companies conduct predictive analysis by combining historical data with machine learning, data mining techniques, and statistical modeling.**
- **These help them to quickly locate patterns and predict risks and opportunities in the future.**
- **This type of analytics works with algorithms and methodologies (like a linear or logistic regression model).**
- **Different algorithms exist for different scenarios, and a mismatch would result in erroneous results.**
- **Without predictive analysis, companies would be prone to making mistakes in the future that they might never recover from.**

4. **Prescriptive** data analytics

- Prescriptive data analytics **involves selecting the best solution for a problem from available options.**
- This type of data analytics **examines results from other analytics and gives guidance on how to reach a specific answer.**

- Prescriptive data analytics **is used in recommendation engines, loan approval engines, dynamic pricing models, machine repair scheduling, and similar tools to examine any decision options and to personalize the process.**
- These options could be in a yes/no or a list.
- These tools illustrate the consequence of every option, and also provide better options.

Companies can use prescriptive analysis to automate decision-making and hasten complex approvals.

5. **Real-time** data analytics

- Real-time data analytics involves using **data immediately when entered into the database.**
- Unlike other types of data analytics that use data from past events (historical data), **this type analyses new data from customers or external sources on the go**.

- Some technologies employed in real-time data analytics include **edge computing**, in-database analytics, in-memory analytics, data warehouse appliances, parallel programming, etc.
- This type of data analytics would be **most helpful if it is used in high availability and low-response time applications.**
- **Companies use it to identify trends and benchmarks faster than their competitors.**
- They can also track and analyze their competitor's operations instantaneously.

6. **Augmented** data analytics

- **Augmented analytics uses machine language (ML) and natural language processing (NLP) to analyze data.**
- **Incorporating machine learning into analytics helps automate the tedious task of code-based data exploration and make it available to business users.**
- **This reduces the chances of errors and enables the data analyst more time to do other actionable tasks.**
- Most analytics software integrates augmented data analytics tools to leverage the machine language and other outstanding features.
- **Companies include this type of data analytics in their analysis process to interact with data organically** – i.e., ***through English or any other natural language – and identify trends.***

IMPORTANCE OF DATA ANALYTICS



Reduce the cost of operation

Predict future trends

Monitor product performance

Strengthen security

THANK YOU